

Log abundance for compact Kähler spaces under logarithmic Iitaka subadditivity

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Abstract

Assume logarithmic Iitaka subadditivity for surjective morphisms with connected fibers between smooth projective complex varieties with compatible reduced simple normal crossing boundaries. We prove log abundance for normal irreducible compact Kähler spaces in every dimension: for a log canonical pair (X, Δ) with effective rational boundary and $K_X + \Delta$ $\mathbb{Q}$ -Cartier, analytic nefness of $K_X + \Delta$ implies semiampleness.

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1 Introduction

The abundance problem asks when the numerical positivity of a log canonical bundle produces a holomorphic map. More precisely, a nef adjoint should have a positive multiple generated by global sections. On a compact Kähler space, nefness is an analytic condition on a cohomology class, whereas generation is a statement about an actual holomorphic line bundle. The distinction is essential in the nonprojective setting.

We prove log abundance for compact Kähler spaces under logarithmic Iitaka subadditivity for smooth projective varieties with reduced boundary. The argument passes through a stronger birational statement: every pseudo-effective smooth log adjoint has a semiample positive part and an effective rational fixed divisor equal to its analytic divisorial negative part. This form

both supports the induction and retains the line-bundle information needed for generation on the original space.

1.1 The assumption and the theorem

A compact Kähler space is a compact complex analytic space with a Kähler form in the sense of local strictly plurisubharmonic potentials on local embeddings. For a rational line bundle L , analytic nefness means that $c_1(L)$ belongs to the closure of the Kähler cone. Equivalently, fix a Kähler form ω and an integer $r > 0$ for which rL is a line bundle. For every $\varepsilon > 0$, that line has a smooth Hermitian metric h_ε satisfying

$$\frac{\sqrt{-1}}{2\pi r} \Theta_{h_\varepsilon} \geq -\varepsilon \omega.$$

Smooth metrics and forms on a singular space are understood through local embeddings. We call L semiample if an actual positive integral multiple is a holomorphic line bundle generated by its global sections.

Assumption 1.1 (Logarithmic Iitaka subadditivity). Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Let D_X and D_Y be reduced effective simple normal crossing divisors, allowing zero divisors, such that

$$\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X).$$

For a very general smooth fiber F , put $D_F = D_X|_F$. Then

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y).$$

Here D_F is reduced SNC and $K_F + D_F \sim (K_X + D_X)|_F$. The Iitaka dimension is $-\infty$ when all positive integral systems are empty, with $(-\infty) + b = -\infty$ also for $b = -\infty$; the zero divisor on a point has Iitaka dimension zero.

Theorem 1.2 (Log abundance for compact Kähler spaces). *Assume Assumption 1.1. Let X be a normal irreducible compact Kähler complex analytic space, and let $\Delta \geq 0$ be a rational divisor such that (X, Δ) is log canonical and $K_X + \Delta$ is rational Cartier. If $K_X + \Delta$ is analytically nef, then it is semiample. Explicitly, there is an integer $m > 0$ such that $m(K_X + \Delta)$ is Cartier and*

$$H^0(X, \mathcal{O}_X(m(K_X + \Delta))) \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow \mathcal{O}_X(m(K_X + \Delta))$$

is surjective at every point of X .

The theorem includes every finite dimension and the zero-boundary case. Assumption 1.1 is used through the projective good-model theorem stated in Proposition 2.10. The cited proof of that projective theorem uses only its zero-boundary case, on a resolved projective Albanese fibration.

1.2 Context and relation to earlier work

In the minimal model program, abundance complements the construction of minimal models: the nef adjoint on a minimal model should determine a canonical fibration. For minimal projective threefolds, Miyaoka proved the case of numerical dimension one and Kawamata completed canonical

abundance [51, 43]. Keel, Matsuki, and McKernan established log abundance for threefolds [45, 46]. In arbitrary dimension, Birkar, Cascini, Hacon, and McKernan established log terminal models for projective klt pairs with big boundary and pseudo-effective adjoint, and finite generation for big adjoints [6].

The compact Kähler history already points to the importance of simple spaces. Peternell’s threefold theorem isolated the possible case of a simple space not bimeromorphic to a finite quotient of a torus [59]; Demailly and Peternell subsequently proved canonical nonvanishing for nef terminal Kähler threefolds, including that case [23]. Höring and Peternell constructed minimal models for normal $\mathbb{Q}$ -factorial terminal compact Kähler threefolds with pseudo-effective canonical class [42]. Campana, Höring, and Peternell treated canonical abundance [10, 11], and Das and Ou established log abundance for compact Kähler log canonical threefolds [20, Corollary 1.3].

With $D_X = D_Y = 0$, Assumption 1.1 is Iitaka’s classical $C_{n,m}$ inequality $\kappa(X) \geq \kappa(F) + \kappa(Y)$ for algebraic fiber spaces [14, Equation (1.0.1)]. Cao and Păun proved the analogous logarithmic inequality over an abelian base when the pair on the total space has an effective rational boundary and is klt [14, Theorem 1.1]. Fujino derives logarithmic subadditivity from the conjectured equality $\kappa_\sigma(X, K_X + D) = \kappa(X, K_X + D)$ for smooth projective X with reduced SNC D [29, 30]. Hashizume proves the reduced-SNC subadditivity statement when the log canonical divisor of the very general fiber is abundant [41, Theorem 1.2]. Here the implication is used in the other direction: the projective companion [55] derives good models for pseudo-effective projective lc adjoints from Assumption 1.1, and those good models anchor the Kähler induction.

Hacon and Xie’s cone theorem, adjunction, relative positivity, and modified-big preparation [39] provide analytic inputs for our program constructions. Section 3 constructs effective generalized base data on a higher carrier, with an exact adjoint pullback identity and a multiplier-ideal inclusion. Fibrewise L^2 positivity and regularization give this presentation; its Stein-factor consequence supplies the non-klt induction. The section then proves the restricted semiample, proper relative good-model, and finite-model statements needed here, following the dimension-induction strategy of [39]. The nef data remain globally nef on a fixed carrier, and the total generalized boundary is globally modified big. Actual rational lines make the selected ray contractions projective; ordinary projective analytic results [31] then construct their flips and the relative models. This produces a Kähler pullback description of the relevant Bott–Chern class. Generation of the prescribed holomorphic line requires the additional divisorial decomposition and boundary arguments. The fibration comparison and period positivity come from [57]; the boundary and lifting constructions adapt inputs in [54, 53]. The precise imported results are stated at their points of use.

1.3 The proof and its main constructions

For a pseudo-effective $(1, 1)$ -class α , $N(\alpha)$ records the least generic vanishing forced along prime divisors. Given a smooth compact Kähler manifold X and a rational SNC boundary B with $J = K_X + B$ pseudo-effective, the induction constructs a modification $\mu : Y \rightarrow X$ with Y smooth and

$$\mu^* J \sim_{\mathbb{Q}} P + R, \quad P \text{ semiample}, \quad R = N(c_1(\mu^* J)),$$

as an identity of rational holomorphic line bundles, with R an effective rational divisor. Section 2 uses transfer rules, algebraic reduction, and generically finite covers to reduce the induction to projective manifolds, nontrivial fibrations, and simple spaces of algebraic dimension zero. Here simple means that no positive-dimensional proper compact subvariety passes through a very general point.

Section 3 supplies two program results. For the fibration case it contracts a known negative part while preserving the prescribed nef line. For the simple case it proves special termination for a chosen scaling and reaches a nef model when the pseudo-effective adjoint is rationally equivalent to a divisor supported on the reduced boundary, allowing negative coefficients. The termination argument uses lower-dimensional induction to construct one model on which an interval of perturbed adjoints is nef. On a common resolution their negative multiplicities are affine, whereas every nontrivial wall of the restricted program changes a slope. Hence only finitely many such walls occur.

Section 4 turns generation on separate strata into generation on the entire reduced dlt boundary, adapting Fujino’s method of admissible sections [28, Section 4]. When a comparison is needed, a Mori contraction realizes Kollár’s residue comparison at the two coefficient-one points of a general $\mathbb{P}^1$ fiber [47, Definition 13 and Proposition 14]. On a common resolution, the pulled-back restrictions of the ambient Kähler class differ by a real linear combination of Chern classes of line bundles. On a stratum whose boundary dominates the image of the map defined by its semiample adjoint, restriction determines sections. For the remaining strata, we prove finite image for self-comparisons on sufficiently divisible pluricanonical systems using an invariant integral, period and lattice arguments, and a uniform cohomological bound for cyclic covers. Products over these finite images give compatible generating sections, used in both closing arguments.

Section 5 first reduces to fibrations whose very general fiber has log Kodaira dimension zero. On a prepared fibration $g : X \rightarrow W$, a relative generating form gives

$$K_X + B \sim_{\mathbb{Q}} g^*H + A_*, \quad H = K_W + T + M,$$

where T is a rational SNC boundary and M is pulled back from a nef rational line on a projective quotient of W . Fiber induction allows us to subtract the part of A_* dominating W from a positive current for $K_X + B$. The resulting metric descends along connected smooth fibers; a zero among the nonnegative coefficients at primes dominating each base prime allows extension, proving H pseudo-effective. If $a(W) = 0$, the projective quotient is a point, so $M \sim_{\mathbb{Q}} 0$ and lower-dimensional induction makes H rationally equivalent to its negative divisor. For projective W , a reduction via a chosen generalized program derived from Assumption 1.1 either lowers the positive base dimension, handled by secondary induction, or reaches a nef line that is big or torsion. At a stopping case, an intersection argument identifies the full negative divisor upstairs. A torsion positive part completes the decomposition directly. In the big nef case, Section 3 contracts that divisor while preserving the nef line; boundary generation, extension, and a new log canonical place at a hypothetical base locus prove semiampleness.

Section 6 proves, independently of Assumption 1.1, that a positive canonical power has a nonzero meromorphic section on every smooth connected simple compact Kähler manifold of algebraic dimension zero. Ou’s uniruledness and foliation results supply the cotangent slope control [58]. A point-threshold bound for big classes of volume one is contradicted using an auxiliary projective bundle over $X \times X$. The diagonal in its square produces a subsheaf occupying a fixed positive fraction of a symmetric cotangent power of that bundle. A second construction over the diagonal of $X \times X$ forces determinant vanishing which, after restriction to a blowup of X , violates the point bound.

Finally, Section 7 uses the meromorphic section to obtain, after resolving and enlarging to a reduced SNC boundary, a pseudo-effective adjoint with a signed representative supported there. Section 3 gives a nef dlt model, and Section 4 gives generation on its reduced boundary. Pseudo-effectivity of the canonical pullback and an intersection argument isolate positive-dimensional boundary fibers away from components whose signed coefficients are nonpositive. A local root

construction adapted from [55, Proposition 3.3] separates the positive and negative divisor supports. An SNC Hodge-module calculation extends the lifting method of [53, Sections 10–12] to residual poles and kills every finite-order obstruction to lifting such a fiber. Douady space and Artin approximation give compact deformations leaving the boundary; compactness of the cycle-space components and relative compactness of bounded-volume cycles allow a Baire argument to produce a covering family. Simplicity forces the actual nef line to be torsion. The program comparison and Lemma 2.6 then permit subtraction of the added boundary, yielding the inductive decomposition.

2 Birational decompositions and the geometric reduction

The proof will produce a semiample line bundle on a smooth model, together with its entire fixed divisorial part. This section makes that statement precise and reduces the induction to two cases: spaces admitting a nontrivial fibration, and simple spaces of algebraic dimension zero.

2.1 The inductive decomposition and its negative part

We use additive notation for rational holomorphic line bundles. Thus $L \sim_{\mathbb{Q}} L'$ means that mL and mL' are holomorphically isomorphic for some positive integer m . If a rational divisor occurs in such an identity, it denotes its associated rational line bundle. In contrast, $c_1(L)$ and $\{D\} = c_1(\mathcal{O}_X(D))$ denote real Bott–Chern classes. Canonical comparisons are made with the usual local meromorphic canonical identifications. In particular, an identity of lines below contains more information than equality of their Chern classes.

Let X be a smooth compact Kähler manifold and let α be a pseudo-effective real $(1, 1)$ -class. Fix a Kähler form ω . For a prime divisor D , its minimal multiplicity is

$$\nu_D(\alpha) = \liminf_{\varepsilon \downarrow 0} \{ \nu_D(T) : T \in \alpha, T \geq -\varepsilon \omega \}.$$

Here T ranges over closed currents, and $\nu_D(T)$ is its generic Lelong number; one may equivalently use currents with analytic singularities. The limit is independent of ω . Boucksom’s divisorial decomposition is

$$N(\alpha) = \sum_D \nu_D(\alpha) D, \quad Z(\alpha) = \alpha - \{N(\alpha)\}.$$

The sum is a finite effective real divisor, and $Z(\alpha)$ is modified nef: its minimal multiplicity at every prime is zero. We write $N(L)$ and $\nu_D(L)$ for $N(c_1(L))$ and $\nu_D(c_1(L))$. These are the analytic notions, with small Kähler perturbations, throughout the paper. We use the foundational results in [7, Sections 2–3 and 5].

The induction asks for a decomposition that retains both the actual line bundle and this analytic negative divisor.

Definition 2.1. For $n \geq 0$, let $\mathcal{G}_n$ be the following assertion. If X is a connected smooth compact Kähler manifold of dimension n , B is a rational simple normal crossing boundary with coefficients in $[0, 1]$, and $J = K_X + B$ is pseudo-effective, then there is a smooth compact Kähler modification $\mu : Y \rightarrow X$ and an actual rational line identity

$$\mu^* J \sim_{\mathbb{Q}} P + R, \quad P \text{ semiample}, \quad R = N(\mu^* J) \geq 0, \quad (2.1)$$

where R is a rational divisor.

The next lemmas make this decomposition stable under further resolutions and allow exceptional resolution errors to be removed. They also provide the fixed-section statement needed to descend the case in which the positive part is torsion.

Lemma 2.2 (Negative-part calculus). *Let α be pseudo-effective on a smooth compact Kähler manifold.*

(i) *Every positive current in α contains the divisorial current $[N(\alpha)]$. If $0 \leq F \leq N(\alpha)$, then*

$$\alpha - \{F\} \text{ is pseudo-effective,} \quad N(\alpha - \{F\}) = N(\alpha) - F.$$

(ii) *Minimal multiplicities are homogeneous and subadditive. In particular, if β is nef, then $N(\alpha + \beta) \leq N(\alpha)$.*

(iii) *If $\mu : Y \rightarrow X$ is a smooth modification and D' is the strict transform of a prime D , then*

$$\nu_{D'}(\mu^* \alpha) = \nu_D(\alpha).$$

(iv) *If γ is modified nef and $j : \tilde{D} \rightarrow X$ is a resolution of a prime divisor, then $j^* \gamma$ is pseudo-effective.*

Proof. The first assertion about currents follows from the definition of minimal multiplicity and Siu decomposition. Homogeneity and subadditivity follow by scaling and adding testing currents. The subtraction identity is the corresponding property in the big cone, followed by a small Kähler perturbation. More explicitly, for $\alpha_\varepsilon = \alpha + \varepsilon\{\omega\}$, subtract $F_\varepsilon = \min\{F, N(\alpha_\varepsilon)\}$ coefficientwise. In the big cone all positive currents contain this divisor, so subtracting it translates each minimal multiplicity by its coefficient. As $\varepsilon \downarrow 0$, $F_\varepsilon \rightarrow F$. The class of $F - F_\varepsilon$ can be absorbed in a Kähler perturbation tending to zero (there are only finitely many components). This gives the upper bound for the asserted identity; subadditivity applied after adding F gives the reverse bound. Weak compactness of positive currents gives pseudo-effectivity of the limit. This is also the subtraction property of the divisorial decomposition in [7, Section 3].

For the strict-transform formula, pull almost-positive testing currents to Y . Near the generic point of D' , the map is an isomorphism, so the generic order is unchanged. This proves one inequality. Conversely, push a testing current on Y to X . To control its negative error, write the error as a small multiple of the fixed positive current $\mu_* \omega_Y$, and add a fixed smooth Kähler representative of $C\omega_X - \{\mu_* \omega_Y\}$ for C sufficiently large. The result is a positive test in a Kähler perturbation tending to zero. The pushforward of ω_Y has no divisorial order at the generic point of D , where μ is an isomorphism. Thus this test has the same generic order as the original one at D' . Regularization, if needed, returns to tests with analytic singularities. This proves the other inequality and the formula. It is the analytic form of the strict-transform comparison in [7, Lemma 5.8].

Finally, choose analytic-singularity tests for a modified-nef class whose generic order at the chosen prime tends to zero. Subtract that generic divisorial order before restricting to a resolution of the prime. The remaining singular set does not contain its generic point, so restriction gives an almost-positive current there. The subtracted multiple and the Kähler error tend to zero. Taking the limit proves the last assertion; see also [7, Propositions 2.4 and 3.8]. $\square$

For a rational line J on a normal compact Kähler space X whose pullback to a smooth resolution is pseudo-effective, we also use minimal multiplicities for divisorial places. If Q appears on a smooth resolution $p : Y \rightarrow X$, set

$$\nu_Q(J) := \nu_Q(p^* J).$$

The strict-transform formula on a common smooth refinement makes this independent of the chosen resolution. When a real class is already on a fixed smooth model, ν_Q continues to denote its coefficient in the negative part on that model.

Lemma 2.3 (Exceptional translation). *Let $p : Y \rightarrow X$ be a proper bimeromorphic morphism from a smooth compact Kähler manifold to a normal compact Kähler space. Let M be a real $(1,1)$ -class on X represented by smooth local potentials, and let $E \geq 0$ be a real p -exceptional divisor. If $\alpha = p^*M + \{E\}$ is pseudo-effective, then p^*M is pseudo-effective and*

$$N(\alpha) = N(p^*M) + E.$$

Proof. It suffices by Lemma 2.2 to prove $E \leq N(\alpha)$. Write

$$E - N(\alpha) = U - V, \quad U, V \geq 0,$$

with no common component, and suppose $U \neq 0$. Its components are p -exceptional. Put $n = \dim Y$ and $\ell = \max\{\dim p(U_i) : U_i \text{ a component of } U\}$. Normality gives $\ell \leq n - 2$. Let η be the pullback of a Kähler form on X , let ω be Kähler on Y , and set

$$q(\gamma, \delta) = \int_Y \gamma \delta \eta^\ell \omega^{n-\ell-2}.$$

The restriction property in Lemma 2.2 gives $q(Z(\alpha), U) \geq 0$. The term $q(p^*M, U)$ is zero: on each component of U , it contains $\ell + 1$ factors pulled back from an image of dimension at most ℓ . Distinct effective divisors have nonnegative intersection against the semipositive and Kähler factors in q . Hence

$$0 \leq q(Z(\alpha), U) = q(U, U) - q(V, U) \leq q(U, U).$$

On the other hand, $q(U, \eta) = 0$ by the same dimension count, whereas $q(\eta, \eta) > 0$. The mixed Hodge index theorem, applied first with Kähler factors and then by a semipositive limit, says that q is negative semidefinite on $\eta^\perp$. It follows that $q(U, U) = 0$ and that U is in the radical of q : the radical assertion follows from negative semidefiniteness on $\eta^\perp$, and then from $q(U, \eta) = 0$. But

$$q(U, \omega) = \sum_i u_i \int_{U_i} \eta^\ell \omega^{n-\ell-1} > 0.$$

Indeed, a component with image dimension ℓ has strictly positive integrand on a dense open. This is a contradiction. The mixed Hodge index statement used here is the mixed Hodge–Riemann relation [24, Theorems A and C]; the limit preserves the assertion that there is at most one positive direction. $\square$

Lemma 2.4 (Pulling up a known decomposition). *Let α be a pseudo-effective real $(1,1)$ -class on a smooth compact Kähler manifold, and suppose*

$$\alpha = \beta + \{R\}, \quad \beta \text{ nef}, \quad R = N(\alpha) \geq 0.$$

For every smooth modification $\mu : Y \rightarrow X$,

$$N(\mu^*\alpha) = \mu^*R.$$

In particular, this applies to the Chern classes of an actual rational line decomposition with a nef rational positive part.

Proof. Subadditivity and nefness give $N(\mu^*\alpha) \leq \mu^*R$. By the strict-transform formula, the difference $U = \mu^*R - N(\mu^*\alpha)$ is effective and exceptional. Moreover $Z(\mu^*\alpha) = \mu^*\beta + \{U\}$ is modified nef. Apply Lemma 2.3 to this class. It gives $U \leq N(Z(\mu^*\alpha)) = 0$, proving the assertion. $\square$

Lemma 2.5 (Fixed sections). *Let L be a pseudo-effective rational line on a smooth compact Kähler manifold. Every section s of an integral multiple mL satisfies*

$$\text{ord}_D(s) \geq m\nu_D(L)$$

at every prime D . If $L \sim_{\mathbb{Q}} P + R$, with P semiample and $R = N(L)$ rational, then in every sufficiently divisible degree multiplication by the canonical section of mR identifies $H^0(X, mP)$ with $H^0(X, mL)$.

Proof. The divisor current $[\text{div}(s)]/m$ is a positive current in $c_1(L)$, so Lemma 2.2 gives the order bound. In the stated decomposition, every section therefore divides holomorphically by the canonical section of mR . Conversely, multiplication gives a section of mL . The actual rational line identity makes these operations inverse after clearing denominators. $\square$

Lemma 2.6 (Uniqueness when the positive part is torsion). *Suppose $L \sim_{\mathbb{Q}} R$, where $R = N(L)$ is an effective rational divisor on a smooth compact Kähler manifold. Then the only positive current in $c_1(L)$ is $[R]$. If $F \geq 0$ is a rational divisor and $L - F$ is pseudo-effective, then*

$$F \leq R, \quad L - F \sim_{\mathbb{Q}} R - F = N(L - F).$$

The same conclusions hold if $L \sim_{\mathbb{Q}} P + R$ with P torsion.

Proof. Write X for the manifold. The assertion is immediate if $\dim X = 0$, so assume $\dim X > 0$. Every positive current T in $c_1(L)$ contains $[R]$. The residual $T - [R]$ is positive with zero cohomology class, so its mass against a Kähler form to the power $\dim X - 1$ is zero. Thus it vanishes. For the second assertion, add $[F]$ to any positive current in $c_1(L - F)$. Uniqueness gives $F \leq R$, and subtraction in Lemma 2.2 gives the asserted negative part. A torsion line has zero Chern class and becomes trivial after taking a positive multiple, so the last statement is the same argument. $\square$

2.2 Birational transfer of the decomposition

All modifications resolving spaces, maps, or ideals can be chosen projective. A projective modification of a compact Kähler space is Kähler; graphs between compact Kähler models can likewise be resolved by smooth compact Kähler manifolds. We use these standard analytic resolution and flattening results without further mention.

For log discrepancies our convention is

$$a(E; X, \Delta) = 1 + \text{ord}_E(K_Y - p^*(K_X + \Delta))$$

on a smooth model $p : Y \rightarrow X$. In particular, on a log resolution of an lc pair, giving each new exceptional divisor coefficient one produces an effective exceptional error. The next proposition explains why this convention is harmless.

Proposition 2.7 (Resolution and descent). *Let (X, Δ) be a normal compact Kähler lc pair with rational boundary and rational Cartier adjoint J . Let $p : Y \rightarrow X$ be a log resolution, and put*

$$B_Y = p_*^{-1}\Delta + \sum_{E \text{ exceptional}} E.$$

If p^*J is pseudo-effective and the conclusion of Definition 2.1 holds for (Y, B_Y) , then there is a smooth compact Kähler modification $r : V \rightarrow X$ with an actual rational line identity

$$r^*J \sim_{\mathbb{Q}} P + R, \quad P \text{ semiample}, \quad R = N(r^*J) \geq 0,$$

where R is a rational divisor. If J is analytically nef, it is semiample on X .

Proof. With compatible canonical choices there is an actual rational divisor identity

$$K_Y + B_Y = p^*J + E_p, \quad E_p = \sum_{E \text{ exceptional}} a(E; X, \Delta)E \geq 0.$$

Suppose $q : V \rightarrow Y$ realizes the decomposition for the left side. Lemma 2.3, applied over X , gives

$$N(q^*(K_Y + B_Y)) = N(q^*p^*J) + q^*E_p.$$

Cancelling q^*E_p in the actual rational line identities proves the first assertion. This argument also shows that further resolutions using the reduced-exceptional convention do not change the assertion $\mathcal{G}_n$.

If J is nef, its pullback is nef, so its negative part is zero. The semiample line in the decomposition is therefore q^*p^*J itself, up to rational line isomorphism. Choose an actual Cartier multiple generated on V . Normality gives $(pq)_*\mathcal{O}_V = \mathcal{O}_X$, and projection formula identifies its sections with the sections downstairs. A base point downstairs would make every pullback section vanish on its nonempty fiber. There is no such point, so that multiple of J is globally generated. $\square$

The second transfer concerns a torsion positive part. Its proof uses the following local construction for finite maps, which will also be used for meromorphic sections and for covers of spaces of algebraic dimension zero.

Lemma 2.8 (Finite analytic norms and characteristic polynomials). *Let $\nu : Z \rightarrow X$ be a finite surjective morphism of normal irreducible complex spaces, of degree $d > 0$, and let L be a holomorphic line bundle on X . A nonzero meromorphic section s of ν^*L has a nonzero meromorphic norm $\text{Nm}_\nu(s)$ in $L^{\otimes d}$, with*

$$\text{div}(\text{Nm}_\nu(s)) = \nu_* \text{div}(s).$$

If s is holomorphic, its norm is holomorphic. If, in addition, s is nonzero at every point of $\nu^{-1}(x)$, then its norm is nonzero at x .

For a global meromorphic function f on Z , there is a monic polynomial $\chi_f(T)$ of degree d with global meromorphic coefficient functions on X such that $\chi_f(f) = 0$ after pulling the coefficients to Z .

Proof. At $x \in X$, choose a holomorphic frame of L and put

$$R = \mathcal{O}_{X,x}, \quad K = \text{Frac}(R), \quad A = (\nu_*\mathcal{O}_Z)_x.$$

The generic algebra $A \otimes_R K$ is a product of finite field extensions of K , of total dimension d . The local meromorphic coefficient a of s belongs to this algebra. Since Z is irreducible and s is not identically zero, its restriction is nonzero on every generic local component. Thus a is nonzero in every field factor, and the determinant of multiplication by a is nonzero. Under a change of frame by a unit g , this determinant changes by g^{-d} . The determinants therefore glue to the asserted

meromorphic section. The valuation formula for a norm over a discrete valuation ring gives the displayed divisor identity at every prime of the normal space X .

If s is holomorphic, then $a \in A$ is integral over R . Its norm is integral over R and belongs to K , hence belongs to R by normality. This does not require A to be locally free. If s is nonzero at every point over x , then a is a unit in the finite semilocal algebra A ; applying the same argument to a^{-1} shows that its norm is a unit in R .

For a meromorphic function f , use instead the determinant of T minus multiplication by its local coefficient. These monic polynomials agree on overlaps, and Cayley–Hamilton shows that they annihilate f . All these constructions use fractions of local analytic germs, so they remain available when the global meromorphic functions on X are constant. $\square$

We now descend a decomposition whose positive part is torsion, the form needed for a covering family on a space of algebraic dimension zero.

Proposition 2.9 (Descent of a purely negative decomposition). *Let $e : Y \rightarrow X$ be a proper generically finite surjective morphism between connected smooth compact Kähler manifolds. Let L be a pseudo-effective rational line on X , and let $F \geq 0$ be a rational divisor on Y . Suppose that on a smooth model over Y , the line $e^*L + F$ is rationally linearly equivalent to its rational negative part. Then there is an effective rational divisor D on X such that*

$$L \sim_{\mathbb{Q}} D = N(L).$$

Proof. Replace Y by the smooth model in the hypothesis and pull F back. Write $e^*L + F \sim_{\mathbb{Q}} R = N(e^*L + F)$. Pull back a positive current in $c_1(L)$ and add $[F]$. Lemma 2.6 gives $F \leq R$. Put $D_Y = R - F$. Subtraction in Lemma 2.2 gives

$$e^*L \sim_{\mathbb{Q}} D_Y = N(e^*L) \geq 0.$$

In particular $[D_Y]$ is the unique positive current in $c_1(e^*L)$.

Choose m so that mL and mD_Y are integral and the latter is the divisor of a holomorphic section s of $e^*(mL)$. Factor e through its normal Stein space:

$$Y \xrightarrow{h} Z \xrightarrow{\nu} X, \quad \nu \text{ finite of degree } d = \deg e.$$

The connected birational map h descends s to a holomorphic section s_Z of $\nu^*(mL)$, by normality and projection formula. Lemma 2.8 gives a holomorphic section of mdL whose divisor is

$$\operatorname{div}(\operatorname{Nm}_{\nu}(s_Z)) = \nu_* \operatorname{div}(s_Z) = me_* D_Y.$$

Thus $D = d^{-1}e_* D_Y$ is effective and satisfies $L \sim_{\mathbb{Q}} D$.

Both e^*D and D_Y are positive divisor currents in $c_1(e^*L)$, so uniqueness gives $e^*D = D_Y$. We already know $N(L) \leq D$. If this inequality were strict at a prime $A \subset X$, choose analytic-singularity tests for L whose generic orders at A tend to $\nu_A(L)$, and pull them to Y . For any prime A' dominating A , the generic orders are multiplied by $\operatorname{mult}_{A'} e^*A$. The pulled-back Kähler errors can be enlarged to Kähler errors upstairs tending to zero. Consequently

$$\nu_{A'}(e^*L) \leq \operatorname{mult}_{A'}(e^*A) \nu_A(L) < \operatorname{mult}_{A'}(e^*A) \operatorname{coeff}_A(D),$$

contrary to $N(e^*L) = D_Y = e^*D$. This proves $D = N(L)$. $\square$

We will also use Proposition 2.9 when e is a modification and $F = 0$. Once the positive part of a decomposition is known to be torsion, it gives $L \sim_{\mathbb{Q}} N(L)$ on the original smooth space; Lemma 2.4 then identifies the negative divisor upstairs with the pullback of $N(L)$.

2.3 The projective anchor and the two geometric cases

Proposition 2.10 (Projective good models). *Under Assumption 1.1, every projective lc pair over $\mathbb{C}$ with rational boundary and pseudo-effective rational Cartier adjoint has a good log minimal model. In particular, $\mathcal{G}_n$ holds for projective manifolds in every dimension, and every nef rational lc adjoint on a projective variety is semiample as an actual rational line.*

Proof. The good-model assertion is the rational-boundary consequence of the full argument in [55, Proposition 2.5, Theorem 9.6, and Section 10]. That argument establishes the additional inductive and nonvanishing premises of its Proposition 2.5 before applying it, and proves a stronger real-boundary induction. Its only use of Assumption 1.1 is in its Lemma 6.1, on a resolved projective Albanese fibration with both boundaries zero. Thus its premise is precisely available here.

For clarity, its good-model conclusion gives on a common smooth projective resolution

$$u^*(K_X + B) = v^*(K_{X_m} + B_m) + F, \quad F \geq 0 \text{ and } v\text{-exceptional},$$

as an actual rational divisor identity; the line on X_m is semiample. This is the comparison in [55, Lemma 2.2]. Algebraic nefness of a rational line on a projective variety implies analytic nefness, by adding arbitrarily small ample rational classes. Lemma 2.3 therefore identifies F with the analytic negative part of the left side. On a smooth projective variety, analytic pseudo-effectivity of a divisor class agrees with algebraic pseudo-effectivity [8, Proposition 1.2]; hence the good-model assertion applies to every projective instance of $\mathcal{G}_n$. This proves the stated form of $\mathcal{G}_n$. If the original adjoint is nef, the same comparison and normal descent give its semiamplicity. $\square$

The remainder of the paper proves the following two propositions. They are stated here so that the global induction can be completed before its technical components are developed. A compact space is *simple* if no positive-dimensional proper compact analytic subvariety passes through a very general point. We write $a(X)$ for its algebraic dimension.

Proposition 2.11 (The fibration case). *Assume Assumption 1.1. Fix $n > 0$ and assume $\mathcal{G}_j$ for $j < n$. Let X be a connected smooth compact Kähler manifold of dimension n , let B be a rational SNC boundary with coefficients in $[0, 1]$, and suppose $K_X + B$ is pseudo-effective. If X admits a dominant meromorphic map to an irreducible compact space Y in Fujiki class $\mathcal{C}$, with $0 < \dim Y < n$, then the conclusion of $\mathcal{G}_n$ holds for (X, B) .*

Proposition 2.12 (The simple case). *Fix $n > 0$ and assume $\mathcal{G}_j$ for $j < n$. Let X be a connected smooth simple compact Kähler manifold of dimension n with $a(X) = 0$. For a rational SNC boundary B with coefficients in $[0, 1]$ and $J = K_X + B$ pseudo-effective, there is a smooth compact Kähler modification $\mu : Y \rightarrow X$ such that*

$$\mu^* J \sim_{\mathbb{Q}} N(\mu^* J),$$

and this negative part is a rational divisor. In particular, the conclusion of $\mathcal{G}_n$ holds with torsion positive part.

Proposition 2.11 is proved in Section 5; it reduces the fibration to relative log Kodaira dimension zero and then descends the adjoint to its base. Proposition 2.12 is proved in Section 7, using the meromorphic nonvanishing theorem of Section 6. Both propositions use $\mathcal{G}$ only in dimensions below n .

Theorem 2.13 (Good divisorial decomposition). *Under Assumption 1.1, the assertion $\mathcal{G}_n$ holds for every finite n .*

Proof. We use Propositions 2.11 and 2.12, whose proofs occupy the rest of the paper. The assertion is immediate in dimension zero. Assume it in smaller dimensions and let (X, B) be as in Definition 2.1 in dimension $n > 0$.

If $a(X) = n$, the Kähler Moishezon theorem makes X projective, and Proposition 2.10 applies. If $0 < a(X) < n$, the algebraic reduction of X is a dominant meromorphic map to a projective model of dimension $a(X)$ [12, Theorem and Definition 3.1]. Proposition 2.11 applies.

It remains to consider $a(X) = 0$. If X is simple, use Proposition 2.12. Otherwise Campana’s maximal covering-family theorem supplies a generically finite evaluation map from an incidence space which itself has a nontrivial fibration; see [13, Lemma 17 and Corollary 18]. After resolving the incidence, the graph, and the base, we obtain a smooth compact Kähler source X' and maps

$$e : X' \longrightarrow X, \quad f : X' \longrightarrow Y', \quad 0 < \dim Y' < n,$$

where e is proper generically finite and Y' is in class $\mathcal{C}$ [27, (1.1.1), (1.5)(3)–(4), (2.2)(1),(3), and (2.3)]. The algebraic dimension of X' is zero. Indeed, a meromorphic function on X' descends through the birational part of the normal Stein factorization of e . Lemma 2.8 gives its characteristic polynomial over the finite part, with meromorphic coefficients on X . These coefficients are constant because $a(X) = 0$, so the function is constant on the irreducible space X' . Bimeromorphic modifications do not change this conclusion.

Choose a reduced SNC divisor B' containing the inverse image of $\text{Supp } B$, after a further resolution. Pullback of logarithmic differentials gives the actual rational identity

$$K_{X'} + B' = e^*(K_X + B) + F, \quad F \geq 0.$$

For example, this follows locally by pulling back logarithmic top forms for the reduced support of B ; decreasing its coefficients to those of B' only increases the error. Thus the adjoint upstairs is pseudo-effective. Proposition 2.11 applies to (X', B') . Its semiample positive part is torsion, since a nonconstant semiample map would give a nonconstant meromorphic function on X' . Proposition 2.9 now gives the required decomposition for $K_X + B$. This exhausts the cases and proves the induction. $\square$

Proof of Theorem 1.2. Apply Theorem 2.13 on a log resolution of the given lc pair, with the reduced-exceptional boundary. The adjoint is pseudo-effective because the original one is nef. Proposition 2.7 then gives generation of an actual Cartier multiple at every point of the original normal space. $\square$

3 Programs with a fixed nef part and special termination

Fix a dimension n , and assume $\mathcal{G}_d$ for every $d < n$. This section supplies two program constructions used in the induction. An ordinary dlt adjoint that is already the sum of a nef rational line and its divisorial negative part has a nef model, reached by contracting that negative part while preserving the nef line. A suitably chosen scaling of a pseudo-effective ordinary dlt adjoint starting on a smooth space has special termination. Its proof uses $\mathcal{G}_d$ only on proper log canonical strata and compares two descriptions of negative multiplicities as the scaling parameter tends to zero. Along the way, cohomology transport supplies both the generic scaling directions and the exceptional splitting carried by the terminal models used in Section 4.

The restricted generalized model input is stated first for use in these ordinary constructions. Its proof occupies the last three subsections: polarization and descent, an already-projective relative construction, and the dimension induction. That proof is independent of the assumptions $\mathcal{G}_d$ used for ordinary special termination.

3.1 Program inputs and descent of actual lines

A normal compact Kähler space is *globally $\mathbb{Q}$ -factorial* if every global Weil divisor has a positive Cartier multiple. It is *globally strongly $\mathbb{Q}$ -factorial* if every global coherent rank-one reflexive sheaf has an invertible reflexive power. The strong property implies the first one by applying it to the sheaf associated to a Weil divisor. Both properties concern global objects; the strong property makes no assertion about reflexive sheaves defined only on arbitrary analytic open subsets. For a rational line J we write $\{J\} = c_1(J)$, and use the same braces for the class of a rational Cartier divisor. A trace of a rational line under a bimeromorphic map means its reflexive transform. Bott–Chern transforms will be used only along the detected birational steps described below; on first Chern classes they agree with reflexive transforms whenever the step is ordinary. All resolutions in the compact arguments below are smooth compact Kähler spaces, and their indicated maps to the spaces being resolved are projective. A common resolution is projective over each indicated model.

Definition 3.1 (A resolution adapted to log canonical strata). An effective rational dlt pair (T, B) on a normal compact Kähler space satisfies the *lc-strata resolution convention* if it admits a projective log resolution $r : W \rightarrow T$ with smooth compact Kähler source such that the strict boundary and the exceptional support together have distinct smooth components forming a simple normal crossing divisor. Writing

$$K_W + B_W^{\text{cr}} = r^*(K_T + B)$$

with compatible canonical choices, every r -exceptional coefficient of B_W^{cr} is strictly less than one, and r is an isomorphism at the general point of every log canonical center of (T, B) .

This convention asserts the existence of one such resolution. It imposes no condition on later resolutions chosen for other purposes. The ordinary dlt models supplied below for the applications in Sections 4, 5, and 7 admit one by choosing a defining dlt resolution that preserves the simple normal crossing open set meeting the general points of all log canonical centers, and resolving any remaining data away from that open set.

An *ordinary negative step* for $J = K_T + B$ is a projective bimeromorphic divisorial contraction, or a diagram

$$T \xrightarrow{f} Z \xleftarrow{f^+} T^+$$

of projective small bimeromorphic morphisms, with connected fibers and normal base, such that $-J$ is f -ample and, in the small case, the trace J^+ is f^+ -ample. The contracted curves on T span one nonzero ray for degrees of global rational lines. We require this ray condition on the contracting side only. All spaces occurring in these steps are compact Kähler, and the working spaces are globally strongly $\mathbb{Q}$ -factorial. The boundary is pushed forward in a divisorial step and strictly transformed in a small step.

On a common smooth compact Kähler resolution, projective over T and the next space with maps p and q , the natural meromorphic comparisons of canonical bundles give the actual rational-line comparison

$$p^*J \sim_{\mathbb{Q}} q^*J^+ + F, \quad F \geq 0, \quad q_*F = 0. \quad (3.1)$$

Here and below these adjoint identities use the local meromorphic canonical identifications. In particular they do not choose a global meromorphic frame for an arbitrary line bundle. Negativity proves Equation (3.1); moreover, its support contains the full inverse image of the non-isomorphism locus in the contraction base. Thus discrepancies increase strictly for a place whose center maps into that locus, and do not decrease elsewhere. This preserves klt and dlt singularities. These assertions, including strictness, are proved in [54, Lemma 7.3]. Their proof is local over the contraction base and has no dimension restriction.

For use on strata, we record why the full support assertion holds for a detected small step. Choose a sufficiently divisible integer $r > 0$ and the evaluation ideal

$$\mathrm{im}(f^*f_*\mathcal{O}_T(rJ) \longrightarrow \mathcal{O}_T(rJ)) = \mathcal{I} \otimes \mathcal{O}_T(rJ).$$

Outside the exceptional locus of f , evaluation is an isomorphism. On a positive-dimensional projective fiber, every section vanishes: rJ has negative degree on every fiber curve, and curves cover every positive-dimensional fiber component. Connectedness gives the same vanishing on the whole nontrivial fiber. Hence the zero set of $\mathcal{I}$ is exactly the exceptional locus. On a common resolution principalizing $\mathcal{I}$, relative generation on the positive side identifies rF in Equation (3.1) with the divisor of $\mathcal{I}\mathcal{O}_W$. Thus $\mathrm{Supp} F$ is the full inverse image of that locus. On a common resolution of a longer negative program, discrepancy monotonicity makes the cumulative comparison dominate this first-step divisor. These are support statements for the actual comparison, stronger than effectiveness and exceptionality alone.

We next state the restricted model inputs used in this section. A nef b-class is a Bott–Chern class that is globally nef on one fixed smooth compact Kähler carrier, with its linear traces on the other models. The generalized adjoint on T is $\alpha = \{K_T + B\} + \beta_T$. Generalized klt, abbreviated gklt, means that all generalized log discrepancies, computed on that carrier, are positive. The modified-bigness hypothesis below concerns the whole boundary-plus-nef trace $\{B\} + \beta_T$. We use the convention of [39, Definition 2.8]: a modified-big trace is the pushforward of a big Bott–Chern class on a modification. The trace itself may be a current; the generalized adjoint is required to define a Bott–Chern class. For a proper map $f : T \rightarrow S$, let q_f be the quotient map to $H_{\mathrm{BC}}^{1,1}(T, \mathbb{R})/f^*H_{\mathrm{BC}}^{1,1}(S, \mathbb{R})$. Relative pseudo-effectivity means $q_f(\alpha) \in \overline{q_f(\mathrm{Psef}(T))}$, where $\mathrm{Psef}(T)$ is the absolute pseudo-effective cone. Equivalently, for Kähler classes ω_T, ω_S , the class α is pseudo-effective over S if for every $\epsilon > 0$ some $c_\epsilon \geq 0$ makes $\alpha + \epsilon\omega_T + c_\epsilon f^*\omega_S$ big. This imposes no projectivity hypothesis on f . For the actual relative line classes of an already projective map, this is the usual relative pseudo-effective cone.

Proposition 3.2 (Restricted Kähler model inputs). *Let $(T, B + \beta)$ be an effective compact Kähler gklt pair of dimension d , with T globally strongly $\mathbb{Q}$ -factorial. Suppose that its nef b-data are globally nef on a fixed smooth carrier and that $\{B\} + \beta_T$ is globally modified big. Write $\alpha = \{K_T + B\} + \beta_T$.*

- (i) *If α is nef, there is a proper morphism with connected fibers $g : T \rightarrow Z$ to a normal compact Kähler space and a Kähler class ω_Z such that $\alpha = g^*\omega_Z$.*
- (ii) *For a proper morphism $\pi : T \rightarrow V$ to a normal compact Kähler space, if α is pseudo-effective over V , there is a chosen good log terminal model over V . It is nonextracting, compact Kähler and globally strongly $\mathbb{Q}$ -factorial. Its adjoint is nef over V , and on a common projective resolution its comparison with α is effective and exceptional over the new model, with strictly positive coefficient at every prime contracted from T . The construction retains compatible forward Bott–Chern traces of the specified data on a common smooth carrier.*

(iii) A compact polytope of these data on one fixed carrier has finitely many marked relative canonical models and relative weak log canonical models with normal compact Kähler targets.

The relative assertion applies to proper morphisms, without assuming that π is projective. It is an assertion about a chosen model, not termination of every generalized flip sequence.

Proof. These are the semiample, proper relative model and finite-model assertions of Theorem 3.55, proved below by dimension induction independently of $\mathcal{G}$. Finite marked weak-model geography is recorded in Lemma 3.54. Projective analytic model constructions used in that induction are supplied by Proposition 3.25. None of these statements promotes an undetected generalized-ray contraction to a projective map. $\square$

We use separately the analytic cone theorem [39, Theorem 1.3]: an α -negative analytic extremal ray of an effective gklt adjoint has a rational curve generator C with $0 < -\alpha \cdot C \leq 2d$. Existence of the supporting contraction in the applications below follows from the nef assertion above. When an ordinary rational adjoint is negative on that ray, Lemma 3.16 and Proposition 3.18 give its projective contraction and ordinary flip. We call these *detected ordinary steps*.

The Bott–Chern transform through such a birational step has a concrete description which does not require the source to be smooth. Write the step as $T \xrightarrow{f} Z \xleftarrow{f^+} T^+$, with $f^+ = \text{id}$ for a divisorial contraction, and let J be its negative rational adjoint. For $\gamma \in H_{\text{BC}}^{1,1}(T, \mathbb{R})$, choose the unique real number c for which $\gamma - c\{J\}$ annihilates the contracted analytic ray. Every contracted curve is on that ray. Lemma 3.15 therefore gives a unique class δ on Z with $\gamma - c\{J\} = f^*\delta$, and we put

$$\gamma^+ = (f^+)^*\delta + c\{J^+\}.$$

The rational-singularity and vanishing hypotheses of that lemma hold: slightly lowering the rational Cartier floor gives a klt adjoint still antiample over Z , and relative vanishing gives rational singularities on the base. On a common resolution, the pullback difference is c times the adjoint comparison in Equation (3.1); it is exceptional over the new model, and over both models in a flip. This defines a linear transform. For the Chern class of a rational line it agrees with its reflexive trace, by the coherent line comparison [54, Lemma 7.2] and exceptional negativity. The already-projective relative program below only needs a ray for degrees of global rational lines; this Bott–Chern construction is used for its subsequent detected analytic-ray programs.

When such a class difference lies in an exceptional divisor span, we write it as $\{F\}$ for the representing real exceptional divisor F . This divisor is unique: an exceptional divisor with zero class has zero degree on every contracted curve, and exceptional negativity applied to both signs makes it zero.

We also use local projective analytic results. For a projective analytic morphism over a Stein neighborhood of a compactum with Fujino’s property (P), relative klt base point freeness says that, if L is a relatively nef Cartier line and $aL - (K + \Gamma)$ is relatively ample for some positive integer a , then all sufficiently high powers of L are relatively generated after shrinking [31, Theorems 6.2 and 6.5]. We use the finite negative-ray truncation in the relative cone theorem in the same setting. Finally, the multigraded adjoint finite-generation theorem applies to a projective morphism from a smooth space, for simultaneous effective SNC klt boundaries with a common relatively ample rational summand [18, Theorem 3.1]. Finitely many smaller Stein neighborhoods suffice over a compact base.

Lemma 3.3 (Descent without changing a Cartier exponent). *Let $f : T \rightarrow Z$ be a projective bimeromorphic contraction with normal target. Suppose an ordinary rational klt adjoint is f -antiample. If a Cartier line L has degree zero on every contracted curve, then f_*L is an invertible sheaf and evaluation is an isomorphism*

$$f^*(f_*L) \simeq L.$$

The same conclusion holds for an ordinary dlt adjoint whose floor is rational Cartier and which is f -antiample. In a flip, a line with zero contracted degree therefore has the same Cartier exponent on both sides, and the two lines are pullbacks of a line on the common base.

Proof. For dlt input, decrease the floor by a sufficiently small positive rational multiple. The resulting pair is klt and remains f -antiample. For this klt pair the base point free hypothesis holds for L , because L has zero fiber degrees and the negative adjoint is relatively ample. Over a smaller base neighborhood every sufficiently high power of L is therefore generated. The associated morphism is constant on each connected fiber: its tautological line has degree zero on every fiber curve, and every positive-dimensional projective image contains a curve. Its graph then factors through Z ; the graph projection is finite bimeromorphic and Z is normal. Descend two consecutive generated powers L^k, L^{k+1} to lines M_k, M_{k+1} . The line $M_{k+1} \otimes M_k^{-1}$ pulls back to L . Projection formula identifies this quotient with f_*L , so these local descents and their evaluation maps agree on overlaps. Pull the descended line to the positive side of a flip. This is the argument of [54, Lemma 7.4]. $\square$

3.2 Relative klt programs and small models

The local finite-generation theorem also supplies an ordinary klt program over a compact base. We use it to obtain globally strongly $\mathbb{Q}$ -factorial small models of strata. The construction keeps track of degrees of global rational lines, which are the degrees needed for exact Cartier descent.

Proposition 3.4 (Relative klt programs over a compact base). *Let $\pi : T \rightarrow V$ be projective bimeromorphic, where V is a normal compact Kähler space and T is normal, compact Kähler, and globally strongly $\mathbb{Q}$ -factorial. Let (T, B) be an effective rational klt pair with rational-line adjoint $J = K_T + B$. There is a finite sequence of ordinary J -negative steps over V , all projective over V , whose last adjoint is curve-nef over V , meaning that it has nonnegative degree on every curve contracted over V . Every working space is compact Kähler and globally strongly $\mathbb{Q}$ -factorial. On a common resolution the comparison from the first adjoint to the last is effective and exceptional over the last space.*

Proof. We give the reduction to the local finite-generation input, including why it yields a single finite program over the compact base. Let $\mathcal{N}_{\mathbb{R}}$ be the finite-dimensional space of degrees of global rational lines on curves over V , and put $r = \dim \mathcal{N}_{\mathbb{R}}$. If $r = 0$, then J is already curve-nef. Assume $r > 0$. Choose rational lines $H_1, \dots, H_r$ whose degree classes form a basis and such that both H_i and $J + H_i$ are relatively ample. Such a basis exists because the relatively ample classes whose sum with J is also relatively ample form a nonempty open set containing sufficiently positive classes. Define the rational simplex of adjoints

$$\Pi = \text{conv}\{J, J + H_1, \dots, J + H_r\}.$$

We will choose a general direction $H = \sum_i \theta_i H_i$ with $\theta_i > 0$ and $\sum_i \theta_i = 1$. Its entire scaling segment lies in this simplex:

$$J + tH = (1 - t)J + \sum_i t\theta_i(J + H_i) \in \Pi \quad (0 \leq t \leq 1),$$

and it lies in the relative interior for $0 < t < 1$.

We produce the klt representatives for these exact vertices on each sufficiently small Stein neighborhood separately. A relative Cartier line and its inverse have local effective representatives there: their proper direct images are coherent of generic rank one, so Cartan generation supplies nonzero local sections. This uses that π is bimeromorphic. Represent each H_i by a divided general free effective divisor $G_i \sim_{\mathbb{Q}} H_i$, using a sufficiently divisible relatively generated multiple. These finitely many general divisors preserve klt with B . Choose a relatively ample integral line A , and local effective representatives $A^+ \sim A$, $A^- \sim -A$ by the preceding argument. For one sufficiently small positive rational ϵ , all boundaries

$$\Delta_0 = B + \epsilon(A^+ + A^-), \quad \Delta_i = B + G_i + \epsilon(A^+ + A^-)$$

are effective and klt. Their adjoints represent J and $J + H_i$, respectively, and ϵA^+ is a common effective relatively ample summand. In particular, every rational adjoint in Π is relatively big.

On this Stein member, choose one simultaneous projective log resolution of the finite family. Give each exceptional prime a coefficient in $(\max\{0, c_0, \dots, c_r\}, 1)$, where the c_j are its crepant coefficients for the vertex pairs. If $E \geq 0$ is resolution-exceptional with $-E$ resolution-ample, then subtracting a sufficiently small multiple of E from the pullback of ϵA^+ leaves a relatively ample rational divisor. A sufficiently small positive rational multiple of this divisor can be removed from every boundary while leaving all coefficients effective and below one; the strict exceptional margins ensure this along the exceptional primes. A divided general free representative of this common summand preserves the simultaneous SNC condition. The smooth multigraded theorem now applies to these vertices. Exceptional corrections leave the local adjoint rings unchanged by projection to a normal space; clearing the finitely many line identifications simultaneously makes them multiplicative. This is the reduction in [54, Section 7.2], using [18, Theorem 3.1].

Here are the two consequences of finite generation that we need. The normalized main relative Proj of any rational adjoint $\Theta \in \Pi$ extracts no prime divisor, and its trace is a relatively ample rational line. Indeed, resolve the degree-one base ideal after a common Veronese and write

$$p^*(m\Theta) \sim q^*H_{\Theta} + G, \quad G \geq 0,$$

with H_{Θ} tautological and relatively ample. Generation says that all sections in degree k vanish at least along kG . If a component of G were not q -exceptional, relative generation of $q_*\mathcal{O}(G) \otimes H_{\Theta}^k$ would supply a section with smaller vanishing there. If a p -exceptional prime Q were not q -exceptional, the same argument with $q_*\mathcal{O}(Q) \otimes H_{\Theta}^k$ would contradict $p_*\mathcal{O}(Q) = \mathcal{O}_T$. This proves nonextraction. There are also only finitely many *marked* normalized main Proj models, where the marking records their common bimeromorphic open over V . To see this, choose finitely many homogeneous generators locally. Proj charts of each diagonal ring use homogeneous monomials as denominators; the supports of those monomials in the finite set of generators determine the degree-zero localizations and their gluing. Only finitely many support patterns occur. Normalization and taking the main component preserve this finiteness. A finite cover of the compact base gives finitely many global marked models, since marked local identifications agree on the common dense open and glue uniquely. These are [54, Lemmas 7.5 and 7.6].

For completeness, we construct the scaling program controlled by this finite list. On a working model, let A_0 be a relatively ample line. The closed curve cone in the dual of its global degree space has compact slice $A_0 \cdot z = 1$: for every line D , both $kA_0 + D$ and $kA_0 - D$ are relatively ample for k large, which bounds every coordinate. Local cone truncations on the finite Stein cover express the negative part of this slice, after any positive ample truncation, using finitely many

actual curve classes. A negative extremal ray can therefore be separated by a rational nef support annihilating just that ray. A positive multiple of the support minus the klt adjoint is relatively ample. Relative base point freeness contracts exactly the ray. If the contraction is divisorial, its single exceptional prime and Lemma 3.3 prove the global strong property downstairs. If it is small, the relative Proj of the adjoint is its small positive model. For any global line D on the negative model, killing its ray degree and applying the same lemma gives an actual rational identity

$$D \sim_{\mathbb{Q}} cJ + f^*D_Z, \quad c \in \mathbb{Q}.$$

It transforms to the positive side, proving the global strong property there. This is the continuation construction of [54, Proposition 7.7]; it also preserves projectivity over V and compact Kählerness.

The transforms of $H_1, \dots, H_r$ continue to span the degree spaces: use pullback in a divisorial step and the preceding decomposition of each line in a small step. A line of zero global relative degrees stays so by Lemma 3.3; curves in the contraction bases can be lifted through projective morphisms. There are only countably many possible finite sequences of curve rays, and each determines its contractions and flips uniquely. We may therefore choose H in the relative interior of $\text{conv}\{H_1, \dots, H_r\}$ so that it avoids all ties between independent ray degrees in advance. At a positive nef wall, write $w = J + tH$ for the traces on the current model, and choose a preceding parameter $s > t$ for which $J + sH$ is relatively ample. On the zero face of the normalized compact curve slice,

$$J = -\frac{t}{s-t}(J + sH)$$

in degrees. Thus J is uniformly negative on that face. Choose the positive ample truncation so that its compact remainder K misses the face. The finite rational polyhedral test and the general choice of H make the face a single ray R . The wall class has a positive minimum on K and is positive on every other truncated curve generator. Hence a sufficiently small open neighborhood of w inside the annihilator $R^\perp$ consists of nef classes with zero face exactly R .

Suppose first that $R^\perp$ has positive dimension. It is a rational hyperplane because R is represented by an integral curve. Choose a rational simplex of support classes in that neighborhood containing the degree class of w in its relative interior, and represent each vertex by a rational line N . For each such N , compactness gives an integer $a > 0$ for which $aN - J$ is relatively ample. Relative base point freeness contracts exactly R and descends N to a relatively ample rational line on the contraction base. These bases agree because the contracted curves agree. The difference between w and the corresponding convex combination of these rational lines has zero relative degree. In the finite rational span of the lines involved, the degree map has rational kernel, since rational lines have rational degrees on integral curves. The difference is therefore a real combination of rational lines of zero relative degree. Lemma 3.3 descends these lines, and lifting curves from the base shows that their descents still have zero degree over V . They do not affect relative ampleness. Convexity now shows that w descends to a relatively ample real line class over V . If $R^\perp = 0$, the same zero-degree descent applies to w ; the contraction base has no curves over V , hence is finite over V , and relative ampleness is automatic.

After a flip, let w^+ be the pullback of this wall class to the positive side. For sufficiently small $\delta > 0$,

$$J^+ + (t - \delta)H^+ = \left(1 - \frac{\delta}{t}\right)w^+ + \frac{\delta}{t}J^+$$

is relatively ample over V : the wall is pulled back from a relatively ample class on the contraction base, and J^+ is relatively ample over that base. After a divisorial contraction, openness of the ample cone on the base gives the same nonempty interval of relative ampleness.

Each working model is consequently one of the marked ample models already counted. Indeed, for a fixed finite prefix and an interior parameter $0 < t < 1$, replace the direction and parameter by nearby rational ones. They remain in Π , and all strict signs in that prefix and the final relative ampleness persist; the associated relative section ring has this working model as its Proj. A repeated marked working model would have the same trace of B , because all steps are nonextracting, and hence the same discrepancies. This contradicts the strict increase in Equation (3.1) at an intervening nontrivial step. The finite list therefore forces termination. The last adjoint is curve-nef, since the continuation construction applies whenever it is not. Composing Equation (3.1) proves the final comparison. $\square$

Corollary 3.5 (Small strong models). *Let (S, Δ) be an effective rational dlt pair on a normal compact Kähler space, with rational-line adjoint. Suppose that an effective rational Cartier divisor D satisfies $\text{Supp } D = \text{Supp } [\Delta]$. There is a projective small morphism $h : Y \rightarrow S$, with Y globally strongly $\mathbb{Q}$ -factorial and compact Kähler, which is crepant for the full adjoint and is an isomorphism over the smooth locus of S . The transformed full pair is dlt. The same assertion for an ordinary klt pair allows $D = 0$.*

Proof. Choose a small rational $\epsilon > 0$ for which $(S, \Delta - \epsilon D)$ is effective and klt. On a projective log resolution $r : W \rightarrow S$, use the strict lowered boundary and give every exceptional prime a coefficient strictly between its crepant coefficient and 1, and at least zero. The resulting klt adjoint has the actual form

$$K_W + \Gamma \sim_{\mathbb{Q}} r^*(K_S + \Delta - \epsilon D) + F, \quad F \geq 0,$$

where every r -exceptional prime has positive coefficient in F . Apply Proposition 3.4 over S . On its endpoint the trace F_Y is effective, exceptional over S , and relatively nef. Negativity makes $F_Y = 0$. Since all initially exceptional primes had positive coefficient, $Y \rightarrow S$ is small and the lowered adjoint is crepant. Restoring ϵD gives the full crepant identity.

A projective small morphism to a smooth germ is an isomorphism: transport a relatively ample line to the smooth germ, where it is Cartier; smallness makes the original line its pullback, contradicting relative ampleness on a nontrivial fiber. Thus h is an isomorphism over an SNC open meeting all lc centers of the dlt pair. The full crepant comparison then proves dlt on Y . $\square$

3.3 Scaling toward zero and keeping a nef rational line

The next construction runs every positive truncation of an ordinary dlt scaling. It also explains why a sufficiently large multiple of a prescribed nef rational line forces every step to be trivial for that line. The integer that clears the line will be the same at every step.

For the detected ordinary scalings constructed below, let $\alpha_0(\lambda) = \{H\} + \lambda\omega$ on the first space T_0 , with initial parameter 1, and let $\alpha_i(\lambda)$ be its linear transform on a working model T_i . Put $\lambda_{-1} = 1$. If the next step occurs at threshold λ_i , its ray is annihilated by $\alpha_i(\lambda_i)$ and is negative for $\alpha_i(0)$. The trace $\alpha_i(\lambda)$ is nef on $[\lambda_i, \lambda_{i-1}]$, which we call its *working interval*. If $\alpha_i(0)$ is nef, we set $\lambda_i = 0$ and stop. Working intervals are allowed to be degenerate; strict decrease of positive thresholds will be arranged only in the proof of special termination.

Lemma 3.6 (Positive truncations and a fixed nef line). *Let (T, B) be an effective rational dlt pair on a globally strongly $\mathbb{Q}$ -factorial compact Kähler d -fold, and put $J = K_T + B$.*

- (i) *For a sufficiently large Kähler class ω , there is an ordinary J -program with scaling of ω , starting with $\{J\} + \omega$ Kähler. Every positive truncation is finite. If J is pseudo-effective, the*

program either reaches a nef adjoint or has positive thresholds tending to zero. If J is not pseudo-effective, the program is finite and ends with a Mori fiber contraction.

- (ii) Let P be a nef rational line with mP Cartier, and choose $b \in \mathbb{Q}$ with $b > 4dm$. The same alternatives hold for $H = J + bP$, according to pseudo-effectivity of H , with ω enlarged if necessary. Every birational step is an ordinary J -negative step on which P is trivial, and the final Mori ray, if present, is also P -trivial. The same line mP descends through every birational contraction and pulls back to its positive side. Its traces remain analytically nef; if P is semiample, its same generated multiple and associated morphism are preserved, also through the final Mori contraction.

Proof. Write $H = J$ in the first case and $H = J + bP$ in the second. Fix ω with $\{H\} + \omega$ Kähler. For each $0 < \eta < 1$, choose a positive rational ϵ so small that, with $F = \lfloor B \rfloor$,

$$(T, B - \epsilon F) \text{ is klt,} \quad \eta\omega + \epsilon\{F\} \text{ is Kähler.}$$

For $\lambda \in [\eta, 1]$ use the presentations

$$\{H\} + \lambda\omega = \{K_T + B - \epsilon F\} + (b\{P\} + \lambda\omega + \epsilon\{F\}),$$

omitting $b\{P\}$ in the first case. Their b-data are globally nef on the fixed initial carrier, and their boundary-plus-nef traces are globally modified big. These are effective gklt data.

We first construct a step whenever the unscaled working trace is not nef. Choose a positive shift strictly below the current wall and use the gklt presentation of its current trace A . The ray at the wall is A -negative. On a compact slice of the analytic curve cone, the cone theorem makes the A -negative part locally polyhedral after a sufficiently small positive Kähler truncation. Separating the chosen extremal ray from the other finitely many rays in that truncation gives a Kähler class ω_R such that $A + \omega_R$ is nef and its zero face is precisely this ray; see the supporting construction in Proposition 3.48. Add the pullback of this new Kähler class to the fixed nef carrier. The resulting adjoint still has globally nef data and modified-big boundary-plus-nef trace, so the nef assertion of Proposition 3.2 supplies its supporting contraction. This uses a new supporting direction on the current space; it does not require the trace of the original scaling direction to be Kähler. We verify a rational ordinary detector before asserting projectivity or constructing a flip. In the second case nefness of P makes J negative on the ray; the first case has this property directly. Lower the floor by a small rational amount, retaining negativity of J_ϵ , so that $-J \cdot R \leq 2(-J_\epsilon \cdot R)$. The ordinary klt cone length bound gives a rational curve generator C with

$$0 < -J \cdot C \leq 4d.$$

In the second case, if $P \cdot C > 0$, integrality of mP gives $P \cdot C \geq 1/m$, contradicting

$$H \cdot C \geq -4d + b/m > 0.$$

Thus in that case the ray is P -trivial. The lowered ordinary adjoint is a global rational detector. The detected projectivity and flip constructions in Lemma 3.16 and Proposition 3.18 supply the projective ordinary step. The difference of J and a positive rational multiple of the lowered adjoint has zero contracted degree; exact descent in Lemma 3.3 shows that restoring the floor preserves the positive-side sign. This is the ordinary replacement of Proposition 3.19, and the full pair remains dlt by Equation (3.1).

In the second case, Lemma 3.3 descends the actual Cartier line mP without changing m . Analytic nefness descends and pulls back under these projective contractions [39, Lemma 2.44]. A generated multiple descends as a generated line by projection formula and pulls back as one. Thus the same length estimate and the same b apply at the next step. The corresponding assertion on a final Mori ray follows from exactly the same estimate. For a semiample P , the map of its fixed generated multiple is constant on each connected projective contracted fiber: otherwise its image contains a curve of positive degree. The map therefore factors through the normal contraction base, including in the fiber-type case.

The traces used in this construction are linear by the degree-zero descent description preceding Lemma 3.3. At a wall the scaled class is pulled back from its base. The comparison for an earlier parameter is a nonnegative multiple of the ordinary adjoint comparison. Consequently the gklt presentations on any fixed positive segment remain gklt along its program, and every working model is a weak model for a member of that segment. The finite marked weak-model assertion in Proposition 3.2 applies. A marked model cannot recur: its boundary trace would be the same, whereas a nontrivial intervening ordinary step strictly increases a discrepancy. There are therefore only finitely many steps with threshold at least η .

For decreasing cutoffs, any already-constructed finite prefix remains valid. The floor perturbations cancel in the displayed classes and continue to cancel under their linear transforms; their nonpositive comparisons preserve the new gklt data. Starting at its last nef wall therefore extends the prefix to the smaller cutoff. This gives one compatible scaling program. A positive limiting threshold would lie in a finite positive truncation, so an infinite program has limit zero. Pseudo-effectivity excludes a negative Mori endpoint. If H is not pseudo-effective, its pseudo-effective threshold in the initial Kähler direction is positive. Every nef working trace, together with its effective exceptional comparison, makes the initial scaled class pseudo-effective, so the scaling thresholds are bounded below by that positive number. The program is therefore finite. Its endpoint cannot be nef, since that comparison would make H pseudo-effective; it is the stated Mori fiber contraction. $\square$

We call these programs *detected ordinary scalings*. Their steps are ordinary negative steps of one analytic extremal ray, with the linear Bott–Chern transform described above. The next lemma connects their nef working traces to the analytic negative part, including its limit at parameter zero.

Lemma 3.7 (Negative parts along a detected ordinary scaling). *In a scaling from Lemma 3.6, write $H = J$ in its first case and $H = J + bP$ in its second, and use the notation $\alpha_0(\lambda) = \{H\} + \lambda\omega$ above. For a parameter λ in the working interval of T_i , a common smooth resolution $p : W \rightarrow T_0$, $q : W \rightarrow T_i$ gives*

$$p^*\alpha_0(\lambda) = q^*\alpha_i(\lambda) + \{E_i(\lambda)\}, \quad E_i(\lambda) \geq 0, \quad q_*E_i(\lambda) = 0. \quad (3.2)$$

Here $E_i(\lambda)$ is an actual real exceptional divisor, and

$$N(p^*\alpha_0(\lambda)) = E_i(\lambda).$$

For the limit statement, let H be any rational line on a normal compact Kähler space T whose pullback to a smooth resolution is pseudo-effective, and let A be a Kähler class on T . On any fixed smooth resolution $r : U \rightarrow T$ and for every prime $Q \subset U$,

$$\lim_{\lambda \downarrow 0} \nu_Q(r^*({H} + \lambda A)) = \nu_Q(r^*{H}).$$

Thus the same convergence holds for every divisorial place, after choosing a resolution on which it appears.

Proof. For λ in the working interval of T_i , every earlier threshold is at least λ . Each earlier step is therefore negative or trivial for this parameter. Composing its comparisons gives Equation (3.2). Since $\alpha_i(\lambda)$ is nef, Lemma 2.3 identifies the exceptional divisor with the negative part.

On the fixed resolution U , choose a Kähler class Ω and $c > 0$ such that $c\Omega - r^*A$ is Kähler. Monotonicity of each minimal multiplicity under addition of a nef class gives

$$\nu_Q(r^*\{H\} + \lambda c\Omega) \leq \nu_Q(r^*(\{H\} + \lambda A)) \leq \nu_Q(r^*\{H\}).$$

The left side tends to the right side by the definition using small Kähler perturbations. This proves the convergence on U , and the same argument applies on a resolution representing any chosen divisorial place. $\square$

Proposition 3.8 (Contracting a known negative part). *Let (T, B) be an effective rational dlt pair on a globally strongly $\mathbb{Q}$ -factorial compact Kähler space, and put $J = K_T + B$. Suppose there is an actual rational-line identity*

$$J \sim_{\mathbb{Q}} P + E,$$

where P is analytically nef, E is an effective rational Cartier divisor, and, on one smooth compact Kähler resolution $\pi : W \rightarrow T$ projective over T ,

$$N(\pi^*\{J\}) = \pi^*E.$$

There is a finite ordinary J -negative program to a nef model T' which contracts precisely the prime components of E among the primes of T . Every step is P -trivial, with a fixed Cartier exponent as in Lemma 3.6, and

$$J_{T'} \sim_{\mathbb{Q}} P_{T'}.$$

If P is semiample, then so is this actual last adjoint.

Proof. Lemma 2.4 makes the stated equality valid on any higher resolution. Choose b as in Lemma 3.6. Adding bP leaves the same negative part. Indeed, on a smooth resolution write $\alpha = \pi^*\{J\}$, $p = \pi^*\{P\}$, and $e = \pi^*E = N(\alpha)$. Subadditivity and homogeneity of negative multiplicities give

$$\begin{aligned} N(\alpha + bp) &\leq e, \\ (1 + b)e &= N((1 + b)\alpha) \leq N(\alpha + bp) + bN(\{e\}) \leq N(\alpha + bp) + be. \end{aligned}$$

Thus $N(\alpha + bp) = e$.

Run the scaling for $H = J + bP$. Let Q be the strict transform on a fixed resolution of a prime component of E . Its multiplicity in $N(\pi^*\{H\})$ is positive. The limit in Lemma 3.7 makes its multiplicity positive for every sufficiently small positive parameter. If that prime still survived on the corresponding working model, Equation (3.2) would make its multiplicity zero, because the comparison there is exceptional over that model. It must therefore be contracted after finitely many steps. The same conclusion holds if a nef endpoint is reached before taking the limit. There are only finitely many components of E , so after a finite prefix all have disappeared.

Throughout the program P descends and remains nef. The actual identity $J_i \sim_{\mathbb{Q}} P_i + E_i$, with E_i the codimension-one pushforward of E , follows from the reflexive trace convention. Once $E_i = 0$, we have $J_i \sim_{\mathbb{Q}} P_i$; since P_i is nef, the program stops. Conversely, a divisorial negative step

can contract only a prime of E_i : its ray is P_i -trivial, so the effective divisor E_i has negative degree on every contracted curve, and those curves are contained in its support. This proves the exact assertion about contracted primes. $\square$

The next lemma records the cohomology carried by a detected ordinary scaling that begins on a smooth space. Its surjectivity will let us choose one generic initial scaling direction for all later models. Its exceptional splitting will also carry the span of line classes to the terminal models used in Section 4.

Lemma 3.9 (Cohomology through ordinary steps). *Let a finite prefix of one of the detected ordinary scalings constructed in Lemma 3.6 start on a smooth compact Kähler space, and let V be any working space. For every smooth compact Kähler resolution $p : U \rightarrow V$ projective over V ,*

$$H^2(U, \mathbb{R}) = p^* H^2(V, \mathbb{R}) \oplus \bigoplus_{Q \text{ exceptional for } p} \mathbb{R}\{Q\}, \quad (3.3)$$

$$H^{1,1}(U, \mathbb{R}) = p^* H_{\text{BC}}^{1,1}(V, \mathbb{R}) \oplus \bigoplus_{Q \text{ exceptional for } p} \mathbb{R}\{Q\}. \quad (3.4)$$

The linear transform from the first space is surjective onto $H^2(V, \mathbb{R})$ and $H_{\text{BC}}^{1,1}(V, \mathbb{R})$, respectively. For a small correspondence between working spaces, the difference of the pulled-back classes and their transforms on a common resolution lies in the span of the primes exceptional over both spaces. Under the displayed decompositions, the Chern class of a global holomorphic line on U is the sum of the pullback of a rational holomorphic line class on V and an exceptional rational divisor class. In particular, passage to the exceptional quotient preserves the real span of line Chern classes.

Proof. For a modification of a smooth compact Kähler manifold, the degree-two modification theorem gives these decompositions, and the exceptional classes have type $(1, 1)$. Suppose they hold at one step. Its contraction base and both working spaces have rational singularities: slight lowering of the rational Cartier floor makes the ordinary pair klt, and the detected contraction has a base with rational singularities. These contractions are bimeromorphic, so a common resolution and Leray give $R^i f_* \mathcal{O} = 0$ for $i > 0$; this verifies the additional hypothesis in Lemma 3.15.

The ray of this detected contraction is one ray for Bott–Chern degrees. This also suffices for degree-two classes. Indeed, pull a class $\gamma \in H^2(V, \mathbb{R})$ to a smooth resolution $p : U \rightarrow V$ and take its Hodge decomposition. By the decomposition already proved for this working model in Equation (3.4), its $(1, 1)$ -part is $p^* \beta + \{E\}$ for a Bott–Chern class β and an exceptional real divisor E . The other Hodge components have zero degree on curves. On every p -contracted curve the degree of E is therefore zero, so both signs of negativity make $E = 0$. Lifting any curve of V through the projective resolution now shows that γ and β have the same curve degrees. Thus subtracting a multiple of $\{J\}$ from either a degree-two or a Bott–Chern class can annihilate every curve of the contracted fiber. By Lemma 3.15, a degree-two class, or a Bott–Chern class, with these zero degrees is pulled back from the base. Subtract a multiple of $\{J\}$ to make any given class annihilate that ray, descend the remainder, and use the trace $\{J^+\}$ to define its transform. On a common resolution the difference is a multiple of the exceptional adjoint comparison in Equation (3.1). For a flip the two exceptional prime sets on that resolution are the same. For a divisorial contraction the target resolution has just the one additional exceptional prime of the step. The old decomposition therefore spans the new one and proves surjectivity of the transform.

The sums are direct. If the pullback of a class from V is an exceptional divisor class, that divisor has zero degree on all curves over V . Applying exceptional negativity to both signs makes the divisor zero; injectivity of pullback gives that the class is zero. The pullback injectivity in

degree two and in Bott–Chern cohomology is also Lemma 3.15, applied to the resolution. Passing to higher smooth resolutions proves the assertion for every p .

For the line assertion, let $\mathcal{H}$ be a holomorphic line on U , and put $\mathcal{Q} = (p_*\mathcal{H})^{**}$. Global strong $\mathbb{Q}$ -factoriality gives an invertible sheaf $\mathcal{M} = \mathcal{Q}^{[m]}$ for some $m > 0$. The coherent evaluation comparison of [54, Lemma 7.2] gives

$$\mathcal{H}^{\otimes m} \simeq p^*\mathcal{M} \otimes \mathcal{O}_U(E), \quad E \text{ an integral } p\text{-exceptional divisor.}$$

Indeed $(p_*\mathcal{H}^{\otimes m})^{**} = \mathcal{M}$, since the two sheaves agree at the general points of all divisors of V . Local generators of $p_*\mathcal{H}^{\otimes m}$ in a frame of $\mathcal{M}$ compare their evaluations with the pulled-back coefficients by meromorphic quotients. Those quotients agree on the isomorphism locus and hence glue; their zeros and poles are exceptional. Taking Chern classes proves the assertion without choosing a meromorphic section of $\mathcal{H}$. $\square$

Before returning to special termination, we record the consequence used in the boundary argument: a lower-dimensional manifold of Kodaira dimension zero has a terminal torsion model carrying the preceding exceptional splitting.

Corollary 3.10 (Terminal models in Kodaira dimension zero). *Let X be a smooth compact Kähler d -fold with $d < n$ and $\kappa(X, K_X) = 0$. A finite ordinary K_X -negative program reaches a globally strongly $\mathbb{Q}$ -factorial compact Kähler terminal space $X_{\min}$ whose actual canonical rational line is torsion. For every smooth compact Kähler resolution projective over $X_{\min}$, the decompositions in Equations (3.3) and (3.4) hold.*

Proof. A pluricanonical section makes K_X pseudo-effective. By $\mathcal{G}_d$, on a smooth modification $\mu: \tilde{X} \rightarrow X$ there is an actual identity

$$\mu^*K_X \sim_{\mathbb{Q}} P + E, \quad P \text{ semiample, } E = N(\mu^*\{K_X\}).$$

Lemma 2.5 identifies the divisible section spaces with those of P . Hence $\kappa(P) = 0$; a semiample line of Iitaka dimension zero has a trivial positive multiple. Proposition 2.9, applied with $e = \mu$ and $F = 0$, gives an effective rational divisor $E_X = N(K_X)$ with $K_X \sim_{\mathbb{Q}} E_X$. Lemma 2.4 then gives $N(\mu^*\{K_X\}) = \mu^*E_X$. Apply Proposition 3.8 with $P = 0$. It reaches a model with torsion canonical line. It is terminal: exceptional places over the initial smooth space have log discrepancy greater than 1, and a divisor contracted from that space starts with log discrepancy 1 and acquires a strict increase in Equation (3.1). Finally apply Lemma 3.9. $\square$

3.4 A comparison model for the restricted scaling

We now prepare the lower-dimensional argument that excludes an infinite sequence of small transformations on a log canonical stratum. We first construct a semiample small model when the negative multiplicities have centers away from the floor. We then make one model nef on an entire interval of perturbations. This interval is the precise lower-dimensional conclusion needed below.

Lemma 3.11 (A semiample small comparison model). *Let (S, Δ) be an effective rational dlt pair on a normal compact Kähler space of dimension $d < n$. Suppose $J = K_S + \Delta$ is a pseudo-effective rational line, and that an effective rational Cartier divisor D satisfies $\text{Supp } D = \text{Supp } [\Delta]$. Assume for every divisorial place Q over S that*

$$\nu_Q(J) = 0 \quad \text{if the center of } Q \text{ is a prime of } S, \text{ or meets } \text{Supp } [\Delta]. \quad (3.5)$$

There is a globally strongly $\mathbb{Q}$ -factorial compact Kähler dlt pair (M, Δ_M) , small bimeromorphic with (S, Δ) , whose adjoint J_M is an actual semiample rational line. Over a neighborhood U of the floor in S , the comparison is a projective small crepant morphism $M_U \rightarrow U$.

For every $\gamma \in H_{\text{BC}}^{1,1}(S, \mathbb{R})$ there is a transformed class $\gamma_M \in H_{\text{BC}}^{1,1}(M, \mathbb{R})$ which is its pullback over U . On a common smooth resolution $p : W \rightarrow S$, $q : W \rightarrow M$,

$$q^* \gamma_M - p^* \gamma = \{E_\gamma\},$$

where E_γ is a real divisor supported on primes exceptional over both spaces, and $\text{Supp } E_\gamma$ is disjoint from $p^{-1}(U)$.

Proof. Resolve the dlt pair, giving new exceptional primes coefficient one. The resolved adjoint is the pullback of J plus an effective exceptional divisor. Apply $\mathcal{G}_d$ to this smooth pair. Lemma 2.3 subtracts the exceptional resolution error from its negative part. On a smooth higher model $r : \tilde{S} \rightarrow S$ we obtain the actual identity

$$r^* J \sim_{\mathbb{Q}} P + E, \quad P \text{ semiample}, \quad E = N(r^* \{J\}). \quad (3.6)$$

Every component of E is exceptional over S and has center disjoint from the floor, by Equation (3.5). Choose m divisible enough that mJ, mP, mE are integral and mP is generated. Lemma 2.5 shows that multiplication by the section of mE identifies the complete section spaces. Over a neighborhood of the floor, E is absent upstairs. The line $mr^* J$ is generated there, and these sections descend to S by normality. Thus mJ is generated on a neighborhood U of the entire floor.

Let $\rho_\Gamma : \Gamma \rightarrow S$ be the normalized graph of the map defined by the complete system $|mJ|$, and write $\varphi : \Gamma \rightarrow \mathbb{P}(H^0(S, mJ)^*)$ for its morphism to projective space. The map ρ_Γ is projective and is an isomorphism over U . The rational line

$$P_\Gamma = \frac{1}{m} \varphi^* \mathcal{O}(1)$$

is semiample. The generated moving system $|mP|$ induces a morphism $\psi : \tilde{S} \rightarrow \Gamma$ whose pullback of P_Γ is P . Choose a small positive rational ϵ for which $(S, \Delta - \epsilon D)$ is effective and klt. On a projective log resolution $W_0 \rightarrow \Gamma \rightarrow S$, give every exceptional prime a nonnegative coefficient strictly above its crepant coefficient for the lowered pair and strictly below 1. Its klt adjoint equals the pullback of $J - \epsilon D$ plus an effective error positive on every prime exceptional over S .

Run Proposition 3.4 over Γ , and write $g : Y \rightarrow \Gamma$ for the endpoint morphism and $h = \rho_\Gamma g : Y \rightarrow S$. The surviving error F_Y is effective and has positive coefficient on every prime of Y exceptional over S . Over U , where $\Gamma = S$, it is also exceptional and relatively nef, since the lowered adjoint is relatively nef and its other term is pulled back from S . Local exceptional negativity makes it zero there. Hence h is small over U , and it is an isomorphism over the smooth part of U by the argument in Corollary 3.5. Restore the floor. The full pair on Y is dlt near it, by the crepant small comparison and the SNC open, and is klt elsewhere. No exceptional prime has center meeting the floor, so $h^* D$ is its strict transform and

$$J_Y \sim_{\mathbb{Q}} h^* J + F_Y.$$

Put $P_Y = g^* P_\Gamma$. On a common smooth model $a : \hat{S} \rightarrow \tilde{S}$, $b : \hat{S} \rightarrow Y$ over S , the maps ψa and gb agree on the common dense open and hence everywhere by separatedness of Γ . Thus $a^* P \sim_{\mathbb{Q}} b^* P_Y$. Define $E_Y^0 = b_* a^* E$. This is an effective rational divisor, exceptional over S and supported away from the floor. Pushing Equation (3.6) to Y gives

$$h^* J \sim_{\mathbb{Q}} P_Y + E_Y^0.$$

The global strong property on Y makes E_Y^0 rational Cartier. The divisor $a^*E - b^*E_Y^0$ is b -exceptional and rationally linearly trivial. Exceptional negativity applied to both signs therefore gives $a^*E = b^*E_Y^0$.

Define $E_Y = E_Y^0 + F_Y$. Adding the restored-boundary error gives

$$J_Y \sim_{\mathbb{Q}} P_Y + E_Y, \quad E_Y \geq 0, \quad (3.7)$$

where E_Y is exceptional over S , is supported away from the floor, and contains every prime exceptional for h . It is rational Cartier by the global strong property. On $\hat{S}$, Lemma 2.4 gives

$$N(a^*r^*\{J\}) = a^*E = b^*E_Y^0.$$

Exceptional translation for b^*F_Y over S then gives $N(b^*\{J_Y\}) = b^*E_Y$. The hypotheses of Proposition 3.8 are now satisfied. Each contracted curve lies in $\text{Supp } E_Y$, since its P_Y -degree is zero and its E_Y -degree is negative. Nontrivial connected projective fibers are covered by curves, so this contraction and its positive replacement are unchanged over a neighborhood of the floor. The same assertion persists with the pushed-forward effective divisor at each step. All primes exceptional over S are contracted and no prime coming from S is contracted. The endpoint M is therefore small with S , has the asserted morphism over U , and has $J_M \sim_{\mathbb{Q}} P_M$ semiample.

Pull γ from S to Y . Although Y may be singular, its subsequent program consists of the detected ordinary analytic-ray steps in Proposition 3.8. At step i , choose $c_i \in \mathbb{R}$ so that $\gamma_i - c_i\{J_i\}$ has degree zero on its contracted ray. All curves in a fiber have class on that same analytic ray. The working spaces have rational singularities after slightly lowering the Cartier floor, and relative vanishing gives rational singularities on the contraction base. Lemma 3.15 therefore descends this class uniquely to the base. Pull it to the positive side and add $c_i\{J_{i+1}\}$, defining γ_{i+1} . The pullback difference is c_i times the ordinary adjoint comparison, hence is an actual real exceptional divisor class. No smooth-start cohomology splitting is needed for this construction.

Telescope these comparisons on a common resolution. The original map $Y \rightarrow S$ was a morphism, and γ was pulled back through it. Since S and M match in codimension one, the resulting divisor is exceptional over both. Each step is an isomorphism over the chosen neighborhood U of the floor. There the class remains the pullback from S ; the corresponding comparison divisor has zero class and is exceptional. Negativity applied to both signs makes it zero there. Thus its support is disjoint from the inverse image of U , as claimed. $\square$

Lemma 3.12 (One nef model for an interval). *Under the hypotheses of Lemma 3.11, let A be any Kähler class on S . There are a normal globally strongly $\mathbb{Q}$ -factorial compact Kähler space V , small bimeromorphic with S , a number $\delta > 0$, and transformed classes $\{J_V\}, A_V$ such that*

$$\{J_V\} + tA_V \text{ is nef for every } 0 \leq t \leq \delta. \quad (3.8)$$

On a common smooth resolution of S and V , the differences of pullbacks of J, A and these transforms lie in the common exceptional span.

Proof. Let $P_M = J_M$ denote the semiample rational line from Lemma 3.11, and use its transform A_M . On a common resolution $p : W \rightarrow S$, $q : W \rightarrow M$, let F_A be the real divisor in the common exceptional span defined by

$$q^*A_M - p^*A = \{F_A\}.$$

It is effective: $-F_A$ is q -nef, because p^*A is nef and the other term is pulled back from M ; apply exceptional negativity. Its support is disjoint from $p^{-1}(U)$. In particular A_M is pseudo-effective,

and its negative multiplicity at every prime of M is zero. Indeed, $q^*A_M = p^*A + \{F_A\}$, and Lemma 2.3 applied over S gives $N(q^*A_M) = F_A$.

Let D_M be the rational Cartier trace of D . Choose $\tau > 0$ so small that $A + \tau\{D\}$ is Kähler. For a fixed sufficiently small $t > 0$, consider on M the boundary $\Delta_M - t\tau D_M$ and the nef b-class represented on W by $tp^*(A + \tau\{D\})$. Its trace on M is $t(A_M + \tau\{D_M\})$, so its generalized adjoint is

$$\{P_M\} + tA_M. \quad (3.9)$$

These are effective gklt data. Over U the b-class descends to M , so its generalized discrepancy contribution is zero there, and subtracting $t\tau D_M$ raises every zero log discrepancy of the dlt pair. Away from the floor the ordinary pair is klt; on one fixed log resolution its finitely many subunit coefficients stay subunit for small t . This proves gklt. The generalized boundary is big. On a common resolution the pullback of the trace of $A + \tau\{D\}$ minus $p^*(A + \tau\{D\})$ is effective exceptional by the same negativity argument as for F_A . The latter pullback is nef and big, so the trace is big; the remaining boundary is effective.

Choose an integer $m > 0$ such that mP_M is generated, and let $g : M \rightarrow Z$ be the Stein factor of its morphism. Then Z is normal projective and $mP_M = g^*H$ for an ample generated Cartier line H on Z . Set $L = \{P_M\} + tA_M$. The preceding preparation makes L an effective gklt adjoint with globally nef b-data and globally modified-big boundary-plus-nef trace. It is pseudo-effective because P_M is nef and A_M is modified nef. Apply the proper relative assertion of Proposition 3.2 over Z . This produces a chosen nonextracting good log terminal model $\phi : M \dashrightarrow V$ and a morphism $g_V : V \rightarrow Z$, with L_V nef over Z . The hypothesis here is properness: the semiample morphism $M \rightarrow Z$ need not be projective. Put

$$mP_V = g_V^*H, \quad A_V = (L_V - \{P_V\})/t.$$

The relative construction preserves the actual line pulled back from Z , and its forward Bott–Chern transport preserves the displayed linear relation; these are the transforms of P_M, A_M .

The map ϕ is small. On a common projective resolution $a : W' \rightarrow M$, $c : W' \rightarrow V$, its comparison is

$$a^*L = c^*L_V + \{E\}, \quad E \geq 0,$$

where E is exceptional over V and has positive coefficient at the strict transform of each prime contracted from M . The forward trace L_V is pseudo-effective, so exceptional translation gives

$$N(a^*L) = N(c^*L_V) + E \geq E.$$

But L is modified nef on M ; the coefficient of $N(a^*L)$ at a strict prime of M is zero. Thus no prime of M is contracted. Nonextraction then proves smallness.

Fix $b > 2dm$. We claim that $L_V + b\{P_V\}$ is absolutely nef. Otherwise the cone theorem for the effective gklt adjoint L_V gives an L_V -negative rational extremal generator C such that

$$(L_V + b\{P_V\}) \cdot C < 0, \quad 0 < -L_V \cdot C \leq 2d.$$

Indeed the cone summand on which L_V is nonnegative also has nonnegative P_V -degree, so a negative ray summand must account for any failure of nefness. If $P_V \cdot C = 0$, ampleness of H makes C vertical over Z , contradicting relative nefness of L_V . Otherwise Cartier integrality gives $P_V \cdot C \geq 1/m$, contradicting

$$(L_V + b\{P_V\}) \cdot C \geq -2d + b/m > 0.$$

This proves the claim. Both $\{P_V\}$ and $(1+b)\{P_V\} + tA_V$ are nef. By convexity,

$$\{P_V\} + sA_V \text{ is nef for } 0 \leq s \leq \frac{t}{1+b}.$$

Thus $\delta = t/(1+b)$ works. The actual line P_V has the same generated multiple as P_M , since both are pulled back from H/m . Its class is the transform of J . The relative comparison and linear transport give exceptional-divisor comparisons for P_M, A_M . Since S, M, V agree in codimension one, their final comparison divisors are exceptional over both S and V , as required. $\square$

3.5 Special termination

Theorem 3.13 (Special termination for a chosen dlt scaling). *Assume $\mathcal{G}_d$ for every $d < n$. Let (X, B) be an effective rational dlt pair on a smooth compact Kähler n -fold, and suppose $L = K_X + B$ is pseudo-effective. There is a Kähler scaling direction ω , with $\{L\} + \omega$ Kähler, and a scaling as in Lemma 3.6, such that after finitely many steps the exceptional, flipping, and flipped loci are disjoint from the reduced floors of the working pairs.*

Proof. We first choose a scaling with strict positive walls and Kähler interior classes. We then reduce the transformations on each log canonical stratum to small diagrams away from its smaller strata. Finally, the single nef interval supplied above rules out infinitely many such diagrams.

Choice of scaling. For every possible finite prefix of the detected ordinary truncation scalings from X , Lemma 3.9 makes the transform of $H^{1,1}(X, \mathbb{R})$ onto the working Bott–Chern space surjective. There are countably many such sequences: curve rays belong to countably many integral homology classes, and a ray determines its contraction and its flip uniquely. On each working space, two distinct curve rays C, C' with nonzero L -degrees give a proper hyperplane of initial directions defined by

$$(L \cdot C)(\omega \cdot C') - (L \cdot C')(\omega \cdot C) = 0,$$

where the degrees use the traces on that working space. Whenever both rays define finite walls, this equation says that their wall parameters agree. Avoid all these hyperplanes, and also avoid $\omega \cdot C = 0$ whenever $L \cdot C = 0$. A general Kähler direction, enlarged to make the initial scaled class Kähler, has these properties.

At a wall the scaled class is pulled back from the contraction base. Immediately after a flip, curves in an opposite-side fiber have positive L -degree. Another negative ray at the same wall would satisfy the excluded wall equation with that opposite-side ray. Immediately after a divisorial contraction, lift any putative new wall curve through the projective morphism. Its lifted ray is distinct from the contracted ray and satisfies the same wall equation; if its L -degree were zero, its ω -degree would also be zero. Both possibilities were excluded. Hence consecutive positive thresholds strictly decrease, and each working interval has nonempty interior. Its interior nef class is big: the class on X is a positive Kähler perturbation of the pseudo-effective class L , and bigness is preserved under the birational comparison. No rational curve can have zero interior degree. An L -negative or L -positive such curve would violate nefness at one of the two nearby parameters; the zero- L case was excluded. The gklt presentation of a positive truncation and [38, Theorem 4.3] therefore make the interior class Kähler. By Lemma 3.6, infinitely many thresholds would tend to zero.

Reduction to small transformations on a stratum. Suppose the sequence is infinite. The homological divisor count [54, Lemma 7.8] discards all divisorial ambient steps after a finite prefix. We next reduce the remaining flips to small transformations on a stratum. For a working ambient pair

(X_i, B_i) , ordinary dlt adjunction on the normalization S of an lc center gives an effective dlt pair (S, Δ_S) and the actual rational-line residue identity

$$K_S + \Delta_S \sim_{\mathbb{Q}} (K_{X_i} + B_i)|_S.$$

Its lc centers are the proper nested lc centers. The coefficients of these differentials form a DCC set depending only on the original coefficients and the chain length. At one restriction they have the form

$$1 - \frac{1}{r} + \frac{\sum_j k_j b_j}{r}, \quad 0 \leq \sum_j k_j b_j \leq 1,$$

which preserves DCC under iteration. These assertions, including the actual residue comparisons, are [54, Lemma 7.16]. Restricting Equation (3.1) along a strict adjunction chain gives an effective comparison on strata, strictly positive for a place whose center maps into the ambient exceptional locus [54, Lemma 7.17]. Both lemmas are dimension-free.

Identify a surviving lc center with its strict transform on the common isomorphism open. There are finitely many lc centers at each stage, and their surviving list stabilizes. A new zero-discrepancy place was already a zero-discrepancy place, and strictness keeps the general point of its center outside the surgery. Starting with the smallest centers, induct on their dimensions. Fix a surviving center and suppose the loci already avoid all its proper subcenters. We will prove that the loci eventually avoid this center as well. The induced diagrams on its normalized strata are then isomorphisms near their non-klt loci. On each stratum, let D_S be the sum of the restrictions of the ambient floor components not used in its generic adjunction chain. Strong $\mathbb{Q}$ -factoriality of the ambient space makes D_S rational Cartier, and the nested-center description gives $\text{Supp } D_S = \text{Supp } [\Delta_S]$. Slightly lowering this effective divisor makes the stratum pair klt.

After a further tail the induced transformations on this stratum are small and their boundary transforms agree. Indeed, a prime extracted in a restricted transformation has discrepancy strictly less than 1 before extraction: its new boundary is effective and strict comparison increases its discrepancy. Its center is away from the unchanged non-klt locus. At the start of this tail there are only finitely many such places. On a fixed compact log resolution, the positive SNC weights $1 - b_j$ have a positive minimum, and the Jacobian inequality bounds every place below the strict cutoff 1; include also the finitely many nonexceptional positive-boundary primes. This is [54, Lemma 7.9], with its lc assertion away from the non-klt locus. Discrepancies do not decrease, so all later extractions belong to this same finite list. Repeated extraction of one place would give a strictly decreasing sequence of its boundary coefficients, contrary to the adjunction DCC property.

Once extractions stop, [54, Lemma 7.8] makes divisorial contractions finite. Its dimension-free count is the dimension of the span of prime divisor cycles in H_{2d-2} of the compact d -dimensional stratum. Restriction to the common isomorphism open uses Borel–Moore homology: removing codimension at least two from the target has no effect in this degree, while a lost prime has nonzero class by its positive Kähler volume. The count strictly drops. The coefficients on the finitely many matched boundary primes then stabilize by monotonicity and DCC. Neither morphism of a restricted diagram can still contract a divisor to the common intermediate space, since strict comparison would change the discrepancy of that matched divisor. The diagrams are therefore small, with identical boundary transforms.

We justify the passage from an isomorphic restricted diagram to absence of ambient surgery near the stratum, also in higher codimension. Its identified boundaries give the same adjunction data, not only isomorphic underlying spaces. Choose a common projective log resolution preserving the general points of the surviving lc centers, and restrict its adjoint comparisons along the strict

adjunction chain described above. The generic point of each surviving center lies outside the surgery: a zero-discrepancy place remains such a place, whereas a center contained in the surgery would acquire strictly larger discrepancy. Thus the strict transform S_W of the normalized stratum is not contained in the comparison support. For a zero-dimensional stratum this already excludes any intersection with the surgery, so suppose its dimension is positive. Its restriction is consequently a well-defined effective divisor. Strict adjunction identifies its class with the difference of the two pulled-back stratum adjoints. When the restricted diagram is an isomorphism with the same boundary, that difference is zero. An effective nonzero divisor on the compact Kähler resolution of S_W has positive mass against a Kähler power, so the restricted divisor is zero.

If an ambient step nevertheless met the stratum, take the first such step. The full-support assertion following Equation (3.1), and surjectivity of S_W onto the stratum, force its comparison support to meet S_W . Noncontainment makes this a nonzero effective restriction, and the cumulative comparison dominates it. This contradicts the vanishing just proved. Earlier steps are isomorphisms on a neighborhood of the stratum by the choice of this first step. The ambient diagram is therefore an isomorphism near the whole stratum. This verifies the support conclusion used in the induction on proper subcenters; for a higher-codimension stratum it is noncontainment in the support, not merely being different from a divisor component, that is needed.

Excluding the remaining walls. It remains to show that only finitely many of these small diagrams are nontrivial. Write S for the first stratum in this last tail and $J = K_S + \Delta$ for its adjoint. Let its restricted original scaling be $\{J\} + \lambda H$. Choose an interior parameter $\lambda_0 > 0$ on this working space, above the remaining walls, and set

$$A = \{J\} + \lambda_0 H.$$

It is Kähler, as the restriction of the interior ambient Kähler class. The exact change of parameter is

$$\{J\} + tA = (1+t)(\{J\} + \lambda H), \quad \lambda = \frac{\lambda_0 t}{1+t}, \quad t = \frac{\lambda}{\lambda_0 - \lambda}. \quad (3.10)$$

It applies for $0 \leq \lambda < \lambda_0$. The new walls t_i strictly decrease toward zero, and the traces $\{J_i\} + tA_i$ are nef on their successive restricted working intervals.

On a common resolution of a nontrivial small stratum step, let D_i be the real exceptional divisor defined by

$$p_i^*\{J_i\} - q_i^*\{J_{i+1}\} = \{D_i\}.$$

The strict adjoint comparison makes D_i an effective nonzero divisor exceptional over both strata. Equality of the scaled pullbacks at its wall gives the exact Bott–Chern equality

$$p_i^*(\{J_i\} + tA_i) - q_i^*(\{J_{i+1}\} + tA_{i+1}) = \left(1 - \frac{t}{t_i}\right) \{D_i\}. \quad (3.11)$$

For t in a later working interval all preceding factors on the right are nonnegative. Telescoping on a common resolution expresses the pullback of $\{J\} + tA$ as the pullback of its nef working trace plus an effective divisor exceptional over that working stratum. Lemma 2.3 identifies the latter divisor with the negative part.

Push the resulting positive currents to one fixed smooth resolution of S and let $t \downarrow 0$. Closedness of the pseudo-effective cone proves that J is pseudo-effective. The negative multiplicities converge valuation by valuation by the limit statement in Lemma 3.7. The comparison divisors have no strict prime of S , since the maps are small. Their coefficients also vanish at every place whose

center meets the floor: each finite composition is unchanged on a neighborhood of the floor by the induction on smaller centers. Passing to the limit proves Equation (3.5).

The stratum has dimension less than n , and the divisor D_S constructed above is rational Cartier with support equal to its floor. Lemmas 3.11 and 3.12 give one small model V on which all transformed classes of $\{J\} + tA$ are nef for $0 \leq t \leq \delta$. On a common smooth resolution $p : W \rightarrow S$, $q : W \rightarrow V$, define the real divisor $F(t)$ in the common exceptional span by

$$p^*(\{J\} + tA) - q^*(\{J_V\} + tA_V) = \{F(t)\}. \quad (3.12)$$

Uniqueness of the representing exceptional divisor makes $F(t)$ coefficientwise affine in t . It is effective on $[0, \delta]$: $-F(t)$ is p -nef because the class pulled back from V is nef, so exceptional negativity applies. It is exceptional over V as well. Lemma 2.3 gives

$$N(p^*(\{J\} + tA)) = F(t) \quad (0 \leq t \leq \delta).$$

Every divisorial negative multiplicity is thus affine on this whole interval. This remains true on higher resolutions by Lemma 2.4.

Choose a nontrivial stratum wall $t_i \in (0, \delta)$. On a common resolution of S, V and the adjacent strata, their nef working models compute the same negative part on their respective open intervals. Their two affine expressions agree at t_i . But Equation (3.11) says that, at a prime with coefficient $d_i > 0$ in D_i , their difference is

$$\left(1 - \frac{t}{t_i}\right) d_i.$$

Their slopes differ by $-d_i/t_i$. A single affine function on $[0, \delta]$ cannot agree with both on two open intervals. This contradiction excludes every such wall. Infinitely many walls would tend to zero, so the nontrivial diagrams on this stratum are finite.

There are finitely many surviving strata. Induction on their dimensions makes all sufficiently late ambient flipping and flipped loci disjoint from the floor. This proves the theorem. $\square$

Corollary 3.14 (A signed adjoint supported on the floor). *Assume $\mathcal{G}_d$ for every $d < n$. In the situation of Theorem 3.13, suppose moreover that L is rationally linearly equivalent, as an actual rational line, to a signed rational divisor supported on $[B]$. The scaling can be chosen to reach a nef ordinary dlt model after finitely many steps.*

Proof. The signed representation persists under codimension-one pushforward. A curve disjoint from the transformed floor has degree zero for each component in this representation, and hence for the transformed adjoint. Every negative operation must therefore meet the floor. After the finite prefix supplied by Theorem 3.13 no further negative operation is possible. The continuation in Lemma 3.6 makes the last adjoint nef. $\square$

3.6 Polarizations, ordinary replacements, and descent

The program constructions below use the analytic cone $\overline{NA}(T)$, dual to the nef cone in $H_{BC}^{1,1}(T, \mathbb{R})$ for spaces with rational singularities [18, Proposition 2.4]. Degrees of global line bundles form a possibly smaller numerical space. We pass from the analytic cone to projective geometry by producing an actual relatively ample line. The ensuing descent statements apply on singular working models as well as on smooth initial spaces.

Lemma 3.15 (Degree-zero Bott–Chern descent). *Let $f : T \rightarrow Z$ be a proper surjective morphism with connected fibers between normal compact complex spaces with rational singularities. Suppose either*

(i) f is bimeromorphic and both spaces belong to Fujiki's class $\mathcal{C}$; or

(ii) f is projective and an effective rational boundary Δ makes (T, Δ) klt with $-(K_T + \Delta)$ relatively nef and big.

Pullback is injective on both $H^2(-, \mathbb{R})$ and $H_{\text{BC}}^{1,1}(-, \mathbb{R})$. In either group its image consists exactly of the classes with degree zero on every curve contracted by f .

Proof. These are, respectively, the two cases of [17, Lemma 2.6(1),(2)]. The first case is birational descent between spaces with rational singularities and does not require a projectivity criterion. The second case uses relative vanishing for the ordinary rational klt pair and the projective relative curve space. In particular, it can be applied to a projective log-Fano contraction before constructing generalized data on its base. $\square$

In applications to an ordinary birational negative contraction, the rationality hypotheses are automatic. The source is locally klt. For a dlt pair, first decrease its rational Cartier floor slightly; relative antiampleness persists. Relative vanishing gives $R^i f_* \mathcal{O}_T = 0$ for $i > 0$ [31, Theorem 5.2]. Composing a resolution of T with f , the Leray spectral sequence proves that Z has rational singularities. The same reasoning works when the klt boundary is chosen separately on finitely many Stein base neighborhoods. For fiber-type contractions in case (ii), rationality of the base follows from [18, Lemma 8.8(i)]; thus that application also precedes the construction of generalized data on the base.

Lemma 3.16 (Projectivity from a rational line detecting the ray). *Let $f : T \rightarrow Z$ be a proper contraction between normal compact Kähler spaces, with T having rational singularities. Suppose γ is Kähler on Z and*

$$\alpha = f^* \gamma, \quad \overline{\text{NA}}(T) \cap \alpha^\perp = R$$

is a nonzero ray. If a global rational line bundle D has $D \cdot R < 0$, then $-D$ is relatively ample and f is projective.

Proof. Fix a Kähler class ω on T and normalize $\overline{\text{NA}}(T)$ by $\omega \cdot z = 1$. The resulting slice S is compact. Its unique point on R has negative D -degree, so $d = c_1(D)$ is negative on a neighborhood of that point in S . On the complementary compact set, α has a strictly positive minimum, while d is bounded. For all sufficiently small $s > 0$, the class $\alpha - sd$ is therefore strictly positive on S and is Kähler by cone duality.

Choose an integer $m > 0$ making mD integral. Then

$$c_1(-mD) + f^*((m/s)\gamma) = (m/s)(\alpha - sd)$$

is Kähler. On a neighborhood in the base with a potential for γ , absorb that potential in a smooth metric on the same global line bundle $-mD$. Its curvature is positive on the fibres. The analytic positive-line-bundle criterion and the fibrewise relative-ampleness criterion make $-mD$ relatively ample. The line bundle was global throughout, so no gluing of unrelated local polarizations is required. $\square$

An analytic contraction supplied by a supporting class therefore becomes projective as soon as one rational line has negative degree on its ray. For a small contraction, that same line will define the flip globally. The local ordinary adjoints used to prove finite generation need not glue; their section algebras will be Veronese subalgebras of one algebra.

Lemma 3.17 (Divisor representatives over a Stein base). *Let $\pi : Y \rightarrow U$ be a projective surjective morphism with connected fibres between normal irreducible complex spaces, where U is Stein, and let $\mathcal{L}$ be a holomorphic line bundle on Y . If π is bimeromorphic, $\mathcal{L}$ has a holomorphic section which is not identically zero, and hence an effective Cartier divisor representative. More generally, after shrinking U around any specified point, $\mathcal{L}$ has a nonzero meromorphic section.*

Proof. Suppose first that π is bimeromorphic. Grauert's proper direct-image theorem makes $\mathcal{F} = \pi_*\mathcal{L}$ coherent. On the nonempty open subset U° where π is an isomorphism, it is a line bundle. Choose $u_* \in U^\circ$. Cartan's theorem A says that global sections generate $\mathcal{F}_{u_*}$, so one global section has nonzero image in $\mathcal{F}_{u_*}/\mathfrak{m}_{u_*}\mathcal{F}_{u_*}$. Under

$$H^0(U, \pi_*\mathcal{L}) = H^0(Y, \mathcal{L})$$

it gives a section s nonzero at the unique point above u_* . Its zero divisor is effective Cartier and represents $\mathcal{L}$. The section may vanish elsewhere; only its meromorphic inverse is being used to obtain a meromorphic trivialization.

For the general case, fix $u \in U$ and $y \in \pi^{-1}(u)$, and choose a π -ample line bundle $\mathcal{H}$. Relative Serre generation, after shrinking to a Stein neighbourhood of u , gives an integer n for which both $\mathcal{L} \otimes \mathcal{H}^n$ and $\mathcal{H}^n$ are relatively generated. Their proper direct images are coherent. Cartan's theorem A and the evaluation maps supply sections s and t of these two bundles which are both nonzero at y . The quotient s/t is the required nonzero meromorphic section of $\mathcal{L}$. At no point is Y assumed Stein. $\square$

Proposition 3.18 (A global algebra for a detected flip). *Let $(T, B + \mathbf{M})$ be a gklt pair on a normal compact Kähler space which is globally strongly $\mathbb{Q}$ -factorial, and let $A \in H_{\text{BC}}^{1,1}(T, \mathbb{R})$ be its adjoint class. Let $f : T \rightarrow Z$ be a projective small contraction onto a normal compact Kähler space. Suppose that the classes of all f -vertical curves lie on one ray $R \subset \overline{\text{NA}}(T)$, that $A \cdot R < 0$, and that a global line bundle L satisfies $L \cdot R < 0$.*

*Then $\bigoplus_{m \geq 0} f_*L^m$ is locally finitely generated. The relative Proj of a sufficiently divisible Veronese is a projective small contraction $f^+ : T^+ \rightarrow Z$, where T^+ is compact Kähler and globally strongly $\mathbb{Q}$ -factorial. The reflexive transform L^+ is a rational line bundle, with a positive power equal to the relatively ample tautological bundle. The transformed boundary and the same b-nef datum define a gklt pair on T^+ , with relatively Kähler adjoint A^+ . This is the generalized flip, with the usual strict discrepancy increase at centres in either exceptional locus.*

Moreover, if $t = (A \cdot C)/(L \cdot C) > 0$ for an f -vertical curve C , then there is a unique $\eta \in H_{\text{BC}}^{1,1}(Z, \mathbb{R})$ such that

$$A = t c_1(L) + f^*\eta, \quad A^+ = t c_1(L^+) + (f^+)^*\eta.$$

Every class on T has the corresponding forward transport with an actual exceptional-divisor correction. No assertion of surjectivity on Bott–Chern groups is required.

Proof. Since f has a global relatively ample bundle and all its vertical curves lie on R , numerical invariance of ampleness on the projective fibres shows that $-L$ is f -ample.

A local rational ordinary adjoint. Fix $z \in Z$ and a relatively compact Stein neighbourhood U . Take a projective log resolution $p : V \rightarrow T_U$ carrying $\mathbf{M}$, and put $\rho = fp$. Write its actual structure boundary as B_V , so

$$[K_V + B_V] + [\mathbf{M}_V] = p^*A.$$

Every coefficient of B_V is less than one. Apply Lemma 3.17 to obtain a Cartier representative D_L of L on T_U . The same birational direct-image argument applies to the coherent rank-one reflexive canonical sheaf: it supplies a canonical form nonzero at a chosen point of the common smooth isomorphism locus. Use this meromorphic form on T_U and its pullback to V to choose compatible canonical representatives, and set

$$N = tp^*D_L - K_V - B_V.$$

For every ρ -vertical curve, its degree equals that of $\mathbf{M}_V$. Thus N is ρ -nef.

The relative bigness of N uses the additional fact that $\rho = f \circ p$ is *bimeromorphic*: both the small contraction f and the resolution p are bimeromorphic. It does not follow from relative nefness alone. In fact every real Cartier divisor is relatively big for this projective bimeromorphic map over the Stein neighbourhood. Here is the required ample-plus-effective decomposition explicitly. Write $N = \sum_j r_j N_j$, where $r_j > 0$, $\sum_j r_j = 1$, and the N_j are rational Cartier divisors in its finite Cartier span. Choose a ρ -ample line bundle and a Cartier representative P for it using Lemma 3.17. If $q_j N_j$ is integral, apply the birational case of that lemma to $\mathcal{O}_V(q_j N_j - P)$. Its nonzero holomorphic section has an effective Cartier zero divisor E_j , giving the actual relation $q_j N_j \sim P + E_j$. This use depends on ρ being bimeromorphic; the section is allowed to vanish along exceptional fibres. Therefore

$$N \sim_{\mathbb{R}} \underbrace{\left(\sum_j \frac{r_j}{q_j} \right)}_H P + \underbrace{\sum_j \frac{r_j}{q_j} E_j}_E.$$

Here H is ρ -ample and $E \geq 0$, proving ρ -bigness. For a map with positive-dimensional general fibres the direct image used above can be zero; no such bigness claim is made for that setting. Make the finitely many section choices on a slightly larger Stein neighbourhood, and now shrink U so its closure is compact in that neighbourhood. Properness makes the inverse image of this closure compact. A log resolution of the finitely many resulting divisors therefore gives a uniform positive bound for the following choice. Choose $\epsilon > 0$ small enough that $(V, B_V + \epsilon E)$ is sub-plt. The divisor $(1 - \epsilon)N + \epsilon H$ is relatively ample. Express it as a positive combination of rational ample divisors and take general members of sufficiently high multiples. Relative Bertini [18, Theorem 2.20], applied on a log resolution of $B_V + E$, gives an effective $\Theta \sim_{\mathbb{R}} N$ such that $(V, B_V + \Theta)$ is sub-plt. The high multiples make the added coefficients arbitrarily small. All choices are made near the compact fibre; shrink U once to retain them.

Put $\Delta = p_*(B_V + \Theta) \geq 0$. Write the actual linear equivalence as $\Theta = N + \sum_j a_j \operatorname{div}_V(g_j)$. The bimeromorphic map p identifies meromorphic function fields, so $g_j = p^*h_j$. Pushing down gives $K_{T_U} + \Delta = tD_L + \sum_j a_j \operatorname{div}_{T_U}(h_j)$; in particular this divisor is real Cartier. Pulling back this same identity gives

$$K_{T_U} + \Delta \sim_{\mathbb{R}} tD_L, \quad p^*(K_{T_U} + \Delta) = K_V + B_V + \Theta.$$

Thus (T_U, Δ) is plt. Any finitely many base line bundles in a relative linear equivalence can first be trivialized by shrinking U . Record the resulting equality as

$$K_{T_U} + \sum_i d_i D_i = tD_L + \sum_j b_j \operatorname{div}(g_j), \quad d_i > 0, \quad (3.13)$$

using the positive support of Δ . After a further relatively compact shrinking, the displayed divisors have finitely many prime components: their locally finite supports meet a compact inverse image.

Equality of coefficients in (3.13) is a finite rational affine system in (d_i, t, b_j) . The klt condition is open within this system, as seen on one log resolution; the pullbacks of its adjoints vary linearly by the displayed identity. A nearby rational solution therefore gives a rational effective klt boundary Δ_q and $t_q \in \mathbb{Q}_{>0}$ with

$$K_{T_U} + \Delta_q \sim_{\mathbb{Q}} t_q D_L. \quad (3.14)$$

The individual local primes D_i need not be $\mathbb{Q}$ -Cartier: the identity ensures that the total adjoint is $\mathbb{Q}$ -Cartier.

One global flip algebra. The adjoint in (3.14) is f_U -antiample. Its ordinary flip and local finite generation follow from [31, Theorems 1.14 and 1.18]. Clearing the displayed linear equivalence identifies a Veronese of its canonical algebra with a Veronese of $\mathcal{R} := \bigoplus_{m \geq 0} f_* L^m$ over U . Hence $\mathcal{R}$ is locally finitely generated by [31, Lemma 2.26]. The graded pieces are coherent. Over a connected normal base open the inverse image is irreducible; products of nonzero sections of line bundles are nonzero there. Thus its graded section algebra is an integral domain, without choosing any global meromorphic frame of L . Compactness supplies one sufficiently divisible Veronese $\mathcal{R}^{(r)}$ generated in degree one. Its relative Proj T^+ restricts to the ordinary flip over every such U . It is therefore normal, locally klt, and small over Z , and has the global relatively ample bundle $\mathcal{O}_{T^+}(1)$. On the common big open this bundle is L^r . Reflexivity gives

$$(L^+)^{[r]} \simeq \mathcal{O}_{T^+}(1).$$

This proves that L^+ is a rational line bundle before any assertion of factoriality. Relative positivity over the compact Kähler base makes T^+ compact Kähler.

For any global rank-one reflexive sheaf $\mathcal{F}^+$ on T^+ , take its coherent reflexive transform $\mathcal{F}$ on T using a common resolution. Some $M = \mathcal{F}^{[m]}$ is a line bundle. Choose $c \in \mathbb{Q}$ so that $(M + cL) \cdot C = 0$, and clear its denominator to obtain an integral bundle $J = n(M + cL)$ numerically trivial over f . The local rational klt pairs (3.14) have antiample adjoint, so Lemma 3.3 yields one global line bundle J_Z with $J = f^* J_Z$. On T^+ , pull back J_Z , undo the rational twist ncL^+ , and compare on the common big open. After clearing the already established index of L^+ , reflexivity gives an invertible positive reflexive power of $\mathcal{F}^+$. This proves global strong factoriality.

The original generalized pair. The local log-Fano pairs above give $R^i f_* \mathcal{O}_T = 0$ for $i > 0$ by [31, Theorem 5.2]. They make T locally klt, hence rational. Leray for a projective resolution then shows that Z is rational. Lemma 3.15(i) therefore gives $f^* H_{\text{BC}}^{1,1}(Z, \mathbb{R}) = R^\perp$, with injective pullback. Thus $A = tc_1(L) + f^* \eta$ for a unique η . Define $A^+ = tc_1(L^+) + (f^+)^* \eta$; it is relatively Kähler.

To construct its actual discrepancy divisor, let $\mathcal{I}$ be the ideal defined by

$$\text{im}(f^* f_* L^r \rightarrow L^r) = \mathcal{I} \otimes L^r.$$

Principalize $\mathcal{I}$ on a common resolution $p : W \rightarrow T$, $q : W \rightarrow T^+$ carrying $\mathbf{M}$. If $\mathcal{I}\mathcal{O}_W = \mathcal{O}_W(-F_L)$, the tautological quotient gives the actual bundle identity

$$q^* \mathcal{O}_{T^+}(1) = p^* L^r \otimes \mathcal{O}_W(-F_L).$$

Hence $E_L = F_L/r$ is effective and exceptional over both sides, and $[E_L] = p^* c_1(L) - q^* c_1(L^+)$. If B_W is the original structure boundary, put $B_W^+ = B_W - tE_L$. Then

$$[K_W + B_W^+] + [\mathbf{M}_W] = q^* A^+.$$

Its pushforward is B^+ , and its discrepancies do not decrease. The local ordinary comparison in (3.14) has discrepancy divisor $t_q E_L$, so its strictness proves the stated strictness for tE_L as well. Thus the original pair remains gklt.

Finally, write any $\theta \in H_{\text{BC}}^{1,1}(T, \mathbb{R})$ uniquely as $\theta = f^*\eta_\theta + sc_1(L)$ and set $\theta^+ = (f^+)^*\eta_\theta + sc_1(L^+)$. Its correction is the actual divisor sE_L . For a rational line M , the number s is rational. Applying Lemma 3.3 to a Cartier multiple of $M - sL$ gives $M = sL + f^*M_Z$ as actual rational lines. Reflexive extension on the common big open gives $M^+ = sL^+ + (f^+)^*M_Z$. Consequently the cohomological transport agrees with the reflexive transform on Chern classes. $\square$

3.6.1 Real boundaries and their linear transport

The rational ordinary step constructed below also determines the step for every sufficiently close real boundary. The relevant comparison is an identity of actual rational lines over the contraction base.

Proposition 3.19 (Ordinary real boundaries). *Let T be normal compact Kähler and globally Weil $\mathbb{Q}$ -factorial, with canonical sheaf a rational line bundle. Let $B \geq 0$ be a real boundary with (T, B) klt. Assume the rational ordinary-step result of Proposition 3.48 for T and its subsequent models. Then every $(K_T + B)$ -negative extremal analytic ray has an ordinary step with projective contractions, compact Kähler models, and the expected opposite relative ample signs. Global Weil $\mathbb{Q}$ -factoriality is preserved; global strong $\mathbb{Q}$ -factoriality is preserved when imposed initially.*

Proof. Put $D = K_T + B$ and fix a D -negative extremal ray R . On the finite positive support of B , effectiveness, the klt condition and negativity on R persist under small coefficient changes. Klt openness is checked on one log resolution of this support. Choose nearby effective rational klt boundaries $B_0, \dots, B_m$ and positive real numbers u_j with

$$B = \sum_j u_j B_j, \quad \sum_j u_j = 1, \quad D_j \cdot R < 0, \quad D_j = K_T + B_j.$$

For $B = 0$ take just $B_0 = 0$. Every D_j is a global rational line bundle. The rational-step theorem for (T, B_0) provides a projective contraction $f : T \rightarrow Z$ onto a compact Kähler base, contracting exactly R , with $-D_0$ relatively ample. The base is rational by [18, Lemma 8.8], so Lemma 3.15 gives $f^*H_{\text{BC}}^{1,1}(Z, \mathbb{R}) = R^\perp$; use case (ii) when f is of fiber type. Let C be a rational curve spanning R , and set $a_j = (D_j \cdot C)/(D_0 \cdot C) \in \mathbb{Q}_{>0}$. On every projective fibre, $-D_j$ is numerically equivalent to the positive multiple $-a_j D_0$. Numerical invariance of ampleness on that fibre, followed by the relative-ampleness criterion, makes $-D_j$ relatively ample. The positive combination $-D$ is relatively ample as a real line bundle. This also handles a fibre-type contraction.

Suppose henceforth that f is birational. A Cartier multiple $M_j = n_j(D_j - a_j D_0)$ is numerically trivial over Z and $M_j - D_0$ is relatively ample. Lemma 3.3, applied to the rational klt pair (T, B_0) gives a rational line bundle A_j on Z with the actual identity

$$D_j = a_j D_0 + f^* A_j. \tag{3.15}$$

Here and below an identity of rational line bundles means an isomorphism after a common integral multiple. The descent lemma is used only for birational f .

For a small f , let $f^+ : T^+ \rightarrow Z$ be its rational D_0 -flip. The rational theorem supplies a global relatively ample adjoint D_0^+ and preserves the stated factoriality and canonical-sheaf conditions. The other transformed adjoints D_j^+ are therefore rational line bundles. Transform (3.15) on the common big open and extend by reflexivity to obtain

$$D_j^+ = a_j D_0^+ + (f^+)^* A_j.$$

All D_j^+ are relatively ample, as is $D^+ = \sum_j u_j D_j^+$. Thus this same small map is the required ordinary real-boundary flip. The real ordinary discrepancy comparison proves that (T^+, B^+) is klt and gives the strict increases at exceptional centres.

For a divisorial contraction, write $D_0 = f^*D'_0 + e_0E$ with $e_0 > 0$. Transforming (3.15) on the target and summing yields

$$D - f^*D' = \left(\sum_j u_j a_j \right) e_0 E.$$

This coefficient is positive. The same discrepancy comparison proves the required assertions. The ambient category was already preserved by the rational step in both cases. $\square$

For later use, let $D = K_T + B$ drive one of the birational steps just constructed, and write $f' : T' \rightarrow Z$ for its positive morphism (or the identity for a divisorial contraction). Lemma 3.15 gives the unique decomposition and forward transform

$$\theta = f^*\eta + s\{D\}, \quad \theta' = (f')^*\eta + s\{D'\}. \quad (3.16)$$

On a common resolution $p : W \rightarrow T$, $q : W \rightarrow T'$, the ordinary comparison $p^*D - q^*D' = F$ is an actual effective real divisor, exceptional over T' and over both sides for a flip. Hence $p^*\theta - q^*\theta' = s\{F\}$. On Chern classes this is the reflexive transform, by the line identities in the proof above and Lemma 3.3. This construction gives forward transport without assuming surjectivity onto the next Bott–Chern group.

3.6.2 Actual real lines and relative numerical spaces

A second polarization argument will be used for maps whose fibers are Moishezon. It applies to a real combination of global line bundles, provided that its degrees on the fibers agree with a relatively Kähler class.

Lemma 3.20 (Polarization by an actual real line bundle). *Let $f : T \rightarrow Z$ be a proper Moishezon map, where T is a compact Kähler space. Suppose that $L \in \text{Pic}(T) \otimes \mathbb{R}$ and a relatively Kähler class ω satisfy*

$$L \cdot C = \omega \cdot C \quad \text{for every compact curve } C \text{ contracted by } f.$$

Then L is relatively ample and f is projective.

Proof. Let V be an irreducible positive-dimensional subspace of a reduced fibre. It is Moishezon. Normalize V and take a smooth projective modification $\pi : \tilde{V} \rightarrow V^n$, chosen so that an effective exceptional divisor E has $-E$ relatively ample. The pullback of the Kähler class from V to its normalization is Kähler. Consequently $\pi^*\omega - \delta[E]$ is Kähler for sufficiently small $\delta > 0$. On the projective manifold $\tilde{V}$ the actual real line bundle $\pi^*L - \delta E$ has the same curve degrees as that Kähler class, and is therefore ample. For example, subtract a sufficiently small positive multiple of a fixed ample class from the Kähler class; the corresponding real line bundle is nef by the projective numerical criterion. Also π^*L is nef, and the decomposition

$$\pi^*L = (\pi^*L - \delta E) + \delta E$$

shows that it is big. Thus

$$L^{\dim V} \cdot V = (\pi^*L)^{\dim V} > 0.$$

The real Nakai–Moishezon criterion for proper algebraic spaces [35, Theorem 1.6], applied to the algebraizations of the Moishezon fibres, proves that L is ample on every reduced fibre. The criterion is insensitive to nilpotents and reducible components. In the coefficient space of the finitely many global line bundles occurring in L , choose, for each fibre, a rational simplex containing L in its interior whose vertices restrict to ample rational lines on that fibre. Openness of fibrewise ampleness for each of these finitely many rational lines gives a base neighbourhood on which the entire simplex is relatively ample. Choose finitely many such neighborhoods covering the base and intersect their coefficient neighborhoods of L . Every rational point of the resulting neighborhood is relatively ample globally. A rational simplex in this intersection also expresses L as a positive combination of relatively ample rational lines. Clearing the denominator of one rational point proves projectivity. $\square$

Remark 3.21. The proof uses only positivity of the top self-intersections of L . Equality of curve degrees with ω does not assert equality of their higher intersection numbers or of their Bott–Chern classes.

The projective morphism needed for the relative program can be chosen on a smooth model of the rational quotient. Here the Hodge-theoretic polarization has an elementary construction, independent of any birational projectivity-descent assertion.

Lemma 3.22 (Smooth Hodge-theoretic polarization). *Let $g : W \rightarrow S$ be a surjective morphism with connected fibers between smooth compact Kähler manifolds. If $g^* : H^0(S, \Omega_S^2) \rightarrow H^0(W, \Omega_W^2)$ is an isomorphism, then g is projective. This applies when its general fiber is rationally connected.*

Proof. We use the smooth argument of [15, Theorem 3.1, Step 1]. Let $d = \dim W - \dim S$. Choose a rational class $u \in H^2(W, \mathbb{Q})$ close to a Kähler class. Write its Hodge decomposition as $u = b + g^*v$, where b is Kähler and v has types $(2, 0)$ and $(0, 2)$. Hodge type and the projection formula give

$$c = g_*(u^d) = g_*(b^d) > 0, \quad g_*(u^{d+1}) = g_*(b^{d+1}) + (d+1)cv.$$

Since pushforward is rational, the class

$$\ell = u - g^* \frac{g_*(u^{d+1})}{(d+1)c} = b - g^* \frac{g_*(b^{d+1})}{(d+1)c}$$

is rational and of type $(1, 1)$. Lefschetz’s theorem makes it the Chern class of a rational line. The displayed equality supplies a metric with positive curvature locally over S , by adding a potential for the base class. Thus this line is relatively ample. For rationally connected general fibers, holomorphic two-forms are pulled back from the smooth base, so the hypothesis holds. $\square$

Lemma 3.23 (A rational Hodge correction over a smooth base). *Let $f : T \rightarrow S$ be an already projective surjective morphism of normal compact Kähler spaces. Assume that S is smooth, T has globally strongly $\mathbb{Q}$ -factorial klt singularities, and, for a projective resolution $r : W \rightarrow T$, the morphism $g = f \circ r$ satisfies*

$$g^* : H^0(S, \Omega_S^2) \xrightarrow{\simeq} H^0(W, \Omega_W^2).$$

Then every $\alpha \in H_{\text{BC}}^{1,1}(T, \mathbb{R})$ can be written

$$\alpha = c_1(L) + f^*\gamma,$$

where $L \in \text{Pic}(T) \otimes_{\mathbb{Z}} \mathbb{R}$ is a finite real combination of global line bundles and $\gamma \in H^{1,1}(S, \mathbb{R})$.

Proof. The composition g is projective, since all the spaces are compact. Choose a g -ample line bundle on W , with integral Chern class H . Set $d = \dim W - \dim S$ and

$$c = g_*(H^d) > 0.$$

Thus c is the positive degree of H^d on a general fibre. The cohomological pushforward here is the usual pushforward between the smooth compact manifolds W and S , defined over $\mathbb{Q}$ and of Hodge bidegree $(-d, -d)$.

Define a rational linear operator on degree-two cohomology by

$$\Pi(\eta) = \eta - g^*\left(\frac{g_*(\eta \smile H^d)}{c}\right).$$

Every class of type $(2, 0)$ on W is pulled back from S , and the same holds for type $(0, 2)$. The projection formula therefore shows that Π kills both types. It preserves type $(1, 1)$. Consequently

$$\Pi(H^2(W, \mathbb{Q})) \subset H^2(W, \mathbb{Q}) \cap H^{1,1}(W).$$

By the Lefschetz $(1, 1)$ theorem the image consists of rational Chern classes of global line bundles. Applying Π to $r^*\alpha$ gives

$$r^*\alpha = c_1(L_W) + g^*\gamma, \quad L_W \in \text{Pic}(W) \otimes \mathbb{R}, \quad \gamma = \frac{g_*(r^*\alpha \smile H^d)}{c} \in H^{1,1}(S, \mathbb{R}).$$

For example, express $r^*\alpha$ in a rational basis of $H^2(W, \mathbb{Q})$ and apply Π to each basis vector. This gives the required finite real combination L_W .

Write $L_W = \sum_j a_j L_j$ with actual line bundles L_j on W . For each j , let

$$\mathcal{F}_j = (r_* L_j)^{**}.$$

It is a rank-one reflexive coherent sheaf. Strong $\mathbb{Q}$ -factoriality gives an integer $m_j > 0$ for which $M_j = \mathcal{F}_j^{[m_j]}$ is a line bundle on T . The canonical identification over the isomorphism locus of r gives

$$L_j^{\otimes m_j} \simeq r^* M_j \otimes \mathcal{O}_W(E_j)$$

for an integral r -exceptional divisor E_j . One may obtain this comparison directly from the evaluation map for $r_* L_j$: the two coherent rank-one sheaves agree away from the exceptional locus, so their invertible transforms differ by an exceptional divisor. No global meromorphic frame of L_j or of the canonical sheaf is required.

Set $L = \sum_j (a_j/m_j) M_j$ and $E = \sum_j (a_j/m_j) E_j$. The preceding identities yield

$$r^*(\alpha - c_1(L) - f^*\gamma) = [E].$$

The divisor E is r -exceptional and is numerically trivial on every r -contracted curve. The exceptional negativity lemma applied in both signs gives $E = 0$. Injectivity of pullback by a resolution, or pushforward of the equality of currents, now gives $\alpha = c_1(L) + f^*\gamma$. $\square$

The hypothesis of Lemma 3.23 persists on each projective globally strongly $\mathbb{Q}$ -factorial klt model over this fixed smooth base: on a common smooth resolution, holomorphic two-forms are invariant under modifications. Thus the lemma applies separately on every working model. It supplies real combinations of global lines; Lemma 3.17 supplies their Cartier-divisor representatives on sufficiently small Stein base neighborhoods.

Lemma 3.24 (The projective relative cone in Bott–Chern cohomology). *Let $f : T \rightarrow S$ be projective between normal compact Kähler spaces with rational singularities, and assume*

$$H_{\text{BC}}^{1,1}(T, \mathbb{R}) = c_1(\text{Pic}(T) \otimes \mathbb{R}) + f^* H_{\text{BC}}^{1,1}(S, \mathbb{R}).$$

Let $N_1^{\text{line}}(T/S)$ denote the space of real combinations of f -contracted curves modulo degrees of global line bundles, and let $\overline{\text{NE}}^{\text{line}}(T/S)$ be their closed effective cone. Their analytic classes give an isomorphism onto the span of the face

$$\mathcal{F}_f = \overline{\text{NA}}(T) \cap (f^* \omega_S)^\perp,$$

where ω_S is any Kähler class on S ; this isomorphism identifies $\overline{\text{NE}}^{\text{line}}(T/S)$ with $\mathcal{F}_f$. In particular, relative extremal rays in the line space are extremal rays of the full analytic cone.

Proof. The assumed decomposition makes analytic equivalence and line-degree equivalence identical on combinations of vertical curves. Their map into the analytic numerical space is therefore well-defined and injective. We must show that its closed effective cone is the entire face, including its classes represented by positive currents.

Fix $z \in \mathcal{F}_f$. For every $\gamma \in H_{\text{BC}}^{1,1}(S, \mathbb{R})$, both $a\omega_S + \gamma$ and $a\omega_S - \gamma$ are Kähler for large a . Positivity then shows that z annihilates $f^*\gamma$. Fix a global relatively ample line H . If a real line D has nonnegative degree on every vertical curve, the projective relative numerical criterion makes $D + \epsilon H$ relatively ample for every $\epsilon > 0$. Its Chern class becomes Kähler after adding a sufficiently large multiple of $f^*\omega_S$. Therefore

$$(c_1(D) + \epsilon c_1(H)) \cdot z \geq 0, \quad c_1(D) \cdot z \geq 0.$$

Applying this to both signs of a relatively numerically trivial line shows that pairing with z factors through the finite-dimensional space of relative line degrees. It is nonnegative on its nef cone. By duality for this projective relative space, it is represented by an element of $\overline{\text{NE}}^{\text{line}}(T/S)$. The decomposition in the statement makes its analytic image equal to z . Conversely every vertical effective curve lies in $\mathcal{F}_f$, and finite-dimensional injectivity preserves closedness. This proves the cone equality and the extremality claim. $\square$

3.7 Good models for already projective relative adjoints

Proposition 3.25 (Projective relative completion). *Let $f : X \rightarrow S$ be an already projective surjective morphism of normal compact Kähler spaces, with X smooth. Let $(X, B + \mathbf{M})$ be an effective generalized klt pair with adjoint A , carried by a projective resolution $p : W \rightarrow X$ with smooth W , so that*

$$p^*A = c_1(K_W) + [B_W] + \mathbf{M}_W.$$

Suppose that there are an actual real line bundle $J \in \text{Pic}(X) \otimes \mathbb{R}$ and $\beta \in H_{\text{BC}}^{1,1}(S)$ such that

$$A = c_1(J) + f^*\beta, \quad N := p^*J - K_W - B_W \text{ is nef and big over } f p.$$

Assume that J is pseudoeffective over S . Then there is a finite chosen A -negative projective program over S , with globally strongly $\mathbb{Q}$ -factorial compact Kähler working spaces, and a nonextracting endpoint X_m such that

$$J_m \sim_{\mathbb{R}} g^*H, \quad A_m = g^*(c_1(H) + \rho^*\beta).$$

Here J_m is the actual real-line transform of J , $g : X_m \rightarrow Z$ is a projective connected-fibre morphism, $\rho : Z \rightarrow S$ is projective, and H is an actual real line bundle ample over S . Thus the class on Z in this formula is relatively Kähler over S . The generalized pair on X_m is gklt. On a common resolution the comparison with A is effective and exceptional over X_m , with strictly positive coefficient at the strict transform of every prime contracted from X .

Proof. We reduce to fixed ordinary pairs, and construct one global program. Choose finitely many Stein open sets $U_\ell \subset S$ and semianalytic Stein compact subsets $W_\ell \subseteq U_\ell$ whose interiors cover S . Such compacts are obtained by intersections with closed polydiscs in local embeddings of S . In particular they satisfy condition (P) of [31]: the source is normal, the base is Stein, the compact is Stein, and its intersection with any analytic subset defined near it has finitely many connected components. This remains true on every normal projective model. Shrinking U_ℓ around W_ℓ preserves the finite cover.

Actual ordinary replacements. On each chart represent the finitely many line bundles in the data by Cartier divisors. For a projective map to a Stein space this can be done by twisting a line bundle and its prospective meromorphic frame by sufficiently positive relative lines and using nonzero sections. The generic-rank-one direct-image argument is an alternative only when the map is bimeromorphic. Relative bigness gives $N \sim_{\mathbb{R}} P + E$, with P relatively ample and $E \geq 0$. For sufficiently small $\epsilon > 0$, $(W, B_W + \epsilon E)$ is sub-klt near the compact inverse image. The line $(1 - \epsilon)N + \epsilon P$ is relatively ample. General divided relative sections give an effective representative T such that

$$\Theta := \epsilon E + T \sim_{\mathbb{R}} N, \quad (W, B_W + \Theta) \text{ is sub-klt.}$$

All supports and coefficient margins are fixed near that compact. With compatible canonical representatives, push the finite principal-divisor expression for $\Theta - N$ down and pull it back again. For $\Delta_\ell = p_*(B_W + \Theta)$ this gives

$$\Delta_\ell \geq 0, \quad K_X + \Delta_\ell \sim_{\mathbb{R}} J|_{X_{U_\ell}}, \quad p^*(K_X + \Delta_\ell) = K_W + B_W + \Theta.$$

In the last equality the chosen principal correction is included in the representative of $K_X + \Delta_\ell$. Thus $(X_{U_\ell}, \Delta_\ell)$ is ordinary klt and its boundary is relatively big. Lemma 11.15 of [31], with its big part equal to Δ_ℓ and its remaining part zero, replaces it, in its actual real linear equivalence class, by a klt boundary with an effective general relatively ample rational summand. Its condition on non-klt centres is vacuous. Choose the reserved ample summand to be $\mathbb{Q}$ -linearly equivalent to a small rational multiple of a fixed global f -ample line. Its later strict transform is consequently $\mathbb{Q}$ -Cartier by the transport of that global line, without a local factoriality claim.

We make these choices for a finite rational polytope of actual global line data. Start with a finite rational span containing J , and add relatively ample rational lines spanning the global relative degree space. The above construction is valid in a neighbourhood of J : retain a small part of the ample summand and use openness of ampleness and the strict klt inequalities. It is also valid along a segment from J to a sufficiently positive adjoint, by adding divided general ample divisors. Use finitely many such representatives and one log resolution on each chart. More explicitly, represent a finite simplex inside the open ample neighbourhood used above; varying its positive coefficients represents all nearby global line parameters on the same finite support. Form the joint finite affine system consisting of the divisor identities on all charts, with common global line coordinates and separate boundary and principal-divisor coordinates. This system is rational. Allow all boundary coefficients to vary, including those originally contributed by the real boundary B ; fix only the reserved rational ample part and the zero-support constraints. Its projection to the global line coordinates contains the just constructed neighbourhood, so it has a

rational affine section. Choose that section sufficiently close to the constructed real affine lift, by rational approximation in its affine space of sections, so that the strict klt and effectiveness margins are retained along the compact segment. In the joint solution space the strict klt and effectiveness margins permit a rational polytope containing the scaling segment. This supplies on every chart a rational polytope of ordinary klt boundaries with a fixed effective relatively ample rational summand, representing the restrictions of the same global line parameters. These boundaries, including the scaling representatives, are fixed before the program starts. Their divisors need not agree on different charts.

One global scaling construction. Apply the global degree-space construction in the proof of Proposition 3.4 to J and a general ample scaling direction in the chosen span. Here are the modifications that allow real line data and a nonbirational map to S . The global relatively ample line still gives a compact normalized slice of the curve cone. The fixed ordinary chart pairs give finite negative-ray truncations. A general direction avoids equal wall parameters for independent ray-degree functionals, exactly as in that proof. A positive wall therefore selects a single ray in the global degree space. A rational nef support is represented by a global rational line; a large multiple minus the chartwise ordinary adjoint is relatively ample. Ordinary relative base point freeness contracts the ray using the evaluation of this one global line.

If the contraction is small, choose a global rational line having negative ray degree and construct its flip by its one global graded algebra. The local ordinary proof of Proposition 3.18 applies to this algebra: all curves of the contraction have proportional degrees for global lines, so the detector and the driving line are positively proportional over the contraction; ordinary rational replacement gives local finite generation. This part of that proof needs only the already projective contraction and these line degrees. It does not require a prior assertion about the full Bott–Chern cone. The relative Proj is the next global model. In a divisorial step the exceptional prime and Lemma 3.3 give the same continuation as in Proposition 3.4. Killing the degree of any global line by a rational multiple of the detector, then applying that lemma, preserves global strong $\mathbb{Q}$ -factoriality in both cases. The global lines continue to span the relative degree spaces. The rational support-simplex argument in that proposition makes the wall a descended relatively ample real line on its contraction base and gives a nonempty interval of ampleness immediately after each step. This argument uses relative line data and projectivity, not birationality of the map to S .

We check explicitly the persistence of the ordinary pairs. Expand the line parameters in finitely many global rational lines. Their reflexive transforms are actual rational lines by the preceding strong property. Pushing forward the fixed principal-divisor identities gives

$$K_{X_i} + \Delta_{\ell,i} \sim_{\mathbb{R}} J_i|_{(X_i)_{U_\ell}}.$$

Hence these ordinary adjoints are $\mathbb{R}$ -Cartier; no local factoriality of individual prime divisors is being assumed. On a common resolution their comparison minus the appropriate positive multiple of the detector comparison is exceptional and relatively numerically trivial. Negativity in both signs makes this difference zero. Every chosen step is consequently negative for the fixed ordinary pairs. They remain effective and klt, and their boundaries remain relatively big. The same equality of comparison divisors, after adding the fixed base pullback, preserves the generalized pair and its discrepancy inequalities.

For termination, every already constructed working model is a weak log canonical model on each chart of a boundary in the fixed polytope: choose a parameter in its nonempty ample scaling interval. Indeed, actual descent at a preceding wall λ_j makes the comparison for $J + tH$ equal to $(1 - t/\lambda_j)$ times its J -comparison. For a parameter in the current interval all preceding λ_j are at least t , so these comparisons are effective. Theorem E of [31] gives finitely many marked models on

each chart. That theorem does not require locally $\mathbb{Q}$ -factorial targets. There are therefore finitely many possible global marked working models. Indeed, isomorphisms between the restrictions of two existing marked models agree on their common dense open, and hence on overlaps. A repeated marked working model would have the same boundary transform and discrepancies, contrary to strict discrepancy increase at an intervening nontrivial step. The global program is finite. Relative pseudoeffectivity excludes a Mori fibre endpoint, so J_m is nef over S . This is a finite-cover argument for a single constructed program; it does not glue independent minimal models.

Semiampleness and the global polarization. On the endpoint the fixed ordinary pair is klt, its effective boundary is relatively big, and its adjoint is nef over W_ℓ . Lemma 11.16 of [31] applies to this already existing weak model. More explicitly, Lemma 11.15 replaces the endpoint boundary by $A_\ell + B_\ell$, where $A_\ell \geq 0$ is relatively ample, $B_\ell \geq 0$, and the pair is klt. Theorem 8.3 then gives semiampleness near W_ℓ : its log canonical, klt-existence, real Cartier, ample-summand and nef-over-compact hypotheses have all been checked.

The actual equivalences identify these local semiample adjoints with restrictions of J_m . Their ample models agree on overlaps by [31, Lemma 11.3]. To retain a global polarization, shrink the initial parameter polytope around J after the finite program: all its finitely many strict step signs persist, so the transformed ordinary boundaries remain klt and big. On each chart the pushforward of its initial common ample rational summand is effective and relatively big. Apply [31, Lemma 11.13] to this common big rational part and the small endpoint boundary polytope. All pairs are klt, so the condition on non-klt centres is vacuous. The lemma gives a common general relatively ample rational summand by one $\mathbb{Q}$ -linearly trivial translation. In particular the rational parametrization is preserved; separate applications of Lemma 11.15 to individual parameters would not suffice here. In this finite rational span Theorem 11.17 of [31] makes the endpoint nef region a rational polytope. Intersect the finitely many inverse images of these regions and express J_m as a positive combination of rational points in its minimal face. The corresponding global rational lines are semiample on every chart by the same ordinary base point free argument. After clearing denominators and taking common sufficiently divisible powers over the finite cover, each is relatively generated globally: generation on the charts is generation of its global coherent relative evaluation map. The product of these finitely many relative morphisms, followed by Stein factorization, gives a projective $g : X_m \rightarrow Z$, a projective $\rho : Z \rightarrow S$, and descended global lines whose positive combination H is relatively ample. It realizes the glued ample model and gives $J_m \sim_{\mathbb{R}} g^*H$. Adding back $f_m^*\beta$ proves the asserted Bott–Chern identity. Composing the effective step comparisons proves the final discrepancy statement. $\square$

Corollary 3.26 (Semiampleness on an existing projective model). *Let $f : T \rightarrow S$ be a projective morphism of normal compact Kähler spaces with T globally strongly $\mathbb{Q}$ -factorial, and let $J \in \text{Pic}(T) \otimes \mathbb{R}$ be nef over S . Suppose that on a finite Stein-compact cover satisfying condition (P), there are effective ordinary klt boundaries Δ_ℓ , big over the base, such that $K_T + \Delta_\ell \sim_{\mathbb{R}} J|_{T_{U_\ell}}$. Then J has the projective relative semiample contraction and actual descended ample real line of Proposition 3.25. In particular this applies to any already existing weak model of the fixed ordinary chart pairs in that proof; one need not run a new program for the adjoint in question.*

Proof. Lemma 11.15 and Theorem 8.3 of [31] give local semiampleness. For an existing weak model this is also Lemma 11.16. To globalize the polarization, take a finite rational span of the global lines defining J . On each chart the ample part of the normalized boundary allows small perturbations in this span. The joint rational affine construction in the preceding proof gives a rational family with a common rational ample part; all boundary coefficients are allowed

to vary. Theorem 11.17 then gives a rational nef region on each chart. The finite-intersection, rational-vertex and global-evaluation argument in that proof gives one projective contraction and its actual descended ample real line. Only its local ample models are identified by uniqueness. $\square$

Remark 3.27 (The two uses of projective completion). For a selective extraction over an already constructed weak model, the carrier map is projective bimeromorphic. The inherited exceptional splitting expresses the nef datum, modulo this base, by actual real lines. Relative nefness follows from the carrier datum, while relative bigness follows from birationality, using the ample-plus-effective construction in the ordinary replacement. The prepared adjoint is a base pullback plus an effective exceptional divisor supported on exactly the unwanted primes. Proposition 3.25 applies. On its endpoint the remaining error is effective, exceptional over the fixed base, and relatively nef; negativity makes it zero. Strict discrepancy improvement prevents contraction of a requested zero-error prime. The chosen extraction boundary must be effective and klt; in particular its prescribed primes have the usual admissible discrepancy range.

For the graph construction, suppose the already projective graph resolution $h : V \rightarrow Y_m$ has smooth source and

$$A_V = h^* A_{Y_m} + [G], \quad G \geq 0.$$

Then the global relative real line is $J = \mathcal{O}_V(G)$, the base class is A_{Y_m} , and its carrier line is $N = G - K_V - B_V$. Its relative degrees equal those of the prepared nef datum. That datum is nef and big on the prepared carrier, for example a pullback of its prepared Kähler class; thus N is nef and big over Y_m . This bigness is a separate hypothesis check, since h can have positive-dimensional general fibres. Effectivity of G gives relative pseudoeffectivity. Apply the proposition over Y_m . The subsequent negativity argument must still remove G_m ; only then is the endpoint adjoint the pullback of A_{Y_m} and good over the original base.

For transport of all classes, verify separately that all Bott–Chern classes on the graph carrier are real-line classes modulo Y_m , using the smooth rationally connected fibration and the inherited exceptional splitting on the lower-dimensional base model. Lemma 3.24 then identifies the relative degree rays with full Bott–Chern rays, so the detected-step transport applies. The driving identity involving G alone does not give this additional property.

Corollary 3.28 (Ordinary projective Mori output). *Let $f : X \rightarrow S$ be an already projective surjective morphism of compact Kähler spaces, with X smooth. Let (X, B) be an effective ordinary real klt pair, and let $D = K_X + B$. If D is not pseudoeffective over S , then its projective program over S , with scaling of a relatively ample actual real line N , terminates with a Mori fibre contraction. The initial scaling parameter is chosen so that $D + TN$ is nef over S . The working spaces are globally strongly $\mathbb{Q}$ -factorial and compact Kähler, and the actual line comparisons and integral descent of Proposition 3.25 apply.*

Proof. The line N is fixed; it need not be a general scaling direction. In particular several extremal rays may have the same wall parameter. Choose $\epsilon > 0$ such that $D + 2\epsilon N$ is still not pseudoeffective over S . On the finite Stein-compact cover choose divided general effective representatives of N so that all the resulting ordinary pairs representing $D + tN$, $0 \leq t \leq T$, are klt. For $\epsilon \leq t \leq T$ reserve a common effective relatively ample rational summand. The joint rational-family construction in Proposition 3.25 puts these fixed pairs in the polytopes required by Theorem E of [31].

Suppose a finite prefix has reached X_i , and put $\mu_i = \lambda_{i-1}$, with $\mu_0 = T$. Its transformed $D_i + \mu_i N_i$ is nef. Define

$$\lambda_i = \min\{t \in [0, \mu_i] : D_i + tN_i \text{ is nef over } S\}.$$

The feasible set is a nonempty closed interval. Inductively $\lambda_i > 2\epsilon$: all previous steps have nonpositive comparison for $D + 2\epsilon N$, so its transform is still not pseudoeffective. A first threshold at most 2ϵ would make this transform nef by convexity between the current and preceding thresholds, a contradiction.

There is a D_i -negative extremal ray R_i with $(D_i + \lambda_i N_i) \cdot R_i = 0$. To see attainment without a generality assumption, let $t < \lambda_i$ tend to λ_i . A ray negative for $D_i + tN_i$ has positive N_i -degree, because $D_i + \mu_i N_i$ is nef. When $t > \lambda_i/2$, it is also negative for $D_i + (\lambda_i/2)N_i$. The latter has fixed ordinary klt big-boundary representatives on the charts, since $\lambda_i/2 > \epsilon$. Normalize each of these boundaries by [31, Lemma 11.15] as $A_\ell + B'_\ell$, with $A_\ell \geq 0$ relatively ample and the new pair klt. The A_ℓ -truncation of the cone theorem for $K + B'_\ell$ gives finitely many rays negative for the entire adjoint in question. The finite cover therefore gives finitely many global negative rays. One of these rays attains the wall, as required.

Choose one such ray, even if the wall vanishes on other rays. The rational-support argument for an individual negative ray in Proposition 3.4 gives a global rational nef support annihilating exactly R_i , and ordinary relative base point freeness contracts it. For the real ordinary boundary one may first choose a nearby rational klt boundary that is still negative on R_i . The contraction therefore has a rational ordinary antiample adjoint. Its flip, when small, is constructed from one global detector algebra as above.

If the contraction is of fibre type, stop. Otherwise the wall $w_i = D_i + \lambda_i N_i$ has degree zero on R_i . In the finite rational span of its actual line factors, this zero-degree subspace is rational: its defining functional has rational values on rational lines. Thus w_i is a real combination of rational lines of zero ray degree. Lemma 3.3 descends those lines to the contraction base. Their combination is nef over S , by lifting curves through the projective contraction. Pulling it to the positive side shows that the transformed wall stays nef. The next threshold therefore satisfies $\lambda_{i+1} \leq \lambda_i$; equality is permitted. No interval of ampleness between successive thresholds is required.

If $E_i \geq 0$ is the ordinary D_i -comparison on a common resolution, this actual wall descent gives, for every real t ,

$$p_i^*(D_i + tN_i) \sim_{\mathbb{R}} q_i^*(D_{i+1} + tN_{i+1}) + (1 - t/\lambda_i)E_i.$$

For $t \leq \lambda_i$ the error is effective. This proves all the nonpositivity and fixed-pair persistence assertions used in the induction, including the lower bound on thresholds.

At every working model use the *wall parameter* $t = \lambda_i$, rather than an interior ample parameter. Since $\lambda_i \leq \lambda_j$ for every earlier step, the displayed comparisons show that X_i is a weak log canonical model of the initial fixed chart pairs representing $D + \lambda_i N$. Its trace is nef by definition. Theorem E counts all these marked weak log canonical models, including the ones at repeated wall parameters. The finite chart cover gives a finite global marked list. Repetition would contradict the strict ordinary D -discrepancy increase at an intervening step. Thus this program with the specified N is finite. Non-pseudoeffectivity excludes a nef endpoint for D , so the last contraction is a Mori fibre contraction. The comparison at $t = 1 = \lambda_i$ is crepant: such a step supplies no strict discrepancy improvement for $D + N$, while it is still strictly negative for D . $\square$

3.8 A restricted dimension induction

All generalized nef data in this subsection are globally nef on a fixed compact Kähler carrier. For the degree arguments we use weak NQC: a positive combination of rational degree-two classes having nonnegative degree on compact curves. Strong NQC implies this condition, and the contraction assertion admits this weak form [39, Definition 2.43]. The nef adjoint itself remains nef in Bott–Chern cohomology throughout. A modified-big boundary means the total trace $B + \mathbf{M}_X$

is globally modified big. Relative assertions retain these absolute hypotheses. A marked model records its bimeromorphic map from the specified source; two markings agree only under an isomorphism commuting with those maps. Weak models below may be arbitrary normal compact Kähler weak log canonical models. Their generalized nef traces are understood as closed currents pushed forward from the fixed nef carrier; only the whole adjoint is required to be a Bott–Chern class. In particular no factoriality, and no separate Bott–Chern class for either $K_Y + B_Y$ or the nef trace, is required of an auxiliary weak target. This category contains both the working models and the adjunction strata used in special termination. Chosen program outputs remain globally strongly $\mathbb{Q}$ -factorial.

For dimension at most d , write $\mathbf{B}_d$ for semiampleness of a nef gklt adjoint with modified-big boundary, including a Moishezon connected contraction onto a normal compact Kähler target. It makes no assertion that every such contraction is projective. Write $\mathbf{C}_d$ for the NQC non-klt contraction assertion of [39, Theorem 1.6], and $\mathbf{C}_{d,\text{big}}$ for its big-adjoint case. The relative assertions are:

- $\mathbf{M}_d$: a relatively pseudo-effective gklt adjoint over any proper map to a normal compact Kähler base S , with the globally nef and modified-big data just specified, has a chosen nonextracting good log terminal model over S . Its working space is globally strongly $\mathbb{Q}$ -factorial and compact Kähler; its connected relative canonical morphism has normal compact Kähler target and is Moishezon. From a smooth common carrier, all Bott–Chern classes have forward traces, and the chosen model has the rational degree-two and Bott–Chern exceptional splittings proved below.
- $\mathbf{F}_d$: a compact polytope of these data on one fixed carrier has a polyhedral relative effective locus, finitely many relative canonical chambers with chosen good terminal models, and finitely many marked weak models, accounting for every weak model in the category specified above.

Passing to the proper image and then to its normal Stein factor allows all base maps to be taken surjective with connected fibres. These operations preserve the compact Kähler base category.

The independent inputs are the analytic cone theorem, the projective results of Proposition 3.25, relative vanishing, adjunction, relative-canonical positivity, and the nef-and-big Kähler criterion. The base presentations and multiplier-ideal descent used in the induction are proved below. The projective constructions use the actual-line lemmas of the preceding subsection. No assertion of termination of an arbitrary sequence of flips is used.

3.8.1 Base presentations and multiplier ideals

The following constructions take place in the category of generalized data specified above. In particular, the whole adjoint is a Bott–Chern class, whereas the boundary and nef traces on a singular space are interpreted as currents. On a smooth carrier the structure boundary is a real divisor; its negative components may be exceptional over the space of the pair. The nef part is specified on that carrier and pulled back to higher models.

Lemma 3.29 (Adjoint-preserving preparation). *Let $(X, B + \mathbf{M})$ be a generalized pair on a normal compact Kähler space, with globally modified-big total boundary $B + \mathbf{M}_X$. Write*

$$A = [K_X + B + \mathbf{M}_X], \quad \mathcal{J}_o = \mathcal{J}(X, B + \mathbf{M}).$$

There are generalized data on X with the same adjoint class, a projective carrier $p : U \rightarrow X$ with U smooth and compact Kähler, an SNC real divisor C whose negative components are p -exceptional, and a Kähler class β such that

$$p^*A = [K_U + C] + \beta, \quad \mathcal{J}_1 = p_*\mathcal{O}_U(-[C]) \supseteq \mathcal{J}_0. \quad (3.17)$$

The carrier may be required to dominate any prescribed projective modification of X , with the structure and exceptional divisors having simultaneous SNC support. The prepared total boundary is globally modified big. In particular, the preparation preserves the gklt condition.

Proof. Choose a common projective log resolution $p_0 : U_0 \rightarrow X$ of the original nef carrier and the prescribed modification, and write

$$p_0^*A = [K_{U_0} + C^0] + \mathbf{M}_{U_0}.$$

The class $\mathbf{M}_{U_0}$ is nef, and the negative part $(C^0)^-$ is exceptional over X . The modified-big preparation [39, Lemmas 2.10 and 2.23] can be made with the prescribed resolution. We give its divisor and ideal details. Choose an effective divisor E_0 with sufficiently large coefficients and full p_0 -exceptional support so that the class

$$[(C^0)^+] + \mathbf{M}_{U_0} + [E_0]$$

is big. Resolve a Kodaira decomposition of this class. On a further projective modification $a : U \rightarrow U_0$ with U smooth one obtains

$$a^*([(C^0)^+] + \mathbf{M}_{U_0} + [E_0]) = \kappa + [D], \quad D \geq 0,$$

where κ is Kähler. The support can be made SNC together with all the other divisors. On any additional blowup, subtract a sufficiently small effective exceptional class from the pulled-back Kähler class and add it to the divisor; an exceptional divisor whose negative is relatively ample makes the resulting class Kähler again.

Put $p = p_0a$ and

$$C^{\text{alt}} = D - a^*(C^0)^- - a^*E_0 - K_{U/U_0}.$$

Then

$$p^*A = [K_U + C^{\text{alt}}] + \kappa,$$

and every negative component of C^{alt} is p -exceptional. Let C^{old} be the crepant structure boundary of the original data on this same model. For $0 < t \ll 1$, set

$$C = (1 - t)C^{\text{old}} + tC^{\text{alt}}, \quad \beta = (1 - t)a^*\mathbf{M}_{U_0} + t\kappa.$$

The class β is Kähler, the adjoint identity is unchanged, and negative coefficients still occur only on exceptional divisors. There are only finitely many coefficients. Thus t can be chosen so that

$$[C] \leq [C^{\text{old}}].$$

Pushing forward the corresponding inclusion of invertible sheaves gives (3.17). The Kähler class β plus the strict transform of the effective trace boundary is big; its pushforward is the prepared total boundary. If the original pair is gklt, its multiplier ideal is the unit ideal, and hence so is $\mathcal{J}_1$. $\square$

Proposition 3.30 (Base presentations for projective contractions). *Let $f : X \rightarrow Z$ be a projective contraction between normal irreducible compact Kähler spaces. Suppose that $(X, B + \mathbf{M})$ has globally modified-big total boundary and that*

$$A = [K_X + B + \mathbf{M}_X] = f^*\gamma, \quad \gamma \in H_{\text{BC}}^{1,1}(Z).$$

There is an adjoint-preserving preparation as in Lemma 3.29, with multiplier ideal $\mathcal{J}_1 \supseteq \mathcal{J}_0$, for which

$$R^i f_* \mathcal{J}_1 = 0 \quad (i > 0), \quad f_* \mathcal{O}_{N_1} = \mathcal{O}_Z / (f_* \mathcal{J}_1), \quad N_1 = V(\mathcal{J}_1). \quad (3.18)$$

If N_1 dominates Z , then $f_ \mathcal{J}_1 = 0$. Otherwise there are generalized data $(Z, B_Z + \mathbf{N})$, a projective carrier $\tau : T \rightarrow Z$ with T smooth and compact Kähler, a real SNC structure divisor C_T , and a Kähler class β_T such that*

$$\begin{aligned} [K_Z + B_Z + \mathbf{N}_Z] &= \gamma, & \tau^* \gamma &= [K_T + C_T] + \beta_T, & \mathbf{N}_T &= \beta_T, \\ B_Z &= \tau_* C_T \geq 0, & B_Z + \mathbf{N}_Z & \text{is globally modified big.} \end{aligned} \quad (3.19)$$

The negative components of C_T are τ -exceptional, and

$$\mathcal{J}_Z := \mathcal{J}(Z, B_Z + \mathbf{N}) = \tau_* \mathcal{O}_T(-\lfloor C_T \rfloor) \supseteq f_* \mathcal{J}_1. \quad (3.20)$$

In particular the base data are gklt wherever $f_ \mathcal{J}_1$ is the unit ideal. If the source data are gklt, the base data are gklt. All adjoint identities are preserved by pullback to higher models.*

Proof. Choose a projective resolution $\mu : W \rightarrow Z$, with W smooth compact Kähler and reduced exceptional divisor F having SNC support. Apply Lemma 3.29 on a carrier dominating the main component of the base change. We obtain projective maps $p : U \rightarrow X$ and $q : U \rightarrow W$ with $fp = \mu q$ and

$$[K_U + C] + \beta = p^* A = q^* \mu^* \gamma, \quad \mathcal{J}_1 = p_* \mathcal{O}_U(-\lfloor C \rfloor). \quad (3.21)$$

Here U is smooth compact Kähler, β is Kähler, and all negative components of C are p -exceptional. The maps are projective by composition and factorization over the separated base W . The map q has connected fibres: this holds over the open set where μ is an isomorphism, and its Stein factorization is finite and bimeromorphic over the normal space W .

The integral divisor $-\lfloor C \rfloor$ has class

$$[-\lfloor C \rfloor] = [K_U + \{C\}] + \beta - (fp)^* \gamma.$$

The pair $(U, \{C\})$ is klt with SNC boundary. Relative Kawamata–Viehweg vanishing, first for p and then for fp , applies because the positive class modulo pullbacks is Kähler [39, Theorem 2.2]. Equivalently, the real line bundle $-K_U - C$ is relatively ample, so the vanishing follows from [32, Theorem 1.2] with the identity as its first morphism. Leray gives $R^i f_* \mathcal{J}_1 = 0$ for $i > 0$. Applying f_* to the sequence defining N_1 , and using $f_* \mathcal{O}_X = \mathcal{O}_Z$, proves (3.18). If N_1 dominates Z , the ideal $f_* \mathcal{J}_1 \subseteq \mathcal{O}_Z$ vanishes generically and therefore vanishes. This proves that alternative.

Suppose from now on that N_1 does not dominate Z . Every q -horizontal coefficient of C is then less than one. Indeed, a component with coefficient at least one forces the multiplier ideal to be proper along its image in X , and its image would make N_1 dominate Z . If Z is a point, the asserted base presentation is the dimension-zero presentation. We therefore assume $b = \dim Z > 0$.

The fibre metric. Choose a smooth representative ρ of $\mu^*\gamma - c_1(K_W)$, a Kähler form ω_W , and a Kähler form representing β , still denoted by β . For some $a > 0$,

$$\beta \geq aq^*\omega_W.$$

Consider the holomorphic line bundle

$$L = \mathcal{O}_U(-K_{U/W} - \lfloor C \rfloor).$$

Equation (3.21) gives a singular Hermitian metric h on L whose curvature current is

$$\Theta_h(L) = \beta + [\{C\}] - q^*\rho. \quad (3.22)$$

Its weights are smooth functions plus the divisorial logarithms with coefficients in $\{C\}$. We normalize dd^c so that $dd^c \log |z|^2 = [z = 0]$ and write squared-norm weights as $e^{-\varphi}$. Let e be the canonical meromorphic section of

$$K_{U/W} \otimes L = \mathcal{O}_U(-\lfloor C \rfloor),$$

so that $\text{div}(e) = -\lfloor C \rfloor$.

Choose an analytic Zariski open set $W^\circ \subseteq W$ on which q is a submersion, the horizontal boundary is relatively SNC, there are no vertical boundary components, μ is an isomorphism onto an open subset of Z , and coherent base change holds in degree zero. On $q^{-1}(W^\circ)$, the divisor $-\lfloor C \rfloor$ is effective and exceptional over X . Normality and the contraction property therefore identify

$$q_*(K_{U/W} \otimes L)|_{W^\circ} = \mathcal{O}_{W^\circ} e.$$

Here $K_{U/W}$ is viewed as the relative canonical line bundle. The fibre norm

$$I(w) = \int_{U_w} |e_w|_h^2$$

is positive and finite. The fractional SNC weights are integrable; any zeros of e_w only improve integrability. Relative SNC coordinates and a partition of unity show that I is smooth: parameter derivatives act on smooth factors multiplying fixed integrable monomial weights.

Set $\psi = -\log I$ on W° . Then

$$\rho + dd^c\psi \geq a\omega_W. \quad (3.23)$$

Here are the details for the singular weights. Locally choose ℓ with $dd^c\ell = \rho - a\omega_W$, and put $h' = he^{-q^*\ell}$. Its weights are psh, and its fibre norm satisfies $I' = e^{-\ell}I$. Restrict the family to a small coordinate line disk Δ_R , with centre 0. The sharp L^2 extension theorem gives a twisted top form extending e_0 and satisfying

$$\int_{q^{-1}(\Delta_R)} |E|_{h'}^2 \leq \pi R^2 I'(0). \quad (3.24)$$

Top-form norms here are normalized to use Euclidean area in the base coordinate. We use the Stein hypersurface form of the optimal extension theorem [37, Theorem 2.2 and §3.5, (3.1)], with defining function the disk coordinate divided by R .

To check applicability, embed the restricted projective family into projective space over the disk and remove a relative hyperplane section that does not contain the central fibre. The complement is Stein. A holomorphic section of L on this Stein space can be chosen nonzero at a point of the

central fibre; deleting its zero divisor trivializes L and leaves a Stein space. The central fibre is smooth and connected, hence irreducible, so both deletions leave a dense open subset of it. For a zero-dimensional fibre choose the deletions to avoid its point. In the resulting trivialization the weight is psh, and the sharp extension theorem applies to canonical forms. The extension extends across the deleted hypersurfaces in reverse order by local L^2 removability: in every original frame the metric is bounded below by a smooth positive metric. The extension retains its norm bound and its prescribed restriction to the central fibre.

After dividing E by the base differential, the rank-one direct image identifies it with $b(t)e$, where b is holomorphic and $b(0) = 1$. Thus the disk average of $|b(t)|^2 I'(t)$ is at most $I'(0)$. Jensen's inequality and the submean inequality for $\log |b|^2$ give

$$\log I'(0) \geq \text{avg}_{\Delta_R} (\log |b(t)|^2 + \log I'(t)) \geq \text{avg}_{\Delta_R} \log I'(t).$$

The same argument applies to every sufficiently small disk at every centre and in every complex line direction. Hence $-\log I' = \psi + \ell$ is psh, which proves (3.23).

Extension over the base. Let D be a prime divisor in $W \setminus W^\circ$. Choose a component Q of q^*D dominating D . When D is not μ -exceptional, choose Q not p -exceptional: take a component over the corresponding prime on Z already on X and then its strict transform. Put

$$u = \text{ord}_Q(q^*D), \quad c = \text{coeff}_Q(C).$$

At a general point of Q , avoiding all other relevant divisors, there are local coordinates with

$$t_1 = z_1^u, \quad t_j = z_j \quad (2 \leq j \leq b).$$

Indeed the map $Q \rightarrow D$ is submersive at a general smooth point, and a local unit in the first equation is absorbed by a holomorphic u th root. Such points can be chosen above every point of a dense analytic open subset of D by generic smoothness and properness. Integration over a fixed small polydisk in the remaining fibre coordinates gives

$$I(t) \geq c_* |t_1|^{-2(c+u-1)/u} \tag{3.25}$$

for some $c_* > 0$ near each such base point. The section and metric contribute the exponent c , while division by the base differential in the relative canonical form contributes $u - 1$.

For a nonexceptional D the chosen Q has $c \geq 0$, so (3.25) bounds ψ from above near its general points. For the finitely many exceptional divisors, the same estimate supplies a nonnegative integer m such that, if k is a smooth metric on $\mathcal{O}_W(F)$ and s_F its divisor section,

$$\varphi = \psi + m \log |s_F|_k^2$$

is locally bounded above near general points of every divisor. Adding local potentials of $\rho + mc_1(k) - a\omega_W$ to φ gives psh weights by (3.23). These weights extend across the divisors and then across the remaining analytic set of codimension at least two, giving

$$S = \rho + mc_1(k) + dd^c \varphi \geq a\omega_W \tag{3.26}$$

on all of W .

A Kähler carrier. Regularize the current in (3.26) in its class, retaining a lower bound for the potentials. More precisely, choose an approximation with analytic singularities such that

$$S' = \rho + mc_1(k) + dd^c \varphi', \quad S' \geq \frac{a}{2} \omega_W, \quad \varphi' \geq \varphi - O(1). \quad (3.27)$$

These simultaneous properties are the Bergman regularization properties of [21]; see also [60, pp. 59–60]. Resolve the analytic singularities and the support of F by a sequence of blowups with smooth centres $r : T \rightarrow W$, with simultaneous SNC support, and write

$$r^* S' = [H] + \zeta, \quad H \geq 0.$$

The residual current ζ has locally bounded potentials. Removing the divisorial part from the positive current $r^* S' - (a/2)r^* \omega_W$ shows that

$$\zeta \geq \frac{a}{2} r^* \omega_W.$$

The difference has vanishing Lelong numbers, so its class is nef by regularization. Choose an effective r -exceptional divisor G with $-G$ relatively ample. For $0 < \eta \ll 1$, the class $(a/2)r^* \omega_W - \eta[G]$ is Kähler. Therefore

$$\begin{aligned} \tau &= \mu r, & \beta_T &= [\zeta] - \eta[G], \\ C_T &= H + \eta G - m r^* F - K_{T/W} \end{aligned} \quad (3.28)$$

has Kähler β_T . If r is an isomorphism, omit G and its correction. The choice of η will be decreased once more below.

The exact class calculation is

$$\begin{aligned} [K_T + C_T] + \beta_T &= r^* c_1(K_W) + [H] + [\zeta] - m[r^* F] \\ &= r^*(c_1(K_W) + [\rho]) = \tau^* \gamma. \end{aligned}$$

All possibly negative terms in C_T are τ -exceptional. Consequently $B_Z = \tau_* C_T$ is effective. Specify the nef b-data by $\mathbf{N}_T = \beta_T$; the displayed identity defines generalized data on Z with adjoint class γ . At the level of currents, pushing the identity gives the required trace identity on Z : exceptional divisor currents push to zero, and a dd^c difference pushes to a dd^c difference. Canonical traces can be computed using local meromorphic canonical frames; a change of frame changes the corresponding Chern potentials by the usual logarithmic transition term. Thus only the whole adjoint, rather than its individual traces, needs to be a Bott–Chern class. The class β_T plus the strict transform of B_Z is big on T , so its pushforward $B_Z + \mathbf{N}_Z$ is globally modified big. These are (3.19).

The multiplier ideal. Take a local holomorphic function $v \in f_* \mathcal{J}_1$. By the pushforward description of $\mathcal{J}_1$,

$$\operatorname{div}(q^* \mu^* v) \geq [C]. \quad (3.29)$$

Thus $q^* \mu^* v e$, tensored with a local base canonical frame, is a holomorphic L -valued top form on U . Its squared norm is locally integrable by the fractional SNC weights in (3.22). Over the submersion locus, fibre integration followed by base integration is precisely this top-form integral. Properness and a finite coordinate covering therefore give

$$\int |\mu^* v|^2 e^{-\psi} < \infty \quad (3.30)$$

locally on W ; the integral is taken where the original fibre metric is defined.

The lower bound on φ' in (3.27) preserves this integrability after replacing ψ by $\varphi' - m \log |s_F|_k^2$. Change variables through r . Along a prime divisor of T , let h, j, k_D, d be respectively the coefficients of $H, \operatorname{div}(\tau^*v), K_{T/W}$, and r^*F . The local integrand has exponent $2(j - h + md + k_D)$, up to a bounded positive factor. Its integrability implies $j - h + md + k_D > -1$, or, since j is integral,

$$j \geq \lfloor h - md - k_D \rfloor.$$

Therefore

$$\operatorname{div}(\tau^*v) \geq \lfloor H - mr^*F - K_{T/W} \rfloor.$$

Choose $\eta > 0$ sufficiently small that adding ηG does not change any of these finitely many round-downs. The resulting inequality is exactly $\operatorname{div}(\tau^*v) \geq \lfloor C_T \rfloor$, proving (3.20) as an inclusion of ideal sheaves. In particular, if $f_*\mathcal{J}_1$ is the unit ideal on an open set, then so is $\mathcal{J}_Z$, and the base data are gklt there.

Finally, the nef data descend to T , and the structure boundaries on higher models are defined by crepant pullback of (3.19). On every common refinement of the source and base carriers, both adjoints are the pullback of γ . Thus the identities remain exact on higher models. $\square$

Lemma 3.31 (Contractions of closed subspaces). *Let D be a compact complex space, possibly nonreduced, and let $\delta \in H^2(D, \mathbb{R})$ have nonnegative degree on compact curves. Use the definitions of a Moishezon morphism and of a contraction defined by a class in [39, Definitions 2.32 and 2.34]. Then the following hold.*

- (i) *If δ is endowed with a Moishezon contraction, so is its restriction to every closed analytic subspace of D .*
- (ii) *Let $g : D \rightarrow V$ be a projective contraction of compact complex spaces, so $g_*\mathcal{O}_D = \mathcal{O}_V$, and suppose $\delta = g^*\epsilon$ for a class $\epsilon \in H^2(V, \mathbb{R})$ having nonnegative degree on compact curves. Then δ is endowed with a Moishezon contraction if and only if ϵ is endowed with one.*

Proof. Write $h : D \rightarrow S$ for a Moishezon contraction defined by δ . For a closed subspace $D_0 \subset D$, take the Stein factorization

$$D_0 \longrightarrow S_0 \longrightarrow S$$

of its restriction. The first map is a contraction, and the second is finite. Thus a curve in D_0 is contracted by the first map exactly when its δ -degree is zero. Choose a projective surjection $P \rightarrow D$ witnessing that h is Moishezon, with $P \rightarrow S$ projective. The closed subspace $P_0 = P \times_D D_0$ is projective over S and projective and surjective over D_0 . It is also projective over S_0 : a relatively ample line for $P_0 \rightarrow S$ remains ample on the fibres of $P_0 \rightarrow S_0$. This proves the first assertion, including its scheme structure.

For the descent direction of the second assertion, every curve in a fibre of g has δ -degree zero. A connected projective complex space is connected by chains of compact curves after passing to its reduction. Therefore h is constant on the underlying set of each fibre of g . It factors as $h = k \circ g$ for a holomorphic map $k : V \rightarrow S$. To see that this factorization holds also for nonreduced spaces, choose a coordinate neighbourhood in S of the image of a fibre. Properness permits restriction to a neighbourhood of its point in V whose inverse image maps into that coordinate neighbourhood. The coordinate functions of h then descend uniquely through $g_*\mathcal{O}_D = \mathcal{O}_V$; their analytic relations descend through the same equality. These local factorizations glue. Furthermore,

$$k_*\mathcal{O}_V = h_*\mathcal{O}_D = \mathcal{O}_S.$$

Every irreducible compact curve $C \subset V$ is dominated by a compact curve $C' \subset D$. Indeed, after base change to the normalization of C , take a component dominating that curve and a projective multisection. The projection formula gives

$$\delta \cdot C' = \deg(C' \rightarrow C) \epsilon \cdot C.$$

Consequently k contracts exactly the curves of ϵ -degree zero. The composite $P \rightarrow D \rightarrow V$ is a projective surjection, and $P \rightarrow S$ is projective, so it witnesses that k is Moishezon.

Conversely, suppose $k : V \rightarrow S$ is a Moishezon contraction defined by ϵ . Then $k \circ g$ is a contraction, and the projection formula verifies its curve criterion, treating separately curves contracted by g . If $Q \rightarrow V$ is a projective surjection witnessing that k is Moishezon, then $D \times_V Q \rightarrow D$ is projective and surjective, while $D \times_V Q \rightarrow Q \rightarrow S$ is projective. This supplies the required witness. $\square$

Lemma 3.32 (Crepant transport of the multiplier closed subspace). *Let $g : X \rightarrow Y$ be a projective bimeromorphic morphism of normal compact Kähler spaces. Suppose effective generalized pair data on X and Y have the same globally nef b-part and are crepant over Y . In particular their adjoint classes A_X and A_Y satisfy $A_X = g^* A_Y$ in Bott–Chern cohomology. Write J_X, J_Y for their multiplier ideals and $N_X = V(J_X)$, $N_Y = V(J_Y)$ for their non-klt closed subspaces. Then*

$$g_* J_X = J_Y, \quad R^i g_* J_X = 0 \quad (i > 0), \quad g_* \mathcal{O}_{N_X} = \mathcal{O}_{N_Y}.$$

The restriction $N_X \rightarrow N_Y$ is a projective contraction. Hence, whenever A_Y is nef, the restriction of A_X to N_X is endowed with a Moishezon contraction if and only if the restriction of A_Y to N_Y is endowed with one.

Proof. Take a projective common log resolution $r : U \rightarrow X$ carrying the nef datum, and put $s = g \circ r$. By crepancy there is one real SNC divisor C such that

$$K_U + [C] + \beta = s^* A_Y = r^* A_X,$$

where β is globally nef. The multiplier ideals are

$$J_X = r_* \mathcal{O}_U(-[C]), \quad J_Y = s_* \mathcal{O}_U(-[C]).$$

Their direct-image equality follows at once. For vanishing, set $L = \mathcal{O}_U(-[C])$ and $\Delta = C - [C]$. The pair (U, Δ) is klt, and the actual real line

$$L - (K_U + \Delta) = -K_U - C$$

has class $\beta - s^* A_Y$. It is relatively nef over Y and relatively big because s is bimeromorphic. The same statements hold over X . Relative Kawamata–Viehweg vanishing, in the form of [32, Theorem 1.2] with the first morphism the identity and the real line $\mathcal{H} = -K_U - C$, therefore gives

$$R^i s_* L = R^i r_* L = 0 \quad (i > 0).$$

The real-line identity in that theorem is the displayed definition of $\mathcal{H}$; its relative positivity is checked by the class identity above. The klt condition means that there are no proper log canonical strata to test for relative log bigness. Leray gives $R^i g_* J_X = 0$ for $i > 0$. Pushing forward $0 \rightarrow J_X \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{N_X} \rightarrow 0$ and using $g_* \mathcal{O}_X = \mathcal{O}_Y$ now proves the asserted structure-sheaf equality. It also identifies the scheme-theoretic image with N_Y , giving the projective contraction between these closed subspaces. Apply Lemma 3.31 to conclude. $\square$

Remark 3.33. The preceding lemma also transports the contraction hypothesis through a crepant projective flip: apply it to the two contractions to their common base, equipped with the descended adjoint and the common nef b-data. On that base the boundary is the effective pushforward of the source boundary, and the nef data remain on the same carrier; the descended Bott–Chern adjoint and its identity on a common resolution define the crepant base data in the whole-adjoint and current-trace category used here. All assertions concern the fixed adjoint and its multiplier ideal. No conclusion about the multiplier ideal of a different driving adjoint is used.

Corollary 3.34 (The contraction step over a lower-dimensional base). *Use the hypotheses and notation of Proposition 3.30. Suppose that $A = f^*\gamma$ is nef and that its restriction to the original non-klt closed subspace $N_o = V(J_o)$ is endowed with a Moishezon contraction. Put*

$$I = f_*J_1, \quad N_* = V(I) \subset Z.$$

Then $\gamma|_{N_}$ is endowed with a Moishezon contraction. If the prepared non-klt closed subspace $N_1 = V(J_1)$ dominates Z , then γ and A are endowed with Moishezon contractions. Otherwise the generalized pair supplied on Z has a non-klt closed subspace endowed with the contraction defined by the restricted class of γ . In this latter case, if γ is nef and NQC and the assertion $\mathbf{C}_{\dim Z}$ is available, then γ and A are endowed with Moishezon contractions.*

Proof. The inclusion $J_o \subset J_1$ gives $N_1 \subset N_o$ as closed analytic subspaces. By Lemma 3.31, the given contraction restricts to N_1 . Proposition 3.30 identifies

$$f_*\mathcal{O}_{N_1} = \mathcal{O}_Z/I = \mathcal{O}_{N_*}.$$

Thus the restriction $N_1 \rightarrow N_*$ is a projective contraction. The restricted adjoint is the pullback of $\gamma|_{N_*}$, so Lemma 3.31 descends the contraction to N_* . Nonnegative degrees on curves of N_* follow from projective multisections and nefness of A .

If N_1 dominates the normal irreducible space Z , then $I = 0$: a local function in I pulls back to zero on the dominating closed subspace and hence vanishes generically on Z . Normality, and in particular reducedness, gives its vanishing. Therefore $N_* = Z$, which proves the assertion in this case.

If N_1 does not dominate Z , the constructed base ideal satisfies $I \subset J_Z$. Hence

$$V(J_Z) \subset V(I) = N_*$$

as closed analytic subspaces. Restriction by Lemma 3.31 provides the required contraction on $V(J_Z)$, with its complete scheme structure. The generalized pair on Z has globally nef carrier data and globally modified-big total boundary by Proposition 3.30. When the stated nef and NQC conditions hold, these facts are precisely the hypotheses of $\mathbf{C}_{\dim Z}$, which gives the contraction on Z . Finally Lemma 3.31 pulls it back through the projective contraction f to X . The case of an empty non-klt closed subspace has the same interpretation, with its contraction condition vacuous. $\square$

3.8.2 A fixed integral grid and cohomology transport

Lemma 3.35 (Integral degree-two descent for a locally log-Fano map). *Let $f : X \rightarrow Z$ be a projective surjective morphism with connected fibres between normal complex analytic spaces. Assume that $f_*\mathcal{O}_X = \mathcal{O}_Z$, that $R^if_*\mathcal{O}_X = 0$ for $i > 0$, and that every point of Z has a neighbourhood U on which there is a klt pair (X_U, Δ_U) with $-(K_{X_U} + \Delta_U)$ ample over U . Then $f^* : H^2(Z, \mathbb{Z}) \rightarrow H^2(X, \mathbb{Z})$ is injective, and its image consists exactly of the integral classes having degree zero on every compact curve contracted by f .*

Proof. Properness, normality and connected fibres identify both $f_*\mathcal{O}_X = \mathcal{O}_Z$ and $f_*\mathcal{O}_X^* = \mathcal{O}_Z^*$. Apply Rf_* to the analytic exponential sequence. The local surjectivity of $\mathcal{O}_Z \rightarrow \mathcal{O}_Z^*$ and the stated coherent vanishing give

$$R^1f_*\mathbb{Z}_X = 0, \quad R^1f_*\mathcal{O}_X^* \simeq R^2f_*\mathbb{Z}_X.$$

Let $\xi \in H^2(X, \mathbb{Z})$ have zero degrees on contracted curves. A germ of its edge image in $R^2f_*\mathbb{Z}_X$ is, under this isomorphism, represented after shrinking U by an actual line bundle L on X_U . Equality of these germs means that, after further shrinking, $c_1(L)$ and $\xi|_{X_U}$ have the same class. In particular L has degree zero on every curve in every fibre above this smaller neighbourhood.

On a sufficiently small Stein base neighbourhood, twisting by a relatively ample line and taking a quotient of two nonzero relative sections gives a meromorphic Cartier representative for L . Thus the Cartier-divisor form of base point freeness applies. The line bundle L is relatively nef, and $aL - (K_{X_U} + \Delta_U)$ is relatively ample for every real a . The projective analytic base-point-free theorem applies to the Cartier line L . Its conclusion is generation for *every* sufficiently large integer power, not only for a divisible subsequence [31, Theorems 6.2 and 6.5]. Thus, over a further relatively compact neighbourhood, both L^m and L^{m+1} are relatively generated. A generated line of degree zero on every curve of a projective connected fibre defines a constant projective map on that fibre: the restriction of the generated line is the pullback of $\mathcal{O}(1)$, and a positive-dimensional projective image contains a curve of positive degree. The corresponding relative image is therefore finite and bimeromorphic over the normal base, and is the base itself. Consequently

$$L^m \simeq f^*M_m, \quad L^{m+1} \simeq f^*M_{m+1}$$

for line bundles on the neighbourhood in Z . Taking their quotient proves $L \simeq f^*(M_{m+1} \otimes M_m^{-1})$. Its germ in $R^1f_*\mathcal{O}_X^*$ is zero, so the edge image of ξ is zero.

The integral Leray spectral sequence, using $R^1f_*\mathbb{Z}_X = 0$, gives

$$0 \longrightarrow H^2(Z, \mathbb{Z}) \xrightarrow{f^*} H^2(X, \mathbb{Z}) \longrightarrow H^0(Z, R^2f_*\mathbb{Z}_X).$$

Exactness at the middle term proves the assertion. The argument is integral throughout and does not discard torsion. The converse follows by restriction of a pulled-back class to a fibre. $\square$

Lemma 3.36 (A fixed integral grid along existing crepant steps). *Let X_0 be a compact Kähler space, and suppose that*

$$\alpha_0 = \sum_{j=1}^r a_j \gamma_{0j}, \quad a_j > 0, \quad \gamma_{0j} \in H^2(X_0, \mathbb{Q}),$$

where each γ_{0j} has nonnegative degree on compact curves. Fix integers $m_j > 0$ such that $m_j \gamma_{0j}$ is the image of an integral cohomology class. Consider an existing sequence of projective birational steps

$$X_i \xrightarrow{f_i} Z_i \xleftarrow{f_i^+} X_{i+1},$$

where f_i^+ is the identity for a divisorial step, f_i satisfies Lemma 3.35, and α_i is trivial on its contracted ray. Assume every f_i -contracted curve has degree proportional to that ray for the classes in the display. Then the classes γ_{ij} descend through f_i and pull back through f_i^+ , the same integers m_j remain integral multiples at every stage, and the transported classes remain nonnegative on curves. In particular, for every integral compact curve $C \subset X_i$,

$$\alpha_i \cdot C > 0 \quad \implies \quad \alpha_i \cdot C \geq \delta, \quad \delta := \min_j \frac{a_j}{m_j} > 0.$$

Proof. Positivity of the coefficients and nonnegativity of the summands imply that every γ_{ij} annihilates the contracted ray. Apply the integral descent lemma to the chosen integral lift of $m_j\gamma_{ij}$ and pull its descended class back to X_{i+1} . This keeps m_j fixed.

To verify nonnegativity, take a curve on X_{i+1} . If it is contracted by f_i^+ , every descended summand has degree zero. Otherwise its image is a curve $\Gamma \subset Z_i$. The projective map f_i has a compact curve mapping onto Γ with positive degree: take a component of its inverse image dominating Γ and intersect with sufficiently many relatively ample hyperplanes. Nonnegativity of the old summand on this curve implies nonnegativity on Γ , hence on the curve on X_{i+1} . This argument uses a positive-degree lift, not a degree-one section. Finally $m_j\gamma_{ij} \cdot C$ is a nonnegative integer, giving the stated lower bound. $\square$

Corollary 3.37 (Uniform crepancy parameter; no termination assertion). *Suppose additionally that the working spaces are d -dimensional klt spaces, that α_i are nef, and that the ordinary canonical negative-ray length bound $0 < -K_{X_i} \cdot C \leq 2d$ is available. For a fixed real number $t > 2d/\delta$, every negative extremal ray chosen in an existing $(K_{X_i} + t\alpha_i)$ -program is α_i -trivial. If the steps are projective ordinary canonical steps, Lemma 3.36 therefore applies inductively with this same value of t .*

If α_0 is not big and $\alpha_0 - c_1(K_{X_0})$ is big, then $K_{X_i} + t\alpha_i$ remains non-pseudo-effective along these crepant steps. Consequently a terminating such program ends with a Mori fibre contraction. This corollary does not assert that the requisite contractions exist or that the program terminates.

Proof. A negative ray is K_{X_i} -negative because α_i is nef. Choose its rational curve generator with the ordinary length bound. If its α_i -degree were positive, then

$$(K_{X_i} + t\alpha_i) \cdot C \geq -2d + t\delta > 0,$$

a contradiction. The step is thus an ordinary canonical step with α_i pulled back from its base, and the preceding lemma applies. On a common resolution the pullback of $\alpha_{i+1} - c_1(K_{X_{i+1}})$ is the pullback of $\alpha_i - c_1(K_{X_i})$ plus the effective canonical comparison. Thus it stays big. Equality of the crepant pullbacks of α_i preserves its nonbigness. The identity

$$(t+1)\alpha_i = (c_1(K_{X_i}) + t\alpha_i) + (\alpha_i - c_1(K_{X_i}))$$

excludes pseudo-effectivity of its first summand. A nef endpoint is therefore impossible. $\square$

Lemma 3.38 (Cohomology splitting along detected steps). *Let X_0 be a smooth compact Kähler manifold. Consider a finite sequence of divisorial contractions and small flips*

$$X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_r$$

between normal globally strongly $\mathbb{Q}$ -factorial compact Kähler spaces with rational singularities. Assume that every negative contraction $f_i : X_i \rightarrow Z_i$ is a projective contraction of one ray of the analytic cone, that Z_i has rational singularities, and that each small flip is the detected flip of a global rational line L_i with $L_i \cdot R_i < 0$. For a divisorial contraction assume the usual single exceptional prime. Then, for every projective smooth resolution $p : U \rightarrow X_i$,

$$\begin{aligned} H^2(U, \mathbb{R}) &= p^*H^2(X_i, \mathbb{R}) \oplus \bigoplus_{E \text{ exceptional for } p} \mathbb{R}[E], \\ H^{1,1}(U, \mathbb{R}) &= p^*H_{\text{BC}}^{1,1}(X_i, \mathbb{R}) \oplus \bigoplus_{E \text{ exceptional for } p} \mathbb{R}[E]. \end{aligned}$$

The first equality also holds over $\mathbb{Q}$. There are compatible forward linear transforms on $H^2(-, \mathbb{R})$ and on Bott–Chern cohomology. They are surjective, are isomorphisms for a small flip, and preserve rational degree-two classes. On a common resolution the pullback difference is an actual exceptional divisor class; for a flip it is exceptional over both spaces.

The exceptional quotient also takes the Chern class of a holomorphic line bundle on U to a rational holomorphic line class on X_i .

Proof. For a modification of a smooth compact Kähler manifold the two splittings are the degree-two modification formula. Suppose that they hold for $X = X_i$. We use only the following birational descent fact: for a proper bimeromorphic map between normal compact spaces in Fujiki’s class with rational singularities, pullback is injective on $H^2(-, \mathbb{R})$ and on Bott–Chern cohomology, and its image in either group consists of the classes having degree zero on every contracted curve. This is the bimeromorphic case of [17, Lemma 2.6(1)].

First note that a degree-two class $\xi \in H^2(X, \mathbb{R})$ has the same degrees on curves as a Bott–Chern class. Indeed, pull it to a smooth projective resolution $a : W \rightarrow X$ and take the $(1, 1)$ -part of its Hodge decomposition. The induction hypothesis writes that part as $a^*\beta + [F]$, with F exceptional over X . On every a -contracted curve, $a^*\xi$ and its $(2, 0)$ and $(0, 2)$ parts have degree zero. Hence F has degree zero on every such curve. Exceptional negativity applied to both signs gives $F = 0$. Projective multisections lifting a curve of X then show that ξ and β have the same degrees on every curve of X .

Suppose first that $X \dashrightarrow X^+$ is a small detected flip. Write $f : X \rightarrow Z$ and $f^+ : X^+ \rightarrow Z$, and fix an integral curve C on its negative ray. For $\xi \in H^2(X, \mathbb{R})$ set

$$s = \frac{\xi \cdot C}{c_1(L) \cdot C}.$$

By the preceding paragraph, $\xi - sc_1(L)$ vanishes on every f -contracted curve. Thus it is $f^*\eta$ for a unique $\eta \in H^2(Z, \mathbb{R})$. Define

$$T\xi = (f^+)^*\eta + sc_1(L^+).$$

For a Bott–Chern class, the same descent takes place in Bott–Chern cohomology, so the two transforms agree. The detected flip construction gives an effective rational divisor E_L on a common projective resolution $a : W \rightarrow X$, $b : W \rightarrow X^+$ such that

$$a^*c_1(L) - b^*c_1(L^+) = [E_L].$$

Consequently

$$a^*\xi = b^*T\xi + s[E_L].$$

If ξ is rational, then s is rational. Since the image of f^* on real degree-two cohomology is the realification of a rational subspace, its unique preimage η is rational as well. This proves rationality of $T\xi$.

The sets of prime divisors exceptional for a and b are identical: the two working spaces are small bimeromorphic. The old splitting and the displayed pullback identity therefore span

$$H^2(W, \mathbb{R}) = b^*H^2(X^+, \mathbb{R}) + \text{span}_{\mathbb{R}}\{[E] : E \text{ is } b\text{-exceptional}\},$$

and similarly in bidegree $(1, 1)$. The sum is direct. If a pullback from X^+ equals an exceptional divisor class, that divisor has degree zero on all b -contracted curves, so both signs of negativity make it zero; pullback injectivity then makes the original class zero. In particular the dimensions

on the two sides agree, T is injective by the old direct sum, and T is surjective. The rational version follows either from the same argument or from realification.

For a divisorial contraction $f : X \rightarrow X^+$, let D be its exceptional prime. Its degree on the contracted ray is negative: otherwise D would be f -nef, contrary to exceptional negativity. For any ξ , subtract

$$\frac{\xi \cdot C}{D \cdot C}[D]$$

and descend the result by the same birational cohomology lemma. This defines $T\xi$. On a common resolution its pullback difference is a multiple of the total transform of D . The target's exceptional prime set is the old one together with the strict transform of D . The same spanning and directness argument gives the target splitting and surjectivity; its kernel is $\mathbb{R}[D]$. Rationality is again immediate from the quotient of rational intersection numbers.

These arguments initially prove the splitting on a chosen common resolution. To obtain it on any projective smooth resolution $U \rightarrow X^+$, dominate that resolution and the chosen one by a smooth common projective resolution. Apply the smooth degree-two modification formula upstairs and push down to U . An exceptional divisor pushes either to zero or to a divisor exceptional over X^+ . This proves spanning on U ; negativity gives directness there.

Finally let $\mathcal{H}$ be a holomorphic line on U and put $\mathcal{Q} = (p_*\mathcal{H})^{**}$. Global strong $\mathbb{Q}$ -factoriality gives an $m > 0$ for which $\mathcal{M} = \mathcal{Q}^{[m]}$ is invertible. The coherent evaluation comparison gives

$$\mathcal{H}^{\otimes m} \simeq p^*\mathcal{M} \otimes \mathcal{O}_U(F)$$

for an integral p -exceptional divisor F . This comparison is obtained from the coherent direct image and evaluation map; it does not assume that $\mathcal{H}$ has a global meromorphic section. Dividing first Chern classes by m proves the line assertion. $\square$

Corollary 3.39 (Stability under projective extraction). *Let $r : Y \rightarrow X$ be a projective bimeromorphic morphism of normal compact Kähler spaces with rational singularities. Suppose that Y is globally strongly $\mathbb{Q}$ -factorial and that X has the splitting property in the preceding lemma. Then Y has that property as well. More precisely, if $E_1, \dots, E_s$ are the prime divisors exceptional for r , then*

$$H^2(Y, \mathbb{R}) = r^*H^2(X, \mathbb{R}) \oplus \bigoplus_{j=1}^s \mathbb{R}[E_j],$$

and the analogous statements hold over $\mathbb{Q}$ and in Bott–Chern cohomology.

Proof. Choose any projective smooth resolution $q : W \rightarrow Y$. The composite $p = rq$ is a projective resolution of X . Its exceptional primes are exactly the q -exceptional primes together with the strict transforms $\tilde{E}_j$ of the E_j . Since every E_j is $\mathbb{Q}$ -Cartier,

$$[\tilde{E}_j] = q^*[E_j] + [F_j]$$

for a rational q -exceptional divisor F_j . Substitute these identities in the splitting for p . The result spans $H^2(W, \mathbb{R})$ by $q^*H^2(Y, \mathbb{R})$ and the q -exceptional divisor classes. Their sum is direct: an exceptional divisor whose class is a pullback has degree zero on every q -contracted curve, and applying exceptional negativity to both signs makes the divisor zero. Pullback injectivity then applies. The same argument in bidegree $(1, 1)$ and over $\mathbb{Q}$ proves the splitting property for Y .

Now expand $q^*\xi$, for $\xi \in H^2(Y, \mathbb{R})$, using the original p -splitting and replace the $[\tilde{E}_j]$ as above. The difference between $q^*\xi$ and a class in $q^*(r^*H^2(X, \mathbb{R}) + \sum_j \mathbb{R}[E_j])$ is q -exceptional, and is

therefore zero by the direct sum just proved. Thus the displayed formula for $H^2(Y, \mathbb{R})$ spans. Its directness follows from exceptional negativity for r and pullback injectivity. The rational and Bott–Chern statements follow by the identical argument. The line bundle assertion of the preceding lemma uses only global strong $\mathbb{Q}$ -factoriality and hence applies to Y as well. $\square$

Support in the special-termination comparison. The following support comparison applies to the detected steps in the dimension induction. Let $\psi : X \dashrightarrow X^+$ be a detected small negative step with maps $f : X \rightarrow Z$ and $f^+ : X^+ \rightarrow Z$, and let L be its negative detector. For a sufficiently divisible $r > 0$, put

$$\mathrm{im}(f^* f_* L^r \rightarrow L^r) = \mathcal{I} \otimes L^r.$$

The zero set of $\mathcal{I}$ is exactly the exceptional locus of f . Outside that locus f is an isomorphism, so the evaluation map is surjective. On a positive-dimensional projective fibre, every section of L^r vanishes: its restriction to each integral curve has negative degree, and those curves cover every positive-dimensional fibre component. Connectedness supplies the whole nontrivial fibre.

On a resolution principalizing $\mathcal{I}$, the detector comparison is a positive multiple of the effective divisor defined by $\mathcal{I}\mathcal{O}_W$. Consequently its support is the full inverse image of the exceptional locus. The driving adjoint differs from a positive real multiple of $c_1(L)$ by a class pulled back from the contraction base. Its comparison divisor is therefore the same positive multiple of the detector comparison: their class difference is exceptional and zero, so both-sign negativity makes it zero as a divisor. The comparison for a longer negative program is at least this first-step divisor, by discrepancy monotonicity on a common resolution.

Suppose the beginning and end restrictions to a surviving normalized lc stratum S are isomorphic and their strict adjunction data agree. The general point of this stratum avoids every intervening surgery: a zero-discrepancy place remains such a place, whereas strict comparison would increase its discrepancy if its center were contained in the exceptional locus. Thus its strict transform S_W on a common resolution is not contained in the comparison support. Iterated strict adjunction identifies the restriction of the effective comparison with the difference of the identified adjoints, so its class is zero. A nonzero effective divisor on a positive-dimensional compact Kähler stratum has positive Kähler mass. The restricted divisor is therefore zero. If the first step meeting the stratum were nontrivial there, the full inverse-image support and surjectivity $S_W \rightarrow S$ would make this restriction nonzero, a contradiction. For a zero-dimensional stratum, generic-point avoidance already excludes an intersection. The whole program is consequently an isomorphism near the stratum. This is the same strict-adjunction support argument used in the proof of Theorem 3.13; it does not use its lower-dimensional abundance assumption.

Proposition 3.40 (A uniform relative NQC bound). *Let $f_0 : U \rightarrow Z$ be an already projective surjective morphism with connected fibres between smooth compact Kähler manifolds, and suppose that*

$$f_0^* : H^0(Z, \Omega_Z^2) \rightarrow H^0(U, \Omega_U^2)$$

is an isomorphism. Let α_0 be a nef class with an expression

$$\alpha_0 = \sum_{j=1}^k r_j \eta_{j,0}, \quad r_j > 0, \quad \eta_{j,0} \in H^2(U, \mathbb{Q}),$$

where $\eta_{j,0}$ has nonnegative degree on every curve vertical over Z . Let $D_0 = K_U + \Delta_U + \mathbf{M}_U$ be a generalized klt adjoint. Consider its projective relative program for $D_0 + \alpha\alpha_0$, under the projective local ordinary reduction described in the proof of Lemma 3.47.

If $\alpha_0 = 0$, triviality is immediate. Otherwise there are positive integers m_j , chosen once on U , and a number

$$\delta = \min_j \frac{r_j}{m_j} > 0$$

such that, for $a\delta > 2 \dim U$, the entire relative program is α_0 -trivial. The same number a works at every step.

Proof. Rational classes modulo the fixed base. Choose an f_0 -ample line bundle with integral Chern class H , and put $e = \dim U - \dim Z$ and $c = f_{0*}(H^e) > 0$. The rational operator

$$\Pi(\eta) = \eta - f_0^* \left(\frac{f_{0*}(\eta \smile H^e)}{c} \right)$$

preserves type $(1, 1)$ and kills types $(2, 0)$ and $(0, 2)$. Indeed these latter classes are pulled back from Z , and the projection formula applies. Thus $\Pi(H^2(U, \mathbb{Q})) \subset H^{1,1}(U) \cap H^2(U, \mathbb{Q})$. By the Lefschetz $(1, 1)$ theorem, choose an actual line bundle $L_{j,0}$ and an integer $m_j > 0$ with

$$\eta_{j,0} = \frac{1}{m_j} c_1(L_{j,0}) + f_0^* \gamma_j, \quad \gamma_j \in H^2(Z, \mathbb{Q}).$$

The original summands $\eta_{j,0}$ need not have type $(1, 1)$. Their base components account for this. Their weighted sum gives

$$\alpha_0 = \sum_j \frac{r_j}{m_j} c_1(L_{j,0}) + f_0^* \Gamma, \quad \Gamma = \sum_j r_j \gamma_j \in H^{1,1}(Z, \mathbb{R}). \quad (3.31)$$

The final assertion follows because f_0^* is an injective Hodge map and the other terms have type $(1, 1)$. The identity is therefore also one of Bott–Chern classes. Each $L_{j,0}$ has nonnegative integral degree on every Z -vertical curve.

Induction across an actual contraction. Suppose at a finite stage $f_i : U_i \rightarrow Z$ we have actual line bundles $L_{j,i}$, nef over Z , and

$$\alpha_i = \sum_j \frac{r_j}{m_j} c_1(L_{j,i}) + f_i^* \Gamma.$$

Assume also that α_i is nef and that the transformed D_i is generalized klt. These assertions hold initially. Let R be a $(D_i + a\alpha_i)$ -negative extremal ray over Z . It is D_i -negative because α_i is nef. Lemma 3.23 and Lemma 3.24 identify the relative curve cone with the corresponding face of the full analytic cone, at the initial model and after every step. The analytic cone theorem therefore supplies a rational generator C with

$$0 < -D_i \cdot C \leq 2 \dim U.$$

All $L_{j,i} \cdot C$ are nonnegative integers. If $\alpha_i \cdot C > 0$, one of them is at least one, and hence $\alpha_i \cdot C \geq \delta$. Therefore

$$(D_i + a\alpha_i) \cdot C \geq -2 \dim U + a\delta > 0,$$

a contradiction. We have $\alpha_i \cdot C = 0$ and $L_{j,i} \cdot C = 0$ for every j . Every curve contracted by the projective extremal contraction $h_i : U_i \rightarrow Y_i$ has class on R when tested by global line bundles, so the same vanishing holds for all of them.

On each fixed Stein chart of Z , choose the effective real ordinary representative of the initial relatively ample nef datum once, before running the program. Its actual real linear equivalence to

the fixed real line-bundle representative persists under pushforward. The Bott–Chern comparison with the generalized adjoint modulo the fixed base class persists separately by exceptional negativity, as detailed below. Thus every h_i is, locally on Y_i , a contraction with an ordinary real klt log-Fano boundary. On a relatively compact base chart, rational approximation in the finite affine system of Cartier adjoint identities gives a rational effective klt boundary with the same antiample sign, as in Proposition 3.18. Lemma 3.3 therefore applies. Consequently

$$M_{j,i} := (h_i)_* L_{j,i} \text{ is a line bundle,} \quad h_i^* M_{j,i} = L_{j,i}.$$

For a divisorial contraction set $L_{j,i+1} = M_{j,i}$. For a flip $h_i^+ : U_{i+1} \rightarrow Y_i$, set $L_{j,i+1} = (h_i^+)^* M_{j,i}$. These are actual line bundles with the *same* integers m_j . The descended Bott–Chern class

$$\alpha_{Y_i} = \sum_j \frac{r_j}{m_j} c_1(M_{j,i}) + f_{Y_i}^* \Gamma$$

satisfies $h_i^* \alpha_{Y_i} = \alpha_i$. It is nef by descent of nefness through the projective surjection h_i ; its pullback to the next model is α_{i+1} . This proves that the step is crepant for α_i . Hence it is also a D_i -negative step, and the original generalized pair remains gklt.

Finally each $M_{j,i}$ is nef over Z . For a Z -vertical curve $B \subset Y_i$, projectivity supplies a curve $B' \subset U_i$ mapping onto it with some positive degree d_B . Then

$$d_B(M_{j,i} \cdot B) = L_{j,i} \cdot B' \geq 0.$$

No assertion $d_B = 1$ is required. Pullback preserves this relative nefness, so the induction applies on U_{i+1} . Its curve degrees are again integers because the transported objects are line bundles, not because any curves were lifted with degree one. This proves the fixed bound through the whole program. $\square$

Remark 3.41 (Scope of the degree grid). The transported rational classes

$$\eta_{j,i} = \frac{1}{m_j} c_1(L_{j,i}) + f_i^* \gamma_j$$

have degrees in $m_j^{-1} \mathbb{Z}_{\geq 0}$ on curves vertical over the fixed smooth base Z . This relative assertion is exactly what the initial projective program requires. It does not assert nonnegativity of these individual classes on all curves of every birational model, and uses neither rational connectedness of resolution fibres nor Graber–Harris–Starr.

3.8.3 The negative-part comparison for a nef adjoint

We write $N_\sigma(\xi)$ for the divisorial negative part of a pseudo-effective class. We use its homogeneity, subadditivity, continuity after adding a vanishing Kähler perturbation, and the identities

$$N_\sigma(r^* \xi + [E]) = N_\sigma(r^* \xi) + E, \quad r_* N_\sigma(r^* \xi) = N_\sigma(\xi)$$

for an effective r -exceptional divisor E . A nef class has zero negative part. These are the divisorial Zariski-decomposition properties of [7, Section 3] and [19, Appendix A]. No identity $N_\sigma(r^* \xi) = r^* N_\sigma(\xi)$ for arbitrary ξ is assumed.

Lemma 3.42 (The absolute negative part after a relative program). *Let $\mu : U \rightarrow X$ be a projective resolution, let α be nef on X , let $c > 0$, and let $F_U \geq 0$ be μ -exceptional. Put*

$$D_U = c\mu^*\alpha + [F_U].$$

*Suppose that $\phi : U \dashrightarrow V$ is a finite D_U -negative program, possibly relative to another base, and denote its transformed class and divisor by D_V and $F_V = \phi_*F_U$. Then*

$$N_\sigma(D_V) = F_V.$$

Proof. Take a common projective resolution $p : W \rightarrow U$, $q : W \rightarrow V$. The negativity comparison for the finite program is

$$p^*D_U = q^*D_V + [E], \quad E \geq 0, \quad E \text{ is } q\text{-exceptional}.$$

This comparison is valid for a relative negative program: its proof uses the negativity of the contracted rays and the positive adjoint on each flipped side, not absolute nefness of the final adjoint. Since p^*F_U is effective and exceptional over X , the exceptional identity and nefness of $(\mu p)^*\alpha$ give

$$N_\sigma(p^*D_U) = p^*F_U = N_\sigma(q^*D_V) + E.$$

Pushing forward by q proves the assertion. In particular, the negative part in this statement is absolute, even when the program that produced V was relative. $\square$

3.8.4 Descent of the effective vertical divisor

Lemma 3.43 (Vertical numerical triviality). *Let $g : V \rightarrow S$ be a projective surjective morphism with connected fibres. Assume that S is globally Weil $\mathbb{Q}$ -factorial. Let $F \geq 0$ be an effective real Cartier divisor on V , vertical over S , such that*

$$F \cdot C = 0 \quad \text{for every curve } C \text{ contracted by } g.$$

Then there is an effective real Cartier divisor F_S on S such that, as actual divisors,

$$F = g^*F_S.$$

Proof. For a prime divisor $P \subset S$, write m_Q for the multiplicity of Q in g^*P , where Q ranges over the prime divisors dominating P . These multiplicities may be computed over the smooth generic locus of P , where P is Cartier. Put

$$b_P = \min_{g(Q)=P} \frac{\text{coeff}_Q F}{m_Q}, \quad F_S = \sum_P b_P P.$$

Only finitely many b_P are nonzero, since any such P is the image of a component of F . Thus F_S is a genuine effective Weil real divisor. Global Weil $\mathbb{Q}$ -factoriality makes this finite divisor real Cartier.

We first check equality over codimension one in S . Work near a general point of P and restrict to a transverse disk. Resolving the source, and cutting by general relative ample hypersurfaces if the relative dimension exceeds one, reduces to a projective surface over that disk with connected fibres. These operations can be performed over a relatively compact Stein neighbourhood; they do not require global sections on S . The intersection matrix of the components of the special fibre is

negative semidefinite, with kernel generated by the full fibre with its multiplicities. The restriction of F has zero intersection with every fibre component. Its coefficients are therefore proportional to those multiplicities. Equivalently, all the ratios in the definition of b_P are equal. When g is birational this assertion over the generic point of P is immediate.

Consequently

$$G = F - g^*F_S$$

is a signed real Cartier divisor supported over a subset of codimension at least two in S . Moreover, G is numerically trivial over S . We check that such a signed exceptional divisor is zero. The assertion is local on S . Over a relatively compact Stein neighbourhood, take $d = \dim V - \dim S$ general relative ample hypersurfaces and resolve their intersection. They give a projective generically finite morphism $H \rightarrow S$. The cuts may be chosen to meet any prescribed component of G in a nonzero divisorial trace. After normalization and Stein factorization, $H \rightarrow S$ factors as a projective birational morphism $H \rightarrow S'$ followed by a finite morphism $S' \rightarrow S$. The restricted divisor $G|_H$ is exceptional for $H \rightarrow S'$ and numerically trivial over S' . Applying the ordinary exceptional negativity lemma to both $G|_H$ and $-G|_H$ gives $G|_H = 0$. The choice of the cuts therefore excludes every nonzero component of G . Hence $G = 0$.

The surface argument above is also the usual proof of the degenerate-divisor negativity statement: after subtracting the minimum multiple of the full fibre, an effective residual fibre divisor omits a component and cannot be numerically trivial. Notice that projectivity of g was a hypothesis; it was not deduced from factoriality. $\square$

3.8.5 The negative part on the lower-dimensional base

Lemma 3.44 (Detecting the negative part by pullback currents). *Let $g : V \rightarrow S$ be a surjective morphism. Suppose that α_S is nef, $F_S \geq 0$ is real Cartier, $c > 0$, and*

$$D_S = c\alpha_S + [F_S], \quad D_V = g^*D_S, \quad N_\sigma(D_V) = g^*F_S.$$

Then $N_\sigma(D_S) = F_S$.

Proof. Nefness gives $N_\sigma(D_S) \leq F_S$. Fix a prime divisor $P \subset S$ and a prime divisor $Q \subset V$ dominating it. Write $m = \text{mult}_Q(g^*P) > 0$ and $b = \text{coeff}_P F_S$. At their generic smooth points, pullback of positive currents satisfies

$$\nu(g^*T, Q) = m\nu(T, P).$$

Indeed, the divisorial term $\nu(T, P)[P]$ pulls back with multiplicity m , and the residual current has zero generic Lelong number there.

Choose Kähler forms ω_S, ω_V and a constant $C > 0$ for which $C\omega_V - g^*\omega_S$ is Kähler. For $\varepsilon > 0$, let T_ε be any positive current in the big class $D_S + \varepsilon[\omega_S]$. Minimality of the multiplicity and monotonicity under adding a nef class give

$$\begin{aligned} m\nu(T_\varepsilon, P) &= \nu(g^*T_\varepsilon, Q) \\ &\geq \nu(D_V + \varepsilon g^*[\omega_S], Q) \\ &\geq \nu(D_V + C\varepsilon[\omega_V], Q). \end{aligned}$$

Taking the infimum over T_ε , and then letting $\varepsilon \downarrow 0$, gives

$$m\nu(D_S, P) \geq \nu(D_V, Q) = \text{coeff}_Q(g^*F_S) = mb.$$

This proves the reverse inequality for every P . All multiplicities are unchanged by resolving away from the generic points in question, so the same argument applies to the normal spaces under consideration.

The ε perturbation is necessary: at a pseudoeffective boundary class, the infimum over its exact positive currents need not equal its minimal multiplicity. The argument uses that equality only for the big perturbed classes. $\square$

Lemma 3.45 (Pullback when the positive part is nef). *Suppose*

$$\xi = P + [F], \quad P \text{ nef}, \quad F \geq 0 \text{ real Cartier}, \quad N_\sigma(\xi) = F.$$

For every projective resolution $p : W \rightarrow S$,

$$N_\sigma(p^*\xi) = p^*F.$$

Proof. Write $N' = N_\sigma(p^*\xi)$. Since p^*P is nef, $N' \leq p^*F$. Birational invariance gives $p_*N' = F$, so $E = p^*F - N' \geq 0$ is p -exceptional. The positive part of $p^*\xi$ is

$$p^*P + [E],$$

and is modified nef. If $E \neq 0$, exceptional negativity in its covering-curve form provides a component of E covered by curves C_t contracted by p , with $E \cdot C_t < 0$ [19, Appendix A, Lemma A.3]. A modified nef class has nonnegative intersection with a general member of a family of curves covering a prime divisor: use currents with arbitrarily small negative part and zero generic divisorial Lelong number, restrict to a general member, and let the negative bound tend to zero. But here

$$(p^*P + E) \cdot C_t = E \cdot C_t < 0,$$

a contradiction. Thus $E = 0$. $\square$

Proposition 3.46 (Any lower-dimensional good model suffices). *Suppose the initial relative program has the data in Lemma 3.42. Assume in addition that its class α_V is nef and agrees with $\mu^*\alpha$ on a common resolution, and that there is an already projective connected-fibre morphism $g : V \rightarrow S$ with*

$$\alpha_V = g^*\alpha_S, \quad D_V = g^*D_S.$$

*Suppose there is a projective small modification $s : S^q \rightarrow S$ such that S^q is globally Weil $\mathbb{Q}$ -factorial; the identity is allowed. Assume that the lower-dimensional adjoint s^*D_S has a good model: there are a normal compact Kähler space S_m , a common projective resolution $p : W \rightarrow S^q$, $q : W \rightarrow S_m$, and a nef semiample class D_m such that*

$$p^*s^*D_S = q^*D_m + [E], \quad E \geq 0, \quad E \text{ is } q\text{-exceptional}.$$

Then α is semiample on X . If the contraction defining semiampleness of D_m is Moishezon, the resulting contraction of α is Moishezon as well. In particular, it is unnecessary to lift a program on S to V , or to require that a chosen program producing S_m preserve α_S at every intermediate step.

Proof. Nefness descends under the surjective morphism g , so α_S is nef. The class of F_V is pulled back from S , so F_V is numerically trivial over S . It is therefore vertical: an effective horizontal component would restrict to a nonzero effective divisor on a general projective fibre, with positive intersection against a suitable power of an ample class, contradicting numerical triviality. For a

birational g , verticality is automatic by dimension. First take a projective resolution $\pi : \tilde{V} \rightarrow V$ of the main component of $V \times_S S^q$. The induced map $\tilde{g} : \tilde{V} \rightarrow S^q$ is projective and has connected fibres. Pull back α_V, D_V, F_V along π , and pull back α_S, D_S along s . All displayed class identities are preserved, and π^*F_V remains an effective vertical divisor. Lemma 3.42 gives $N_\sigma(D_V) = F_V$ before this modification. Because $D_V = c\alpha_V + [F_V]$ with α_V nef, Lemma 3.45 then gives

$$N_\sigma(\pi^*D_V) = \pi^*F_V.$$

We may therefore replace (V, S, g) by $(\tilde{V}, S^q, \tilde{g})$ and suppress the new superscripts in the rest of the proof. The good-model comparison now reads $p^*D_S = q^*D_m + [E]$. No assertion that the crepant subboundary on $\tilde{V}$ is effective is needed: this space is used only for classes, currents and the effective divisor π^*F_V . The lower-dimensional generalized pair is the crepant pullback to the small model S^q .

The class identity $[F_V] = g^*(D_S - c\alpha_S)$ shows that F_V is numerically trivial over S . Lemma 3.43 gives an actual effective real Cartier divisor F_S with $F_V = g^*F_S$. Injectivity of pullback yields

$$D_S = c\alpha_S + [F_S].$$

The established identity $N_\sigma(D_V) = F_V$ and Lemma 3.44 therefore give $N_\sigma(D_S) = F_S$, and Lemma 3.45 gives

$$N_\sigma(p^*D_S) = p^*F_S.$$

On the other hand q^*D_m is nef, so the exceptional-divisor identity applied to the good-model comparison gives

$$N_\sigma(p^*D_S) = E.$$

Thus $E = p^*F_S$ as actual divisors. Subtracting them from the comparison proves the exact class equality

$$q^*D_m = c p^*\alpha_S.$$

This proves the needed preservation of α_S from the final good model, without an assumption about its intermediate models.

Choose a contraction $h : S_m \rightarrow T$ to a normal compact Kähler space and a Kähler class κ on T with $D_m = h^*\kappa$. Taking a resolution of the main component of $V \times_S W$, and then a common projective resolution with U , produces a projective modification $r : R \rightarrow X$ and a holomorphic map $\ell : R \rightarrow T$ satisfying

$$r^*\alpha = \frac{1}{c}\ell^*\kappa.$$

For every curve C in a fibre of r this equality implies $\ell(C)$ is a point, since a Kähler class has positive degree on every nonconstant image curve. The fibres of the projective modification r are connected projective complex spaces, hence are connected by chains of curves. Therefore ℓ is constant on each fibre of r . The factorization theorem for a proper map onto a normal space gives a holomorphic map $f : X \rightarrow T$ with $\ell = f \circ r$. Pushing forward the class equality by r gives

$$\alpha = f^*(\kappa/c).$$

The maps from the main fibre-product component to S_m have connected general fibres; their Stein factorizations are finite birational over the normal space S_m , hence have connected fibres everywhere. Together with the connected fibres of h , this shows that ℓ , and therefore f , has connected fibres. This is the required semiampleness of α .

Finally, $R \rightarrow S_m$ is projective: it is obtained from the projective map g by base change, followed by projective resolutions and the projective map q . If h is Moishezon, choose a projective surjection $H \rightarrow S_m$ such that $H \rightarrow T$ is projective. A component of $R \times_{S_m} H$ dominating R is projective and surjective over X , and its map to T is projective. This is a Moishezon witness for f . $\square$

Lemma 3.47 (The contraction step in the dimension induction). *Assume C_{d-1} , B_{d-1} and M_{d-1} . Then $C_{d,\text{big}}$ and B_d hold.*

Proof. The big non-klt contraction. Apply the construction of [39, Section 3]. On its smooth modification $\nu : U \rightarrow X$ it writes

$$\nu^*\alpha = [D_U] + \eta_U, \quad D_U \geq 0, \quad \eta_U \text{ Kähler.}$$

The induction over the jumping coefficients of the multiplier ideal reduces the new reduced stratum to dimension at most $d - 1$. The given contraction on the old non-klt subspace and C_{d-1} provide the Moishezon maps of those strata. Every map that is promoted to a projective map in that construction contracts exactly the curves on which $\nu^*\alpha$ vanishes. On each such curve the actual real line bundle $-D_U$ has degree $\eta_U \cdot C$. On a further projective common resolution $q : U'' \rightarrow U$, choose an effective exceptional divisor E with $-E$ relatively ample and choose $\varepsilon > 0$ sufficiently small. Use the decomposition

$$q^*\nu^*\alpha = [D''] + \eta'', \quad D'' = q^*D_U + \varepsilon E, \quad \eta'' = q^*\eta_U - \varepsilon[E].$$

Here η'' is Kähler, and the actual real line bundle $-D''$ has degree $\eta'' \cdot C$ on every contracted curve. Restrict this decomposition to the smooth reduced components in the gluing construction. Lemma 3.20 then supplies precisely the relative ampleness needed in place of [39, Lemma 2.42], including the common-resolution step in its Claim 3.5. The exceptional term $-\varepsilon[E]$ makes the positive part η'' Kähler on the common resolution.

The exact sequences of multiplier ideals, relative vanishing, finite pushouts, and extension from sufficiently thickened $\text{Supp } D_U$ in that proof then apply unchanged. They give the birational Moishezon contraction in $C_{d,\text{big}}$. In particular the gluing step uses neither generalized termination nor a projectivity criterion for an undetected ray.

Preparation for both cases of B_d . The modified-big perturbation [39, Lemma 2.23] supplies a nef datum that dominates a Kähler class on its carrier. Subtracting a sufficiently small multiple of the pullback of a Kähler class ω_X still leaves a nef datum on that carrier. Thus α is a gklt adjoint plus $\varepsilon\omega_X$, and the cone argument of [39, Lemma 2.45] makes α NQC. This preparation applies whether or not α is big.

The nef-and-big case of B_d . For a gklt pair the multiplier ideal is the unit ideal, so the preceding big non-klt assertion now applies and gives a birational contraction $h : X \rightarrow Y$. On a projective resolution $p : W \rightarrow X$ chosen projective over Y , relative Kawamata–Viehweg vanishing for the effective exceptional divisor $-[B_W]$ gives $R^i h_* \mathcal{O}_X = 0$ for $i > 0$. Thus Y has rational singularities. Proper birational Bott–Chern descent [18, Lemma 8.7] gives $\alpha = h^*\gamma$. The descended adjoint is gklt, nef and big, and has no trivial curve. The criterion [38, Theorem 4.3] makes γ Kähler.

For completeness, the projectivity input in that criterion is restricted to a map $S \rightarrow Z'$ between smooth compact Kähler manifolds with rationally connected general fibre: S is the chosen smooth divisor on a resolution and Z' is the resolution of the null-locus component. The smooth-source, smooth-base argument in [15, Theorem 3.1, Step 1] applies. Its subsequent relative program starts with this projective map. Thus this application of the criterion uses only the smooth-source, smooth-base projectivity argument and an already-projective relative program.

The non-big case of $\mathbf{B}_d$. Use a projective small factorization and the modified-big perturbation of the pair. The non-pseudo-effectivity of K_X gives an MRC fibration by [58]. Choose a smooth Kähler MRC base Z and a smooth modification $\mu : U \rightarrow X$ such that $U \rightarrow Z$ is projective. Only the smooth case of [15, Theorem 3.1, Step 1] is needed: pullback identifies the holomorphic two-forms since the general fibre is rationally connected. Lemma 3.23 identifies any $(1, 1)$ -class on U , modulo a class pulled back from Z , with an actual real line bundle. Lemma 3.24 identifies its relative rays with rays of the analytic cone; both statements persist along the projective relative construction.

Write, as in [39, Theorem 4.1, Claim 4.1],

$$D_U(a) = K_U + \Delta_U + \mathbf{M}_U + a\alpha_U = (a+1)\alpha_U + F_U, \quad \alpha_U = \mu^*\alpha, \quad \Delta_U, F_U \geq 0,$$

where Δ_U is klt and F_U is μ -exceptional. The modified-big replacement is chosen, as in the cited Claim 4.1, so that the nef datum $\mathbf{M}_U$ on this smooth carrier is Kähler. If a further projective resolution is needed, subtract a sufficiently small effective exceptional divisor from its nef part and add that divisor to the subboundary; this preserves the adjoint and the klt inequalities. Choose a_0 using the uniform bound of Proposition 3.40. It makes the relative $D_U(a_0)$ -program α_U -trivial and preserves that choice through every step. This is a *projective* relative program from its first step: the nef data are represented by real line bundles modulo Z . Lemma 3.23 applies on every intermediate model over this fixed smooth base. More precisely, put $r : U \rightarrow Z$ and choose one actual global real line bundle L with

$$\mathbf{M}_U + a_0\alpha_U = c_1(L) + r^*\gamma.$$

The bundle L is relatively ample, because $\mathbf{M}_U$ is Kähler and α_U is nef. On each of a fixed finite collection of relatively compact Stein charts of Z , choose an effective real divisor Θ representing L , with $(U, \Delta_U + \Theta)$ klt. The representatives can retain a positive relatively ample part in their boundaries. Here the equality $K_U + \Delta_U + \Theta \sim_{\mathbb{R}} K_U + \Delta_U + L$ is an actual real linear equivalence of line bundles; the equality with $D_U(a_0)$ modulo $r^*\gamma$ is separately an equality of Bott–Chern classes.

Fix these ordinary pairs on the initial charts. Their actual linear equivalences and their Bott–Chern comparisons with $D_U(a_0)$ modulo the fixed base class persist separately throughout a global relative program. Proposition 3.25 applies: the source is smooth, its adjoint is represented by an actual global real line modulo Z , its carrier nef part is relatively ample, and $D_U(a_0) = (a_0+1)\mu^*\alpha + [F_U]$ is pseudo-effective. It gives a finite program $U \dashrightarrow V$ and a good relative endpoint. This is one global program controlled by fixed ordinary chart pairs. Its construction uses global line algebras, not a choice of unrelated local minimal models. Proposition 3.40 makes every step α_U -trivial with the same value a_0 .

Here is an explicit ample-model comparison that gives the required descent of α_V . Both $D_V(a_0)$ and α_V are nef over Z . Put $a_1 = a_0 + 1$ and $a_2 = a_0 + 2$. Before the program starts, choose fixed ordinary representatives on the finite chart cover for all three relatively ample nef parts $\mathbf{M}_U + a_j\alpha_U$, $j = 0, 1, 2$, on one simultaneous log resolution. The α_U -crepancy of the constructed program makes its endpoint V a weak model for each of these three fixed ordinary adjoints. Their traces $D_V(a_j) = D_V(a_0) + (a_j - a_0)\alpha_V$ are nef. Corollary 3.26 therefore makes

$$D_V(a_j) = D_V(a_0) + (a_j - a_0)\alpha_V$$

semiample over Z . Their zero curves are exactly the common zero curves of $D_V(a_0)$ and α_V . Their projective ample-model contractions therefore have the same fibres: these fibres are connected by

curves, and each contraction is constant on the fibres of the other. Identify the two normal targets, writing the common contraction as $g : V \rightarrow S$. If $D_V(a_j) = g^*\lambda_j$, with λ_j relatively Kähler over Z , subtraction gives

$$\alpha_V = g^*(\lambda_2 - \lambda_1) = g^*\alpha_S.$$

This is an equality of Bott–Chern classes, not merely of curve degrees. Since the initial program is α_U -trivial, it is also $D_U(a_1)$ -negative. From now on put $a = a_1$ and suppress the argument in $D_U(a)$ and $D_V(a)$.

The maps $V \rightarrow Z$ and $S \rightarrow Z$ are projective. Moreover g is projective: a relatively ample bundle for $V \rightarrow Z$ restricts to an ample bundle on every fibre of g , and hence is g -ample. Each $D_U(a_j)$ is not big globally: pushing a big such class to X would make $(a_j + 1)\alpha$ big. If it were big over the smooth MRC base Z , relative-canonical positivity [39, Theorem 2.48], together with the pseudo-effectivity of K_Z , would make it big globally. It is therefore not big over Z . This property persists to its relative semiample model, so its relative canonical morphism has $\dim S < d$. Apply Proposition 3.30 to g and the whole adjoint D_V . Its generalized data are effective and gklt. The nef datum is represented on the initial carrier by $\mathbf{M}_U + a\alpha_U$, a Kähler class; on a common carrier over V its pullback is nef and big. Adding the strict transform of the effective boundary proves that the total boundary on V is globally modified big. The proposition therefore gives

$$D_S = K_S + B_S + \mathbf{N}_S, \quad D_V = g^*D_S,$$

with effective gklt data on S , a globally nef carrier, and globally modified-big total boundary. Independently, α_S is nef by descent of nefness through the surjective projective map g . The lower-dimensional model assertion will be applied to this whole adjoint D_S .

Take a projective small factorization $s : S^q \rightarrow S$ and resolve the main component of $V \times_S S^q$. The resulting maps $\pi : \tilde{V} \rightarrow V$ and $\tilde{g} : \tilde{V} \rightarrow S^q$ are projective, and $\tilde{V}$ is compact Kähler. Pull back D_V , α_V and F_V to $\tilde{V}$, and D_S and α_S to S^q . The pair on S^q is gklt and its boundary-plus-nef class remains modified big. Lemma 3.42 first gives $N_\sigma(D_V) = F_V$. Lemma 3.45 then gives $N_\sigma(\pi^*D_V) = \pi^*F_V$. The new total space is used only as a carrier for this equality and for pullbacks of positive currents; no effective boundary on it is needed. Rename these spaces V and S . No additional relative program is needed for this modification.

The vertical-divisor descent and negative-part lemmas above now give actual effective divisors F_S, F_V with

$$F_V = g^*F_S, \quad D_S = (a+1)\alpha_S + [F_S], \quad N_\sigma(D_S) = F_S.$$

Apply $\mathbf{M}_{d-1}$ to the generalized pair with adjoint D_S . On a common resolution $p : W \rightarrow S$, $q : W \rightarrow S_m$ of the resulting good model, write

$$p^*D_S = q^*D_m + [E], \quad E \geq 0 \text{ exceptional over } S_m.$$

The negative-part comparison proved above gives $E = p^*F_S$, and hence

$$q^*D_m = (a+1)p^*\alpha_S.$$

Since D_m is semiample, α_S is semiample after descent through p . Pulling back by g and using the α -trivial initial program gives semiampleness of α on X . The target is compact Kähler, and the resulting map is Moishezon, as follows by taking common projective modifications of the maps just constructed. The comparison uses only the chosen lower good model; no lower-dimensional program is lifted to V .

□

Proposition 3.48 (A negative ray with an actual detector). *Assume B_d . Let A be a gklt adjoint in dimension at most d on a globally strongly $\mathbb{Q}$ -factorial compact Kähler space, with globally nef carrier data. Let R be an A -negative extremal ray of the full analytic cone. If an actual global rational line L has $L \cdot R < 0$, the ray has a projective contraction; a small contraction has its detected flip. The working models retain the strong factoriality and compact Kähler properties. In particular this applies to an ordinary rational adjoint $A = c_1(K_X + B)$ with detector $K_X + B$, and to ordinary real boundaries by Proposition 3.19.*

Proof. The local polyhedrality of the negative cone makes R an exposed ray. Choose a nef support λ whose null analytic face is R . Normalize the analytic cone by a Kähler class. On a neighbourhood of the point representing R , the class $-A$ is positive. On the remaining compact part of this slice, λ has a positive minimum. Consequently, for a sufficiently large b , the class $\omega_R = b\lambda - A$ is Kähler. Thus $A + \omega_R = b\lambda$ is nef with precisely the null face R . The gklt presentation with nef carrier datum increased by the pullback of ω_R has modified-big total boundary. Apply B_d to construct its contraction. Lemma 3.16 makes the same global line $-L$ relatively ample, and Proposition 3.18 constructs the flip when needed. For a divisorial contraction the exceptional-prime negativity and integral line descent give strong factoriality on the target: pull back a rank-one reflexive sheaf, remove its degree by a rational multiple of the exceptional divisor, descend the resulting line, and compare on the common big open. Local ordinary log-Fano replacement gives rational singularities by relative vanishing. The supporting target and projective flip are compact Kähler. $\square$

3.8.6 Relative good models and finite geography

Lemma 3.49 (Removing the preparation error). *Let $\mu : W \rightarrow X$ be a projective smooth preparation over S with $A_W = \mu^*A_X + [F]$, where $F \geq 0$ is μ -exceptional and has positive coefficient on every added prime. If $W \dashrightarrow Y$ is a chosen good log terminal model of A_W over S , then it induces a good log terminal model of A_X over S .*

Proof. On a common projective resolution $p : V \rightarrow W$, $q : V \rightarrow Y$, write $p^*A_W = q^*A_Y + [E]$ with $E \geq 0$ exceptional over Y . Put $r = \mu p$ and $D = E - p^*F$. Then

$$[D] = r^*A_X - q^*A_Y, \quad r_*D = r_*E \geq 0.$$

The divisor $-D$ is r -nef because A_Y is nef over S . Negativity gives $D \geq 0$, hence $E \geq p^*F$. Every added prime is therefore exceptional over Y . This proves nonextraction from X and gives its effective exceptional comparison with A_Y . Strictness at any contracted prime of X follows from strictness for the chosen log terminal model of W , since F has coefficient zero there. The same semiample endpoint makes the model good. $\square$

Lemma 3.50 (Recovering the uncontracted original primes). *Let a gklt pair on X have a nonextracting weak good model $X \dashrightarrow Y$ over S . Assume Y is compact Kähler, globally strongly $\mathbb{Q}$ -factorial and has the exceptional splitting of Lemma 3.38. Then a projective crepant extraction $Y' \rightarrow Y$ gives a good log terminal model of the original pair. The exceptional splitting persists on Y' .*

Proof. On a common resolution write the actual comparison $p^*A_X = q^*A_Y + [E]$, where $E \geq 0$ is q -exceptional. There are finitely many primes of X contracted by its marking. Let $\mathcal{P}$ consist of those with coefficient zero in E . Take a projective log resolution $r : W \rightarrow Y$ which contains all these valuations and dominates the fixed nef carrier. In the crepant structure boundary B_W , the coefficient of each member of $\mathcal{P}$ is its original effective boundary coefficient, hence lies in $[0, 1)$.

Keep these coefficients unchanged. Increase each other r -exceptional coefficient, if necessary from a negative value, to a number in $(\max\{0, b_Q\}, 1)$. Keep all nonexceptional coefficients unchanged. The resulting effective klt boundary Γ_W satisfies

$$[K_W + \Gamma_W] + \mathbf{M}_W = r^* A_Y + [F],$$

where $F \geq 0$ has precisely the unwanted exceptional primes as its support. The inherited splitting gives actual real-line representatives for every class modulo Y . Since r is birational, the nef carrier datum is relatively big; the adjoint is relatively represented by F and is pseudo-effective. Apply Proposition 3.25 over Y . Its projective endpoint $r' : Y' \rightarrow Y$ has an effective relatively nef exceptional trace F' , so negativity gives $F' = 0$. A prime outside $\text{Supp } F$ cannot be contracted by an F -negative birational program: on a general curve through its generic point the effective divisor has nonnegative degree. Thus all members of $\mathcal{P}$ survive, and all unwanted exceptional primes disappear. No new prime is extracted by the program. It follows that r' is crepant and that the marking from X contracts exactly primes with strictly positive original comparison coefficient. It is nonextracting and is therefore a log terminal model. The pullback of the semiample adjoint on Y makes it good. Finally apply Corollary 3.39. $\square$

Lemma 3.51 (A detector throughout the big-adjoint construction). *Assume $\mathbf{B}_d$ and $\mathbf{F}_{d-1}$. The big-adjoint construction of [39, Section 5.1] gives a chosen good model in dimension d using only detected steps. It also gives the case of $\mathbf{M}_d$ in which $A + cf^*\omega_S$ is big for some $c > 0$, with the map to S retained throughout.*

Proof. After the smooth preparation in [39, Proposition 5.12], write the adjoint as

$$A = D + \omega, \quad D \geq 0, \quad \omega \text{ Kähler}, \quad \text{Supp } D = N_+(A).$$

Here $N_+(A)$ denotes the reduced divisor with support $\bigcap_{0 < \epsilon \ll 1} \text{Supp } N_\sigma(A - \epsilon\omega)$; this support is constant for sufficiently small positive ϵ and is independent of the chosen Kähler class ω [39, Lemma 5.3]. At a nonterminal scaling step,

$$A_i \cdot R_i < 0, \quad (A_i + t_i \omega_i) \cdot R_i = 0, \quad t_i > 0.$$

Consequently $\omega_i \cdot R_i > 0$ and $D_i \cdot R_i < 0$. A component of the actual effective divisor D_i supplies a global rational Cartier detector. The supporting contraction follows from $\mathbf{B}_d$, and its projectivity and flip follow from Proposition 3.18. This argument does not require ω_i to remain nef.

To apply $\mathbf{B}_d$ to a gdlt presentation, lower its finitely many floor coefficients on the initial smooth carrier and add the same small class to the Kähler nef datum. The adjoint does not change, and the resulting gklt presentation persists through the same negative steps. The original gdlt presentation is retained for adjunction to the floor. Thus the special-termination proof of [39, Lemma 5.8] applies with $\mathbf{F}_{d-1}$ on its smooth lower-dimensional carrier. Its other inputs are adjunction, discrepancy monotonicity, and the local Cartier-index calculation. The last neighbourhood-isomorphism step uses the evaluation-ideal support argument in the [support comparison above](#): the comparison of the first detected flip has support equal to the full inverse image of its exceptional locus, and later comparisons dominate it. One does not infer disjointness of images merely from a vanishing restriction of an arbitrary exceptional divisor. The theta induction of [39, Proposition 5.11] now gives a weak model. Applying $\mathbf{B}_d$ to its nef adjoint makes it good, and Lemma 3.50 gives the log terminal model. All its steps are detected or belong to the already-projective relative extraction in that lemma.

For completeness, the negative-part identity needed in this argument is only

$$N(p^*\xi + E) = N(p^*\xi) + E \quad (E \geq 0 \text{ } p\text{-exceptional}),$$

not the stronger formula $N(p^*\xi) = p^*N(\xi)$ for an arbitrary pseudo-effective class. The comparison with a nef endpoint identifies its exceptional error with the negative part. The formula for N_+ follows by writing a Kähler form upstairs as $p^*\omega - F$, with F positive on every exceptional prime, and applying the displayed identity to $p^*(\xi - \epsilon\omega) + E + \epsilon F$. Pushforward identifies the nonexceptional coefficients; ϵF supplies all exceptional primes. These are exactly the uses in the theta argument.

Now put $P = f^*\omega_S$ and suppose $A + cP$ is big for some c . Fix $\delta > 0$ smaller than the ω_S -degree of every compact integral curve on S and enlarge c so $c > 4d/\delta$. Use the following direct preparation, which keeps the original unshifted globally nef datum. On a smooth projective carrier $\nu : W \rightarrow X$ write

$$\nu^*(A + cP) = [G] + \omega, \quad G \geq 0, \quad \omega \text{ Kähler}, \quad [K_W + B_W] + M_W = \nu^*A,$$

with the non-Kähler support property used in the big-class preparation. Here $\text{Ex}(\nu)$ below denotes the reduced divisor formed by all ν -exceptional primes. Write $B_W = B_W^+ - B_W^-$. For small $\epsilon > 0$ put

$$\begin{aligned} B' &= B_W^+ + \epsilon B_W^- + \epsilon(G + \text{Ex}(\nu)), \\ M'_W &= M_W + c\nu^*P + \epsilon\omega, \\ b &= B_W^+ + \frac{\epsilon}{1 + \epsilon} \text{Ex}(\nu). \end{aligned}$$

The new boundary is klt and M'_W is Kähler. Direct addition gives

$$A' = [K_W + B'] + M'_W = (1 + \epsilon)([K_W + b] + M_W + c\nu^*P).$$

Moreover $B' \geq b$. In every theta branch the cumulative added boundary satisfies $0 \leq C \leq 1 - B'$, so

$$b + \frac{C}{1 + \epsilon} \leq b + \frac{1 - b}{1 + \epsilon} < 1.$$

Thus the unshifted presentation $[K_W + b + C/(1 + \epsilon)] + M_W$ is genuinely gklt with the *original* globally nef datum. This is the presentation used for the horizontal length bound. No base form is subtracted from an arbitrary newly prepared nef datum.

A horizontal negative ray would have P -degree at least δ ; the bound $0 < -A_{\text{unshifted}} \cdot C \leq 4d$ would contradict $c\delta > 4d$. Every constructed step is therefore over S . Its projective connected fibres are contracted by f , so f factors through its base and through the flip. The final finite extraction is also over S .

The shifted nef endpoint is semiample by B_d . Its zero face contains no horizontal curve: decompose such a curve by the unshifted adjoint's cone theorem; nefness of the shifted class forces all summands into the zero face, and a summand of positive P -degree again contradicts the length bound. The shifted class is big, so the connected semiample map is bimeromorphic. Its connected fibres are Moishezon and are connected by chains of compact curves. The preceding zero-face argument makes the map to S constant on each such curve, hence on every fibre. The analytic factorization lemma therefore factors the map to S through the semiample map. Subtracting $c\omega_S$ from the normalized class on its target gives the required relative Kähler class for

the unshifted adjoint. The effective resolution error from the preparation is removed by the usual negativity comparison, giving a model of the original data.

The forward traces and exceptional splittings persist by Lemma 3.38. In the last extraction, the newly extracted prime classes move from the old exceptional span into the pullback span; they are actual rational Cartier classes. Both-sign negativity proves directness as before. $\square$

Lemma 3.52 (Changing the base of a relatively trivial adjoint). *Suppose $g : X \rightarrow Y$ is projective, its general fibre is rationally connected, and a gklt adjoint satisfies $A_X = g^*A_Y$. Assume the prepared nef datum is Kähler on a smooth carrier. Let $Y \dashrightarrow Y_m$ be the chosen nonextracting good model supplied by M_{d-1} over S , with its exceptional cohomology splitting, and let $\dim Y < d$. Then a single projective relative construction gives a weak good model of (X, A_X) over S with that splitting. Lemma 3.50 makes it log terminal. No step of the base program is lifted.*

Proof. Take a common projective smooth resolution $T \rightarrow Y$, $T \rightarrow Y_m$ and resolve the main component of $X \times_Y T$, further dominating the prepared nef carrier; denote the resulting smooth space by V . Its morphism $h : V \rightarrow Y_m$ is projective. Before constructing a program, verify algebraicity for all classes. The projective RC map $V \rightarrow T$ and Lemma 3.23 express every class modulo T by actual real lines. The splitting of the chosen Y_m expresses classes from T , modulo Y_m , by exceptional divisor classes. Hence

$$H_{\text{BC}}^{1,1}(V) = c_1(\text{Pic}(V) \otimes \mathbb{R}) + h^*H_{\text{BC}}^{1,1}(Y_m).$$

Lemma 3.24 identifies the relative cone with the full analytic face. These properties and the exceptional splitting persist along the projective relative steps by Lemma 3.38.

Choose the effective gklt preparation on this final carrier. Its adjoint has the identity

$$A_V = h^*A_{Y_m} + [G], \quad G = G_{\text{base}} + F \geq 0,$$

where G_{base} is pulled back from the actual effective base comparison and F is exceptional over X , with positive coefficient on every added prime. Its nef datum is nef and big on the prepared carrier. Modulo Y_m it is the actual real-line class $[G] - c_1(K_V) - [B_V]$. Thus every hypothesis of Proposition 3.25 holds, with the effective relative adjoint G . Denote its global relative endpoint by V_m and the strict transform of G by G_m .

On a common resolution $p : W \rightarrow V$, $q : W \rightarrow V_m$, the actual adjoint comparison has effective q -exceptional error E . The difference $p^*G - q^*G_m - E$ has zero Bott–Chern class and is q -exceptional, because $q_*p^*G = G_m$. Negativity in both signs makes it zero:

$$p^*G = q^*G_m + E. \tag{*}$$

Choose a big common-isomorphism open $U \subset Y_m$ for $Y \dashrightarrow Y_m$. Such an open exists because that map is nonextracting. On U we have $G_{\text{base}} = 0$. Let $r : W_U \rightarrow X_U$ be the induced projective birational map. The divisor q^*G_m is r -nef, since the endpoint adjoint is nef over U and the base class has zero degree on r -fibres. Equation (*) gives

$$r_*q^*G_m = -r_*E \leq 0,$$

since F is exceptional over X . Negativity yields $q^*G_m \leq 0$. Effectivity yields $G_m|_U = 0$. This step removes the possible *horizontal* components of F ; they were not assumed very exceptional at the start.

Now G_m is supported over codimension at least two in Y_m . It is nef over Y_m , so very-exceptional negativity gives $G_m = 0$. The endpoint adjoint is the pullback of A_{Y_m} , hence is nef and good over S . Its relative canonical morphism is Moishezon: it is the connected factor of the composite of the projective morphism to Y_m and the lower-dimensional Moishezon canonical morphism. Lemma 3.49 removes the added source-exceptional errors and proves nonextraction. All these operations are over S .

The all-class invariant was established before the relative program; its preservation follows at each step from the same cone identification and forward cohomology splitting. If the construction has contracted any original prime crepantly, apply Lemma 3.50 to retain it. $\square$

Lemma 3.53 (The global rational quotient over a fixed base). *Assume M_{d-1} and the case of M_d in which a base shift makes the adjoint big. Then M_d holds when no class $A + cf^*\omega_S$, $c \geq 0$, is big.*

Proof. First take the usual smooth carrier with effective gklt boundary and adjoint $\mu^*A + [F_0]$, where F_0 is effective exceptional with full exceptional support. Its boundary-plus-nef trace is big and its adjoint is not big, so its canonical class is not pseudo-effective. Take its global MRC and resolve the MRC graph and the fixed carrier, obtaining a projective map to a smooth compact Kähler Z with K_Z pseudo-effective. Perform the Kähler-nef-part preparation *after* these graph resolutions, preserving the maps to Z and S . This order ensures that the final scaling datum is Kähler, rather than merely the nef pullback of an earlier Kähler datum. Write the resulting adjoint as

$$A_0 = K_U + B_U + M_U = \mu^*A + [F], \quad F \geq 0 \text{ exceptional}, \quad M_U \text{ Kähler.}$$

The global modified-big hypothesis makes a preliminary base shift unnecessary. Relative pseudo-effectivity is preserved. No base shift of A_0 is big, since pushing a big such class down would make a base shift of A big. The fixed morphism $h : U \rightarrow Z$ is projective and has RC general fibres. All these preparations precede the next, fresh base shift.

Put $P = f_U^*\omega_S$ and choose $c > 2d/\delta$, where every integral curve on S has ω_S -degree greater than $\delta > 0$. Set

$$A_c = A_0 + cP = D + N, \quad D = K_U + B_U, \quad N = M_U + cP.$$

Since A_c is not big globally, relative-canonical positivity [39, Theorem 2.48(2)], together with pseudo-effectivity of K_Z , shows that A_c is not big over Z , whether or not it is globally pseudo-effective. The class N is big over Z . Consequently D cannot be pseudo-effective over Z : otherwise $D + N = A_c$ would be big over Z . The smooth projective RC Hodge projection represents every class modulo Z by an actual real line. The ordinary global klt adjoint D is therefore relatively non-pseudo-effective. The projective ordinary Mori construction of Corollary 3.28, with scaling of the actual relatively ample real line represented by N , terminates in a Mori fibre space. Lemma 3.23 and Lemma 3.24 identify every relative ray with a full analytic ray. Its nef b-data retain the fixed globally nef carrier U .

First suppose A_c is not pseudo-effective globally. The retained relative-canonical positivity theorem on h , together with pseudo-effectivity of K_Z , then says that A_c is not pseudo-effective over Z . The terminating relative program ends in a Mori contraction at a threshold $\tau > 1$. At every step its scaled zero ray has A_c -degree negative. Inductively the map to S is retained: if such a ray were horizontal, its P -degree would exceed δ , while the gklt cone theorem for A_0 gives a rational generator of $-A_0$ -degree at most $2d$, contradicting $A_c \cdot R \leq 0$. Once $P \cdot R = 0$, the step is negative or crepant for A_0 , so its gklt presentation persists. The same argument applies to the

final Mori ray. This would be an A_c -negative Mori contraction over S , contradicting the preserved pseudo-effectivity over S . Hence A_c is pseudo-effective globally.

It is not big globally. Relative-canonical positivity now implies that it is not big over Z . Its pseudo-effective threshold for $D + tN$ over Z is exactly one: for $t > 1$ the class is big, while pseudo-effectivity for $t < 1$ would make A_c big because N is big over Z . The same finite relative program therefore ends in a Mori contraction $g : U' \rightarrow Y$ at threshold one. All birational steps, including any steps at threshold one, and the final zero ray are over S by the preceding length argument with $A_c \cdot R \leq 0$. For a zero ray with positive P -degree, A_0 would again have degree less than $-c\delta$. Thus Y carries a map to S and $\dim Y < d$. The class A_c has zero degree on every g -contracted curve. This is an ordinary projective log-Fano contraction: rationalize its effective real ordinary boundary, preserving klt and relative antiampleness. Relative vanishing gives rational singularities on Y , which is compact Kähler because it is projective over Z . Lemma 3.15 therefore gives an actual equality $A_c = g^*A_Y$ in Bott–Chern cohomology. This descent precedes the base presentation. Apply Proposition 3.30 to g and the whole adjoint A_c . The negative or crepant steps preserve its effective gklt data. Its nef part is the trace of $M_U + cP$ from the fixed carrier; on a common carrier it is nef and big, so its total boundary is globally modified big. The proposition gives effective gklt data for A_Y with globally nef carrier and globally modified-big total boundary. Relative pseudo-effectivity descends through the projective surjection g , so M_{d-1} applies over S , with the base shift retained. For the source data in Lemma 3.52, use Lemma 3.29 to choose a Kähler nef datum on a smooth carrier while keeping A_c fixed. Apply that graph lemma to the chosen lower-dimensional output.

Adjoint-preserving preparation is compatible with the original pair in this comparison. On a common resolution of a nonextracting good output, keep the original nef datum and subtract the effective model comparison divisor from the original structure divisor. The resulting structure divisor has the same adjoint as the output; its trace is effective by nonextraction, and its discrepancies have not decreased. Strictness on contracted original primes is preserved. Thus the prepared presentation supplies a comparison for the original data as well.

Steps at threshold one can be A_c -crepant divisorial contractions. Consequently the composite constructed so far need only be a weak good model of the original prepared adjoint. Apply Lemma 3.50 to extract the finitely many original primes with zero comparison coefficient. This gives the strict log terminal model and preserves the cohomology invariant. The shift changes neither relative comparisons nor relative nefness. Lemma 3.49 removes the full-support preparation error and proves the claimed M_d . $\square$

3.8.7 Finite geography and closing the induction

Lemma 3.54 (Canonical models first, weak models second). *Assume B_d and M_d . Then F_d holds for compact polytopes of globally nef b -data on a fixed carrier and globally modified-big boundary, over the fixed base.*

Proof. First prove, by induction on the parameter-polytope dimension, polyhedrality of the effective locus, finite canonical-model chambers, and finitely many *chosen* terminal models on those chambers, following [39, Theorem 5.15]. The zero-dimensional case uses M_d and uniqueness of the canonical model; it does not yet assert finiteness of all weak models.

At a central point choose $X \dashrightarrow X_m \rightarrow X_c$ over S . The transport invariant in M_d supplies the traces of the whole nearby polytope on X_m . Shrink so the finitely many strict exceptional inequalities remain strict. Add a common pullback from S so that the central canonical class on X_c is Kähler absolutely. On the boundary of the parameter polytope, apply the smaller-parameter

induction over X_c ; its existence input is $M_d(X_c)$, just proved independently. No projectivity of $X_m \rightarrow X_c$ or Hodge projection over the possibly singular X_c is used.

The central adjoint is zero over X_c , so the relative effective locus is the radial cone on its boundary effective locus. Apply the cone-length estimate of [39, Claim 5.1] on the finitely chosen *gklt terminal models* $h_i : Y_i \rightarrow X_c$, all of dimension d . Write $A_{u,i}$ for the boundary-parameter adjoint, nef over X_c , and $\psi_i : Y_i \rightarrow Z_i$ for its relative canonical morphism. The cone theorem is not applied to the possibly lower-dimensional canonical targets Z_i .

Let $H_i = h_i^* \omega_c$ and choose $\delta > 0$ below the ω_c -degree of every compact integral curve in X_c . Fix $\bar{\lambda} < \delta/(\delta + 2d)$ and for $0 < \lambda \leq \bar{\lambda}$ put

$$A_{\lambda,i} = (1 - \lambda)H_i + \lambda A_{u,i}.$$

The same cone estimate shows more than nefness: for a sufficiently small $\epsilon > 0$, uniform on the finite list and the chosen radial interval, $A_{\lambda,i} - \epsilon H_i$ is nef. Indeed on a horizontal $A_{u,i}$ -negative rational ray its degree is greater than $(1 - \lambda - \epsilon)\delta - 2d\lambda > 0$; on the nonnegative part of the cone and on the vertical cone it is nonnegative. It follows for every class $\xi \in \overline{NA}(Y_i)$ that

$$A_{\lambda,i} \cdot \xi = 0 \iff H_i \cdot \xi = A_{u,i} \cdot \xi = 0.$$

In particular its null curves are exactly those contracted by ψ_i . The central data are crepant through these maps over X_c ; thus the convex data defining $A_{\lambda,i}$ remain gklt with globally modified-big boundary. Apply B_d to $A_{\lambda,i}$. Its connected semiample map and ψ_i both have Moishezon fibres, and each connected such fibre is connected by chains of compact curves. Equality of their contracted curves makes each map constant on the fibres of the other. The analytic factorization lemma identifies the two maps. Thus the selected relative canonical model is also the canonical model over S on this radial interval. This proves local polyhedrality and the finite local list; compactness gives the global finite list of canonical models and chosen terminal models.

Uniform preparation preserving every weak model. Write $A_u = [K_X + B_u + M_{u,X}]$ for the adjoints in the parameter polytope. We first record why a weak model of these gklt data extracts no divisor. On a common resolution $p : U \rightarrow X$, $q : U \rightarrow Y$ put $D = p^* A_u - q^* A_{u,Y}$, using the actual structure-boundary comparison. The weak-model discrepancy inequalities give $p_* D \geq 0$, and $-D$ is p -nef because $A_{u,Y}$ is nef over the fixed base. Negativity therefore gives $D \geq 0$. An extracted prime on Y has coefficient one in the weak-model boundary, whereas its coefficient in the structure boundary of the gklt source is less than one; it would give a negative coefficient of D . Hence there is no such prime. Consequently D is q -exceptional and the target is itself gklt.

Fix a parameter u_0 . The boundary plus nef part is globally modified big by hypothesis. The modified-big preparation [39, Lemma 2.23] gives, on a fixed projective smooth carrier $a : W \rightarrow X$, an effective klt boundary Γ , a Kähler class γ_0 , and an effective a -exceptional real divisor F such that

$$[K_W + \Gamma] + \gamma_0 = a^* A_{u_0} + [F]. \quad (3.32)$$

Here is the effectivity detail in this use of the preparation. Its new nef datum dominates a Kähler form on a carrier. After resolving the new structure boundary, choose an effective exceptional divisor whose negative is relatively ample and subtract a sufficiently small multiple from the pulled-back datum to make it Kähler. Add the same multiple to the structure boundary, then replace its negative coefficients by zero and, if needed, add small positive coefficients on the remaining exceptional primes. All coefficients stay below one. Only exceptional coefficients have been increased; their increase is the divisor $F \geq 0$ in (3.32).

On a sufficiently small closed polyhedral neighbourhood of u_0 put

$$\gamma_u = \gamma_0 + a^*(A_u - A_{u_0}).$$

These classes remain Kähler by openness, and the same Γ and the same F satisfy

$$[K_W + \Gamma] + \gamma_u = a^*A_u + [F].$$

Thus this is one affine family of gklt pairs on one smooth carrier, with a fixed effective boundary and Kähler nef data.

Let $\psi : X \dashrightarrow Y$ be *any* weak model at a parameter u in this neighbourhood, and resolve the induced map $W \dashrightarrow Y$ by $r : T \rightarrow W$, $q : T \rightarrow Y$. Its original comparison is

$$(ar)^*A_u = q^*A_{u,Y} + E_u, \quad E_u \geq 0 \text{ is } q\text{-exceptional.}$$

Since ψ is nonextracting, every a -exceptional prime is exceptional over Y . Hence r^*F is also q -exceptional, and

$$r^*([K_W + \Gamma] + \gamma_u) = q^*A_{u,Y} + E_u + r^*F. \quad (3.33)$$

This proves that $W \dashrightarrow Y$ is a weak model of the prepared pair. In detail, subtract the effective exceptional error on the right from its structure boundary on T and push down to Y . The resulting boundary is precisely the transform of Γ , it is effective, the adjoint is the already existing Bott–Chern class $A_{u,Y}$, and all discrepancies are at least those of the prepared gklt pair. This uses generalized data at the level of currents. On T keep the nef datum $r^*\gamma_u$, and subtract the displayed effective q -exceptional error from the prepared structure boundary. Pushing forward by q gives the boundary Γ_Y and the trace of this fixed nef b-class. Their sum with the canonical current represents the already existing locally exact adjoint $A_{u,Y}$; the resolution identity defines its structure boundary. The generalized discrepancies have not decreased. This argument neither requires $K_Y + \Gamma_Y$ to be real Cartier nor asserts that the nef trace is separately a Bott–Chern class. It therefore applies to every Kähler weak target, including the adjunction strata used in special termination.

Every weak model occurs in the finite canonical list. Choose a small box in $H^{1,1}(W, \mathbb{R})$ which spans that vector space and whose addition to the compact family $\{\gamma_u\}$ stays inside the Kähler cone. For any Y above choose a Kähler class η_Y and define $\eta_W = r_*q^*\eta_Y$. Smooth-source cohomology puts η_W in $H^{1,1}(W, \mathbb{R})$ and gives an actual r -exceptional real divisor D_η with

$$r^*\eta_W - q^*\eta_Y = [D_\eta].$$

Its negative is r -nef, so $D_\eta \geq 0$ by exceptional negativity. As $W \dashrightarrow Y$ extracts no divisors, D_η is also q -exceptional. Scale η_Y by a sufficiently small positive number so that η_W lies in the fixed box. Adding this identity to (3.33) yields an effective q -exceptional comparison with target class $A_{u,Y} + \eta_Y$, which is Kähler over the original base. Thus $W \dashrightarrow Y$ is the canonical model of a member of this one enlarged prepared polytope. Its canonical-model list is finite by the first part of the geography proof. The induced list of maps from X is therefore finite as well. A finite cover of the original compact parameter polytope completes the proof for all weak models. $\square$

Theorem 3.55 (The restricted package in every dimension). *The assertions B_d , C_d , M_d and F_d hold in every finite dimension, for the globally nef carrier data and globally modified-big*

boundary traces specified above. All birational steps used to prove them are detected steps or belong to an already-projective relative construction with the stated relative-line algebraicity. A general semiample morphism is asserted to be Moishezon; projectivity is asserted only when the construction provides an actual relatively ample line.

Proof. The assertions in dimensions zero and one follow from degrees and factorization of maps of compact curves. Suppose the package holds in dimensions less than d . Lemma 3.47 first proves $C_{d,\text{big}}$ and then B_d . The former uses C_{d-1} ; the nonbig part of the latter uses only a chosen M_{d-1} model and the explicit negative-part comparison. In particular, no lower program is lifted.

With B_d available, Proposition 3.48 constructs detected negative steps in dimension d . Lemma 3.51, using the already known F_{d-1} , gives the big-shift case of M_d . Lemma 3.53 gives its remaining case, using M_{d-1} and the projective graph construction. This proves M_d over every proper compact Kähler base, with its chosen-model invariant. Lemma 3.54, whose hypotheses are both B_d and M_d , now proves F_d .

It remains to prove C_d when its nef NQC adjoint α is not big. First use Lemma 3.29, preserving the adjoint α and making the nef datum Kähler on a smooth carrier. Its multiplier-ideal inclusion allows the given contraction to restrict to the prepared non-klt closed subspace by Lemma 3.31. Next take the projective dlt preparation in [39, Section 6], keeping its crepant structure data. The working space is klt and globally strongly $\mathbb{Q}$ -factorial. The retained crepant structure boundary need not be lc; it is effective, including the extracted primes of coefficient at least one. On a common carrier the nef datum is nef and big. Consequently the total boundary trace is modified big, and $\alpha - c_1(K_X)$ is a big Bott–Chern class. The contraction on the non-klt closed subspace transports through this preparation by Lemma 3.32. Choose an NQC expression with positive coefficients and integral multiples of its rational degree-two summands. Corollary 3.37 provides a single number t such that each negative ray of $K_X + t\alpha$ is α -trivial. Choose a Kähler scaling class ω with $K_X + t\alpha + \omega$ nef. Every selected negative ray is ordinary K_X -negative; the actual rational canonical line is its detector. Proposition 3.48 supplies its projective contraction or flip, and the fixed integral grid persists. The descended α remains nef by projective descent of nefness.

The driving class is non-pseudo-effective throughout, by the corollary. Its initial pseudo-effective scaling threshold is positive; choose a cutoff $\varepsilon > 0$ below that threshold. On the fixed initial carrier the compact segment of globally nef data

$$t\alpha + [\varepsilon, 1]\omega$$

has modified-big total boundary. At its scaling parameter each working model is a marked weak model of a member of this segment. Each selected step is negative for the fixed driving adjoint, so its discrepancies do not decrease and increase at a valuation affected by that step. An infinite sequence would repeat a marked model by F_d ; the cumulative nonzero effective comparison for the fixed adjoint would then contradict equality of that marking. Hence this program terminates. Non-pseudo-effectivity excludes a nef endpoint, leaving an α -trivial ordinary Mori contraction $f : X' \rightarrow Z$.

The structure data of the prepared contraction problem remain crepant throughout this α -trivial program. Their effective traces and globally nef and big carrier datum therefore persist. Lemma 3.32 transports the contraction on the non-klt closed subspace through each birational step. Lemma 3.15 gives

$$\alpha_{X'} = f^*\gamma$$

in Bott–Chern cohomology. The class γ is nef by projective descent; Lemma 3.35 descends the fixed rational NQC summands, and projective multisections preserve their nonnegative degrees.

Apply Corollary 3.34 to this projective Mori contraction. In its notation the prepared non-klt subspace N_1 has Stein space $V(f_*J_1)$ over Z . If N_1 dominates Z , this Stein space is Z itself and its inherited contraction already supplies the contraction of γ . Otherwise the base presentation has multiplier ideal $J_Z \supset f_*J_1$, so the contraction restricts to $V(J_Z)$ as a closed subspace of $V(f_*J_1)$. Since $\dim Z < d$, the assertion C_{d-1} then gives the contraction of γ . Pullback supplies the contraction on X' . Its pullback to projective common resolutions factors through the original space by Lemma 3.31, because the adjoint classes are crepant. This gives the required Moishezon contraction on the original space. This proves C_d and completes the induction. $\square$

Corollary 3.56 (Finite positive parts of detected scalings). *Let a gklt adjoint with globally nef carrier data be run with scaling of the trace of a Kähler class on a fixed smooth carrier. Suppose that every chosen negative analytic ray has an actual global rational line detector and that the total boundary trace on each positive scaling segment is globally modified big. Every segment whose scaling parameters lie in a fixed interval $[\varepsilon, T]$, with $\varepsilon > 0$, is finite. If the driving adjoint is not pseudo-effective, the scaling terminates with a Mori fibre space.*

Proof. Proposition 3.48 constructs each step and Lemma 3.38 supplies the forward traces from the fixed carrier. Each reached model at a scaling parameter is a marked weak model of that parameter's adjoint. The compact parameter segment satisfies F_d , so there are finitely many such markings. Strict increase for the fixed driving adjoint rules out repetition. If that adjoint is not pseudo-effective, the positive pseudo-effective scaling threshold gives a fixed positive cutoff containing every parameter of a continuing program. The finite program cannot end nef, hence ends with the claimed Mori map. $\square$

4 Generation along the dlt boundary

We prove that the restrictions of a nef adjoint to the separate log canonical strata fit together to generate its restriction to the whole reduced boundary. Semiampleness on the normal components does not by itself supply compatible sections on their union. The proof has three parts. A Mori contraction reduces the obstruction to restriction from a stratum to one comparison of residues. Crepant transport respects all lower residues. Finally a finiteness argument permits us to impose every comparison simultaneously by taking products of sections.

Throughout this section, $n \geq 1$ and $\mathcal{G}_j$ is assumed for every $j < n$. We use ordinary dlt pairs with effective rational boundary. A *stratum* is an irreducible log canonical center; we also allow the ambient space itself when describing adjunction. A *boundary stratum* is a stratum contained in the reduced floor. An *incidence* is an inclusion of strata, and it is *immediate* when the smaller stratum has codimension one in the larger. On a crepant SNC model, a *unit component* is a boundary component of coefficient one, and a *unit stratum* is an irreducible component of an intersection of unit components. All residue comparisons are in a common divisible even adjoint degree.

Theorem 4.1 (Generation on the reduced boundary). *Assume $\mathcal{G}_j$ for every $0 \leq j < n$. Let (V, B) be a normal irreducible compact Kähler dlt n -fold with effective rational boundary B . Suppose that V is globally $\mathbb{Q}$ -factorial, that (V, B) satisfies the lc-strata resolution convention of Definition 3.1, and that the actual $\mathbb{Q}$ -Cartier adjoint $J = K_V + B$ is analytically nef. Then the restriction of an actual positive Cartier multiple of J to the whole reduced space $S = [B]$ is globally generated.*

The assertion is vacuous if $S = \emptyset$. The remainder of the section proves it when the floor is nonempty. The construction of compatible sections follows the admissible-section method of

Fujino [28, Section 4]; the main work here is to establish its restriction and finiteness inputs for strata of arbitrary dimension in the compact Kähler setting.

4.1 Adjunction, the semiample systems, and their comparisons

We first fix precisely which sections have to agree. The following local facts retain the actual line, including the residue identifications at every intersection.

Lemma 4.2 (Strata and descent of residues). *For each stratum $Z \subseteq V$ there is an effective rational boundary B_Z on the normal compact Kähler space Z such that (Z, B_Z) is dlt and, in every sufficiently divisible even degree q ,*

$$\mathcal{O}_Z(q(K_Z + B_Z)) \simeq \mathcal{O}_V(qJ)|_Z. \quad (4.1)$$

This is the actual meromorphic iterated-residue identification. The prime components of $C_Z := [B_Z]$ are precisely the strata $D \subset Z$ in an immediate incidence. Sections of $\mathcal{O}_V(qJ)|_D$ on these components D which have equal residues on every common lower stratum descend uniquely to a section of the restricted line on the entire reduced C_Z . The same statement applies to the components of $S \subseteq V$.

Proof. Choose one resolution $p : \widehat{V} \rightarrow V$ witnessing Definition 3.1, and write $K_{\widehat{V}} + \widehat{B} = p^*(K_V + B)$ using the actual meromorphic identification. Its unit components are the strict transforms $\widehat{S}_i$ of the finitely many components S_i of S . Every component $\widehat{Z}$ of an intersection of distinct $\widehat{S}_i$'s maps birationally onto a stratum Z , because the map is an isomorphism at its general point. Conversely every stratum is obtained this way. The general SNC locus makes both the component and its index set unique. Compactness makes the collection finite. The empty intersection is $\widehat{V}$ itself. Write $p_Z = p|_{\widehat{Z}} : \widehat{Z} \rightarrow Z$, so $p_V = p$.

Here are the adjunction and normality details, including the facts used at deeper intersections. For a chosen set of indices, keep their coefficients one and allow each unused coefficient to be either one or $1/2$. Write $B^e = B - \sum_i (1 - e_i) S_i$. Global $\mathbb{Q}$ -factoriality makes this an actual rational-line operation, and

$$\widehat{B}^e = \widehat{B} - p^* \sum_i (1 - e_i) S_i.$$

The subtracted divisor is effective. All exceptional coefficients remain below one, and the remaining unit components are exactly the retained strict transforms. Successive SNC adjunction on $\widehat{Z}$ gives

$$(K_{\widehat{V}} + \widehat{B}^e)|_{\widehat{Z}} = K_{\widehat{Z}} + \widehat{B}_{\widehat{Z}}^e.$$

Here and below such a restriction equality means equality of the meromorphic residue maps in degree q , and hence equality of the actual lines. If two orders of residue differ by an ordering sign, its q -th power is one.

We induct on the number of chosen indices. At each stratum already constructed, we retain normality, effectivity of every weighted different, and the crepant SNC residue comparison on $\widehat{Z}$. For this chosen comparison the exceptional coefficients are below one, the unit components are the retained strict intersections, and the full-weight pair is dlt. These assertions hold for $Z = V$. Lowering unused unit components preserves dlt; lowering all of them makes the effective pair at the current stratum klt. It has rational singularities and is Cohen–Macaulay by the dimension-free floor-connectedness lemma [53, Lemma 2.1]. If just one unused coefficient is retained, the unit

divisor of the induced SNC boundary is a disjoint union R of smooth divisors on $\widehat{Z}$. The same lemma gives

$$(p_Z)_* \mathcal{O}_R = \mathcal{O}_{p_Z(R), \text{red}}.$$

The fibers are connected, so the images of different connected components of R cannot meet. Each component maps birationally to its image; the displayed equality, factored through finite normalization, makes that image normal. It is compact Kähler by restriction of the local Kähler potentials from V .

For every allowed weight choice, push forward the SNC different to define the rational different on this new stratum. In codimension one, the residue of a local frame of the ambient invertible line identifies its divisorial adjoint with the restricted line. Reflexive extension on the normal stratum gives Equation (4.1) everywhere. Pulling it back agrees with the original SNC residue on a dense open and therefore everywhere, proving crepancy as an actual meromorphic identification.

The different is effective. At a general point of any of its primes, the normal surface-slice calculation of [53, Lemma 6.2], which is stated in every dimension, preserves exactly the residue order and reduces it to adjunction on a normal surface germ with effective boundary and a smooth marked curve. For completeness, on a minimal resolution of that surface, write the crepant boundary as the effective strict transform plus an exceptional divisor E . For each exceptional curve D ,

$$E \cdot D = -(K_{\widehat{T}} + B^{\text{str}}) \cdot D \leq 0.$$

Indeed $B^{\text{str}} \cdot D \geq 0$, and adjunction gives $K_{\widehat{T}} \cdot D = 2p_a(D) - 2 - D^2 \geq 0$; minimality excludes a smooth rational exceptional (-1) -curve. The negative-definite exceptional intersection matrix gives $E \geq 0$: if $E = P - N$ with disjoint nonnegative parts and $N \neq 0$, then $E \cdot N = P \cdot N - N^2 > 0$, contrary to the preceding inequalities. The strict transform is finite birational over the smooth marked curve, and hence isomorphic to it. Adjunction to that smooth strict transform thus has nonnegative coefficient. This is the original coefficient by the surface-slice residue calculation.

The full-weight pair is dlt and has the asserted floor. On an SNC model, at the general point of the center of a divisorial valuation v , take normal parameters $x_1, \dots, x_a$, giving an unmarked parameter boundary weight zero. The SNC discrepancy inequality is

$$a(v; \widehat{Z}, \widehat{B}_{\widehat{Z}}) \geq \sum_{i=1}^a (1 - b_i) v(x_i).$$

Every $v(x_i) > 0$. Thus a zero-discrepancy center is an intersection of unit components. Those intersections meet the isomorphism locus, while every exceptional component has coefficient below one. To produce a dlt resolution for the restricted map, principalize its nonisomorphism locus and resolve together with the ordered SNC boundary. That locus contains no entire log canonical center, so the displayed discrepancy inequality makes every new exceptional coefficient strictly below one. This is the dlt resolution criterion. It also identifies the unit primes downstairs with the next incident strata.

Finally let R_{all} be the whole unit union on $\widehat{Z}$. Sections on its smooth components which agree on all their SNC intersections glue by the elementary equalizer for the coordinate ideals of an SNC union. Even iterated residues make the equalities independent of the chosen chain. Apply the floor-connectedness lemma to the full-weight pair and this full unit union. It gives $(p_Z)_* \mathcal{O}_{R_{\text{all}}} = \mathcal{O}_{C_Z}$. Projection formula therefore descends the glued section uniquely to the restricted invertible line on the reduced C_Z . The proof with $Z = V$ gives the assertion for S . $\square$

Each boundary stratum has dimension less than n . The class of $J_Z := K_Z + B_Z$ is nef, since it is the restriction of the nef adjoint class. Proposition 2.7, applied using $\mathcal{G}_{\dim Z}$ on a

log resolution, makes the actual rational line J_Z semiample. Increase q once for all so that $L_Z := \mathcal{O}_Z(qJ_Z) = \mathcal{O}_V(qJ)|_Z$ is generated for every boundary stratum. Its map and Stein factorization give

$$f_Z : Z \longrightarrow Y_Z, \quad L_Z = f_Z^* N_Z, \quad (4.2)$$

where Y_Z is normal projective, f_Z has connected fibers, and N_Z is ample. These are actual line identities. We call C_Z *vertical* if $f_Z(C_Z) \neq Y_Z$; the empty floor is vertical. If C_Z dominates Y_Z , then for every $k \geq 1$ the restriction

$$H^0(Z, L_Z^k) \longrightarrow H^0(C_Z, L_Z^k|_{C_Z})$$

is injective. Indeed, every section comes from Y_Z ; if its pullback vanishes on C_Z , surjectivity makes it vanish at every point of Y_Z .

Fix a Kähler class c on V , and let c_Z be its restriction to each stratum. For boundary strata Z, Z' of the same dimension, an *arrow* is a proper bimeromorphic comparison with a projective resolution whose source is smooth and compact Kähler,

$$\begin{array}{ccc} & T & \\ p \swarrow & & \searrow p' \\ Z & & Z' \end{array}$$

such that the two pulled-back adjoint lines are equal as invertible subsheaves of meromorphic q -pluricanonical forms on T , and

$$p^* c_Z - (p')^* c_{Z'} \in \text{NS}(T)_{\mathbb{R}}. \quad (4.3)$$

Here $\text{NS}(T)_{\mathbb{R}}$ is the real span of first Chern classes of holomorphic lines, with torsion removed. The degree-two formula for a smooth modification says that its new classes are exceptional divisor classes. It follows that Equation (4.3) is independent of further resolution and is preserved by composition. These arrows therefore form a groupoid on the finite set of strata of each dimension. An arrow $\gamma : Z \dashrightarrow Z'$ transports sections by

$$(p')^*(\gamma_* s) = p^* s.$$

Normality gives existence and uniqueness. Transport commutes with multiplication and identifies the complete systems and their Stein targets in Equation (4.2). The equality of meromorphic lines defines this transport of adjoint sections; the class congruence will control self-arrows when the floor is vertical.

4.2 One residue comparison suffices for restriction

The only obstruction to extending a section from C_Z is its possible variation among the connected components of a general floor fiber. We show that all these components meet one general Mori fiber, whose floor is connected or consists of two points. Thus at most one residue comparison is needed.

Lemma 4.3 (Restriction and a Mori link). *For each positive-dimensional boundary stratum Z , there is either no comparison or one arrow between two components of C_Z , allowing a component to be compared with itself, with the following property. In all sufficiently large divisible degrees k , a section of $L_Z^k|_{C_Z}$ extends to Z if its two residues agree under that arrow. With no arrow,*

every such section extends. The degree bound is uniform over sections. When present, the arrow compares the two coefficient-one points on general $\mathbb{P}^1$ fibers of a projective Mori contraction on a bimeromorphic compact Kähler model of Z .

Proof. We use the dimension-free torsion-free restriction lemma [53, Lemma 7.1]. In its notation, for an effective compact Kähler lc pair (T, G) , klt away from its reduced floor C , and a connected map $f : T \rightarrow P$ to a normal compact base from which an actual positive adjoint multiple is pulled back, it asserts that $R^1 f_* \mathcal{O}_T(-C)$ is torsion-free. The ideal sequence gives

$$\mathcal{Q} := \text{coker}(\mathcal{O}_P \longrightarrow f_* \mathcal{O}_C) \hookrightarrow R^1 f_* \mathcal{O}_T(-C). \quad (4.4)$$

For $T = Z$ and $P = Y_Z$, if C_Z is vertical, $\mathcal{Q}$ has proper support and is zero. Serre vanishing for the kernel of $\mathcal{O}_P \rightarrow f_* \mathcal{O}_C$, tensored with N_Z^k , extends every restriction in a uniform large-degree tail. If C_Z dominates, the same display says that a section extends as soon as its values are constant on the reduced floor fiber over a dense open of Y_Z : its class in the torsion-free $\mathcal{Q} \otimes N_Z^k$ then vanishes, and the local lifts glue. They are unique because C_Z dominates Y_Z , so $\mathcal{O}_{Y_Z} \rightarrow (f_Z)_* \mathcal{O}_{C_Z}$ is injective. This uses neither reduced special scheme fibers nor base change at their points.

Suppose now that C_Z dominates Y_Z . There is an effective rational Cartier divisor D with $\text{Supp } D = \text{Supp } C_Z$: restrict to Z the globally $\mathbb{Q}$ -Cartier floor primes of V not containing Z . For small rational $\epsilon > 0$, $(Z, B_Z - \epsilon D)$ is effective klt. Apply Corollary 3.5 to obtain a projective small strong $\mathbb{Q}$ -factorialization $\rho : Z_0 \rightarrow Z$, crepant also for the full pair. Write Δ_0 for its full boundary and $D_0 = \rho^* D$. The full adjoint

$$P = K_{Z_0} + \Delta_0$$

is the actual semiample rational line pulled back from Y_Z , and the lowered klt adjoint is $J_{\text{low}} = P - \epsilon D_0$. Put $d = \dim Z$, choose $m > 0$ with mP Cartier and generated, and take $b > 4md$. Consider

$$H = J_{\text{low}} + bP = (1 + b)P - \epsilon D_0.$$

On a general fiber of the map to Y_Z , its class is $-\epsilon\{D_0\}$. This is the negative of a nonzero effective divisor because the floor dominates Y_Z , so its pairing with a Kähler power is negative. If H were pseudo-effective, a positive current representing it would restrict to a positive current on almost every resolved smooth fiber, contradicting this pairing. Thus H is not pseudo-effective.

Apply Lemma 3.6 to the lowered klt pair and the nef rational line P . Its H -program with Kähler scaling has ordinary J_{low} -negative steps, is finite, and ends in a projective Mori contraction. Every step, including the final Mori ray, is P -trivial. The same actual Cartier line mP descends at each birational step. Consequently its semiample map to the fixed base Y_Z persists on every working model. That map is constant on each connected final Mori fiber and hence factors through the Mori contraction. We obtain

$$T \xrightarrow{u} W \xrightarrow{g} Y_Z.$$

Both maps have connected fibers. The full adjoint on each model is the trace of P . On a common resolution of a birational step, its natural meromorphic canonical comparison is an exceptional divisor with zero class, since the two actual lines descend from the same line on the contraction base. Exceptional negativity applied to both signs makes that divisor zero. Thus the full pair remains crepant, including its meromorphic adjoint identifications. The full transformed pair (T, G) is effective lc and is klt away from its full floor C^+ : there it agrees with the transformed lowered klt pair. The program is nonextracting, so the transformed divisor D^+ has $\text{Supp } D^+ = \text{Supp } C^+$. Since the full adjoint is pulled back from Y_Z , the lowered adjoint on a u -fiber is $-\epsilon D^+$. Hence D^+ is relatively ample over W .

Compare the connected components of a general floor fiber on the original small model and on T . On a common projective log resolution the full crepant subboundary is the same. Its unit union maps to both floors with connected fibers by [53, Lemma 2.1]. Proper closed surjections with connected fibers preserve connected components, also after restricting over a point of Y_Z . Thus the connected components of the two underlying floor fibers correspond.

Every connected component of the floor fiber on T meets one common general Mori fiber. Indeed $C^+ \rightarrow W$ is surjective because C^+ is the support of the relatively ample D^+ . Apply Equation (4.4) to u . Surjectivity of the floor makes $\mathcal{O}_W \rightarrow u_*\mathcal{O}_{C^+}$ injective; the displayed torsion-free cokernel then makes $u_*\mathcal{O}_{C^+}$ torsion-free. Let

$$C^+ \longrightarrow \overline{C} \longrightarrow W$$

be the Stein factorization. The finite space $\overline{C}$ is reduced, and every irreducible component of $\overline{C}$ dominates W . To see the last implication algebraically, a vertical minimal prime of a finite reduced algebra over a domain would give, by prime avoidance, a nonzero element killed by a nonzero element of the domain. After shrinking Y_Z to a dense open, the fiber-dimension theorem gives every component of $\overline{C}_y$ dimension $\dim W - \dim Y_Z$. The fiber W_y is irreducible of that dimension: a resolution of W still has connected fibers over Y_Z , and a general one is smooth and connected and surjects onto W_y . Finiteness now makes each component of $\overline{C}_y$ dominate W_y . Consequently each connected component of C_y^+ meets $u^{-1}(w)$ for the same general $w \in W_y$.

The reduced general fiber F of u is irreducible by the same resolution argument. If $\dim F \geq 2$, the support of the effective ample Cartier multiple of $D^+|_F$, whose support is $C^+ \cap F$, is connected. One elementary proof, valid even if F_{red} is not normal, cuts by general hyperplanes to an integral projective surface. If the ample divisor split into disjoint nonzero Cartier parts D_1, D_2 , their pullbacks to a resolution would be orthogonal, while the projection formula gives $v^*(D_1 + D_2) \cdot v^*D_i > 0$ for both i . Thus both $(v^*D_i)^2 > 0$, contradicting the surface Hodge index theorem. If $\dim F = 1$, choose the fiber also off the singular, boundary intersection, and ramification loci. It is a smooth connected curve and

$$0 = \deg(K_F + G|_F) = 2g(F) - 2 + \deg(G|_F).$$

There is a unit point, so $F \simeq \mathbb{P}^1$. Its floor has one or two points; if there are two they exhaust the horizontal boundary. In every connected case the preceding common-fiber observation makes the general floor fiber connected, so no comparison is needed. In the remaining case the two unit points represent all its connected components.

Normalize the horizontal floor primes and take their Stein factorizations over W . If there are two primes, each finite Stein map has degree one and is an isomorphism over normal W ; their main fiber product gives a proper bimeromorphic comparison. If there is one prime, its normal Stein space is a double cover of W . The exchange on its general fiber extends over the branch locus: the reduced horizontal non-diagonal component of the double fiber product has finite birational projections to the normal cover and hence both projections are isomorphisms. Its graph lifts to the normalized prime. Composing with the program graphs transports the comparison to two original normal floor primes, possibly the same one. All graph projections are projective, and the primes survive birationally because the program extracts no divisors.

This comparison is crepant as a meromorphic residue identity. On a smooth ruled open, order the markings after an étale local cover, choose a base volume ξ , and put the markings at $z = 0, \infty$. A pulled-back q -pluriadjoint frame has the form

$$a(w)(dz/z)^{\otimes q} \otimes \xi^{\otimes q}.$$

Its two residues are $a(w)\xi^{\otimes q}$ and $(-1)^q a(w)\xi^{\otimes q}$, which agree. This is the two-marking Poincaré-residue comparison of [47, Section 3, Definition 13 and Proposition 14]. On a resolved graph both restricted actual lines are pulled back from Y_Z ; their meromorphic embeddings agree on this dense open and hence everywhere, including along exceptional primes. Equal residues therefore give a constant value on the entire general floor fiber. The horizontal consequence of Equation (4.4) extends the section. Normality descends it from the small model. Its restriction is the prescribed section on the original reduced floor, since every original floor prime is covered birationally and the two restrictions agree after that pullback.

It remains to check Equation (4.3). Let $\hat{T}$ be a common smooth resolution of the parent and Mori models and let $a, b : H \rightarrow \hat{T}$ be the two maps from a resolved branch graph. The resolution of the Mori model is an isomorphism near a general entire $\mathbb{P}^1$ fiber: its centers have codimension at least two, hence cannot dominate the ruling base. Every holomorphic two-form on $\hat{T}$ comes from the base on that ruled open. In fact its vertical one-form terms vanish on $\mathbb{P}^1$, and its remaining coefficients are constant on that compact fiber. Thus $a^* - b^*$ kills $H^{2,0}$, and also $H^{0,2}$ by conjugation. This rational Hodge map on degree two has image of type $(1, 1)$. The Lefschetz $(1, 1)$ theorem puts its real image in $\text{NS}(H)_{\mathbb{R}}$. Apply this to the pulled-back parent class cz . The link is therefore an arrow as defined above. $\square$

4.3 Transport respects every lower residue

An arrow can contract a divisor to a higher-codimension stratum. The next local observation explains how to compare residues even in that case. All valuations in its proof are local monomial valuations in an SNC chart.

Lemma 4.4 (Unit strata above an SNC stratum). *Let $p : T \rightarrow Z$ be a projective log resolution of a dlt pair, with crepant SNC subboundary on the smooth T . Let $R \subseteq T$ be a unit stratum and let $D = p(R)$ be its center. Then D is a dlt stratum. There is a unit stratum $R' \subseteq R$ such that $p(R') = D$ and $R' \rightarrow D$ is birational. In the iterated residue comparison along R' , the normal logarithmic Jacobian is ± 1 .*

Proof. The center is a dlt stratum by the SNC discrepancy calculation in the proof of Lemma 4.2. Work over the generic SNC part of D , of codimension a , with normal coordinates $x_1, \dots, x_a$ for the unit divisors. The zero-discrepancy divisorial valuations centered there are exactly the primitive rational monomial rays in $\mathbb{R}_{>0}^a$. Here is a local verification that also fixes their normalization. The discrepancy inequality in the proof of Lemma 4.2 says on every log-smooth model that the center of a zero-discrepancy valuation is exactly a unit stratum: an additional unmarked normal parameter, or a parameter with coefficient below one, would give positive discrepancy. Blow up that stratum. If the positive normal orders of the valuation are $w_1, \dots, w_b$, the chart indexed by a smallest order w_k replaces these by w_k and the positive members of $w_i - w_k$. The exceptional divisor is again unit, and the center is exactly the new unit stratum. While there is more than one positive order their sum decreases. Eventually the center is the generic point of one unit divisor, where a normalized divisorial valuation has order one. Reversing these ordinary toric blowups proves monomiality and primitive normalization. Conversely this subtraction algorithm for any primitive positive integer vector terminates at order one, and reversing it extracts that monomial valuation. Comparison on a common graph gives uniqueness of the valuation with those weights and of its center on any model.

For a unit component E meeting the inverse image of the generic SNC part of D , put

$$v_E = (\text{ord}_E x_1, \dots, \text{ord}_E x_a) \in \mathbb{Z}_{\geq 0}^a.$$

If the unit components through a unit stratum Q meeting that inverse image are $E_1, \dots, E_b$, define

$$\phi_Q : \mathbb{R}_{\geq 0}^b \longrightarrow \mathbb{R}_{\geq 0}^a, \quad (t_1, \dots, t_b) \longmapsto \sum_{j=1}^b t_j v_{E_j}, \quad \sigma_Q = \phi_Q(\mathbb{R}_{\geq 0}^b).$$

The source carries its standard lattice $\mathbb{Z}^b$. For the fixed resolution T , these cones subdivide the orthant over its interior. On rational rays, ϕ_Q sends a monomial valuation on T to the same normalized valuation on Z , so it is injective on each cone. Interiors of different cones are disjoint, because the center and the weights of a monomial valuation on T are unique. Every rational ray in the orthant occurs: view its valuation on T and use the same SNC description there. There are only finitely many cones over the generic part of D ; density of rational rays therefore gives coverage. Taking subsets of unit components gives the faces, and uniqueness of valuations also on faces identifies intersections as common faces. Consequently σ_R is a face of an a -dimensional cone $\sigma_{R'}$ with the same downstairs center, for a unit stratum $R' \subseteq R$.

Let $E_1, \dots, E_a$ be the unit components through R' . The integer matrix $A = (\text{ord}_{E_j} x_i)$ of this full cone is unimodular. Indeed every positive primitive integral vector in the source cone is the normalized vector of a divisorial valuation, so its image is primitive. If a prime divided $\det A$, a nonzero kernel vector modulo that prime, lifted to a positive primitive integral vector, would have nonprimitive image. Hence $\det A = \pm 1$. The strata R' and D have the same dimension. We next prove that their generically finite map has degree one. In local coordinates at a general point of R' ,

$$x_i = u_i(y, z) \prod_{j=1}^a y_j^{A_{ij}},$$

where the u_i are units, the y_j are normal coordinates, and z are coordinates along the stratum. The tangential map is generically étale in characteristic zero. Since A^{-1} is integral, replace y_j by $\tilde{y}_j = y_j \prod_i u_i^{(A^{-1})_{ji}}$. Then $x_i = \prod_j \tilde{y}_j^{A_{ij}}$, and the map consisting of $\tilde{y}$ and the downstairs tangential coordinates is locally biholomorphic at a general étale point of $R' \rightarrow D$. Two distinct points over a general point of D would therefore give two lifts of nearby target points whose normals approach with any fixed positive rate vector in the interior of this cone. Varying the nonzero leading coefficients fills an open set of nearby torus points, so such a target can be chosen in the locus where p is an isomorphism. This contradiction proves that $R' \rightarrow D$ is birational. Finally, after taking residue along R' , the normal coefficient of $\bigwedge_i dx_i/x_i$ is $\det A$ times that of $\bigwedge_j dy_j/y_j$; terms differentiating the units vanish in that residue. Since $\det A = \pm 1$, the even pluriresidue removes the sign. $\square$

Lemma 4.5 (Preservation of lower restrictions). *Fix $0 \leq d < n$ and $k \geq 1$. For each boundary stratum Z of dimension at most d , let $s_Z \in H^0(Z, L_Z^k)$. Suppose these sections agree under the residue restriction for every incidence, and that every arrow in dimensions below d carries its source section to its target section. For any d -arrow $\gamma : Z \dashrightarrow Z'$ and every component D' of $C_{Z'}$,*

$$(\gamma_* s_Z)|_{D'} = s_{D'}.$$

Moreover verticality of C_Z is invariant under d -arrows.

Proof. The assertions are immediate for $d = 0$, when every floor is empty. For $d > 0$, choose a common log resolution $p : T \rightarrow Z$, $p' : T \rightarrow Z'$ of the arrow. For a target floor prime D' , start with its strict transform R . It is a unit stratum and maps birationally to D' . Its source center

$p(R)$ is a stratum by Lemma 4.4. If $R \rightarrow p(R)$ is not birational, that lemma replaces R by a unit stratum inside it which maps birationally onto the same source center. The target center may then shrink. If the map to that center is not birational, apply the lemma on the target side, and continue alternately. Each necessary replacement strictly decreases dimension. The process therefore ends with one unit stratum mapping birationally to lower strata D_0, D'_0 on the two sides.

The image in the common system base remains unchanged throughout. Indeed the two maps from the resolved graph to the Stein targets are identified by the arrow, and a replacement keeps its entire image on the side then being treated. Initially the image was $f_{Z'}(D')$. The terminal comparison of D_0 and D'_0 is an arrow: restricting the actual meromorphic identification to iterated residues gives crepancy, with the logarithmic determinant sign removed by Lemma 4.4; restricting the line Chern classes in Equation (4.3) gives the same condition modulo $\text{NS}_{\mathbb{R}}$. Lower invariance thus identifies the transported and assigned residues on D'_0 .

This equality determines the entire section on D' . The Stein map of $L_{D'} = L_{Z'}|_{D'}$ is the Stein factor of $f_{Z'}|_{D'}$, because the line induced on that Stein space is ample; its target is finite over $f_{Z'}(D')$. The image of D'_0 in this target is a closed irreducible subset of full dimension, since it still maps onto $f_{Z'}(D')$, and therefore is the whole target. Both sections on D' come from it. Equality after restriction to D'_0 proves equality on D' .

The same construction shows that the image of every target floor prime in the system base is the image of a lower source stratum, and the reverse statement follows from the inverse arrow. A floor dominates the system base on one side exactly when it does on the other. This proves the last assertion. $\square$

4.4 Finite actions on vertical strata

Once lower-dimensional sections are compatible and invariant, Lemma 4.5 preserves their restrictions under every arrow. If C_Z dominates Y_Z , injectivity of restriction forces any such transported extension to equal the assigned extension. Thus only vertical strata require the following argument for finite image.

Proposition 4.6 (Finite image of self-arrows). *Let Z be a boundary stratum with C_Z vertical. For all sufficiently large divisible k , the self-arrows of Z have finite image on $H^0(Z, L_Z^k)$.*

For $m = qk$, a pluricanonical eigenform $s \in H^0(Z, L_Z^k)$ with $\int_Z |s|^{2/m} < \infty$ becomes the m -th power of a holomorphic top form on a smooth resolution of a cyclic root cover, so the auxiliary proposition below makes its eigenvalue a root of unity. A uniform bound on the middle Betti numbers of selected resolutions of these covers in the fixed degree will then bound the possible orders; boundedness of the representation will give finite image. The auxiliary proposition uses the polarization congruence in Equation (4.3).

4.4.1 Top-form scalars

Proposition 4.7 (A polarized top-form scalar). *Let U be a smooth connected compact Kähler manifold of dimension $d < n$, and let $g : U \dashrightarrow U$ be bimeromorphic. Suppose that on a common smooth resolution the two pullbacks of some Kähler class on U differ by an element of $\text{NS}_{\mathbb{R}}$. If $0 \neq \alpha \in H^0(U, K_U)$ and $g^*\alpha = \mu\alpha$, then μ is a root of unity.*

We first control the canonical Iitaka base; this step uses $\mathcal{G}_d$ and the top form, with no polarization congruence. To specify that base, let U be a smooth connected compact Kähler d -fold

with $d < n$ and $\kappa(U, K_U) \geq 0$. A pluricanonical section makes K_U pseudo-effective, so $\mathcal{G}_d$ gives a smooth modification $\pi : U' \rightarrow U$ and

$$\pi^* K_U \sim_{\mathbb{Q}} P + E_U, \quad P \text{ semiample}, \quad E_U = N(\pi^* K_U).$$

Choose $\ell > 0$ sufficiently divisible that the connected semiample map and its Stein target satisfy

$$f : U' \longrightarrow Y, \quad \mathcal{O}_{U'}(\ell P) = f^* N,$$

where Y is normal projective and N is ample, and that Lemma 2.5 identifies every graded piece in

$$R_\ell(U) := \bigoplus_{j \geq 0} H^0(U, j\ell K_U) \simeq \bigoplus_{j \geq 0} H^0(Y, N^j).$$

This is the full section ring of the ample line N , so $Y \simeq \text{Proj } R_\ell(U)$. Passing to a further divisible Veronese gives the same Proj. We call Y the *canonical Iitaka base*. A bimeromorphic self-map of U acts on this graded ring and therefore induces an automorphism of Y .

Lemma 4.8 (Finite action on the canonical Iitaka base). *Let U be a smooth connected compact Kähler manifold of dimension $d < n$, let $0 \neq \alpha \in H^0(U, K_U)$, and let $g : U \dashrightarrow U$ be bimeromorphic with $g^* \alpha = \mu \alpha$. The induced automorphism h of the canonical Iitaka base $Y = \text{Proj } R_\ell(U)$, for a sufficiently divisible ℓ as above, has finite order.*

Proof. The top form gives $\kappa(U, K_U) \geq 0$, so the preceding construction applies. The assertion is immediate for a point base. Otherwise use $f : U' \rightarrow Y$ above, and write α also for its pullback to U' . For each positive integer k the canonical integral $s \mapsto \int_U |s|^{2/k}$ on $H^0(U, kK_U)$ is positive, continuous, and invariant under bimeromorphic change of variables. Choose j such that N^j is very ample. The powers of g and their inverses are bounded on the corresponding degree $j\ell$, which defines Y . This linear action is semisimple with eigenvalues of absolute value one. Also $|\mu| = 1$, by integration of $\alpha \bar{\alpha}$. Hence the finite measure

$$\nu = f_*(i^{d^2} \alpha \wedge \bar{\alpha})$$

is invariant under h ; exceptional sets have measure zero here.

Suppose that h has infinite order. In the projective linear group of this degree, the algebraic closure of a power of h is then a positive-dimensional torus T , acting faithfully on Y . Indeed a bounded linear operator is diagonalizable, and the connected algebraic closure of a cyclic diagonal group is a torus. We will obtain a contradiction from any nontrivial one-parameter subgroup of T .

Let Y° be a dense Zariski open in the smooth locus where f is smooth and the relative top form obtained from α is nonzero. Put $r = \dim U' - \dim Y$. A general fiber F of an Iitaka map has $\kappa(F) = 0$. It has a nonzero holomorphic top form here, so $h^0(F, K_F) = 1$: two such forms would have a nonconstant ratio and give positive Iitaka dimension. If ξ is a local holomorphic volume frame on Y° and $\sigma = \alpha/\xi$ is the relative top form, fiber integration writes the measure as

$$\nu = \|\sigma\|_{\text{Hdg}}^2 i^{(\dim Y)^2} \xi \wedge \bar{\xi}. \quad (4.5)$$

The coefficient is smooth and strictly positive after this shrink. We include $r = 0$, when the top Hodge line is the one-dimensional degree-zero line.

Set

$$Y^* = \bigcup_{j \in \mathbb{Z}} h^j(Y^\circ).$$

Invariance of ν under h makes the smooth positive densities agree on overlaps, extending the density in Equation (4.5) to Y^* . This open is invariant under T . Its complement is the intersection of the closed algebraic sets $h^j(Y \setminus Y^\circ)$, which is closed algebraic by Noetherianity. Its algebraic stabilizer contains the cyclic group generated by h , and therefore contains its algebraic closure. Choose a one-parameter subgroup $\mathbb{C}^* \subset T$ which acts nontrivially, and a general whole orbit in Y^* . The T -invariance of the domain supplies this whole orbit. We now construct a horizontal period metric on Y^* and prove that its restriction to the orbit vanishes.

For a smooth Kähler family a total Kähler class polarizes the primitive real variation in middle cohomology. On a simply connected open of Y° , take the smallest real subvariation of the full middle cohomology containing the line F^r of top forms and its conjugate. It is the intersection of the flat real Hodge subspaces with this property; finite dimensionality reduces the intersection to a finite one. This construction commutes with restriction and continuation: in a flat trivialization the condition of being a Hodge subspace, and of containing the indicated line, is a real analytic equality for the Hodge projections, so equality on one open continues on the simply connected cover. The primitive variation is one such subvariation, so the smallest one lies in primitive middle cohomology. Its polarization is the middle cup pairing with the usual sign, independent of the total Kähler class, because no power of that class occurs in the middle primitive pairing. The horizontal period metric of this polarized real variation is therefore intrinsic.

This possibly degenerate metric extends consistently to Y^* . To compare the metrics on an overlap of two translates of Y° , shrink the overlap further so that a resolution of the fiberwise birational graph is a smooth family over it. The two pullbacks on middle cohomology are flat Hodge embeddings into the cohomology of this common resolution and carry the top lines to the same line. The smallest subvariation there is the pullback of the smallest subvariation on either side: it is contained in both images, and applying the inverse embedding gives the reverse inclusion. Projection formula identifies their middle cup polarizations. Their horizontal metrics therefore agree on this smaller open and, by continuity, on the overlap.

We claim that the period metric restricts to zero on every whole orbit of this one-parameter subgroup in Y^* ; constant orbits are immediate. For a nonconstant orbit, pull it back under $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$. The negative upper bound for holomorphic sectional curvature in horizontal directions of a period domain, together with the curvature inequality for a pulled-back curve metric, applies to its local period representations; it holds for real polarizations as well [36, Theorem 9.1]. On a disk of radius R centered at an arbitrary point, translate the center to $z = 0$ and write the pulled-back metric as $\chi(z)|dz|^2$, continuous and smooth where positive, with curvature $-2\chi^{-1}\partial_z\partial_{\bar{z}}\log\chi \leq -\kappa < 0$. Compare it with

$$\chi_R(z) = \frac{4R^2}{\kappa(R^2 - |z|^2)^2}.$$

Direct calculation gives $\partial_z\partial_{\bar{z}}\log\chi_R = \kappa\chi_R/2$. If the maximum of χ/χ_R exceeded one, it would occur where $\chi > 0$ in the interior, since χ_R diverges at the boundary and χ is bounded on the closed disk. At that point,

$$0 \geq \partial_z\partial_{\bar{z}}\log(\chi/\chi_R) \geq \frac{\kappa}{2}(\chi - \chi_R) > 0,$$

a contradiction. Thus the Ahlfors–Schwarz comparison gives $\chi(0) \leq 4/(\kappa R^2)$ [2]. Letting $R \rightarrow \infty$ forces the metric to vanish on $\mathbb{C}$. The argument is local and allows zeros of the induced metric, so the local representations just constructed suffice. Consequently their period maps are locally constant along each such orbit. In any holomorphic volume frame the logarithm of the coefficient

in Equation (4.5) is harmonic along each orbit: the top Hodge line is flat there, and σ is a nonzero holomorphic multiple of a flat generator.

This is incompatible with finite mass. Diagonalize the one-parameter action in projective coordinates, and choose two coordinates nonzero at a general point with different weights, of difference $a \neq 0$. A small smooth transverse slice Q on which their ratio is one gives a holomorphic local biholomorphism onto its image

$$\Phi : \mathbb{C}^* \times Q \longrightarrow Y^*$$

with fibers of size at most $|a|$: the ratio at $\Phi(z, t)$ is z^a . The entire $\mathbb{C}^*$ factor is allowed because Y^* is invariant. In the product frame $(dz/z) \wedge dt_1 \wedge \cdots \wedge dt_{\dim Q}$, write $\Phi^*\nu$ with coefficient $\rho(z, t) > 0$. For fixed t , $\log \rho(z, t)$ is harmonic on $\mathbb{C}^*$. Its circular mean at $|z| = e^u$ is $A(t)u + B(t)$. Jensen's inequality therefore gives, with an irrelevant positive normalization,

$$\int_{\mathbb{C}^*} \rho(z, t) \frac{i dz \wedge d\bar{z}}{|z|^2} \geq c \int_{\mathbb{R}} e^{A(t)u+B(t)} du = \infty.$$

Tonelli's theorem contradicts the finite mass of $\Phi^*\nu$, which is at most $|a|\nu(Y)$ by the bounded covering degree. This proves that h has finite order. $\square$

After a power, the scalar problem can thus be restricted to a fiber with Kodaira dimension zero. The polarization congruence then passes to that fiber; the next two lattice arguments control its nonprojective factors. In the next lemma, the hypothesis on the origin of the model is what provides the degree-two exceptional splitting; rational singularities alone are not used for that splitting.

Lemma 4.9 (The scalar on a symplectic factor). *Let Y be a terminal compact Kähler primitive symplectic space of dimension $2e < n$, obtained from a smooth model by the program of Corollary 3.10. Let $g : Y \dashrightarrow Y$ be bimeromorphic and let $0 \neq \eta \in H^0(Y, \Omega_Y^{[2]})$. Suppose there is a class $c \in H^{1,1}(Y, \mathbb{R})$ whose two pullbacks on a common resolved graph of g differ by real line Chern classes, and such that on some projective resolution $p : R \rightarrow Y$ with smooth compact Kähler source*

$$p^*c = [\beta] + \{E\},$$

where E is an exceptional real divisor and β is a smooth closed semipositive $(1, 1)$ -form, strictly positive on a nonempty open. Then the scalar in $g^\eta = \lambda\eta$ is a root of unity.*

Proof. We use the pure Hodge structure on $H^2(Y, \mathbb{Q})$ and its rational Beauville–Bogomolov form q_Y , up to positive rational scale. This form is positive on the real symplectic plane and Lorentzian on $H^{1,1}(Y, \mathbb{R})$; see [5, Lemma 2.1, Definition 5.2, and Lemmas 5.3 and 5.7]. The pullback and $(1, 1)$ identifications for these rational singularities, extension of the two-form [44, Corollary 1.8], and Lemma 3.9 give

$$H^2(R, \mathbb{Q}) = p^*H^2(Y, \mathbb{Q}) \oplus \langle [p\text{-exceptional primes}] \rangle_{\mathbb{Q}}. \quad (4.6)$$

Indeed that program lemma gives the real $(1, 1)$ equality with a direct exceptional sum; the $(2, 0)$ and $(0, 2)$ summands compare by form extension, and all the summands in the resulting equality are rational.

Let $\mathcal{E}_R$ be the rational span of the p -exceptional prime classes. Identify $H^2(R, \mathbb{Q})/\mathcal{E}_R$ with $H^2(Y, \mathbb{Q})$ by Equation (4.6), and define the lattice

$$\Lambda_Y = \text{im}(H^2(R, \mathbb{Z})/\text{tors} \longrightarrow H^2(R, \mathbb{Q})/\mathcal{E}_R) \subset H^2(Y, \mathbb{Q}).$$

This amounts to quotienting the integral lattice by its saturated exceptional sublattice. The lattice is independent of a refinement $v : R' \rightarrow R$: integral Gysin satisfies $v_*v^* = 1$, and the smooth degree-two modification formula says that $x' - v^*v_*x'$ is exceptional and that the new exceptional span is v^* of the old one plus the v -exceptional span. A common smooth refinement therefore identifies the quotient lattices from any two resolutions over Y .

The form η^e trivializes K_Y . On a resolution R_g of the graph of g , with projections p_1, p_2 , the divisors of the two pulled-back volume forms agree. Terminality says that their positive components are exactly the exceptional primes. Thus the exceptional lists for the two projections are identical; in particular g is small in both directions. Let $\mathcal{E}$ be their common exceptional rational span, and let

$$\bar{p}_i^* : H^2(Y, \mathbb{Q}) \xrightarrow{\sim} H^2(R_g, \mathbb{Q})/\mathcal{E}.$$

Both maps carry Λ_Y onto the image of integral graph cohomology in this quotient. Hence $G = (\bar{p}_1^*)^{-1}\bar{p}_2^*$ is a rational Hodge automorphism of $H^2(Y, \mathbb{Q})$ preserving Λ_Y .

It also preserves q_Y . The two symplectic forms pull back up to the scalar λ , and integration of their top powers gives $|\lambda| = 1$. Use the usual positive normalization of $W = (p_1^*\eta p_1^*\bar{\eta})^{e-1} = (p_2^*\eta p_2^*\bar{\eta})^{e-1}$. For $(1, 1)$ classes x, y , the differences between p_1^*Gx and p_2^*x are exceptional for both projections. Expand $p_1^*Gx p_1^*Gy - p_2^*x p_2^*y$ into two terms, each with one such exceptional factor and one factor pulled back by the appropriate projection. Projection formula makes both pairings with W zero. This is precisely the $(1, 1)$ part of the Beauville–Bogomolov formula. The symplectic plane is preserved isometrically because $|\lambda| = 1$, and the Hodge decomposition is orthogonal. Thus G is a q_Y -isometry.

The class c in the statement has $q_Y(c) > 0$. Since β is nef over Y , exceptional negativity applied to the divisor E , with $\{E\} = p^*c - [\beta]$, gives $E = \sum_i e_i E_i \geq 0$. Pair with the positive normalization $W_R = (p^*\eta p^*\bar{\eta})^{e-1}$. Exceptional classes pair to zero with p^*c by projection formula, and hence for some constant $b_e > 0$

$$q_Y(c) = b_e \left(\int_R \beta^2 W_R + \sum_i e_i \int_{E_i} \beta W_R \right) > 0. \quad (4.7)$$

Every term is nonnegative. The first is strictly positive on the common open where β is positive and the symplectic form is nondegenerate.

Set $H_{\mathbb{Q}} = H^2(Y, \mathbb{Q})$ and $D_{\mathbb{Q}} = H_{\mathbb{Q}} \cap H^{1,1}(Y, \mathbb{C})$. Lefschetz $(1, 1)$ on R , together with the line-trace assertion of Lemma 3.9, identifies $D_{\mathbb{Q}}$ with the rational span of line Chern classes. It is G -invariant because G is a rational Hodge automorphism. The same line-trace assertion applied to the graph congruence gives $Gc - c \in D_{\mathbb{R}}$. The line $\mathbb{C}\eta$ survives in $H_{\mathbb{C}}/D_{\mathbb{C}}$; our immediate aim is to prove that all eigenvalues on this quotient have absolute value one. Write $\Sigma = (H^{2,0} \oplus H^{0,2})_{\mathbb{R}}$, the positive real two-plane. There are three possibilities for the restriction of the Lorentz form to $D_{\mathbb{R}}$.

If $D_{\mathbb{R}}$ is negative definite, its orthogonal projection c_0 of c to $D_{\mathbb{R}}^{\perp}$ is fixed by G and has positive square by Equation (4.7). On $D_{\mathbb{R}}^{\perp}$, the invariant space $\Sigma \oplus \mathbb{R}c_0$ is positive definite and its complement is negative definite. All eigenvalues there have absolute value one. If $D_{\mathbb{R}}$ contains a positive vector, it is nondegenerate with one positive direction. Its orthogonal complement is the positive plane Σ plus a negative-definite space, giving the same conclusion. In either case $H_{\mathbb{R}}/D_{\mathbb{R}} \simeq D_{\mathbb{R}}^{\perp}$.

In the remaining case $D_{\mathbb{R}}$ is negative semidefinite with radical a rational isotropic line ℓ . The lattice action on ℓ is ± 1 . On $\ell^{\perp}/\ell$ the form is the positive plane Σ plus a negative-definite $(1, 1)$ space, and the quotient $H_{\mathbb{R}}/\ell^{\perp}$ is dual to ℓ . The invariant flag $0 \subset \ell \subset \ell^{\perp} \subset H_{\mathbb{R}}$ therefore shows that all eigenvalues on $H_{\mathbb{R}}$ have absolute value one; this argument also allows unipotent parts. In

all cases the eigenvalues on $H_{\mathbb{C}}/D_{\mathbb{C}}$ have absolute value one. The rational quotient $H_{\mathbb{Q}}/D_{\mathbb{Q}}$ carries the preserved lattice

$$\overline{\Lambda}_Y = \Lambda_Y / (\Lambda_Y \cap D_{\mathbb{Q}}),$$

where the intersection is saturated. Thus λ is an algebraic integer and all its algebraic conjugates have absolute value one. Kronecker's theorem makes it a root of unity. $\square$

Lemma 4.10 (The scalar on a torus). *Let A be the integral Hodge automorphism of $H^1(T, \mathbb{Z})$ induced by an automorphism of a compact complex torus T . Suppose a positive Hermitian class $c \in H^{1,1}(T, \mathbb{R})$ satisfies $(\wedge^2 A)c - c \in \text{NS}(T)_{\mathbb{R}}$. Then $\det(A|H^{1,0}(T))$ is a root of unity.*

Proof. Put $V = H^1(T, \mathbb{Q})$ and $D_{\mathbb{Q}} = \text{NS}(T)_{\mathbb{Q}}$. For each eigenvalue θ of $A_{\mathbb{C}}$, let V_{θ} be its generalized eigenspace and set $d_{\theta} = \dim V_{\theta}$, $p_{\theta} = \dim(V_{\theta} \cap H^{1,0}(T))$. Complex conjugation gives $p_{\theta} + p_{\bar{\theta}} = d_{\theta} = d_{\bar{\theta}}$.

For $|\theta| \neq 1$, project c to the block $V_{\theta} \wedge V_{\bar{\theta}}$, with $\wedge^2 V_{\theta}$ understood for real θ . This projection preserves $D_{\mathbb{Q}}$. Indeed the Chinese-remainder projectors onto the generalized spaces are polynomials in A with algebraic coefficients. Apply them to the two tensor slots separately and then alternate. The resulting maps are algebraic linear combinations of rational Hodge maps on $\wedge^2 V$; these preserve rational $(1, 1)$ classes, hence $D_{\mathbb{Q}}$ by Lefschetz $(1, 1)$. Separate slot projections avoid any collision of products of eigenvalues in degree two.

On this block $A_2 = \wedge^2 A$ has the single generalized eigenvalue $\theta\bar{\theta} = |\theta|^2 \neq 1$. The inverse of $A_2 - 1$ there is a finite polynomial in its nilpotent part. Project the congruence for c to the block and apply this inverse. It follows that its block $c_{\theta, \bar{\theta}}$ belongs to $D_{\mathbb{C}}$. Positivity says that this block is a perfect tensor between V_{θ} and $V_{\bar{\theta}}$, pairing their opposite Hodge types: in a basis of $H^{1,0}$ the relevant matrices are positive-definite principal Hermitian blocks. For real θ the corresponding alternating tensor is nondegenerate.

The intersection of this block with $D_{\mathbb{C}}$ is defined over $\overline{\mathbb{Q}}$. Nondegeneracy is the nonvanishing of a determinant, so it contains a nondegenerate tensor with algebraic coefficients. For any Galois automorphism γ , the conjugate tensor remains in $D_{\overline{\mathbb{Q}}}$, hence is still of type $(1, 1)$ for the given Hodge decomposition, and is perfect between $V_{\gamma(\theta)}$ and $V_{\gamma(\bar{\theta})}$. Opposite-type perfection yields

$$p_{\gamma(\theta)} + p_{\gamma(\bar{\theta})} = d_{\theta} \quad (|\theta| \neq 1). \quad (4.8)$$

This does not replace $\gamma(\bar{\theta})$ by $\overline{\gamma(\theta)}$, and assumes no Galois invariance of the Hodge decomposition.

The top-form scalar is the algebraic integer $z = \prod_{\theta} \theta^{p_{\theta}} = \det(A|H^{1,0})$, also when generalized eigenspaces are nontrivial. For any Galois automorphism δ , reindex by the image roots and pair complex conjugates:

$$\begin{aligned} \log |\delta z| &= \frac{1}{2} \sum_{\theta} (p_{\delta^{-1}(\theta)} + p_{\delta^{-1}(\bar{\theta})}) \log |\theta| \\ &= \frac{1}{2} \sum_{\theta} d_{\theta} \log |\theta| = \frac{1}{2} \log |\det A| = 0. \end{aligned}$$

For $|\theta| \neq 1$ the second equality uses Equation (4.8) with $\gamma = \delta^{-1}$; the other terms vanish. The last equality is unimodularity of the integral action. Kronecker's theorem now proves the assertion. $\square$

Proof of Proposition 4.7. Apply Lemma 4.8. After a power, g acts trivially on the canonical Iitaka base. Use the modification $\pi : U' \rightarrow U$ and the map $f : U' \rightarrow Y$ in its construction, and restrict over a general point where the graph and the fibration are smooth and the fiber meets the isomorphism locus of π . Dividing α by a local base volume gives a nonzero top form on the smooth

compact fiber F , which has $\kappa(F) = 0$; its returning map scales the form by the corresponding power of μ . A point fiber gives scalar one. For a positive-dimensional fiber, let ω_U be a Kähler form representing the class in the proposition and put $\beta_F = (\pi|_F)^*\omega_U$. This is a smooth semipositive form, strictly positive on the dense open where π is an isomorphism. Pulling the original graph congruence to U' and restricting to the fiber shows that the two pullbacks of $[\beta_F]$ on a common resolved graph of the returning map of F differ by real line classes. This weaker positivity suffices below.

Since $\dim F \leq d < n$, Corollary 3.10, using $\mathcal{G}_{\dim F}$, supplies a terminal compact Kähler minimal model M of F with torsion canonical line. The descended nonzero reflexive top form trivializes K_M itself. Indeed a power of that form, in a trivialization of a multiple of K_M , is a nonzero holomorphic function on the compact connected M , hence a nonzero constant. On a graph resolution of the returning map the two divisors of this volume form agree; terminality makes their positive components exactly the exceptional primes. The returning map on M is therefore small in both directions.

The Kähler Beauville–Bogomolov decomposition [4, Theorem A] supplies a finite cover, étale in codimension one, of M of the form

$$P = T \times \prod_i H_i \times C,$$

where T is a torus, the H_i are primitive symplectic factors, and C is the product of the strict Calabi–Yau factors of dimension at least three. One-dimensional factors are elliptic curves and are absorbed into T ; two-dimensional factors are placed among the symplectic ones. An absent factor is a point and contributes scalar one. Reflexive forms on these klt spaces can be computed on resolutions by [44, Corollary 1.8]. A power of the small returning map lifts bimeromorphically to P . To justify this, purity makes the restriction of the cover over M_{reg} finite étale [4, Section 3.2, before Lemma 3.7]. This restriction is connected and, after choosing base points, determines a finite-index subgroup of $\pi_1(M_{\text{reg}})$, well-defined up to conjugacy. Remove the codimension-two exceptional sets from the smooth loci. Removing an analytic subset of complex codimension at least two from a manifold does not change its fundamental group. The compact normal connected M is reduced and paracompact, with bounded local tangent dimension. Its analytic pair (M, M_{sing}) therefore has a real analytic embedding [1, Theorem 1] and a compatible locally finite triangulation [50, Section 3, Theorem 1]. Compactness makes the triangulation finite; barycentric subdivision gives the smooth locus finite CW homotopy type, so its fundamental group is finitely generated. There are only finitely many subgroups of the fixed index of the cover. A power preserves the conjugacy class of its covering subgroup, so it lifts on this open, and normalization over the proper graph extends the lift bimeromorphically. We may take further powers throughout.

We explain why the factors can be considered separately. All holomorphic one-forms on P come from T . Their pullbacks show on the smooth open that the torus coordinate of the map depends only on the torus coordinate. Modulo the ideal generated by positive-degree torus forms, the two-forms have basis the symplectic forms η_i . Write $\dim H_i = 2e_i$. The highest nonzero power of $\sum a_i \eta_i$ has order $\sum_{a_i \neq 0} e_i$; thus the locus where its full power vanishes is the union of the coordinate hyperplanes. The map permutes these hyperplanes and hence the lines $\mathbb{C}\eta_i$, matching their nilpotence orders $e_i + 1$. No torus two-form can be added to an image of η_i , since wedging that term with the e_i -th power of the corresponding target symplectic form would contradict that nilpotence order. The kernels of the forms now show that each symplectic coordinate depends only on the corresponding symplectic factor. Take a power to remove the permutations.

If C is positive-dimensional, it has no holomorphic form of degree $\dim C - 1$: the form algebra of a strict factor has only degrees zero and its top degree, and no factor has dimension one. In the

pullback of the volume form of C , the component with one differential outside C and all the others in C therefore vanishes. The derivative in the C directions has full rank, because the whole map is bimeromorphic and the other coordinates have just been separated. Its cofactors then force the derivative of the C coordinate in outside directions to vanish. Thus that coordinate also depends only on C . These conclusions on a dense smooth open give factor bimeromorphic maps by taking their proper graphs. Their degrees are one because their product has degree one.

The scalar for C is a root of unity by projective pluricanonical finiteness. In detail, $H^{2,0}$ of a smooth compact Kähler resolution of C vanishes by the strict-factor form algebra and reflexive extension. Rational approximation of a Kähler class and Kodaira's theorem make this resolution projective. Thus C is Moishezon; it is projective by the Kähler projectivity criterion for rational singularities [52, Corollary 6']. Its canonical line is trivial and its returning comparison is crepant, as seen from the volume form. Chow's theorem algebraizes its proper analytic graph. The projective log pluricanonical representation theorem [33, Theorem 1.1] gives finite scalar image.

We spell out how the polarizing class reaches the other factors without asserting a cohomology splitting for the singular product P . Choose projective resolutions of its factors with smooth compact Kähler sources, and form the smooth compact Kähler product

$$X_P = T \times \prod_i \tilde{H}_i \times \tilde{C}.$$

Resolve the map $P \dashrightarrow F$ together with $X_P \rightarrow P$. This gives a smooth compact Kähler space R , a modification $v : R \rightarrow X_P$, and a morphism $r : R \rightarrow F$. Put $\beta = r^* \beta_F$. It is smooth semipositive and strictly positive on a dense open, since r is generically finite. Put $a = v_*[\beta]$. The smooth modification formula gives $[\beta] = v^* a + \{E_v\}$ for an exceptional real divisor E_v . The graph congruence for $[\beta_F]$ pulls to the cover and, by Gysin pushforward, gives the graph congruence for a modulo real line classes; the exceptional differences are themselves line classes. Pass to a common refinement and use the product of resolutions of the separated factor graphs to compute this congruence; refinements change it only by exceptional line classes. The restriction a_i of a to one factor slice is independent of the other coordinates, by Künneth on the smooth product X_P . Restricting the product graph therefore compares a_i to itself modulo line classes on the factor graph.

Choose the factor slice generally so that it meets the open where β is strictly positive and meets each v -exceptional center in codimension at least two, or avoids that center. This is possible by the fiber dimension theorem applied to each center and the projection to the other factors. On the resolved strict transform of this slice, the restriction β_i is still semipositive and strictly positive on an open, and the restriction of E_v is exceptional over the factor. Thus the pullback of a_i differs from $[\beta_i]$ only by the class of an exceptional real divisor. This gives both the required positivity and the congruence on each smooth factor model.

For a symplectic factor, its symplectic volume trivializes its canonical line. Canonical singularities preserve the canonical ring on resolution, so a smooth resolution has Kodaira dimension zero. Its dimension is $\dim H_i \leq \dim F < n$. Corollary 3.10, using $\mathcal{G}_{\dim H_i}$, runs from that resolution to a terminal model. It is still primitive symplectic: the unique two-form extends, and its top power trivializes the canonical line and is nondegenerate on the smooth locus; irregularity and the form algebra are unchanged. Take a common projective resolution of this terminal model carrying the pullback of β_i . Lemma 3.9 decomposes its class as the pullback of a class c on the terminal model plus the class of an exceptional real divisor. The line-trace assertion of that lemma sends the line Chern classes in the factor graph congruence to rational line classes on the terminal model. Pushing through the exceptional quotient therefore shows that the returning graph compares c to

itself modulo line classes. These are the hypotheses of Lemma 4.9. That lemma makes the scalar on its two-form, and therefore on its top form, a root of unity.

For the torus, push the semipositive form to the torus as a positive current and average it over translations. The average is a smooth invariant positive Hermitian representative: for every nonzero tangent direction its average is strictly positive because the original form is strictly positive on an open. A bimeromorphic map of a torus is an affine holomorphic automorphism; its linear part preserves the integral H^1 lattice. The comparison modulo $\text{NS}_{\mathbb{R}}$ and Lemma 4.10 make its top-form scalar a root of unity. The product of the factor scalars is the appropriate power of μ . It is a root of unity, and hence so is μ . $\square$

4.4.2 Finite image on vertical strata

Proof of Proposition 4.6. Put $d = \dim Z$ and $m = qk$, so sections of L_Z^k are meromorphic m -pluricanonical forms. Let

$$E_m = \left\{ s \in H^0(Z, mJ_Z) : \int_Z |s|^{2/m} < \infty \right\}.$$

The integral is computed on the regular locus, or on any resolution by change of variables. This is a vector space: for $2/m \leq 1$ the elementary power inequality preserves integrability under sums. On a log resolution an adjoint section is locally

$$h(z) \frac{(dz_1 \wedge \cdots \wedge dz_d)^{\otimes m}}{\prod_i z_i^{mb_i}}, \quad b_i \leq 1.$$

It is integrable if h vanishes on every unit component; smaller coefficients cause no obstruction. Every unit valuation has center in C_Z . Thus E_m contains pullbacks of base sections vanishing along $f_Z(C_Z)$. For large k these define Y_Z birationally: multiply one nonzero section of a high ample power vanishing on that proper subset by a complete very ample system. If Y_Z is a point, verticality forces $C_Z = \emptyset$ and a nonzero section is integrable. In particular $E_m \neq 0$.

The function $s \mapsto \int |s|^{2/m}$ is continuous on E_m and positive away from zero. In an integrable basis, the densities of bounded linear combinations are dominated by a constant times the sum of the basis densities, proving continuity by dominated convergence. Its invariance under arrows makes the action and its inverse uniformly bounded in any norm on E_m . Moreover this representation detects the representation on all of $H^0(Z, mJ_Z)$. If an arrow acts identically on E_m , it acts identically on Y_Z because that system is birational. For $0 \neq s \in E_m$ and any other section t , the ratio t/s comes from the rational function field of Y_Z . The arrow fixes both this ratio and s , and hence fixes t .

Take an eigenform $s \in E_m \setminus \{0\}$ for the transport γ_* of one self-arrow, with eigenvalue λ . In the orientation fixed above, $\gamma_* = (\gamma^{-1})^*$; put $g = \gamma^{-1}$, so $g^*s = \lambda s$. Form the full normalized analytic cyclic root cover of this meromorphic pluriform and take a projective resolution of all its components. Locally, if $s = a\eta^{\otimes m}$ in a meromorphic canonical frame, the cover is the normalization of $t^m = a$, and its tautological top form is $\alpha = t\eta$. These descriptions glue when the frame changes. The finite analytic normalization and this full-root construction are also described in the proof of [53, Lemma 7.1]. On the smooth compact Kähler resolution U_s of the full cover, let $\pi_s : U_s \rightarrow Z$ be the composite map. Change of variables gives

$$\int_{U_s} |\alpha|^2 = (\deg \pi_s) \int_Z |s|^{2/m} < \infty.$$

A meromorphic top form with finite local L^2 norm has no pole, by the one-variable integral transverse to a putative pole. Thus α is holomorphic. Choosing $\mu^m = \lambda$ lifts g bimeromorphically to the full cover, possibly permuting components, with $\tilde{g}^*\alpha = \mu\alpha$.

The lift preserves a Kähler class modulo $\text{NS}_{\mathbb{R}}$. A sufficiently large multiple of the pullback of c_Z , plus a relatively ample line class for the finite cover and its projective resolution, is Kähler on U_s . Pulling up Equation (4.3) compares the first summand, and the changes of the relatively ample classes are line classes. After a power fixes one connected component, that component has dimension $d = \dim Z < n$, so Proposition 4.7 applies there. It follows that μ , and hence λ , is a root of unity.

The possible orders are bounded in this fixed degree m . We give the parameter argument because pointwise torsion alone would not make a bounded group finite. Fix a smooth resolution X of Z and an effective divisor D clearing the poles of a basis of E_m . Put $Q_m = \mathbb{P}(E_m^\vee)$, the projective parameter space of lines of forms under our quotient convention. The universal section on $X \times Q_m$ belongs to

$$(K_X(D))^{\otimes m} \boxtimes \mathcal{O}_{Q_m}(1).$$

Pull the parameter space back by the finite coordinate-power map $[z_0 : \cdots : z_b] \mapsto [z_0^m : \cdots : z_b^m]$. Its pullback of $\mathcal{O}(1)$ is $\mathcal{O}(m)$, so the displayed line now has an m -th root $\mathcal{L}$. The cyclic algebra $\bigoplus_{i=0}^{m-1} \mathcal{L}^{-i}$, with multiplication defined by the universal section, defines a finite locally free family $\mathcal{C}$ of rank m . Every parameter is a nonzero form. Each fiber is therefore reduced: its algebra is torsion-free over the smooth X and is generically squarefree. Every fiber component meets the inverse image $W \subset \mathcal{C}$ of the nonvanishing locus of the universal section; there the cover is finite étale. Normalize $\mathcal{C}$ and resolve the normalization projectively. The composite $\rho : \tilde{\mathcal{C}} \rightarrow \mathcal{C}$ can be chosen to be an isomorphism over W . Put $E_\rho = \rho^{-1}(\mathcal{C} \setminus W)$. The locus $\{t : \dim(E_\rho)_t \geq d\}$ is proper analytic by the proper fiber-dimension theorem: every component of the total family meets W , and its general fiber has dimension d . Remove this locus and the critical values of the resolved map. Over the remaining dense open the family is smooth and proper, and each fiber is the disjoint union of smooth resolutions of the corresponding root-cover components, because every component meets the locus where ρ is an isomorphism. Proper smooth transport makes the sum of their middle Betti numbers constant there. Resolve the finitely many irreducible components of the proper analytic complement. On each such resolution pull back the original finite cyclic family, take its reduction, and normalize and resolve anew; restricting the previous normalization is not required. The parameter dimension decreases, so finitely many such steps cover all parameters. There is consequently a uniform bound B_m for the middle Betti number of the entire disjoint union of selected smooth resolutions of every full cover. This does not assert a bound for arbitrary further resolutions.

On the disjoint union of the selected resolutions, let p_1, p_2 be the projections of a smooth resolved graph of the lift. The integral Gysin endomorphism $(p_1)_!p_2^*$ on middle cohomology has μ as an eigenvalue: the nonzero holomorphic top form has a nonzero cohomology class, $p_2^*[\alpha] = \mu p_1^*[\alpha]$, and $(p_1)_!p_1^* = \text{id}$. If the order of μ is a , its cyclotomic minimal polynomial therefore has degree $\varphi(a) \leq B_m$. There are only finitely many such a . Thus the eigenvalues $\lambda = \mu^m$ on E_m lie in a fixed finite set.

The bounded group on E_m has compact closure in $\text{GL}(E_m)$. Every element of that closure still has its eigenvalues in this finite set, by continuity of characteristic polynomials. On the identity component the characteristic polynomial must be that of the identity. A compact linear group is unitarizable, so an element all of whose eigenvalues are one is the identity. The identity component is trivial; a compact Lie group with this property is finite. The image on E_m , and therefore on the full section space, is finite. $\square$

4.5 Compatible sections on the whole boundary

We now use restriction, transport, and finite self-arrow image to construct sections simultaneously on all boundary strata.

For $d \geq 0$ and $k \geq 1$, a system of tuples in degree k through dimension d means a vector subspace

$$\mathcal{V}_d(k) \subseteq \bigoplus_{\substack{Z \subset S \text{ a stratum} \\ \dim Z \leq d}} H^0(Z, L_Z^k).$$

It is *compatible* if every tuple agrees under the residue restrictions for every incidence, and *invariant* if every arrow carries its source component to its target component. It *generates* if for every point of every stratum some tuple has nonzero value there. Compatibility and invariance are linear conditions. If such a system generates, the span of componentwise a -th powers of its tuples gives a compatible invariant generating system in degree ak . Thus we may always pass to a sufficiently large divisible later degree.

Proposition 4.11 (Generating tuples). *There is a degree $k > 0$ and a compatible invariant generating system of tuples through dimension $n - 1$.*

Proof. At the point strata use the same scalar in the canonical zero-form generator 1. The even residue convention identifies these generators along every path and under every point arrow. This gives the required system in dimension zero; if there are no points the empty system is understood.

Suppose the system has been constructed through dimension $d - 1$. Replace it by powers and their span in a common degree k chosen so that Lemma 4.3 applies on every d -stratum, Proposition 4.6 applies on every vertical d -stratum, and $\mathcal{I}_{f_Z(C_Z)} \otimes N_Z^k$ is generated for every d -stratum Z . These conditions hold in a common divisible tail: there are finitely many strata, and the last condition is Serre's theorem. For a d -stratum Z , Lemma 4.2 glues the lower tuple to a section on its whole C_Z . Its residues satisfy the one link in Lemma 4.3, if that link is present, because the lower tuple is invariant. That lemma extends the section to Z .

Let $\mathcal{P}_d(k)$ be the vector space of all tuples through dimension d whose lower part is in $\mathcal{V}_{d-1}(k)$ and whose new components restrict to that lower part. Call its elements *pre-tuples*. Its projection to $\mathcal{V}_{d-1}(k)$ is surjective, since the finitely many extensions can be chosen independently. This system already generates. To check a point $z \in Z$, put $y = f_Z(z)$. If $y \in f_Z(C_Z)$, choose a floor point over y and a lower tuple nonzero there. Every extension is the pullback of a section of N_Z^k , so it is nonzero at every point over y , including z . If $y \notin f_Z(C_Z)$, the chosen generation of $\mathcal{I}_{f_Z(C_Z)} \otimes N_Z^k$ supplies a base section nonzero at y . It pulls back to a section vanishing on C_Z ; combine it with the zero lower tuple and zero components on the other new strata. This also treats an empty floor. Surjectivity to the lower system preserves generation at lower points.

We impose invariance in dimension d . For a dominating floor, injectivity of restriction and Lemma 4.5 already make every pre-tuple invariant on that stratum. Verticality is invariant under arrows, so the remaining orbits consist entirely of vertical strata.

For each such orbit choose a representative Z , and let G_Z be the finite isotropy image on $H^0(Z, L_Z^k)$ given by Proposition 4.6. For a pre-tuple with component s_Z , form the norm

$$P_Z(s_Z) = \prod_{g \in G_Z} g(s_Z) \in H^0(Z, L_Z^{k|G_Z|}).$$

Choose a common positive integer a divisible by all $|G_Z|$, use $P_Z(s_Z)^{a/|G_Z|}$, and transport it to each member of that orbit. This is independent of the chosen arrow: changing that arrow by an

isotropy arrow merely permutes the factors in degree k . No finiteness assertion in degree ak is needed. On strata with dominating floor use s_Z^a , and on all lower strata use the componentwise a -th power of the assigned tuple.

The resulting tuple is compatible. By Lemma 4.5, every transported factor in a norm has exactly the prescribed lower restriction. The norm and its indicated power therefore restrict to the a -th power of that restriction, the same value used on all lower strata and in all other orbits. It is invariant in dimension d by construction, and remains invariant below it.

These tuples still generate. Fix a point z in an orbit member. For each of the finitely many transported factors, choose a point over z on a resolution of its comparison graph. Equality of the pulled-back invertible adjoint lines identifies the fiber values. The factor is nonzero at z exactly when the representative component s_Z is nonzero at the corresponding point of Z . Evaluation at each such point is a nonzero linear functional on the generating space $\mathcal{P}_d(k)$. A finite union of their proper kernels cannot cover a complex vector space. One pre-tuple makes all factors nonzero and gives a norm nonzero at z . At a lower point a nonzero assigned value stays nonzero after its a -th power. Taking the linear span of all the constructed tuples therefore gives a compatible invariant generating system in degree ak . This completes the induction on d . $\square$

Proof of Theorem 4.1. Take the system in Proposition 4.11. Its components on the prime components of S agree on every common lower stratum, so Lemma 4.2 descends each tuple uniquely to a section of the actual line $\mathcal{O}_V(qkJ)|_S$. At any $x \in S$, choose a prime component through x and a tuple nonzero there. It is the pullback of the descended value in the same invertible line fiber at x , so that value is nonzero. The finite-dimensional span of these descended sections generates at every point of the reduced S . This is the claimed global generation of an actual Cartier multiple. $\square$

5 Descent along a fibration

Throughout this section we assume Assumption 1.1.

We prove Proposition 2.11. Its geometric reduction leads to a fibration whose very general fiber has logarithmic Kodaira dimension zero. In sufficiently divisible degrees the log pluricanonical systems on that fiber are one-dimensional, and a relative generating form determines an adjoint line on the base. We prove the resulting rank-one proposition by a secondary induction on the positive base dimension: after proving pseudo-effectivity on the base, we either lower its dimension or lift a nef line and its entire fixed divisor. The last step proves generation of the remaining nef line.

5.1 Reduction to the rank-one case

Proposition 5.1 (The rank-one case). *Assume Assumption 1.1. Fix $n > 0$ and assume $\mathcal{G}_d$ for every $d < n$. Let X be a smooth connected compact Kähler manifold of dimension n , let B be a rational SNC boundary, and suppose that $J = K_X + B$ is pseudo-effective. Suppose there are smooth compact Kähler manifolds $\tilde{X}, W$, a modification $\pi : \tilde{X} \rightarrow X$, and a proper surjective holomorphic map $g : \tilde{X} \rightarrow W$ with connected fibers such that*

$$0 < k := \dim W < n, \quad W \text{ is projective or } a(W) = 0.$$

Assume that $\tilde{B} = \pi_^{-1}B + \text{Exc}(\pi)_{\text{red}}$ has SNC support and that, on a very general smooth fiber F of g ,*

$$\kappa(F, K_F + \tilde{B}_F) = 0.$$

Then (X, B) satisfies $\mathcal{G}_n$.

Proof of Proposition 2.11, assuming Proposition 5.1. We use the reduced-exceptional boundary after every source modification. The effective discrepancy identity in Proposition 2.7 preserves pseudo-effectivity, and that proposition transfers the resulting decomposition back to the original pair. A dominant meromorphic map to a space in class $\mathcal{C}$ can be resolved on smooth compact Kähler models; taking Stein factorization gives connected fibers. These operations preserve the positive dimensions of the base and the fiber.

We first suppose $a(X) > 0$. If $a(X) = n$, then X is Moishezon and Kähler, hence projective, and Proposition 2.10 applies. Otherwise resolve the algebraic reduction as a fibration $g : X' \rightarrow W$ to a smooth projective variety of dimension $a(X)$. Put $J' = K_{X'} + B'$ for the modified log adjoint. Its restriction to a very general smooth fiber is pseudo-effective. Indeed, one may choose a sequence of positive currents in $c_1(J') + \varepsilon_j[\omega]$, with analytic singularities and $\varepsilon_j \downarrow 0$, and restrict all of them outside a countable union of proper analytic subsets of the base. By $\mathcal{G}_{\dim F}$, the restricted adjoint has nonnegative Kodaira dimension.

That dimension must be zero. If it were positive, generic coherent base change would give a divisible q for which $\mathcal{E} = g_*\mathcal{O}_{X'}(qJ')$ has a fiber system with a positive-dimensional image on very general fibers. A sufficiently large ample twist of the coherent sheaf $\mathcal{E}$ is generated at the generic point of the projective W . Evaluation would therefore give two sections of $\mathcal{O}_{X'}(qJ') \otimes g^*\mathcal{O}_W(A)$, for one ample A , whose ratio is nonconstant on a general g -fiber. This ratio is a meromorphic function on X' . Every meromorphic function on X' factors through its algebraic reduction, a contradiction. Proposition 5.1 now applies.

Suppose next that $a(X) = 0$, and resolve the fibration in the hypothesis as $f : (X', B') \rightarrow Y$. The base may be replaced by a smooth compact Kähler model, and $a(Y) = 0$, since its meromorphic functions pull back to X' . Write $d = \dim X' - \dim Y > 0$. The same restriction argument and $\mathcal{G}_d$ show that

$$\kappa(F, K_F + B'_F) \geq 0$$

on very general smooth fibers. If this integer is $j < d$, the relative Iitaka construction of [57, Lemma 2.6] gives, after the same source modifications, a factorization

$$X'' \xrightarrow{g} W \xrightarrow{h} Y$$

with connected fibers, $\dim W = \dim Y + j$, and $\kappa(G, K_G + B''_G) = 0$ on a very general g -fiber. The cited lemma applies to rational SNC boundaries on compact manifolds in class $\mathcal{C}$; its conclusion concerns the restricted log adjoint, and does not require finite generation. A Kähler model of W has algebraic dimension zero because it is dominated by X'' . Thus $0 < \dim W < n$, and Proposition 5.1 applies again.

It remains to exclude the possibility that $K_F + B'_F$ is big. The stable-family comparison below requires a horizontal boundary with coefficients strictly less than one, so we first lower the boundary while keeping the restricted adjoint big. Let B'_{hor} be the sum of the components of B' dominating Y . There is a rational $0 < \lambda < 1$ such that $K_F + \lambda B'_{\text{hor}, F}$ is big for very general F . To justify the uniform choice, exclude at once the generic base-change and image-dimension exceptional sets for all rational λ and all divisible degrees. On a fiber outside this countable union, openness of the big cone gives one such λ , and a single degree has full-dimensional image. Generic base change makes the same choice work very generally. The boundary $\lambda B'_{\text{hor}}$ is horizontal, rational SNC, and has all coefficients strictly less than one. The stable-family comparison [57, Lemma 5.2] therefore supplies a proper generically finite cover $Y' \rightarrow Y$, a surjection $Y' \rightarrow R$ to a

smooth projective variety, and a projective stable family $\mathcal{V} \rightarrow R$ with positive-dimensional general fiber, such that the main transform of X' is bimeromorphic to $\mathcal{V} \times_R Y'$.

The main transform still has algebraic dimension zero. Normalize it and factor its proper generically finite map to X' through the normal Stein space. A meromorphic function descends through the birational part. Lemma 2.8 gives its characteristic polynomial over the finite part, with meromorphic coefficients on X' . Those coefficients are constant because $a(X') = 0$, so irreducibility makes the function constant. Normalization and modifications preserve meromorphic functions. It follows also that $a(Y') = 0$, so the surjection from Y' to the projective R forces R to be a point. But then the main transform is bimeromorphic to the product of Y' with a positive-dimensional projective variety. Rational functions on that factor contradict algebraic dimension zero. This excludes the big case and completes the reduction. $\square$

5.2 The adjoint line on the base

We prepare the rank-one fibration and record exactly what the relative generating form supplies. The construction of the Hodge line and its positivity are those of [57, Proposition 2.7 and Theorem 3.1]. The relative injection and the last assertion below are consequences of the local order calculation in that proof; they will be needed after the base is changed.

Lemma 5.2 (Prepared comparison). *Let $g_0 : (X_0, B_0) \rightarrow W_0$ be a fibration with connected fibers from a smooth compact Kähler manifold to a smooth compact manifold in class $\mathcal{C}$. Assume B_0 is a rational SNC boundary and $\kappa(F, K_F + B_{0,F}) = 0$ on very general smooth fibers. After modifications there is a diagram*

$$\begin{array}{ccc} X & \xrightarrow{\pi} & X_0 \\ g \downarrow & & \downarrow g_0 \\ W & \xrightarrow{\rho} & W_0 \end{array}$$

with X, W smooth compact Kähler, g a fibration with connected fibers, and $B = \pi_*^{-1} B_0 + \text{Exc}(\pi)_{\text{red}}$ SNC. Every prime of X whose g -image has codimension at least two is π -exceptional. There are a rational SNC boundary T on W , a surjection $p : W \rightarrow S$ with connected fibers to a smooth projective variety, and a nef rational line P_S on S . Put

$$M = p^* P_S, \quad H = K_W + T + M, \quad J = K_X + B.$$

If $\dim S > 0$, then $K_S + a_0 P_S$ is big for some rational $a_0 > 0$; if S is a point, $M \sim_{\mathbb{Q}} 0$. There is a rational divisor A_* on X giving an actual rational line identity

$$J \sim_{\mathbb{Q}} g^* H + A_*. \quad (5.1)$$

These data satisfy the following properties.

- (i) The horizontal part $A_{*,\text{hor}}$ is effective. On a very general smooth fiber it is the zero divisor of a relative generating section of a divisible multiple of $K_F + B_F$, divided by that multiple.
- (ii) For each prime $D \subset W$, let E run over the primes of X dominating D , and put $a_E = \text{ord}_E(g^* D)$. Then

$$(A_*)_E \geq 0, \quad \min_{E \rightarrow D} \frac{(A_*)_E}{a_E} = 0. \quad (5.2)$$

There is no assertion here about the sign at a prime whose image has codimension at least two.

(iii) For all degrees q divisible by one positive integer, multiplication by the relative generating form identifies

$$H^0(W, qH) \simeq H^0(X, qJ)$$

and preserves ratios of sections. In the same degrees, for every proper holomorphic $h : W \rightarrow V$, there is a natural injection

$$(hg)_* \mathcal{O}_X(qJ) \hookrightarrow h_* \mathcal{O}_W(qH). \quad (5.3)$$

(iv) Let $\tau : W' \rightarrow W$ be a further smooth modification included in another diagram with the preceding properties over the same reference X_0 , using the same source-boundary convention. Let H' be the base line produced by that diagram. Then, as actual rational lines,

$$H' \sim_{\mathbb{Q}} \tau^* H + E_H, \quad E_H \geq 0 \text{ } \tau\text{-exceptional}. \quad (5.4)$$

Proof. The flattening construction of [57, Lemma 2.4] gives the stated reference property, and preserves it after any finite sequence of further base modifications. Its source boundary is the one in [57, Lemma 2.1]. We may resolve the discriminant and the horizontal divisor of the relative generator so that all relevant strata are smooth over the dense smooth log locus. Proposition 2.7 of [57] then gives T and a rational parabolic Hodge line M . A positive multiple of that line is the highest rank-one Hodge step of a complex summand of a pure real-polarizable variation with an integral lattice. Theorem 3.1 of [57] gives the factorization $M = p^* P_S$ and the stated positivity, after a further modification and a positive rational rescaling. These are identities in $\text{Pic} \otimes \mathbb{Q}$, not merely identities of classes.

Here is the order computation that gives Equations (5.1)–(5.3). Choose a degree m for which $g_* \mathcal{O}_X(m(K_{X/W} + B))$ has generic rank one and all boundary denominators are cleared. Its reflexive hull is a line on the smooth W . Let s be a meromorphic relative generating form in a local frame of this line. At the generic point of a prime D , choose a nonvanishing ordinary base volume ω_W , and set

$$l_E = \frac{1}{m} \text{ord}_E(s \wedge g^* \omega_W^m), \quad \delta_E = B_E, \quad \alpha_D = \min_{E \rightarrow D} \frac{l_E + \delta_E}{a_E}.$$

The relative-times-base identification in this formula is taken in the m -th canonical tensor power. The source is resolved over the generic point of D , so all the primes needed for the divisorial test occur in this minimum. In the notation of the proof of [57, Proposition 2.7], the parabolic order is

$$\beta_D = \min_{E \rightarrow D} \frac{l_E + 1 - a_E}{a_E}, \quad T_D = \alpha_D - \beta_D.$$

Thus the corresponding local frame of $T + M$ has order α_D . Comparing its pullback with the log adjoint form defines A_* , with

$$(A_*)_E = l_E + \delta_E - a_E \alpha_D. \quad (5.5)$$

Changing the generating form multiplies it by a meromorphic base function. This changes both α_D and β_D by its normalized base order and leaves the comparison invariant. The local comparisons therefore give the rational divisor and the line identity globally. Equation (5.5) proves Equation (5.2). Horizontally the comparison is precisely the zero divisor of the fiberwise log section, so it is effective and has the asserted restriction.

The same order computation proves the relative injection. In a divisible degree q , a meromorphic base coefficient φ gives a regular base section at D exactly when

$$\text{ord}_D(\varphi) + q\alpha_D \geq 0.$$

Its corresponding upstairs orders at the primes $E \rightarrow D$ are

$$a_E \operatorname{ord}_D(\varphi) + q(l_E + \delta_E).$$

Their simultaneous nonnegativity is equivalent to the preceding inequality. Conversely, an upstairs section restricts to a multiple of the generator on general connected fibers, so its ratio with the generator descends meromorphically to W . The order inequalities make it regular outside codimension two on W , and normality extends it. Consequently

$$g_* \mathcal{O}_X(qJ) \hookrightarrow \mathcal{O}_W(qH).$$

Applying the left exact functor h_* proves Equation (5.3). The reverse global comparison follows from the reference property: its only initially untested poles are on g -contracted primes, all exceptional over X_0 ; pushing to X_0 , extending in codimension two, and pulling back with the reduced-exceptional boundary removes them. This is the global comparison in [57, Proposition 2.7].

Finally consider $\tau : W' \rightarrow W$. The parabolic rational line pulls back under SNC modifications, as in the construction preceding [57, Theorem 3.1]. The coefficient T_D at an old strict transform is unchanged. Indeed the sheaf of regular relative adjoint forms inside the common meromorphic generator line is unchanged over the generic point of D by the log modification formula. Equivalently, the old inequalities above already test all vertical primes there; in SNC coordinates logarithmic pullback preserves those inequalities, so new source exceptional primes impose no stronger one. This keeps α_D unchanged in compatible frames, while parabolic pullback keeps β_D unchanged. Over a new τ -exceptional prime every source prime is exceptional over X_0 : a pre-existing one had image of codimension at least two on W , and a new one is exceptional by construction. All therefore have boundary coefficient one. The formula for $T_D = \alpha_D - \beta_D$ then gives coefficient one on that new base prime. Thus

$$T' = \tau_*^{-1} T + \operatorname{Exc}(\tau)_{\operatorname{red}}.$$

The usual log discrepancy formula for the SNC pair (W, T) now gives $K_{W'} + T' = \tau^*(K_W + T) + E_H$, with E_H effective and exceptional. Adding the pulled-back parabolic line proves Equation (5.4). $\square$

Lemma 5.3 (Pseudo-effectivity of the base line). *In the setting of Lemma 5.2, suppose $\dim X = n$, $0 < \dim W < n$, J is pseudo-effective, and $\mathcal{G}_d$ holds for $d < n$. Then*

$$A_{*,\operatorname{hor}} \leq N(J) \quad \text{as divisors,} \quad H \text{ is pseudo-effective.}$$

Proof. On a very general smooth fiber F , use the model supplied by $\mathcal{G}_{\dim F}$ to write $\mu_F^* J_F \sim_{\mathbb{Q}} P_F + N(\mu_F^* J_F)$. Lemma 2.5 identifies the divisible section spaces with those of P_F , so $\kappa(P_F) = \kappa(F, J_F) = 0$; a semiample line of Kodaira dimension zero is torsion. Proposition 2.9, applied to the modification μ_F with zero added divisor, gives $J_F \sim_{\mathbb{Q}} N(J_F)$ on F , with $N(J_F)$ rational. Lemma 2.5 then identifies the normalized zero divisor of the prepared relative generator with $N(J_F)$. Thus

$$A_{*,\operatorname{hor}}|_F = N(J_F).$$

We compare these fiber multiplicities with those on X . For each of the finitely many components E of $A_{*,\operatorname{hor}}$, choose a countable sequence of small-Kähler-perturbation currents with analytic singularities whose generic orders at E approach $\nu_E(J)$. Choose a common very general F for these sequences and these components. The currents restrict positively after the same perturbations, and analytic singularities ensure that their orders along the components of $E|_F$ are their generic

orders along E . Each restricted order is at least the corresponding perturbed minimal multiplicity on F . Passing to the limit gives $(A_*)_E \leq \nu_E(J)$. This proves

$$A_{*,\text{hor}} \leq N(J). \quad (5.6)$$

Choose a positive curvature current for the rational line J . Every such current contains its divisorial negative part, hence subtracting $[A_{*,\text{hor}}]$ leaves a positive current. On the inverse image of a dense smooth open W° , with all vertical data removed, Equation (5.1) identifies the resulting singular metric with one on g^*H . In a frame pulled back from W° its weight is plurisubharmonic. It is constant on each compact connected smooth fiber, by the maximum principle (with the value $-\infty$ allowed). Local holomorphic sections of the submersion show that these values form a plurisubharmonic weight on $H|_{W^\circ}$.

This weight extends across the missing divisors. At the generic point of any such divisor D , choose $E \rightarrow D$ for which $(A_*)_E = 0$, using Equation (5.2). At a general point of E away from the other comparison divisors, the comparison after subtracting the horizontal part has neither a zero nor a pole. We can choose local coordinates, with the remaining vertical coordinates denoted by w , in which

$$g(u, z_2, \dots, z_k, w) = (u^{a_E}, z_2, \dots, z_k).$$

Indeed $g|_E$ has maximal tangential rank generically and a local equation of g^*D is a unit times u^{a_E} ; the unit has a local a_E -th root. A smaller such chart covers a neighborhood of the base point. The upstairs plurisubharmonic weight is locally bounded above there, and the comparison unit is bounded. Hence the descended weight is locally bounded above near the generic point of D . The removable singularity theorem for plurisubharmonic functions extends it across D away from codimension two, and the Hartogs extension for plurisubharmonic functions extends it across the remaining analytic subset. The extensions respect the line transitions because they agree on the dense open. They give a positive singular metric on H , which proves pseudo-effectivity. $\square$

Lemma 5.4 (A base of algebraic dimension zero). *Under the assumptions of Lemma 5.3, if $a(W) = 0$, then, on W ,*

$$H \sim_{\mathbb{Q}} E_W = N(H) \quad \text{with } E_W \geq 0. \quad (5.7)$$

Proof. The projective quotient S in Lemma 5.2 is a point: otherwise its rational functions pull back nontrivially to W . Thus $M \sim_{\mathbb{Q}} 0$ and $H \sim_{\mathbb{Q}} K_W + T$. By Lemma 5.3 and $\mathcal{G}_{\dim W}$, its pullback to a smooth higher model is a semiample rational line plus its negative divisor. Every semiample line on a space of algebraic dimension zero is torsion, since its generated system has zero-dimensional image. Proposition 2.9, applied to this modification with zero added divisor, therefore gives Equation (5.7), including rationality of the divisor. $\square$

5.3 Projective programs under the assumption

The argument for a projective base uses a terminating program to reach a nef adjoint and, after decreasing the nef data, a program ending in a Mori fiber space. The conditional input is termination in the pseudo-effective case. We first state the exact program result and then derive that case from the projective good models in Proposition 2.10.

Proposition 5.5 (A conditional generalized program). *Assume Assumption 1.1. Let $(V, B + \mathbf{M})$ be a projective generalized dlt rational pair, where V is $\mathbb{Q}$ -factorial, $B \geq 0$, and the nef b -divisor $\mathbf{M}$ is determined by a nef rational Cartier divisor on a projective birational model. Put $D = K_V + B + M_V$.*

There is a choice of D -MMP with ample scaling which terminates. If D is pseudo-effective, its endpoint V_m has nef adjoint D_m . If D is not pseudo-effective, its endpoint has a Mori fiber contraction to a projective variety of smaller dimension. The birational steps and their endpoints stay $\mathbb{Q}$ -factorial generalized dlt, and the forward birational map extracts no divisors. In the pseudo-effective case, on a common resolution one has an actual rational divisor identity

$$u^*D = v^*D_m + F, \quad F \geq 0 \text{ and } v\text{-exceptional.}$$

For the pseudo-effective alternative, we first derive the relative weak Zariski decompositions used by the minimal-model existence theorem.

For a projective morphism, first replace its image by its normal Stein factor. We use the relative convention of [64, Section 2]: a divisor is pseudo-effective over the base when its restriction to a very general fiber of this surjective morphism is pseudo-effective. An NQC weak Zariski decomposition of D over the base is a birational equality $h^*D \equiv_Z P + N$, where P is a nonnegative real combination of relatively nef rational Cartier divisors and $N \geq 0$. The decomposition below has rational Cartier parts.

Lemma 5.6 (Relative canonical decomposition). *Assume Assumption 1.1. Let V be a smooth quasi-projective complex variety and let $f : V \rightarrow Z$ be projective, with Z normal and quasi-projective. If K_V is pseudo-effective over Z , then on a normal variety Y with a projective birational morphism $h : Y \rightarrow V$ there are rational Cartier divisors P, N such that*

$$h^*K_V \equiv_Z P + N, \quad P \text{ nef over } Z, \quad N \geq 0.$$

Proof. We use the construction in the proof of [56, Proposition 3.3]. Its absolute input is a nef-plus-effective rational decomposition for the canonical divisor of a smooth projective variety with pseudo-effective canonical class. This is supplied here by Proposition 2.10 and its common-resolution comparison.

Replace the image of f by its normal Stein factor. This does not change relative numerical classes or contracted curves, and now f is surjective with connected fibers. Its very general fiber is smooth projective with pseudo-effective canonical divisor, hence is non-uniruled by [8, Theorem 0.2 and Corollary 0.3]. Compactify the projective morphism and resolve away from V . We obtain $\bar{f} : \bar{V} \rightarrow \bar{Z}$, with $\bar{V}$ smooth projective and $\bar{Z}$ normal projective, unchanged over Z . If $\dim Z = 0$, V is already projective and the absolute input proves the lemma.

Suppose $\dim Z > 0$, and let $r : \tilde{Z} \rightarrow \bar{Z}$ be a smooth projective resolution. Choose a sufficiently positive very ample divisor A_Z on $\bar{Z}$ and a general $B \in |2A_Z|$. The pulled-back systems on both $\tilde{Z}$ and $\bar{V}$ are base point free. Bertini makes r^*B and $\bar{f}^*B$ smooth nonempty reduced divisors. Their double covers

$$\pi_Z : Z^\sharp \rightarrow \tilde{Z}, \quad \pi : V^\sharp \rightarrow \bar{V}$$

are smooth integral projective varieties. The canonical formulas are

$$K_{Z^\sharp} \sim_{\mathbb{Q}} \pi_Z^*(K_{\tilde{Z}} + r^*A_Z), \quad K_{V^\sharp} \sim_{\mathbb{Q}} \pi^*(K_{\bar{V}} + \bar{f}^*A_Z).$$

The first line is big when A_Z is sufficiently positive. The rational map $V^\sharp \dashrightarrow Z^\sharp$ has the same very general fibers as f . A covering family of rational curves on $V^\sharp$ would either dominate a covering family of rational curves on $Z^\sharp$, or cover its very general fibers by vertical rational curves. Both are impossible. Thus $V^\sharp$ is non-uniruled and $K_{V^\sharp}$ is pseudo-effective by BDPP.

Apply the absolute decomposition to $V^\sharp$. Moving the rational principal difference in the canonical formula into the nef part gives an actual rational divisor decomposition for a pullback of

$\pi^*(K_{\bar{V}} + \bar{f}^*A_Z)$. Take a common equivariant resolution of that model and its conjugate under the covering involution. On this smooth projective resolution $g : T \rightarrow V^\sharp$, average the two decompositions. We obtain

$$g^*\pi^*(K_{\bar{V}} + \bar{f}^*A_Z) = \bar{P} + \bar{N}, \quad \bar{P} \text{ nef}, \quad \bar{N} \geq 0,$$

where $\bar{P}$ and $\bar{N}$ are invariant rational Cartier divisors.

Let $q : T \rightarrow \bar{Y}$ be the finite quotient by the involution. The invariant composite πg induces a projective birational morphism $\bar{h} : \bar{Y} \rightarrow \bar{V}$, with $\bar{Y}$ normal and projective. An invariant rational divisor descends by dividing its coefficient at an upstairs prime by the ramification index. The descended divisor is rational Cartier when the original divisor is: after clearing denominators, choose one Cartier equation on a semilocal neighborhood of the finite orbit above a point (a Cartier divisor is principal on a semilocal ring), and multiply all its translates. The product is invariant and has divisor equal to the group order times the original divisor. It descends to a local equation for a multiple of the downstairs divisor. This argument includes stabilizers and ramification primes.

Accordingly $\bar{P} = q^*P$ and $\bar{N} = q^*N$ for rational Cartier divisors on $\bar{Y}$. Lifting curves through the finite map shows that P is nef, and coefficientwise descent shows that N is effective. Finite pullback is injective on rational divisors, so the displayed identity descends. Restrict it to $Y = \bar{h}^{-1}(V)$. The term pulled back from A_Z is numerically trivial over Z , giving $h^*K_V \equiv_Z P + N$, as required. $\square$

Proof of Proposition 5.5. Lemma 5.6 and [64, Theorem 5.2], applied to the generalized pair $(U/Z, 0 + 0)$ with U smooth, prove relative minimal-model existence for smooth varieties by dimension induction. The hypothesis of that theorem is relative smooth minimal-model existence one dimension lower; the lemma supplies its other hypothesis, an NQC weak Zariski decomposition. The induction begins in dimension zero. Then [64, Theorem 5.4] supplies minimal models for every pseudo-effective NQC generalized lc pair in the relevant dimension. Their Theorem 2.7 gives existence in the Birkar–Shokurov sense.

The nef data in the statement are NQC, because a nef rational Cartier divisor is a positive multiple of a nef Cartier divisor. In the pseudo-effective case the preceding paragraph therefore gives the Birkar–Shokurov model required by [64, Theorem 4.2]. In the other case, the non-pseudo-effective alternative of that theorem applies directly. To meet its scaling hypothesis, take an effective sufficiently positive ample rational divisor A whose general components have sufficiently small coefficients. On a fixed log resolution carrying $\mathbf{M}$, these components are transverse to the boundary and preserve generalized log canonicity, while their ample class can be chosen large enough that $D + A$ is nef. The theorem supplies a terminating MMP starting on V .

We verify the step types inductively. Suppose V_i is $\mathbb{Q}$ -factorial generalized dlt. By [64, Remark 2.15], each birational step has the form

$$V_i \xrightarrow{g_i} Z_i \xleftarrow{h_i} V_{i+1},$$

where g_i contracts a negative extremal ray and h_i is small and projective. Suppose g_i is divisorial, and write R for its ray. An exceptional prime E has $E \cdot R < 0$: otherwise it would be effective, exceptional, and g_i -nef, contrary to negativity. There is only one exceptional prime. Indeed, a combination of two distinct exceptional primes can be chosen to have degree zero on R ; negativity applied to both signs of that combination would make it zero.

We show directly that Z_i is $\mathbb{Q}$ -factorial. For a prime divisor T on Z_i , let $\tilde{T}$ be its strict transform, choose a curve C spanning R , and put

$$a = -\frac{\tilde{T} \cdot C}{E \cdot C} \in \mathbb{Q}.$$

Choose a positive integer m for which $m(\tilde{T} + aE)$ is Cartier. Its line has degree zero on R , so [66, Theorem 1.5] gives an actual Cartier line $\mathcal{L}_T$ on Z_i and an isomorphism

$$\mathcal{O}_{V_i}(m(\tilde{T} + aE)) \simeq g_i^* \mathcal{L}_T.$$

On an open set trivializing $\mathcal{L}_T$, the Cartier divisor $m(\tilde{T} + aE)$ is principal upstairs. Pushing this principal divisor identity down through the birational g_i gives a principal divisor mT on that open set. Thus T is rational Cartier. It follows that h_i is an isomorphism: the pushforward of an h_i -ample Cartier divisor is rational Cartier on Z_i , and its pullback is the original divisor because h_i is small. It has degree zero on every contracted curve, so relative ampleness rules out positive-dimensional fibers; a finite birational map to the normal Z_i is an isomorphism. If g_i is small, the relatively ample canonical model h_i is the flip of [40, Definition 3.2]. Thus the chosen birational steps are divisorial contractions and flips. The underlying variety of a generalized dlt pair is klt, and these steps preserve the $\mathbb{Q}$ -factorial generalized dlt category [40, Remark 2.3 and Lemma 3.7]. They introduce no prime divisors on the new models. For completeness, on a common resolution of one step $V_i \dashrightarrow V_{i+1}$ carrying $\mathbf{M}$, taking the difference of the two generalized discrepancy formulas cancels their common nef divisor. The resulting difference of the adjoint pullbacks is exceptional over the new model and anti-nef over it. The negativity lemma makes this difference effective. Composing these actual rational divisor comparisons gives $u^*D = v^*D_m + F$ with F effective and exceptional over the endpoint. The endpoints in the two cases are the nef and Mori fiber endpoints supplied by the terminating program theorem. $\square$

5.4 Reducing a projective base

The nef line M supplied by the Hodge construction need not be semiample. A first projective program replaces H by a nef transform. Positive Iitaka dimension gives either a smaller base or a big line. In nonpositive Iitaka dimension, rationally trivial nef data give a trivial line. In the remaining case, decreasing M leads to a second program that lowers the base dimension while preserving the first nef line.

Lemma 5.7 (Projective base reduction). *Assume Assumption 1.1 and the setting of Lemma 5.3, with W projective of dimension k . Then one of the following holds.*

- (i) *After a smooth source modification with the reduced-exceptional boundary, there is a fibration to a smooth projective variety V with $0 < \dim V < k$ and logarithmic Kodaira dimension zero on very general smooth fibers.*
- (ii) *There are a normal $\mathbb{Q}$ -factorial projective variety W_m , a birational contraction $W \dashrightarrow W_m$ extracting no divisors, and a nef rational line H_m on W_m , either big or rationally trivial, such that on a common smooth resolution $\mu : W' \rightarrow W$, $\nu : W' \rightarrow W_m$,*

$$\mu^*H \sim_{\mathbb{Q}} \nu^*H_m + E, \quad E \geq 0 \text{ } \nu\text{-exceptional.} \quad (5.8)$$

Here all the line comparisons are actual rational line identities.

Proof. The smooth pair with boundary T and nef data $\mathbf{M}$ determined by M on W is a rational $\mathbb{Q}$ -factorial generalized dlt pair. Its adjoint H is pseudo-effective by Lemma 5.3. Proposition 5.5 gives a chosen terminating program with scaling to a $\mathbb{Q}$ -factorial generalized dlt model W_{nef} with nef transformed adjoint

$$H_{\text{nef}} = K_{W_{\text{nef}}} + T_{\text{nef}} + M_{\text{nef}}.$$

Here M_{nef} is the trace of the nef b-divisor whose nef representative is M on W . On a common smooth resolution $\mu : W' \rightarrow W$, $\sigma : W' \rightarrow W_{\text{nef}}$, the actual comparison is

$$\mu^* H \sim_{\mathbb{Q}} \sigma^* H_{\text{nef}} + E_{\text{nef}}, \quad E_{\text{nef}} \geq 0 \text{ } \sigma\text{-exceptional.} \quad (5.9)$$

The program extracts no divisors. We use this particular model also when the nef data are rationally trivial.

If $\kappa(H) > 0$, Equation (5.9) preserves the divisible section spaces. For $\kappa(H) = k$ the nef H_{nef} is big, and the second alternative holds with $W_m = W_{\text{nef}}$ and $H_m = H_{\text{nef}}$. If $0 < \kappa(H) < k$, the exact global comparison in Lemma 5.2 gives $\kappa(J) = \kappa(H)$. Take the Iitaka fibration of J on the source. Its projective image, followed by Stein factorization and a projective resolution, has dimension $\kappa(H)$. The log adjoint on very general fibers has Kodaira dimension zero by the relative Iitaka construction [57, Lemma 2.6], including its persistence under the reduced-exceptional source convention. This is the first alternative.

Suppose $\kappa(H) \leq 0$ and $M \sim_{\mathbb{Q}} 0$. We may choose zero nef data in this rational equivalence class. The same generalized dlt model W_{nef} is then an ordinary dlt model and $H_{\text{nef}} \sim_{\mathbb{Q}} K_{W_{\text{nef}}} + T_{\text{nef}}$. Its adjoint is nef, so Proposition 2.10 makes H_{nef} semiample as an actual rational line. Since $\kappa(H_{\text{nef}}) = \kappa(H) \leq 0$, a generated multiple has constant image and is trivial. Thus the second alternative holds with $W_m = W_{\text{nef}}$ and $H_m = H_{\text{nef}} \sim_{\mathbb{Q}} 0$.

It remains to suppose

$$\kappa(H) \leq 0, \quad M \not\sim_{\mathbb{Q}} 0. \quad (5.10)$$

Here $\dim S > 0$. For a rational $c \in (0, 1)$, put $H_c = K_W + T + cM$. We claim

$$H_c \text{ is not pseudo-effective} \quad (0 < c < 1, c \in \mathbb{Q}). \quad (5.11)$$

Suppose otherwise for one c , and choose an ample rational line U on S . We will find $\delta > 0$ for which $H - \delta p^* U$ has an effective representative, forcing $\kappa(H) \geq \dim S > 0$. Choose a rational $a > \max\{1, a_0\}$ and a small rational $\lambda > 0$ such that $K_S + aP_S - \lambda U$ is big. This is possible because P_S is nef and $K_S + a_0 P_S$ is big. Choose

$$0 < \eta < \lambda \frac{1-c}{a-1} \quad \text{rational.}$$

The rational line $cP_S + \eta U$ is ample. A sufficiently high general member divided by its degree pulls back to a rational boundary transverse to T with subunit coefficient. Applying the nonvanishing part of Proposition 2.10 to this ordinary projective log pair gives

$$E_c \sim_{\mathbb{Q}} H_c + \eta p^* U, \quad E_c \geq 0.$$

A nonzero section defining E_c restricts nontrivially to very general p -fibers in a fixed divisible degree. The hypotheses of the weak-effectivity statement [57, Lemma 3.3] are now satisfied: the source is smooth in class $\mathcal{C}$, T is rational SNC, the base S is smooth projective, and a relative log system is nonzero. Applied to the big line $K_S + aP_S - \lambda U$, it gives

$$E_a \sim_{\mathbb{Q}} K_W + T + aM - \lambda p^* U, \quad E_a \geq 0.$$

For

$$\theta = \frac{1-c}{a-c}, \quad \delta = \theta\lambda - (1-\theta)\eta > 0,$$

the rational interpolation is

$$(1-\theta)E_c + \theta E_a \sim_{\mathbb{Q}} H - \delta p^*U.$$

Multiplying pullbacks of sections of U by a section of the effective left-hand side yields $\kappa(H) \geq \kappa(p^*U) = \dim S > 0$. This contradicts Equation (5.10) and proves Equation (5.11).

We next run a program to fiber type while descending the actual nef line H_{nef} through every contraction. Choose the common resolution in Equation (5.9) to determine the nef data. The negativity lemma gives

$$F_M := \sigma^*M_{\text{nef}} - \mu^*M \geq 0 \quad \sigma\text{-exceptional}.$$

Indeed its negative is σ -nef and exceptional. For every rational $0 < e < 1$, Equation (5.9) gives

$$\mu^*H_{1-e} \sim_{\mathbb{Q}} \sigma^*(H_{\text{nef}} - eM_{\text{nef}}) + E_{\text{nef}} + eF_M. \quad (5.12)$$

If $H_{\text{nef}} - eM_{\text{nef}}$ were pseudo-effective, this equality and effectivity of the last two terms would make μ^*H_{1-e} , and hence H_{1-e} , pseudo-effective, contrary to Equation (5.11).

Fix rational $0 < \varepsilon < 1$, choose an integer $r > 0$ with rH_{nef} Cartier, and choose an integer $d_0 > 2kr$. This coefficient will force every ray of the second program to have zero degree for the descended nef line. Decreasing the nef data to $(1-\varepsilon)M$ preserves generalized dlt singularities: on the displayed resolution the crepant boundary changes by $-\varepsilon F_M$, so discrepancies do not decrease. Add d_0H_{nef} , determined as nef data on W_{nef} . It changes no discrepancies, and the resulting adjoint is

$$(1+d_0)H_{\text{nef}} - \varepsilon M_{\text{nef}} = (1+d_0) \left(H_{\text{nef}} - \frac{\varepsilon}{1+d_0} M_{\text{nef}} \right).$$

It is not pseudo-effective by the preceding paragraph. The non-pseudo-effective case of Proposition 5.5 therefore gives a chosen terminating program to a Mori fiber contraction in the $\mathbb{Q}$ -factorial generalized dlt category.

At any stage of this second program, let H_i be the descended rational line, assuming inductively that it is nef and rH_i is Cartier, and write

$$G_i = K_{W_i} + T_i + (1-\varepsilon)M_i.$$

An extremal ray negative for $G_i + d_0H_i$ is negative for G_i . The pair with adjoint G_i is still generalized dlt: the added nef b-divisor d_0H_i descends to the current model, so removing it changes no discrepancy. The generalized length bound [40, Proposition 3.13] gives a curve C on that ray with $0 < -G_i \cdot C \leq 2k$. If $H_i \cdot C > 0$, Cartier integrality would give

$$\frac{d_0}{r} \leq d_0H_i \cdot C < -G_i \cdot C \leq 2k,$$

a contradiction. Thus $H_i \cdot C = 0$. The exact contraction statement [66, Theorem 1.5] descends the Cartier line rH_i along this contraction as an actual Cartier line. In a flip we pull that same line to the flipped side. The descended line is nef: every curve on the contraction base has a curve above it mapping with positive degree, so its degree is nonnegative; pullback preserves nefness. Consequently r and nefness persist, and the argument applies inductively to all steps and to the final contraction. Each step is crepant for H_i .

Let W_m be the last birational model of this second program and let H_m be the descended line on it. Crepancy for the H_i , together with the absence of extraction, turns Equation (5.9) into Equation (5.8) on a common resolution, with an effective divisor exceptional over W_m . The final contraction $h : W_m \rightarrow V$ has connected fibers and, by the same exact descent,

$$H_m \sim_{\mathbb{Q}} h^* H_V$$

for a rational line H_V on the normal projective V . If V is a point, the second alternative holds with $H_m \sim_{\mathbb{Q}} 0$.

Assume $\dim V > 0$. To obtain the first alternative, it remains to show that the log adjoint on the new very general fibers has Kodaira dimension zero. Resolve the base program by $\mu : W'' \rightarrow W$ and $\nu : W'' \rightarrow W_m$, and prepare the source over W'' , calling it $g'' : (X'', B'') \rightarrow W''$. The diagram we use is

$$\begin{array}{ccccccc} X'' & \xrightarrow{g''} & W'' & \xrightarrow{\nu} & W_m & \xrightarrow{h} & V \\ & & \mu \downarrow & & & & \\ & & W & & & & \end{array}$$

with composite $f = h\nu g''$. Equation (5.4) and Equation (5.8) give

$$H'' \sim_{\mathbb{Q}} \nu^* H_m + E'', \quad E'' \geq 0 \text{ } \nu\text{-exceptional.}$$

To see the exceptionality, the error in Equation (5.8) is already exceptional over W_m , and every μ -exceptional prime is exceptional over W_m as well: otherwise W_m would extract a divisor over W . For every degree divisible by one fixed integer, Equation (5.3), exceptional descent, and projection formula now yield

$$\begin{aligned} f_* \mathcal{O}_{X''}(q(K_{X''} + B'')) &\hookrightarrow (h\nu)_* \mathcal{O}_{W''}(qH'') \\ &= h_* \mathcal{O}_{W_m}(qH_m) = \mathcal{O}_V(qH_V). \end{aligned}$$

Here $\nu_* \mathcal{O}_{W''}(qE'') = \mathcal{O}_{W_m}$ by normality and $h_* \mathcal{O}_{W_m} = \mathcal{O}_V$. Generic coherent base change, simultaneously for the countably many divisible degrees, bounds the dimension of every such system on a very general f -fiber by one. The log adjoint on that fiber is pseudo-effective by restriction of a countable sequence of perturbed currents with analytic singularities. Its dimension is less than n , so the lower-dimensional $\mathcal{G}$ supplies a nonzero section in some divisible degree. Taking a common multiple shows that the fiber log Kodaira dimension is zero. All maps in the composite have connected fibers; resolving V and the source keeps this property and gives a smooth projective base of dimension $0 < \dim V < k$. This is the first alternative. $\square$

5.5 The negative divisor upstairs

The next lemma lifts either base decomposition and determines the whole negative divisor on the source, including primes over subsets of codimension at least two in the base that the local comparison did not test.

Lemma 5.8 (Lifting the decomposition). *Assume the setting of Lemma 5.3. Suppose one of the following additional conditions holds:*

- (i) $a(W) = 0$ and $H \sim_{\mathbb{Q}} E_W = N(H)$, with $E_W \geq 0$;

(ii) there is a birational morphism $\nu : W \rightarrow W_m$ to a normal $\mathbb{Q}$ -factorial projective variety and an identity

$$H \sim_{\mathbb{Q}} \nu^* H_m + E_W, \quad E_W \geq 0 \text{ } \nu\text{-exceptional},$$

where H_m is a nef rational line, either big or rationally trivial.

Then there is an effective rational divisor A on X and an actual rational line identity

$$J \sim_{\mathbb{Q}} P_0 + A, \quad A = N(J) \geq 0, \quad (5.13)$$

where $P_0 \sim_{\mathbb{Q}} 0$ in the first case and in the rationally trivial part of the second case, and $P_0 = (\nu g)^* H_m$ in the big part of the second case.

Proof. Use the chosen base identity in Equation (5.1). In a rationally trivial case fix a rational trivialization of the positive line and put $P_0 = 0$; in the big case put $P_0 = (\nu g)^* H_m$. Define the rational divisor

$$A := A_* + g^* E_W.$$

The composed actual line identities give $J \sim_{\mathbb{Q}} P_0 + A$. The divisor A may at first have signed coefficients. We first prove $A \geq 0$.

In a rationally trivial case a divisible multiple of the base line has a section whose normalized zero divisor is exactly E_W . Its section under the exact global comparison has normalized zero divisor A . Regularity of this section proves $A \geq 0$, also on the initially untested primes. In the big case choose a Kodaira decomposition

$$H_m \sim_{\mathbb{Q}} A_0 + G_0, \quad A_0 \text{ ample}, \quad G_0 \geq 0.$$

For rational $0 < t < 1$,

$$H_m \sim_{\mathbb{Q}} \underbrace{(1-t)H_m + tA_0}_{\text{ample}} + tG_0.$$

Fix a prime $\Gamma \subset X$. A sufficiently divisible section of the ample summand can be chosen not to vanish at the generic point of $(\nu g)(\Gamma)$; its pullback has zero order at Γ . Multiplication by the section of tG_0 and then the global comparison gives a regular upstairs section. Its normalized order is

$$A_{\Gamma} + t \operatorname{ord}_{\Gamma}((\nu g)^* G_0) \geq 0.$$

Letting $t \downarrow 0$ proves $A_{\Gamma} \geq 0$. This applies to every prime in A .

Nefness of P_0 now implies $N(J) \leq A$. Set $Q = A - N(J) \geq 0$, an effective real divisor. By Equation (5.6), Q is vertical. Moreover

$$\{P_0 + Q\} = \{J\} - \{N(J)\}$$

is modified nef. We prove $Q = 0$.

Let $b : X \rightarrow Y$ denote $g : X \rightarrow W$ in the first case and $\nu g : X \rightarrow W_m$ in the second. Thus $\dim Y = k$, and b has connected fibers. If $Q \neq 0$, let

$$l = \max\{\dim b(\Gamma) : \Gamma \text{ a component of } Q\} \leq k - 1.$$

Choose a Kähler class η on Y (an ample class when Y is projective) and a Kähler form ω on X . On $H^{1,1}(X, \mathbb{R})$ define

$$\langle \alpha, \beta \rangle_l = \int_X \alpha \beta (b^* \eta)^l \omega^{n-l-2}.$$

In this pairing a divisor or rational line denotes its cohomology class. The exponent is nonnegative since $l \leq k-1 \leq n-2$. The restriction of a modified-nef class to a resolution of every prime is pseudo-effective. Applying this to $\{P_0 + Q\}$ and pairing on each component of Q gives $\langle P_0 + Q, \Gamma \rangle_l \geq 0$. The P_0 -term vanishes by dimension, since it is a pullback from Y and $\dim b(\Gamma) \leq l$. Summing with the coefficients of Q gives

$$\langle Q, Q \rangle_l \geq 0. \quad (5.14)$$

We now separate images of codimension at least two, where the mixed Hodge index theorem gives strict negativity, from divisorial images, where only a multiple of the whole fiber can have square zero.

If $l < k-1$, the mixed Hodge index theorem contradicts this inequality. In this form $b^*\eta$ has positive square because $l+2 \leq k$, while $\langle Q, b^*\eta \rangle_l = 0$ by image dimension. The form is a limit of the mixed Hodge index forms obtained by replacing $b^*\eta$ with $b^*\eta + \epsilon[\omega]$; it therefore has at most one positive direction. In the orthogonal complement of a positive-square vector it is negative semidefinite, with equality only in the radical of the whole form. But $\langle Q, \omega \rangle_l > 0$, since at least one component of Q has image dimension l . Thus Q is not in the radical, and its square is strictly negative, contrary to Equation (5.14).

It remains to treat $l = k-1$. Terms from primes whose image has smaller dimension vanish in this form, as do cross terms from different divisorial images. Fix a prime divisor $D \subset Y$ and take all the primes $E_1, \dots, E_s \subset X$ dominating it. Write $a_i = \text{ord}_{E_i}(b^*D) > 0$ and $C_{ij} = \langle E_i, E_j \rangle_{k-1}$. The divisor D is rational Cartier, also when $Y = W_m$. Consequently

$$C_{ij} \geq 0 \ (i \neq j), \quad \sum_j C_{ij} a_j = 0.$$

The first assertion is positivity of the proper intersection of distinct effective primes. The second pairs E_i with the pulled-back divisor D ; the additional base factor makes the intersection vanish by dimension. Components of b^*D mapping into a proper subset of D make no contribution. The graph with edges $C_{ij} > 0$ is connected. Indeed over a very general point of D the fiber is connected, all its points lie in the components E_i , and an intersection contributes a positive entry precisely when it dominates D .

For any real vector $x = (x_i)$, the two displayed properties give the exact identity

$$x^\top C x = - \sum_{i < j} C_{ij} a_i a_j \left(\frac{x_i}{a_i} - \frac{x_j}{a_j} \right)^2.$$

Thus the block is negative semidefinite with kernel spanned by (a_i) . Equation (5.14) forces every nonzero block of Q to equal a positive scalar multiple of (a_i) . In particular Q then contains every prime over that divisorial image D .

In the second case this is impossible. The morphism $\nu : W \rightarrow W_m$ is an isomorphism over the generic point of D , and E_W is exceptional. Equation (5.2) therefore supplies a component over D whose A -coefficient is zero. Its Q -coefficient is zero as well, contradicting the preceding conclusion.

In the first case, the same zero minimum shows that each divisorial image of a nonzero block of Q is a component of $E_W = N(H)$. Here $\{Q\}$ itself is modified nef. The class

$$b_*({Q}[\omega]^{n-k})$$

is also modified nef on the smooth W . To see this directly, choose small-perturbation positive currents for $\{Q\}$ with zero generic divisorial Lelong numbers, wedge them with ω^{n-k} , and push

forward. Such a current has no trace mass on a divisor, so its push has no mass on a base divisor: the inverse image is a union of divisors and subsets of higher codimension, all of zero trace mass. The pushed currents therefore have zero generic divisorial Lelong numbers. Their classes tend to the displayed class; a fixed smooth representative for the vanishing error converts this into the defining small-Kähler-perturbation condition for modified nefness. On the other hand, pushing the actual divisor Q in this formula kills the components with smaller image and gives a nonzero positive linear combination of the components of $N(H)$. Boucksom's exceptionality of the components of a divisorial negative part says that no such combination is modified nef [7, Definition 3.10 and Theorem 3.12]. This final contradiction proves $Q = 0$, and hence Equation (5.13). $\square$

5.6 Generation from a big nef line

The remaining nef part is pulled back from a big nef line on a projective variety. We use generation along the log canonical boundary to prove that the adjoint itself is semiample.

Proposition 5.9 (Generation from a big nef base line). *Assume $\mathcal{G}_d$ for $d < n$. Let (Z, Δ_Z) be a globally $\mathbb{Q}$ -factorial dlt compact Kähler pair of dimension n satisfying the resolution condition of Definition 3.1, with effective rational boundary and analytically nef adjoint $L = K_Z + \Delta_Z$. Suppose there are a smooth compact Kähler manifold U , a projective resolution $r : U \rightarrow Z$, a proper surjection $f : U \rightarrow Y$ to a normal projective variety, and a big nef rational line H_Y on Y such that*

$$r^*L \sim_{\mathbb{Q}} f^*H_Y \quad (5.15)$$

as actual rational lines. Then L is semiample.

Proof. Put $D_Z = \lfloor \Delta_Z \rfloor$. We first prove the following assertion for every pair and diagram satisfying the hypotheses of this proposition:

$$\mathbf{B}(L) \cap D_Z = \emptyset. \quad (5.16)$$

Here $\mathbf{B}(L)$ is the intersection of the base loci of all positive Cartier multiples of L . Once this is proved, a log canonical threshold at any remaining stable base locus will create a new floor there and give a contradiction.

Take a higher log resolution if needed and put

$$K_U + F = r^*L =: \ell, \quad C = F^{-1}, \quad E = C - \lfloor F \rfloor, \quad \Delta_F = F - \lfloor F \rfloor.$$

The equality defining F uses the canonical meromorphic identification. The divisor F is an SNC subboundary. The integral divisor E is effective, r -exceptional, and has no component in common with C : a nonexceptional coefficient of F is a coefficient of the effective Δ_Z , while every coefficient of F is at most one. The image of C is contained in D_Z . By Theorem 4.1, a divisible multiple of $L|_{D_Z}$ is globally generated on the whole reduced floor.

Suppose first that a component C_i dominates Y . Pulling the floor sections to C_i shows that $(f|_{C_i})^*H_Y \sim_{\mathbb{Q}} \ell|_{C_i}$ is semiample. Semiampleness descends under a proper surjection to a normal space in this situation. Factor $C_i \rightarrow Y$ through its normal Stein factor Y_i . Projection formula descends generation through the connected-fiber map. For the finite map $Y_i \rightarrow Y$, at each $y \in Y$ choose a section of the generated pullback line avoiding every point of the finite fiber; finitely many evaluation kernels cannot cover the section space. Lemma 2.8 gives a norm section of a fixed power of the line on Y , nonzero at y . These sections generate a fixed multiple of H_Y . Equation (5.15) and normal birational descent then give semiampleness of L .

We may therefore assume that C is vertical over Y . Choose a sufficiently divisible integer $u > 1$ for which $uL|_{D_Z}$ is generated, and consider the integral line

$$\mathcal{N} := \mathcal{O}_U(u\ell + E - C) = \mathcal{O}_U(K_U + \Delta_F + (u-1)\ell). \quad (5.17)$$

Although the expression on the right uses rational divisors, its sum is the integral line on the left. The restriction sequence is

$$0 \longrightarrow \mathcal{N} \longrightarrow \mathcal{O}_U(u\ell + E) \longrightarrow \mathcal{O}_C(u\ell + E) \longrightarrow 0.$$

The direct image of the last sheaf is supported on the proper analytic subset $f(C)$, so it is torsion. Consequently, the two assertions

$$R^1 f_* \mathcal{N} \text{ torsion-free,} \quad H^1(Y, f_* \mathcal{N}) = 0 \quad (5.18)$$

will give surjectivity on global sections: torsion-freeness makes the connecting map to $R^1 f_* \mathcal{N}$ zero, and the H^1 -vanishing then removes the obstruction to lifting a global section.

We prove Equation (5.18) using analytic injectivity. Equation (5.15) identifies $\mathcal{N} \otimes K_U^{-1}$ with $\Delta_F + (u-1)f^*H_Y$ in $\text{Pic}(U) \otimes \mathbb{Q}$; this comparison need not be an isomorphism of the corresponding integral lines. Fix a very ample line A_Y on Y and its Fubini–Study metric. A Kodaira decomposition of the big nef H_Y , given arbitrarily small weight as in the proof of Lemma 5.8, writes $(u-1)H_Y$ as a positive rational multiple of A_Y plus an effective rational divisor whose fixed singular contribution is arbitrarily small and whose remaining contribution is a general divided ample member. Choose the small weight below the log canonical threshold of its pullback relative to the klt SNC boundary Δ_F , and choose the ample member generally. Then the resulting boundary on U is klt. Choose a positive integer q_0 clearing all its denominators and realizing both the chosen Kodaira decomposition and the preceding rational-line comparison as line isomorphisms, in particular

$$(\mathcal{N} \otimes K_U^{-1})^{\otimes q_0} \simeq \mathcal{O}_U(q_0 \Delta_F) \otimes f^* \mathcal{O}_Y(q_0(u-1)H_Y).$$

The divisor metrics and the Fubini–Study metric define a singular Hermitian metric on $(\mathcal{N} \otimes K_U^{-1})^{\otimes q_0}$. Its q_0 -th root is a metric on the actual line $\mathcal{N} \otimes K_U^{-1}$, with multiplier ideal $\mathcal{O}_U$ and curvature dominating a positive multiple of the pullback Fubini–Study form. The same statements hold after every nonnegative twist by f^*A_Y .

The injectivity theorem [34, Theorem A] applies on the compact Kähler U : multiplication by the pullback of any nonzero section of a positive power of A_Y is injective on

$$H^1(U, \mathcal{N} \otimes f^*A_Y^j) \quad (j \geq 0)$$

into the correspondingly higher twist. Its assumptions are exactly the semipositive smooth metric on the multiplying line, the preceding positive lower curvature bound, and the trivial multiplier ideal. This injectivity implies Equation (5.18). For the second assertion, the Leray edge map injects $H^1(Y, f_* \mathcal{N})$ into $H^1(U, \mathcal{N})$. Multiplication by a section of a sufficiently high A_Y -power sends this subspace to zero, since Serre vanishing kills $H^1(Y, f_* \mathcal{N} \otimes A_Y^j)$ for large j . Injectivity upstairs forces the original subspace to be zero. For the first assertion, suppose $R^1 f_* \mathcal{N}$ has a nonzero torsion subsheaf. A section of a sufficiently high power of A_Y annihilates a nonzero such subsheaf. After a further sufficiently large twist it has a nonzero global section, and Serre vanishing and Leray identify that section with a nonzero class in $H^1(U, \mathcal{N} \otimes f^*A_Y^j)$. Multiplication annihilates this class, again contradicting injectivity. This proves both assertions.

The resulting surjectivity extends the floor sections as follows. Pull any section of $uL|_{D_Z}$ to C , multiply by the canonical section of $E|_C$, and extend by this surjectivity. The extended section descends to Z , since $r_*\mathcal{O}_U(E) = \mathcal{O}_Z$ by exceptionality and normality. On the strict transforms of the components of D_Z , division by the canonical section of E recovers the prescribed section. The descended restriction therefore equals it on the whole reduced D_Z , since equality holds generically on every component. Since $uL|_{D_Z}$ is generated, this proves Equation (5.16).

We finish by showing that this stable base locus is empty. The base loci of factorial Cartier multiples form a descending sequence of compact analytic subsets and hence stabilize. Sections exist because H_Y is big and Equation (5.15) descends their pullbacks. Choose a divisible degree v whose base locus is $\mathbf{B}(L)$, and let $\mathfrak{b}$ be its base ideal. If $\mathbf{B}(L) \neq \emptyset$, the ideal is a unit near D_Z , while (Z, Δ_Z) is klt outside D_Z . Its log canonical threshold

$$t = \text{lct}_{(Z, \Delta_Z)}(\mathfrak{b})$$

is therefore a positive rational number, attained by a divisor whose center is contained in $\mathbf{B}(L)$. This follows directly on a simultaneous log resolution of the pair and the ideal: the threshold is the minimum of the positive rational discrepancies divided by the positive integral ideal orders.

Choose an integer $s > t$ and general divisors $D_1, \dots, D_s \in |vL|$. On the same resolution their fixed part is the divisor of $\mathfrak{b}$; their free transforms are jointly transverse to the SNC data. Hence

$$\left(Z, \Delta_Z + \frac{t}{s} \sum_{i=1}^s D_i \right)$$

is log canonical: the fixed part is at its log canonical threshold and the free components have coefficients $t/s < 1$. There is a new lc place centered in $\mathbf{B}(L)$, and the new adjoint satisfies the actual identity

$$L_{\text{new}} \sim_{\mathbb{Q}} (1 + tv)L.$$

Apply [38, Theorem 3.3] to this compact Kähler lc rational pair. It gives a projective strongly $\mathbb{Q}$ -factorial dlt modification $d : Z' \rightarrow Z$. Since the pair is lc, all extracted divisors have log discrepancy zero in our convention, so the modification is crepant. The source is again compact Kähler. Choose its defining dlt log resolution, with exceptional crepant coefficients strictly below one and an isomorphism at the general point of every lc stratum; resolving the remaining data away from those points gives the resolution required by Definition 3.1. The floor on Z' has a point over the center of this lc place, hence over $\mathbf{B}(L)$. The new nef adjoint is d^*L_{new} . On a common resolution with U , it satisfies Equation (5.15) with the big nef line $(1 + tv)H_Y$. The previously proved assertion (5.16) therefore makes its stable base locus disjoint from its floor. On the other hand, normality and projection formula identify all divisible sections under d , and give

$$\mathbf{B}(d^*L_{\text{new}}) = d^{-1}\mathbf{B}(L).$$

This locus contains the stated floor point, a contradiction. Thus $\mathbf{B}(L) = \emptyset$; factorial stabilization provides an actual globally generated multiple of L . $\square$

5.7 Completion of the rank-one case

Proof of Proposition 5.1. By Proposition 2.7 we can begin on the resolved source in the statement. We use a secondary induction on the positive base dimension k . Apply Lemma 5.2 and then Lemma 5.3 to obtain the prepared data and the pseudo-effective line H .

If $a(W) = 0$, Lemma 5.4 and the first case of Lemma 5.8 give $J \sim_{\mathbb{Q}} N(J)$, which is already the required decomposition with trivial positive part. If W is projective, apply Lemma 5.7. Its first alternative has a strictly smaller positive base dimension, so the secondary induction and birational transfer finish the proof. In its second alternative, take a common smooth base resolution in Equation (5.8) and prepare the source over it. Equation (5.4) adds an effective divisor exceptional over the old base. It is also exceptional over W_m , because the base program extracted no divisors. Thus the second case of Lemma 5.8 applies.

We have obtained

$$J \sim_{\mathbb{Q}} P_0 + A, \quad A = N(J) \geq 0,$$

with P_0 nef. A rationally trivial P_0 finishes the proof. In the remaining case $P_0 = b^*H_m$, where $b : X \rightarrow W_m$ is proper and H_m is big and nef on the projective W_m . Apply Proposition 3.8 to this nef-plus-negative presentation. It gives a globally strongly $\mathbb{Q}$ -factorial dlt compact Kähler model (Z, Δ_Z) with nef adjoint L_Z . The ordinary dlt endpoint has the resolution property of Definition 3.1. On a common smooth resolution $\mu : U \rightarrow X$, $r : U \rightarrow Z$ its exact positive-line comparison is

$$r^*L_Z \sim_{\mathbb{Q}} \mu^*P_0 = (b\mu)^*H_m.$$

Proposition 5.9 makes L_Z semiample. Hence μ^*P_0 is semiample. Lemma 2.4 identifies $\mu^*A = N(\mu^*J)$, so this is precisely $\mathcal{G}_n$ on U . Finally Proposition 2.7 transfers the decomposition back through all the prepared source modifications. $\square$

6 Meromorphic nonvanishing on simple spaces

The simple case of the induction needs a divisor representing a multiple of the canonical bundle; its coefficients need not be nonnegative. The following result supplies precisely this starting point.

Theorem 6.1. *Let X be a smooth connected compact Kähler manifold. Suppose that $a(X) = 0$ and that no positive-dimensional proper compact analytic subvariety passes through a very general point of X . Then $K_X^{\otimes m}$ has a nonzero meromorphic section for some integer $m > 0$.*

The proof compares two ways of measuring poles. After normalizing a big class on X to have volume one, its pole order at a very general point is bounded. We construct a projective bundle over X^2 whose volume grows linearly with a parameter q . Restriction to the exceptional divisor of the diagonal in the square of this bundle then produces a high-rank subsheaf of a symmetric cotangent power. A second diagonal forces substantial vanishing of its determinant; descending that determinant produces a point pole of order comparable to $q^{1/(2 \dim X + 1)}$, contradicting the point bound. The argument uses only meromorphic sections of line bundles; it does not require nonconstant meromorphic functions on X .

For the proof, suppose that the conclusion fails. Write

$$n = \dim X, \quad \mathcal{L} = K_X, \quad L = c_1(\mathcal{L}) \in H_{\text{BC}}^{1,1}(X, \mathbb{R}).$$

We may assume $n \geq 2$: a compact complex curve is projective, and the assertion for a point is immediate. Throughout this Section, an inequality between real $(1, 1)$ -classes is in pseudo-effective order: $\alpha \leq \beta$ means that $\beta - \alpha$ is pseudo-effective. For a divisor D , $\{D\}$ denotes $c_1(\mathcal{O}(D))$. Pullbacks of classes will occasionally be suppressed when the map is evident.

6.1 Line subsheaves and point poles

We first record the consequence of simplicity that controls all the determinant lines used below.

Lemma 6.2. *Under the contrary hypothesis above, $H^1(X, \mathcal{O}_X) = 0$, the group $\text{Pic}(X)$ is countable, and $L \geq 0$. If $\mathcal{H}$ is a holomorphic line bundle and $\mathcal{H} \rightarrow \Omega_X^{\otimes k}$ is nonzero, where $k \geq 0$, then*

$$c_1(\mathcal{H}) \leq kL. \quad (6.1)$$

There is a complement of a countable union of proper analytic subsets of X with the following further property. For every point x in this complement, let

$$a : Y = \text{Bl}_x X \rightarrow X, \quad F = a^{-1}(x).$$

Every nonzero line map $\mathcal{H}_Y \rightarrow (a^\Omega_X)^{\otimes k}$ satisfies*

$$c_1(\mathcal{H}_Y) \leq ka^*L. \quad (6.2)$$

Proof. If the Albanese map of X is nonconstant, simplicity makes its image n -dimensional: a positive-dimensional general fiber of smaller dimension would be a forbidden subvariety. At a point where the Albanese map has rank n , some n invariant one-forms have nonzero wedge after pullback. This gives a nonzero holomorphic section of K_X , contrary to our assumption. Thus $H^1(X, \mathcal{O}_X) = 0$. The exponential sequence embeds $\text{Pic}(X)$ into the countable group $H^2(X, \mathbb{Z})$.

The manifold X is not uniruled, by simplicity. Ou's non-pseudo-effectivity criterion for the canonical class therefore gives $L \geq 0$ [58, Theorem 1.1]. We recall how the slope and foliation results in the same paper give the more precise inequality in Equation (6.1). Let γ be any class in the full dual of the pseudo-effective cone, and use the slope

$$\mu_\gamma(\mathcal{E}) = \frac{c_1(\mathcal{E}) \cdot \gamma}{\text{rk } \mathcal{E}}$$

for torsion-free sheaves. If $\mu_{\gamma, \min}(\Omega_X) < 0$, the first Harder–Narasimhan piece $\mathcal{T} \subset T_X$ has strictly positive slope. It is saturated and semistable. The tensor slope inequality shows that the bracket $\wedge^2 \mathcal{T} \rightarrow T_X/\mathcal{T}$ is zero: the minimum slope of its source is at least $2\mu_\gamma(\mathcal{T})$, whereas the maximum slope of its target is smaller than $\mu_\gamma(\mathcal{T})$. Consequently $\mathcal{T}$ is a foliation. Its dual has strictly negative maximum slope and is not pseudo-effective by [58, Proposition 4.5]. Ou's foliation theorem [58, Theorem 1.4] makes this foliation the tangent foliation of a meromorphic fibration with compact general leaf closures. Its rank is strictly between zero and n : rank n would give $\mu_\gamma(T_X) = -L \cdot \gamma/n \leq 0$. The general leaf closure is therefore a positive-dimensional proper subvariety through a general point, again contradicting simplicity.

We have proved $\mu_{\gamma, \min}(\Omega_X) \geq 0$. In its Harder–Narasimhan filtration all quotient slopes are now nonnegative, so $\mu_{\gamma, \max}(\Omega_X) \leq L \cdot \gamma$. The tensor slope inequality [58, Lemma 4.4] gives

$$c_1(\mathcal{H}) \cdot \gamma \leq \mu_{\gamma, \max}(\Omega_X^{\otimes k}) \leq kL \cdot \gamma.$$

Separation by the dual cone proves Equation (6.1). For $k = 0$ this is also immediate from the effective zero divisor of a nonzero map $\mathcal{H} \rightarrow \mathcal{O}_X$.

It remains to choose x uniformly. For any vector bundle $\mathcal{V}$, independent global sections are generically pointwise independent. Indeed, choose a maximal pointwise independent subfamily on a dense open set. Expressing any other section in that family by minors gives meromorphic function coefficients. They are constant because $a(X) = 0$, so maximality among a linearly independent

family forces all its members to be pointwise independent. Thus evaluation on $H^0(X, \mathcal{V})$ is injective away from a proper analytic subset (unless that vector space is zero, in which case there is no restriction). Apply this to $\mathcal{V} = \mathcal{H}^{-1} \otimes \Omega_X^{\otimes k}$ for every $\mathcal{H} \in \text{Pic}(X)$ and $k \geq 0$. There are countably many such bundles.

For x outside the resulting exceptional set, write any line bundle on $Y = \text{Bl}_x X$ uniquely as $a^*\mathcal{H} \otimes \mathcal{O}_Y(mF)$, $m \in \mathbb{Z}$. A line map from this bundle to $(a^*\Omega_X)^{\otimes k}$, restricted away from F , extends over x to a section of $\mathcal{H}^{-1} \otimes \Omega_X^{\otimes k}$ by Hartogs' theorem. If $m > 0$, this section vanishes at x , since $a_*\mathcal{O}_Y(-mF) = \mathcal{I}_x^m$. The choice of x rules this out. Thus $m \leq 0$, and Equation (6.1) yields

$$c_1(a^*\mathcal{H} \otimes \mathcal{O}_Y(mF)) \leq ka^*L + m\{F\} \leq ka^*L.$$

If a target has a finite filtration whose graded pieces are direct sums of the indicated cotangent tensors, take the first nonzero associated graded component of the line map and then a nonzero direct-sum component. The preceding argument applies with the tensor order of that component. $\square$

We will use several standard facts about volumes of real $(1,1)$ -classes. Our normalization is $\text{vol}(\alpha) = \int \alpha^e$ for a nef class on an e -fold. Volume is continuous, homogeneous, monotone in pseudo-effective order, invariant under modification, and its e -th root is concave on the big cone. Analytic Fujita approximation computes it by Kähler parts on smooth projective modifications. Here a *smooth projective modification* means a projective modification whose source is smooth; the sources used here are compact Kähler manifolds obtained by resolving coherent analytic ideals. For a smooth irreducible divisor D and a big class α , write $\text{vol}_{|D}(\alpha)$ for the numerical restricted volume. It is zero when D is contained in the non-Kähler locus [65, Theorem 1.1]; otherwise it is the supremum of the masses on D of restrictions of Kähler currents with analytic singularities that are not generically singular on D [16, Lemma 2.7]. The divisorial derivative and its continuity on the big cone are

$$\frac{d}{du} \text{vol}(\alpha - u\{D\}) = -e \text{vol}_{|D}(\alpha - u\{D\}); \quad (6.3)$$

see [65, Theorem 1.1]. We apply this formula only on smooth manifolds while the varying class is big.

Here is a useful precise form of the approximation in the restricted volume formula. Resolve the log-ideal singularities of a current restricted to D . Its pullback is an effective real divisor plus a positive residual current with locally bounded potentials. The latter dominates a positive multiple of the pulled-back Kähler form. A current with locally bounded potentials has zero Lelong numbers, so Demailly regularization makes its class, after subtracting that multiple, nef [21, Theorem 1.1]. On a projective modification there is an effective exceptional divisor whose negative is relatively ample. Subtracting a sufficiently small multiple of this divisor from the residual class therefore makes that class Kähler; add the same multiple to the divisor part. Round all divisor coefficients slightly upwards to rational numbers. Openness of the Kähler cone preserves the Kähler property. These changes can be arbitrarily small in top intersections, which compute the original mass by the bounded-potential product formula. Thus we may approximate a restricted mass by

$$h^*(\alpha|_D) = \beta + \{D'\}, \quad \beta \text{ Kähler}, \quad D' \geq 0 \text{ a rational divisor.} \quad (6.4)$$

Moreover, D' has at least the log-ideal orders of the original restriction on every further resolution. We will use this last property to turn poles into vanishing conditions. Only D' is made rational; the class β and the horizontal part of α remain real.

Lemma 6.3 (A point pole bound). *Let W be a smooth compact Kähler manifold of dimension $e \geq 2$, α a big real $(1, 1)$ -class, and $z \in W$. Suppose a smooth projective modification $\mu : W' \rightarrow W$, which is an isomorphism near z , admits a decomposition*

$$\mu^* \alpha = K + \{D\}, \quad K \text{ Kähler}, \quad D \geq 0,$$

where D misses the point z' over z . If the ordinary analytic Seshadri constant $\epsilon(K, z')$ is at least $\eta > 0$, then, on the blowup $b : \widehat{W} = \text{Bl}_z W \rightarrow W$ with exceptional divisor G ,

$$\sup\{u \geq 0 : b^* \alpha - u\{G\} \geq 0\} \leq \frac{\eta}{2} + 2^{e-1} \text{vol}(\alpha) \eta^{-(e-1)}. \quad (6.5)$$

Proof. Set $u_0 = \eta/2$. Blowing up z' , the class $K - u_0\{G'\}$ is Kähler. The Fujita decomposition is unchanged near G' , so it supplies a Kähler current for $\alpha_{u_0} = b^* \alpha - u_0\{G\}$ that is smooth near G . Its restriction there has class $u_0 c_1(\mathcal{O}_{\mathbb{P}^{e-1}}(1))$. The restricted volume is consequently u_0^{e-1} : the current gives this lower bound, and the volume of the restricted class gives the opposite bound.

Let τ denote the left side of Equation (6.5). The classes $\alpha_u = b^* \alpha - u\{G\}$ are big for $0 \leq u < \tau$, since α_0 is big and the pseudo-effective cone is convex. The concave function $f(u) = \text{vol}(\alpha_u)^{1/e}$ satisfies, by Equation (6.3),

$$f'(u_0) = -\frac{u_0^{e-1}}{f(u_0)^{e-1}}.$$

Its tangent line at u_0 must remain positive up to τ . Therefore

$$\tau \leq u_0 + \frac{f(u_0)^e}{u_0^{e-1}} \leq u_0 + \frac{\text{vol}(\alpha)}{u_0^{e-1}},$$

which is Equation (6.5). $\square$

Fix a Kähler class ω on X . The class L is not big: a big holomorphic line bundle would make X Moishezon, contrary to $a(X) = 0$. For $t > 0$ define

$$P = r(L + t\omega), \quad r = \text{vol}(L + t\omega)^{-1/n}. \quad (6.6)$$

Then P is a big real class, $\text{vol}(P) = 1$, $L \leq P/r$, and $r \rightarrow \infty$ as $t \downarrow 0$. Choose a smooth Fujita model $\mu : Y_P \rightarrow X$ with Kähler part P' satisfying

$$\mu^* P = P' + \{D_P\}, \quad v := \int_{Y_P} (P')^n > \frac{1}{2}. \quad (6.7)$$

At a very general point y of this model there is no positive-dimensional proper subvariety: its image would be one through a very general point of X , and y avoids the exceptional locus. The ordinary Seshadri formula for a Kähler class,

$$\epsilon(K, y) = \inf_{\substack{W \ni y \\ \dim W > 0}} \left(\frac{\int_W K^{\dim W}}{\text{mult}_y W} \right)^{1/\dim W},$$

therefore gives $\epsilon(P', y) = v^{1/n} \geq 2^{-1/n}$. This is the ordinary nef/Kähler formula [63, Theorem 2.8]; see also [16, Equation (1.5)] and the Kähler cone criterion of [22]. We use this ordinary formula in both applications below. Lemma 6.3, with $\text{vol}(P) = 1$, now yields a constant C_n depending only on n such that, for a very general x ,

$$\tau(P, x) := \sup\{u \geq 0 : a^* P - u\{F\} \geq 0\} \leq C_n, \quad a : \text{Bl}_x X \rightarrow X. \quad (6.8)$$

Here and below positive constants denoted c_n, C_n may be decreased or increased from one occurrence to the next. They are independent of all parameters and choices of currents and modifications.

6.2 A projective bundle with controlled volume

We now produce a class whose volume grows, while its point pole threshold grows much more slowly. On X^2 , let $\mathcal{L}_i$ and L_i denote the pullbacks of $\mathcal{L}$ and L from the i -th factor. Use the quotient convention and put

$$\pi : Z = \mathbb{P}_{X^2}(\mathcal{L}_1 \oplus \mathcal{L}_2) \longrightarrow X^2, \quad \xi = c_1(\mathcal{O}_Z(1)), \quad d = \dim Z = 2n + 1.$$

Thus $\pi_* \mathcal{O}_Z(m) = \text{Sym}^m(\mathcal{L}_1 \oplus \mathcal{L}_2)$ for $m \geq 0$. The zero divisor A_i of $\pi^* \mathcal{L}_i \rightarrow \mathcal{O}_Z(1)$ has class $\xi - L_i$; it is the section on which ξ restricts to L_{3-i} . In particular $\xi = \{A_i\} + L_i \geq 0$.

Lemma 6.4. *For X and Z above, assume that $K_X^{\otimes m}$ has no nonzero meromorphic section for every integer $m > 0$. Through a very general point of X^2 , the only positive-dimensional proper compact irreducible analytic subvarieties are the two factor slices. Through a very general point of Z , the only positive-dimensional compact irreducible analytic subvarieties are a fiber of π , the full inverse images of the two factor slices, and Z itself.*

Proof. Let $C \subset X^2$ be irreducible through a pair whose coordinates are very general in X . Each projection image is a point or all of X , by properness and simplicity. If exactly one is a point, C is the corresponding full slice. If both projections are surjective, test their fibers at a general point of C whose coordinates are still very general. A positive-dimensional fiber must be the full other factor. Thus either $C = X^2$, or $\dim C = n$ and both projections are generically finite.

We show that subvarieties of the latter kind cannot sweep X^2 . The space of compact n -cycles on a compact Kähler manifold has countably many irreducible components, each compact [49, Theorem 1.1]. On a component containing integral cycles generically finite over both factors, integrality holds on a dense open set and the two projection degrees remain positive, as can also be tested by intersection with pulled-back Kähler forms. Suppose the incidence over one such component dominates X^2 . Resolve the parameter space and the dominating incidence component, obtaining maps

$$h : \mathcal{Y} \longrightarrow \mathcal{S}, \quad F_i : \mathcal{Y} \longrightarrow X$$

between smooth spaces. For general $t \in \mathcal{S}$, the fiber $Y_t = h^{-1}(t)$ is smooth, compact, and bimeromorphic to the integral cycle, hence irreducible. Its maps $f_i = F_i|_{Y_t} : Y_t \rightarrow X$ are generically finite. Choose t also so that Y_t meets the dense open set where $d(F_1, h)$ is invertible and $d(F_1, F_2)$ has rank $2n$. The first assertion follows from generic finiteness of f_1 , and the second from the assumed dominance in X^2 .

Our immediate aim is to turn this dominance into a meromorphic frame of $f_2^* T_X$. Its determinant would meromorphically trivialize $f_2^* K_X^{-1}$, and hence also $f_2^* K_X$. On Y_t , the equal-rank bundle map

$$d(F_1, h)|_{Y_t} : T_{\mathcal{Y}|Y_t} \longrightarrow f_1^* T_X \oplus (\mathcal{O}_{Y_t} \otimes T_t \mathcal{S})$$

has a meromorphic inverse, given by its adjugate and determinant. For fixed $v \in T_t \mathcal{S}$, apply this inverse to $(0, v)$ and then apply dF_2 . The result is a meromorphic section of $f_2^* T_X$. At a general point where $d(F_1, F_2)$ has rank $2n$, the resulting map $T_t \mathcal{S} \rightarrow f_2^* T_X$ is surjective: in local coordinates (F_1, h) , it is the derivative of F_2 in the parameter directions with F_1 fixed. Choose n fixed vectors v_i whose values there are independent. Their wedge is a nonzero meromorphic section of $f_2^* K_X^{-1}$ on the irreducible Y_t ; its inverse trivializes $f_2^* K_X$ meromorphically.

Factor the proper generically finite map f_2 as

$$Y_t \longrightarrow \bar{Y}_t \xrightarrow{\nu} X$$

by Stein factorization. Here $\bar{Y}_t$ is normal and irreducible, the first map is a modification, and ν is finite of degree $N > 0$. Meromorphic sections descend across a modification of a normal space, so the section descends to a nonzero meromorphic section of ν^*K_X . Lemma 2.8 gives a nonzero meromorphic section of $K_X^{\otimes N}$. This norm is defined in local finite analytic fraction algebras, so it remains available even though the global meromorphic function field of X is just $\mathbb{C}$. It contradicts our assumption.

Consequently the incidence image of each such cycle component is a proper compact analytic subset of X^2 . There are only countably many components, so their union misses a very general pair. This proves the assertion for X^2 .

Now let $W \subset Z$ be irreducible through a very general point. Its image is a point, a full factor slice, or X^2 . If its generic relative dimension is one, it is the full inverse image of that image. Otherwise it is a multisection of the restricted $\mathbb{P}^1$ -bundle. Over a slice or over X^2 , a multisection is a divisor with line bundle $\mathcal{O}(k) \otimes \pi^*\mathcal{H}$, $k > 0$. Its homogeneous equation has coefficients in

$$H^0(S, \mathcal{H} \otimes \mathcal{L}_1^{\otimes i} \otimes \mathcal{L}_2^{\otimes (k-i)}), \quad 0 \leq i \leq k,$$

where S is the relevant slice or X^2 , and a fixed-factor line is understood as constant. A single nonzero monomial cuts out only the axis sections and vertical divisors. A multisection not on an axis therefore has two nonzero coefficients. Their quotient is a meromorphic section of a nonzero power of $\mathcal{L}_1 \otimes \mathcal{L}_2^{-1}$. Over a slice this is, up to inversion and a constant line, a positive power of K_X . Over X^2 , restriction to a general factor slice gives the same conclusion. This is impossible. The axes themselves miss a very general point of Z , proving the claim. $\square$

Let q tend to infinity through positive integers. For each sufficiently large q , choose $t = t(q)$ in Equation (6.6) so that $r = r(q) \geq q^2$, and use the resulting normalized class $P = P(q)$. Define the real class and the scale

$$M = P_1 + P_2 + q\xi, \quad A_q = q^{1/d}. \quad (6.9)$$

Lemma 6.5. *For these choices, M is big and*

$$c_n q \leq \text{vol}(M) \leq C_n q, \quad \tau(M, z) := \sup\{u \geq 0 : b_z^* M - u\{F_z\} \geq 0\} \leq C_n A_q \quad (6.10)$$

at a very general point $z \in Z$, where $b_z : \text{Bl}_z Z \rightarrow Z$ has exceptional divisor F_z .

Proof. Pull Z to the Fujita model Y_P^2 from Equation (6.7), and put $H = P'_1 + P'_2$ on this pulled-back bundle. There is a further smooth projective modification $\nu : \widehat{Z} \rightarrow Y_P^2$ ($\mu^*\mathcal{L}_1 \oplus \mu^*\mathcal{L}_2$) and a Fujita decomposition of the pullback of M whose Kähler part is exactly

$$K = \frac{1}{2}\nu^*H + \Theta, \quad (6.11)$$

where Θ is Kähler and has degree q on a general vertical line. The modification and the divisor part miss that line.

Here is the construction. It suffices to decompose $H/2 + q\xi$ as Θ plus an effective divisor, clean on a whole general vertical line. Since $\mathcal{O}(1)$ is relatively ample, a small positive class of the form $\delta(\xi + cH)$ is Kähler for some $c > 0$. Choose $\delta > 0$ so small that $\delta < q$ and the remaining horizontal part of $H/2$ is Kähler. Represent the remaining ξ first as $\{A_1\} + L_1$ and then as $\{A_2\} + L_2$. Regularize the pseudo-effective classes L_i with analytic singularities, paying their arbitrarily small negative errors from that horizontal Kähler part. This gives two Kähler currents in $H/2 + q\xi$, each smooth off its own axis and a proper horizontal analytic set. The maximum of their potentials

is locally bounded along an entire general vertical line, since the two axes are disjoint. Analytic regularization preserving a smaller Kähler lower bound [7, Theorem 2.1(ii)] gives a Kähler current with analytic singularities missing that line. Resolve its singularities and make the small Kähler adjustment described before Lemma 6.3. The divisor still misses a general vertical line, so the residual class Θ has degree exactly q there. Adding the other half of H and the pullbacks of the divisor parts D_P gives Equation (6.11). Its two summands are nef, with Θ Kähler, so every mixed intersection used in the following bounds is nonnegative.

At a very general point of $\widehat{Z}$, Lemma 6.4 lists the possible positive-dimensional subvarieties: the vertical line, the strict transforms of the two full bundles over slices, and $\widehat{Z}$. They have multiplicity one there. By Equation (6.11) and nonnegativity of mixed intersections of nef classes, their top intersections are respectively bounded below by

$$\begin{aligned} K \cdot (\text{vertical line}) &= q, \\ \int_{\text{slice}} K^{n+1} &\geq \frac{n+1}{2^n} qv, \\ \int_{\widehat{Z}} K^d &\geq \frac{d}{2^{2n}} \binom{2n}{n} qv^2. \end{aligned} \tag{6.12}$$

For example, the middle line is the term containing one factor Θ and n factors from the varying P' ; pushforward of Θ along a general vertical line is q . The last line is the term with one Θ and $2n$ horizontal factors. These intersections give $\text{vol}(M) \geq c_n q$. The ordinary Seshadri formula and $q \geq 1$ give $\epsilon(K, z') \geq c_n q^{1/d} = c_n A_q$ at a very general z' : each possible dimension is at most d , and every numerator in that formula is at least $c_n q$.

For the upper bound, subtract the axis A_1 . For $0 \leq u \leq q$ put

$$M_u = M - u\{A_1\} = P_1 + P_2 + uL_1 + (q - u)\xi.$$

The endpoint M_q is pulled back from X^2 and has zero volume on the d -fold Z . It is pseudo-effective, and M_0 is big by the preceding construction, so M_u is big for $u < q$. The restriction of M_u to $A_1 \simeq X^2$ is

$$(P + uL)_1 + (P + (q - u)L)_2.$$

The restricted volume along A_1 is at most the volume of this restriction. For big classes α_i on manifolds of dimensions e_i ,

$$\text{vol}(\text{pr}_1^* \alpha_1 + \text{pr}_2^* \alpha_2) = \binom{e_1 + e_2}{e_1} \text{vol}(\alpha_1) \text{vol}(\alpha_2). \tag{6.13}$$

One can see this directly from the non-pluripolar product formula [9]: the envelope with minimal singularities of a sum on a product is the sum of the two envelopes, by testing the defining inequality on successive slices. Its top product has only the indicated binomial term. Since $L \leq P/r$ and $r \geq q^2$, monotonicity and Equation (6.13) bound the restriction volume by

$$\binom{2n}{n} (1 + u/r)^n (1 + (q - u)/r)^n \leq C_n.$$

Integrating Equation (6.3) from 0 to q gives $\text{vol}(M) \leq C_n q$. Finally apply Lemma 6.3 with $e = d$, $\eta = c_n A_q$, and $\text{vol}(M) \leq C_n q$. Both terms on the right of Equation (6.5) are at most $C_n A_q$, since $A_q^d = q$. This proves Equation (6.10). $\square$

6.3 A direct-image estimate

The next lemma bounds a Kähler volume upstairs in terms of a class on the base. Its determinant formula will let Lemma 6.2 control that base class. The base need not be projective, and the horizontal class is allowed to be real.

Lemma 6.6. *Let $f : U \rightarrow Z_0$ be a surjective projective morphism of smooth connected compact Kähler manifolds, with $\dim Z_0 = b_0 \geq 1$ and $\dim U = b_0 + e$, $e \geq 1$. Suppose*

$$\beta = f^*G + c_1(\Lambda)$$

is a Kähler class, where $G \in H_{\text{BC}}^{1,1}(Z_0, \mathbb{R})$ and $\Lambda \in \text{Pic}(U) \otimes \mathbb{Q}$. Put

$$w = \int_{U_z} \beta^e > 0$$

on a general fiber. For sufficiently large divisible integers j , let $\mathcal{F}_j = f_\mathcal{O}_U(j\Lambda)$ and $R_j = \text{rk } \mathcal{F}_j$. Here $j\Lambda$ is an actual line bundle and $c_1(\mathcal{F}_j)$ means $c_1(\det \mathcal{F}_j)$. Then*

$$R_j = \frac{w}{e!} j^e + O(j^{e-1}), \quad (6.14)$$

$$B_0 := G + \lim_j \frac{c_1(\mathcal{F}_j)}{j R_j} = \frac{f_*\beta^{e+1}}{(e+1)w} \geq 0, \quad (6.15)$$

$$\int_U \beta^{b_0+e} \leq (e+1)^{b_0} w \text{ vol}(B_0). \quad (6.16)$$

The limit in Equation (6.15) is a limit of real Bott–Chern classes.

Proof. The restriction of $c_1(\Lambda)$ to each fiber is represented by the restriction of the Kähler form β . The fiberwise criterion for relative ampleness makes Λ relatively ample after clearing denominators. Relative Serre vanishing then kills the higher direct images for all sufficiently large divisible j . Analytic Grothendieck–Riemann–Roch [48], in degrees zero and two, gives

$$R_j = \frac{f_*c_1(\Lambda)^e}{e!} j^e + O(j^{e-1}), \quad c_1(\mathcal{F}_j) = \frac{f_*c_1(\Lambda)^{e+1}}{(e+1)!} j^{e+1} + O(j^e).$$

The degree-zero pushforward is w . Expanding $\beta = f^*G + c_1(\Lambda)$ shows that

$$f_*\beta^{e+1} = f_*c_1(\Lambda)^{e+1} + (e+1)wG,$$

because terms with at least two horizontal factors have negative fiber degree after pushforward. This proves the equality in Equation (6.15); the cohomological GRR equality is an equality in Bott–Chern cohomology by the $\partial\bar{\partial}$ -lemma on compact Kähler manifolds. Pushforward of the positive form β^{e+1} is a positive closed $(1,1)$ -current, so B_0 is pseudo-effective.

To prove Equation (6.16), we need a nef class on the base. The class B_0 is presently only known to be pseudo-effective. We flatten f to obtain a nef class above the base and compare it with the pullback of B_0 . Take a smooth projective flattening modification $p : Z'_0 \rightarrow Z_0$, the equidimensional main transform $\bar{U}$ of $U \times_{Z_0} Z'_0$, and a resolution $U' \rightarrow \bar{U}$. Write $f' : U' \rightarrow Z'_0$ and $q : U' \rightarrow U$ for the maps and $\beta' = q^*\beta$. The class

$$B'_0 = \frac{f'_*(\beta')^{e+1}}{(e+1)w}$$

is nef. It is enough to show that the positive pushforward current representing this class has zero Lelong numbers. This current can be computed by integration on the cycle $\bar{U}$ of the smooth form pulled back from U . At a base point, cover the compact fiber by finitely many coordinate neighborhoods, each embedded in a product of base coordinates and ambient coordinates. In the mass over a base ball of radius δ , after wedging with a base Euclidean form to power $b_0 - 1$, the integrand is bounded by a finite sum of projection volume forms using $b_0 - 1$ base coordinates and $e + 1$ generic linear combinations of all coordinates of the ambient product, including the remaining base direction. These projections can be chosen finite on the neighborhoods: fixing the $b_0 - 1$ base coordinates leaves local dimension at most $e + 1$, by equidimensionality, and generic ambient coordinates finish a finite projection. After shrinking to compact subneighborhoods their degrees are bounded. Change of variables therefore bounds each integral by $O(\delta^{2(b_0-1)})$ times the measure of the remaining coordinate range. On each compact subneighborhood these ranges, taken over closed base balls, are nested compact sets whose intersection is the image of the central fiber. That image has measure zero in the $e + 1$ coordinates, because the fiber has dimension e . Continuity of finite measure from above shows that the range measures tend to zero. Thus the mass is

$$o(\delta^{2(b_0-1)}).$$

This also covers $b_0 = 1$, when it asserts absence of an atom. The pushforward current has zero Lelong numbers at every point, and Demailly regularization [21, Theorem 1.1] proves that its class B'_0 is nef.

Pushforward under p gives $p_* B'_0 = B_0$. For a modification between smooth compact Kähler manifolds, the kernel of pushforward on real Bott–Chern $(1, 1)$ -classes is generated by the classes of its exceptional prime divisors. Since $p_* p^* B_0 = B_0$, it follows that $B'_0 - p^* B_0$ is an exceptional real divisor class. It is p -nef because B'_0 is nef. Apply relative negativity to this exceptional real divisor. Locally over the base, the usual proof for a projective modification cuts by general hyperplanes to a surface and uses the negative definite intersection matrix of exceptional curves; it forces every coefficient of a relatively nef exceptional divisor to be nonpositive. Thus

$$B'_0 = p^* B_0 - \{D_0\}, \quad D_0 \geq 0 \text{ } p\text{-exceptional}.$$

In particular

$$\int_{Z'_0} (B'_0)^{b_0} = \text{vol}(B'_0) \leq \text{vol}(p^* B_0) = \text{vol}(B_0). \quad (6.17)$$

Choose a Kähler class ω' on Z'_0 and put $C = B'_0 + \varepsilon \omega'$. For $0 \leq k \leq b_0$, set

$$I_k = \int_{U'} (\beta')^{e+k} (f'^* C)^{b_0-k}.$$

These are mixed intersections of nef classes. The first two satisfy

$$I_0 = w \int_{Z'_0} C^{b_0}, \quad I_1 = (e+1)w \int_{Z'_0} B'_0 C^{b_0-1} \leq (e+1)w \int_{Z'_0} C^{b_0}.$$

The mixed nef inequalities make the sequence I_k log-concave. Every I_k is positive: on the dense open where q is a local biholomorphism and f' is a submersion, β' is positive definite and $f'^* C$ has rank b_0 , so the defining top form is strictly positive. The successive ratios are therefore at most $I_1/I_0 \leq e + 1$. Thus

$$\int_U \beta^{b_0+e} = I_{b_0} \leq (e+1)^{b_0} w \int_{Z'_0} C^{b_0}.$$

Letting $\varepsilon \downarrow 0$ and using Equation (6.17) proves Equation (6.16). $\square$

We will also use the determinant formula without its volume bound.

Corollary 6.7. *Let Y be a smooth compact Kähler manifold and $\mathcal{E}$ a holomorphic vector bundle of rank at least two. Write $\pi_{\mathcal{E}} : \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ and $\zeta_{\mathcal{E}} = c_1(\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1))$. Suppose that a real class D on Y satisfies*

$$c_1(\mathcal{H}) \leq kD \quad \text{whenever} \quad 0 \neq (\mathcal{H} \rightarrow \mathcal{E}^{\otimes k})$$

for a line bundle $\mathcal{H}$ and integer $k \geq 0$. For a real class A on Y and a real number $b \geq 0$,

$$\pi_{\mathcal{E}}^* A + b\zeta_{\mathcal{E}} \geq 0 \quad \implies \quad A + bD \geq 0. \quad (6.18)$$

Proof. Choose a real class H_0 on Y so that $\zeta_{\mathcal{E}} + \pi_{\mathcal{E}}^* H_0$ is Kähler; relative ampleness of $\mathcal{O}(1)$ permits such a choice. Take numbers $\delta \downarrow 0$ with $b + \delta > 0$ rational. Adding $\delta(\zeta_{\mathcal{E}} + \pi_{\mathcal{E}}^* H_0)$ to the pseudo-effective class in Equation (6.18) makes it big. A Fujita decomposition on a smooth projective modification $h : U \rightarrow \mathbb{P}_Y(\mathcal{E})$, with the divisor coefficients rounded upwards as in Equation (6.4), has the form

$$\beta = f^*(A + \delta H_0) + c_1(\Lambda), \quad \Lambda = (b + \delta)h^*\mathcal{O}(1) - \mathcal{O}_U(D') \text{ in } \text{Pic}(U) \otimes \mathbb{Q},$$

where $f = \pi_{\mathcal{E}} h$, β is Kähler, and $D' \geq 0$ is rational. For divisible j ,

$$f_* \mathcal{O}_U(j\Lambda) \subseteq \text{Sym}^{j(b+\delta)} \mathcal{E}.$$

Its determinant of rank R_j consequently maps into $\mathcal{E}^{\otimes j(b+\delta)R_j}$. The assumed line inequality gives

$$\frac{c_1(\mathcal{F}_j)}{jR_j} \leq (b + \delta)D.$$

Lemma 6.6 makes $A + \delta H_0 + \lim c_1(\mathcal{F}_j)/(jR_j)$ pseudo-effective. Adding the preceding pseudo-effective difference shows that $A + \delta H_0 + (b + \delta)D$ is pseudo-effective. Let $\delta \downarrow 0$. This proves the corollary. In this argument only $b + \delta$ and the coefficients of D' are rational; A , H_0 , and D remain real classes. $\square$

6.4 Restriction to the diagonal

We now use the volume of M to obtain a high-rank subsheaf of a symmetric cotangent power on Z . Distinguish the two copies of Z by bracketed indices and blow up their diagonal:

$$b : B = \text{Bl}_{\Delta_Z}(Z \times Z) \rightarrow Z \times Z, \quad E = b^{-1}(\Delta_Z).$$

Thus $E = \mathbb{P}_Z(\Omega_Z)$; its points are normal lines in T_Z . The dimensions are $\dim B = 2d$ and $\dim E = 2d - 1$, and $E \rightarrow Z$ has relative dimension $d - 1$. Put $\zeta = c_1(\mathcal{O}_E(1))$, so that $\{E\}|_E = -\zeta$. For $s \geq 0$ define

$$C_s = b^*(M_{[1]} + M_{[2]}) - s\{E\}, \quad s_{\max} = \sup\{s \geq 0 : C_s \geq 0\}.$$

The class C_0 is big, so $s_{\max} > 0$ and C_s is big for $0 \leq s < s_{\max}$. Restricting a Kähler current with analytic singularities to the fiber of the first projection at a very general $z \in Z$ gives the class $b_z^* M - s\{F_z\}$. The current can be restricted for a general such z , and Lemma 6.5 therefore gives

$$s_{\max} \leq C_n A_q. \quad (6.19)$$

Write $v_E(s) = \text{vol}|_E(C_s)$ for $0 < s < s_{\max}$. The restriction class is

$$C_s|_E = 2M + s\zeta. \quad (6.20)$$

Lemma 6.8. *There are constants $c_n, C_n > 0$ such that, for every sufficiently large q , one can choose a rational number*

$$c_n A_q \leq s \leq C_n A_q$$

and a Kähler current T in C_s with analytic singularities, not generically singular on E , with the following property. On a smooth projective modification $h : U \rightarrow E$, its restricted mass has a rational-divisor approximation

$$h^*(2M + s\zeta) = \beta + \{D'\}, \quad \beta \text{ Kähler}, \quad D' \geq 0 \text{ rational}, \quad (6.21)$$

which retains all log-ideal orders of $T|_E$. If $f : U \rightarrow Z$ is the natural map and

$$w = \int_{U_z} \beta^{d-1}$$

on a general fiber, then

$$w \geq c_n s^{d-1}. \quad (6.22)$$

Proof. First let s be any rational number in $(0, s_{\max})$ with $v_E(s) > 0$, and use the approximation in Equation (6.4) for any current used to compute this restricted volume. In Lemma 6.6, the data for Equation (6.20) are

$$b_0 = d, \quad e = d - 1, \quad G = 2M, \quad \Lambda = sh^* \mathcal{O}_E(1) - \mathcal{O}_U(D') \text{ in } \text{Pic}(U) \otimes \mathbb{Q}.$$

For large divisible j , the associated sheaves satisfy

$$\mathcal{F}_j = f_* \mathcal{O}_U(j\Lambda) \subseteq \text{Sym}^{js} \Omega_Z, \quad R_j = \text{rk } \mathcal{F}_j, \quad 0 < w \leq s^{d-1}. \quad (6.23)$$

The inclusion follows by pushing $\mathcal{O}_U(-jD') \subseteq \mathcal{O}_U$ through h . For the last inequality restrict the decomposition to a general $\mathbb{P}^{d-1}$ -fiber: monotonicity of volume bounds the Kähler volume there by $\text{vol}(s\zeta) = s^{d-1}$.

We claim that the class B_0 of Equation (6.15) satisfies

$$0 \leq B_0 \leq C_n M. \quad (6.24)$$

Here is the determinant calculation, including its vertical sign. The equality $H^1(X, \mathcal{O}_X) = 0$ and the Künneth decomposition give $\text{Pic}(X^2) = \text{pr}_1^* \text{Pic}(X) \oplus \text{pr}_2^* \text{Pic}(X)$. Hence, for some lines $\mathcal{Q}_{j,i}$ on X and integer ℓ_j ,

$$\det \mathcal{F}_j = \pi^*(\mathcal{Q}_{j,1} \boxtimes \mathcal{Q}_{j,2}) \otimes \mathcal{O}_Z(\ell_j), \quad c_1(\det \mathcal{F}_j) = Q_j + \ell_j \xi,$$

where $Q_j = c_1(\mathcal{Q}_{j,1})_1 + c_1(\mathcal{Q}_{j,2})_2$. Taking the determinant of the inclusion in Equation (6.23) gives a nonzero line map into $\Omega_Z^{\otimes k}$, $k = jsR_j$. It is first defined where $\mathcal{F}_j$ is locally free and extends over the codimension-two complement. Filter this tensor using

$$0 \longrightarrow \pi^* \Omega_{X^2} \longrightarrow \Omega_Z \longrightarrow \mathcal{O}_Z(-2) \otimes \pi^*(\mathcal{L}_1 \otimes \mathcal{L}_2) \longrightarrow 0.$$

A nonzero associated graded component has i relative factors and k_1, k_2 cotangent factors from the first and second factors of X^2 , respectively, where $0 \leq i \leq k$ and $k_1 + k_2 = k - i$. Its relative degree is $-2i$. Pushing the component to X^2 forces

$$m_j := -2i - \ell_j \geq 0, \quad \ell_j \leq -2i \leq 0.$$

A nonzero homogeneous monomial in $\text{Sym}^{m_j}(\mathcal{L}_1 \oplus \mathcal{L}_2)$, with exponent $0 \leq m_{j,1} \leq m_j$ in the first factor, gives, on the two general slices,

$$c_1(\mathcal{Q}_{j,1}) \leq (i + m_{j,1} + k_1)L, \quad c_1(\mathcal{Q}_{j,2}) \leq (i + m_j - m_{j,1} + k_2)L$$

by Lemma 6.2. Each coefficient on the right is at most $k - \ell_j$. Since $L \geq 0$, we obtain

$$\ell_j \leq 0, \quad Q_j \leq (jsR_j - \ell_j)(L_1 + L_2). \quad (6.25)$$

Lemma 6.6 supplies a limit of the entire determinant class divided by jR_j . The projective-bundle decomposition of Bott–Chern cohomology gives separate limits Q and ℓ of the two displayed components. Since

$$B_0 = 2(P_1 + P_2) + (2q + \ell)\xi + Q \geq 0,$$

testing on a general vertical line gives $2q + \ell \geq 0$. Equation (6.25) gives $\ell \leq 0$ and $Q \leq (s - \ell)(L_1 + L_2)$. Use $\xi \geq 0$, $-\ell \leq 2q$, $L \leq P/r$, $r \geq q^2$, and $s \leq C_n A_q \leq C_n q$. They give

$$B_0 \leq 2(P_1 + P_2) + 2q\xi + (s + 2q)(L_1 + L_2) \leq C_n M,$$

which proves Equation (6.24).

Volume monotonicity, Lemma 6.5, and Equation (6.16) now imply

$$\int_U \beta^{2d-1} \leq C_n w q, \quad v_E(s) \leq C_n q s^{d-1}. \quad (6.26)$$

For the second inequality, approximate the mass of each current arbitrarily closely by β and use $w \leq s^{d-1}$, then take the supremum over currents. This proves it for rational s ; continuity of divisorial restricted volume extends it to every $s \in (0, s_{\max})$.

At $s_{\max}$ the class is on the boundary of the pseudo-effective cone, so its volume is zero. The product formula in Equation (6.13), modification invariance, and the derivative formula in Equation (6.3) yield

$$2d \int_0^{s_{\max}} v_E(s) ds = \text{vol}(C_0) = \binom{2d}{d} \text{vol}(M)^2 \geq c_n q^2. \quad (6.27)$$

Choose a fixed small $\theta_n > 0$. The contribution of $0 < s < \theta_n A_q$, by Equation (6.26), is at most $C_n \theta_n^d q^2$. Fix θ_n small enough that this is less than half the last lower bound. The remaining interval has length at most $C_n A_q$, by Equation (6.19). Continuity therefore permits a rational $s \in [\theta_n A_q, s_{\max})$ such that

$$v_E(s) \geq c_n q^2 / A_q = c_n q A_q^{d-1} \geq c_n q s^{d-1}.$$

Choose a current with restricted mass at least three quarters of $v_E(s)$, and then an approximation as in Equation (6.21) with $\int_U \beta^{2d-1} \geq v_E(s)/2$. The first inequality of Equation (6.26) gives $w \geq c_n s^{d-1}$, as claimed. $\square$

For the selected s, T, h, β, D' and w , retain the natural map $f : U \rightarrow Z$ and put

$$\Lambda = sh^* \mathcal{O}_E(1) - \mathcal{O}_U(D'), \quad \mathcal{F}_j = f_* \mathcal{O}_U(j\Lambda), \quad R_j = \text{rk } \mathcal{F}_j,$$

where j is sufficiently large and divisible. Here Λ is a rational line, and Equation (6.23) gives $\mathcal{F}_j \subseteq \text{Sym}^{js} \Omega_Z$. With q and these selected data fixed, Equations (6.14) and (6.22) give

$$\text{rk Sym}^{js} \Omega_Z = \binom{js + d - 1}{d - 1}, \quad \lim_j \frac{R_j}{\text{rk Sym}^{js} \Omega_Z} = \frac{w}{s^{d-1}} \geq c_n. \quad (6.28)$$

The limit is through divisible integers. Thus the first diagonal has produced a subsheaf occupying a fixed positive proportion of the symmetric power for all sufficiently large divisible j . The remaining proof carries this rank bound to a modification of Z and converts it into a large common order of vanishing for the determinant map on that model.

6.5 A pole along the incidence

We next locate a large pole of the selected current T along a smooth incidence submanifold of B . This will impose vanishing conditions on the symmetric-power subsheaf in Equation (6.23).

Let $U_0 = \pi^{-1}(\Delta_X) \subset Z$, where Δ_X is the diagonal in X^2 . Since $\mathcal{L}_1|_{\Delta_X} = \mathcal{L}_2|_{\Delta_X} = K_X$, there is a canonical identification

$$U_0 = X \times \mathbb{P}^1.$$

For $\lambda \in \mathbb{P}^1$ write $D_\lambda = X \times \{\lambda\} \subset Z$. Inside $Z \times Z$ consider

$$\mathcal{V}_0 = \{((x, x, \lambda), (y, y, \lambda)) : x, y \in X, \lambda \in \mathbb{P}^1\} \simeq X^2 \times \mathbb{P}^1.$$

It meets Δ_Z in $\Delta_X \times \mathbb{P}^1$. The strict transform $V \subset B$ is $\text{Bl}_{\Delta_X \times \mathbb{P}^1} \mathcal{V}_0$; it is smooth, of dimension $d = 2n + 1$. Its first projection is a smooth family over U_0 . Over $z = (x, x, \lambda)$ its fiber is

$$Y = \text{Bl}_x D_\lambda \simeq \text{Bl}_x X,$$

and its intersection with E in that fiber is the projective space of lines in $T_z D_\lambda$, inside the projective space of lines in $T_z Z$. These assertions follow either from the blowup of the section $y = x$ in this family or from the coordinates used below.

Let $\rho_V : \text{Bl}_V B \rightarrow B$ have exceptional divisor G_V . Denote by $b_V \geq 0$ the generic divisorial pole of $\rho_V^* T$ along G_V . Equivalently, it is the generic log-ideal order of T along V , with the coefficient of the logarithm included.

Lemma 6.9. *The pole just defined satisfies*

$$b_V \geq s - C_n. \quad (6.29)$$

Proof. Remove $b_V[G_V]$ from $\rho_V^* T$. Its residual positive current can be restricted to G_V : for analytic singularities removal of the generic divisorial pole leaves a potential that is not identically $-\infty$ on that divisor. Choose $z = (x, x, \lambda) \in U_0$ general enough for all restrictions and for Lemmas 6.2 and 6.3. On the bundle

$$G_V|_Y = \mathbb{P}_Y(N_{V/B}^*|_Y), \quad Y = \text{Bl}_x X,$$

this restriction is a positive current in the class

$$a^*(2P + qL) - s\{F\} + b_V \zeta_V, \quad (6.30)$$

where $\zeta_V = c_1(\mathcal{O}_{\mathbb{P}(N_{V/B}^*|_Y)}(1))$. Indeed $M|_{D_\lambda} = 2P + qL$, the first projection to Z is constant on Y , and $E|_Y = F$. Also $G_V|_{G_V} = -\zeta_V$, explaining the positive sign of the last term.

The conormal bundle in Equation (6.30) has exact sequences

$$0 \longrightarrow \mathcal{O}_Y^{\oplus n} \longrightarrow N_{V/B}^*|_Y \longrightarrow a^*N_{D_\lambda/Z}^* \otimes \mathcal{O}_Y(F) \longrightarrow 0, \quad (6.31)$$

$$0 \longrightarrow \Omega_X \longrightarrow N_{D_\lambda/Z}^* \longrightarrow \mathcal{O}_X \longrightarrow 0. \quad (6.32)$$

The first constant term is the conormal of U_0 in the first copy of Z , evaluated at z . For the last term in Equation (6.31), the conormal of the strict transform of D_λ in $\text{Bl}_z Z$ is $a^*N_{D_\lambda/Z}^* \otimes \mathcal{O}_Y(F)$: in a blowup chart, a normal coordinate is divided by an exceptional coordinate, giving precisely the twist by F for conormals. Equation (6.32) is the conormal sequence for $D_\lambda \subset U_0 \subset Z$; the diagonal in X^2 has conormal Ω_X , and the λ -normal direction is constant on X .

Filter a k -fold tensor of $N_{V/B}^*|_Y$ by these sequences. A graded term has the form

$$\mathcal{O}_Y(hF) \otimes (a^*\Omega_X)^{\otimes k'} \otimes (\text{a constant vector space}), \quad 0 \leq k' \leq h \leq k.$$

For any nonzero line map into that tensor, take a nonzero graded component. Lemma 6.2 gives

$$c_1(\mathcal{H}_Y) \leq h\{F\} + k'a^*L \leq k(\{F\} + a^*L),$$

since both $\{F\}$ and a^*L are pseudo-effective. Apply Corollary 6.7 to Equation (6.30), with $D = \{F\} + a^*L$. We obtain

$$a^*(2P + (q + b_V)L) - (s - b_V)\{F\} \geq 0. \quad (6.33)$$

If $b_V \geq s$, the conclusion is immediate. Otherwise $b_V < s \leq C_n A_q \leq C_n q$, and $L \leq P/r$ gives

$$\left(2 + \frac{q + b_V}{r}\right) a^*P - (s - b_V)\{F\} \geq 0.$$

Use the point bound in Equation (6.8). Since $r \geq q^2$, its consequence

$$s - b_V \leq C_n \left(2 + \frac{q + b_V}{r}\right)$$

is bounded by a dimensional constant. This proves Equation (6.29). $\square$

6.6 Determinant vanishing and cancellation

Blow up U_0 in the base of E :

$$p_+ : Z^+ = \text{Bl}_{U_0} Z \longrightarrow Z, \quad J_+ = p_+^{-1}(U_0).$$

This is the bundle Z pulled to $\text{Bl}_{\Delta_X}(X^2)$, since $U_0 = \pi^{-1}(\Delta_X)$. In particular Z^+ is smooth. Put

$$E^+ = E \times_Z Z^+ = \mathbb{P}_{Z^+}(p_+^*\Omega_Z).$$

Extend the incidence ideal from B to E , and then to E^+ :

$$\mathcal{J} = (\mathcal{I}_V \cdot \mathcal{O}_E) \cdot \mathcal{O}_{E^+}.$$

The products denote extension of ideals under the indicated maps. Figure 1 locates $V \cap E$ and this extension of the incidence ideal to E^+ .

The next lemma turns the pole bound in Equation (6.29) into a common vanishing order for the coefficients of each determinant map.

$$\begin{array}{ccc}
V & \hookrightarrow & B \\
\text{Bl}_{\Delta_X \times \mathbb{P}^1} \downarrow & & \downarrow \text{Bl}_{\Delta_Z} \\
\mathcal{V}_0 & \hookrightarrow & Z \times Z
\end{array}
\qquad
\begin{array}{ccc}
E^+ & \longrightarrow & E = \mathbb{P}_Z(\Omega_Z) \\
\downarrow & & \downarrow \\
Z^+ & \xrightarrow{p_+ = \text{Bl}_{U_0}} & Z
\end{array}$$

$$\begin{aligned}
V_z &= \text{Bl}_x X, \\
(V \cap E)_z &= \mathbb{P}(T_z^* D_\lambda) \\
&\subset E_z = \mathbb{P}(T_z^* Z).
\end{aligned}
\qquad
\begin{aligned}
J_+ &= p_+^{-1}(U_0), \quad U_0 = X \times \mathbb{P}^1, \\
\mathcal{J} &= (\mathcal{I}_V \cdot \mathcal{O}_E) \cdot \mathcal{O}_{E^+}.
\end{aligned}$$

Figure 1: Two geometric constructions used in the determinant estimate. On the left, the strict transform $V \subset B$ meets E_z in the directions tangent to D_λ . Under the quotient convention, the displayed projectivized cotangent spaces parametrize tangent lines. On the right, E^+ is the Cartesian base change over $Z^+ = \text{Bl}_{U_0} Z$, whose exceptional divisor is J_+ . The restricted incidence ideal extends to $\mathcal{J}$ on E^+ . The ensuing local calculation identifies the normal parameter of J_+ among its generators.

Lemma 6.10. *On a smooth projective modification $h_+ : U^+ \rightarrow E^+$ one can choose an approximation*

$$h_+^*(2p_+^*M + s\zeta) = \beta^+ + \{D^+\}, \quad \beta^+ \text{ K\"ahler}, \quad D^+ \geq 0 \text{ rational}, \quad (6.34)$$

retaining the log-ideal orders of $T|_E$, such that the general fiber volume w^+ of β^+ over Z^+ satisfies $w^+ \geq w/2$. Let $f_+ : U^+ \rightarrow Z^+$ be the natural map and put

$$\Lambda^+ = sh_+^* \mathcal{O}_{E^+}(1) - \mathcal{O}_{U^+}(D^+), \quad \mathcal{F}_j^+ = (f_+)_* \mathcal{O}_{U^+}(j\Lambda^+), \quad R_j^+ = \text{rk } \mathcal{F}_j^+.$$

For every sufficiently large divisible j , write

$$\iota_j : \mathcal{F}_j^+ \hookrightarrow p_+^* \text{Sym}^{js} \Omega_Z, \quad \varphi_j : \det \mathcal{F}_j^+ \longrightarrow \bigwedge^{R_j^+} (p_+^* \text{Sym}^{js} \Omega_Z)$$

*for the inclusion and its nonzero determinant map. Here $\det \mathcal{F}_j^+ = (\bigwedge^{R_j^+} \mathcal{F}_j^+)^{**}$, and the determinant map extends across the complement of the locally free locus, which has codimension at least two. Localize a coordinate local ring of Z^+ at the height-one prime of J_+ , obtaining a discrete valuation ring R with uniformizer u . In local frames over R , define*

$$h_j = \min\{\text{ord}_u(c) : c \text{ is a nonzero coefficient of } \varphi_j\}.$$

Equivalently, h_j is the minimum order of the maximal minors of ι_j ; it is independent of the frames. If σ_{J_+} is the canonical section of $\mathcal{O}_{Z^+}(J_+)$, then φ_j factors as

$$\det \mathcal{F}_j^+ \xrightarrow{\cdot \sigma_{J_+}^{h_j}} \det \mathcal{F}_j^+(h_j J_+) \xrightarrow{\tilde{\varphi}_j} \bigwedge^{R_j^+} (p_+^* \text{Sym}^{js} \Omega_Z),$$

where $\tilde{\varphi}_j$ is holomorphic. The common order satisfies

$$h_j \geq c_n j R_j^+ s, \quad (6.35)$$

provided q is sufficiently large.

Proof. Blow up $\mathcal{J}$ and take a common smooth projective resolution with $U \rightarrow E$ from Equation (6.21). Pull back β and D' . Subtracting a sufficiently small rational multiple of an effective exceptional divisor whose negative is relatively ample makes the pulled-back Kähler class Kähler on this resolution; add that multiple to the effective part. Further upward rational rounding may be arbitrarily small. This gives Equation (6.34) with all original orders retained. Its general fiber intersection is as close to w as desired, so arrange $w^+ \geq w/2$. Lemma 6.6 and the effective divisor give

$$\mathcal{F}_j^+ \subseteq p_+^* \text{Sym}^{js} \Omega_Z, \quad R_j^+ = \frac{w^+}{(d-1)!} j^{d-1} + O(j^{d-2}). \quad (6.36)$$

We spell out the local order imposed on the polynomials in this inclusion. Near $z = (x, x, \lambda)$, use coordinates $(\mathbf{x}, \mathbf{a}, \lambda)$ on the first Z , where $\mathbf{x}$ and $\mathbf{a}$ each have n components and $U_0 = (\mathbf{a} = 0)$. Write the second coordinates as $(\mathbf{x} + \mathbf{h}, \mathbf{a} + \mathbf{k}, \lambda + \mu)$, with $\mathbf{h}, \mathbf{k}$ each having n components. Then

$$\mathcal{V}_0 = (\mathbf{a} = \mathbf{k} = \mu = 0), \quad \Delta_Z = (\mathbf{h} = \mathbf{k} = \mu = 0).$$

In a blowup chart meeting the lines tangent to D_λ , take

$$h_1 = v, \quad h_i = v\alpha_i \ (2 \leq i \leq n), \quad k_i = vy_i \ (1 \leq i \leq n), \quad \mu = vy_{n+1}.$$

Here $E = (v = 0)$, while the strict transform is $V = (a_1 = \cdots = a_n = y_1 = \cdots = y_{n+1} = 0)$. In particular,

$$\mathcal{I}_V|_E = (a_1, \dots, a_n, y_1, \dots, y_{n+1}). \quad (6.37)$$

On a chart of the blowup of U_0 , write $a_1 = u$ and $a_i = u\tau_i$ for $i \geq 2$. Along the general point of $J_+ = (u = 0)$, the ideal $\mathcal{J}$ is exactly

$$(u, y_1, \dots, y_{n+1}). \quad (6.38)$$

Here u measures vanishing along J_+ , while the y_i measure transverse fiber directions. Thus a term of transverse degree ℓ can contribute at most ℓ to the incidence order; the remaining order must come from its coefficient in u . We now make this statement precise.

Suppose locally that the singularities of T are $c \log |\mathbf{a}| + O(1)$, with the logarithmic normalization for which a divisorial coefficient is c times the ideal order. If $k = \text{ord}_V(\mathbf{a})$, then $b_V = ck$. The normal Taylor coefficients of degree less than k vanish on a dense open set of V , hence on V locally. Thus $\mathfrak{a} \subseteq \mathcal{I}_V^k$ also near a general point of $V \cap E$. Restricting to E and pulling to E^+ gives the corresponding order in Equation (6.38). At its general center the displayed generators are regular coordinates. If $H_{\mathcal{J}}$ is the exceptional divisor of their blowup, its valuation is the ordinary ideal-adic order:

$$\text{ord}_{H_{\mathcal{J}}}(g) = \max\{k : g \in (u, y_1, \dots, y_{n+1})^k\}.$$

The common resolution dominates this blowup. Pullback preserves the effective difference between D' and the resolved log divisor, and the later adjustments only increase the effective part. Thus D^+ has coefficient at least b_V at this valuation. Every polynomial image of a section of $\mathcal{F}_j^+$ has integral valuation at least $\lceil jb_V \rceil$. Locally bounded remainders have zero order at this valuation.

There are n homogeneous coordinates along $T_z D_\lambda$ and $n+1$ transverse homogeneous coordinates in $T_z Z$. Denote these two groups by $H_1, \dots, H_n$ and $Y_1, \dots, Y_{n+1}$. For $m = js$, a local polynomial in $\text{Sym}^m(p_+^* \Omega_Z)$ has the form

$$\sum_{|\alpha|+|\gamma|=m} c_{\alpha,\gamma}(u) H^\alpha Y^\gamma.$$

Work over the discrete valuation ring R used to define h_j , choosing the chart parameter u as uniformizer, and let κ be its residue field. The coefficients $c_{\alpha,\gamma}$ lie in R . Equation (6.38) implies, monomial by monomial,

$$\text{ord}_u(c_{\alpha,\gamma}) \geq \max\{0, [jb_V] - |\gamma|\}. \quad (6.39)$$

Indeed, on the chart $H_1 \neq 0$, group the dehomogenized polynomial as

$$\sum_{\gamma} P_{\gamma}(H_2/H_1, \dots, H_n/H_1)(Y/H_1)^{\gamma}, \quad P_{\gamma} \in R[H_2/H_1, \dots, H_n/H_1].$$

Put $b_j = [jb_V]$. At the general center of $(u, Y/H_1)$, order at least b_j forces P_{γ} to be divisible by $u^{b_j-|\gamma|}$ whenever $|\gamma| < b_j$. Otherwise its first nonzero reduction would be a nonzero polynomial over κ , which stays nonzero in $\kappa(H_2/H_1, \dots, H_n/H_1)$; in the associated graded ring it would give a nonzero term of total $(u, Y/H_1)$ -degree less than b_j . The transverse monomials there are independent. Comparing the internal polynomial coefficients over the base residue field κ now gives Equation (6.39).

We compare the number of monomials with the rank estimate in Equation (6.36). The ambient rank and the number of monomials of transverse degree ℓ are

$$N_m = \binom{m+2n}{2n}, \quad N_{m,\ell} = \binom{\ell+n}{n} \binom{m-\ell+n-1}{n-1}.$$

For fixed $\delta \in (0, 1)$, let $H_m(\delta)$ count the monomials with $\ell > (1-\delta)m$. Summing instead over their internal degree gives

$$H_m(\delta) \leq \binom{m+n}{n} \binom{[\delta m] + n}{n}, \quad \limsup_{m \rightarrow \infty} \frac{H_m(\delta)}{N_m} \leq \binom{2n}{n} \delta^n.$$

On the other hand, $w^+ \geq w/2$ and Equation (6.36) show that the approximation on E^+ retains at least half the rank fraction in Equation (6.28):

$$\lim_{j \rightarrow \infty} \frac{R_j^+}{N_{js}} = \frac{w^+}{s^{d-1}} \geq \frac{w}{2s^{d-1}} \geq c_n > 0.$$

Fix $\delta > 0$, depending only on n , so small that, for each fixed q and its selected data, fewer than $R_j^+/2$ monomials have transverse degree greater than $(1-\delta)js$ for all sufficiently large divisible j . For each remaining monomial, Equation (6.39) and Lemma 6.9 give

$$\text{ord}_u(c_{\alpha,\gamma}) \geq j(s - C_n) - (1-\delta)js = j(\delta s - C_n) \geq \frac{\delta js}{2}.$$

The last inequality holds once q is large, since $s \geq c_n A_q \rightarrow \infty$.

Over the same discrete valuation ring R , the torsion-free sheaf $\mathcal{F}_j^+$ becomes a free module of rank R_j^+ . Write its inclusion into the symmetric power as a matrix with the monomials as rows. Every maximal minor uses at least $R_j^+/2$ of the rows whose coefficients have order at least $\delta js/2$. All maximal minors therefore have order at least $\delta j R_j^+ s/4$. By the definition of h_j , this proves Equation (6.35). Dividing the determinant map by $\sigma_{J_+}^{h_j}$ gives a holomorphic map at every point of codimension one: the coefficients have the required order along J_+ , and its local equation is a unit away from J_+ . Hartogs' theorem extends the divided map across the remaining set of codimension at least two, giving the stated factorization through $\det \mathcal{F}_j^+(h_j J_+)$. $\square$

Completion of the proof of Theorem 6.1. Fix q sufficiently large for the preceding lemmas, with its chosen P, s, T and modifications. Choose a very general point x in the first factor of X^2 . The slice of Z^+ above x is

$$S = \mathbb{P}_Y(\mathcal{O}_Y \oplus a^*\mathcal{L}), \quad a : Y = \text{Bl}_x X \longrightarrow X,$$

and $J_+|_S$ is the pullback of F . We choose x so that Equation (6.2), Equation (6.8), the restrictions of the currents below, and all determinant maps for divisible j can be tested on this slice. These exclude at most countably many proper analytic sets and the measure-zero exceptional sets for current restrictions.

Write the restricted determinant line as

$$(\det \mathcal{F}_j^+)|_S = \pi_S^* \mathcal{Q}_j^+ \otimes \mathcal{O}_S(\ell_j^+), \quad \mathcal{Q}_j^+ = c_1(\mathcal{Q}_j^+).$$

Factoring the common zero of order h_j from the determinant map gives a nonzero map whose source on S is

$$\pi_S^*(\mathcal{Q}_j^+ \otimes \mathcal{O}_Y(h_j F)) \otimes \mathcal{O}_S(\ell_j^+).$$

The plus sign of $h_j F$ follows because division of the map by the local equation of $J_+^{h_j}$ enlarges its source from $\det \mathcal{F}_j^+$ to $\det \mathcal{F}_j^+ \otimes \mathcal{O}(h_j J_+)$. Its target embeds into the jsR_j^+ -fold tensor of the restriction of $p_+^* \Omega_Z$. In particular we use the cotangent bundle of the original Z , with its restricted filtration

$$0 \longrightarrow \pi_S^*(\mathcal{O}_Y^{\oplus n} \oplus a^* \Omega_X) \longrightarrow (p_+^* \Omega_Z)|_S \longrightarrow \mathcal{O}_S(-2) \otimes \pi_S^* a^* \mathcal{L} \longrightarrow 0.$$

The same homogeneous-monomial calculation as in Equation (6.25), now applying Equation (6.2) to the second factor of X^2 , gives

$$\ell_j^+ \leq 0, \quad \mathcal{Q}_j^+ + h_j \{F\} \leq (jsR_j^+ - \ell_j^+) a^* L. \quad (6.40)$$

More explicitly, if the chosen graded term has i relative factors, the pushed polynomial has degree $-2i - \ell_j^+ \geq 0$. Its exponent in $a^* \mathcal{L}$, together with the i relative factors and the cotangent order from the second factor of X^2 , is at most $jsR_j^+ - \ell_j^+$. The cotangent factors from the first factor of X^2 are constant. This proves the displayed inequality using the tensors of $a^* \Omega_X$, without introducing Ω_Y or Ω_{Z^+} .

Apply Lemma 6.6 to the decomposition in Equation (6.34). Equation (6.15) shows that the limits of the restricted determinant components divided by jR_j^+ exist; denote them by Q^+ and ℓ^+ . Divide Equation (6.40) by jR_j^+ , use Equation (6.35), and pass to the closed pseudo-effective cone. We obtain

$$\ell^+ \leq 0, \quad Q^+ + c_n s \{F\} \leq (s - \ell^+) a^* L. \quad (6.41)$$

The pseudo-effective class in Equation (6.15), restricted to the very general slice S , is

$$\pi_S^*(Q^+ + 2a^*P) + (\ell^+ + 2q)\xi \geq 0. \quad (6.42)$$

Testing on a general vertical line gives $\ell^+ + 2q \geq 0$. Let $A \subset S$ be the axis of $\mathbb{P}_Y(\mathcal{O}_Y \oplus a^* \mathcal{L})$ with divisor class $\{A\} = \xi$. On $A \simeq Y$, its normal class is $\{A\}|_A = \xi|_A = a^* L \geq 0$. Add an arbitrarily small Kähler class to the class in Equation (6.42), and choose a Kähler current with analytic singularities in the resulting big class. Remove its generic divisorial pole along A and restrict the residual current to A . Adding back the removed nonnegative multiple of $a^* L$ preserves pseudo-effectivity. Passing to the limit gives

$$Q^+ + 2a^*P + (\ell^+ + 2q)a^*L \geq 0. \quad (6.43)$$

The slope bound in Equation (6.41) supplies the pseudo-effective class $(s - \ell^+)a^*L - Q^+ - c_ns\{F\}$. Adding it to Equation (6.43) cancels the entire normalized determinant class restricted to the axis, $Q^+ + \ell^+a^*L$, leaving

$$2a^*P + (s + 2q)a^*L - c_ns\{F\} \geq 0.$$

Since $L \leq P/r$, the point bound in Equation (6.8) implies

$$\frac{c_ns}{2 + (s + 2q)/r} \leq C_n. \quad (6.44)$$

The denominator is bounded independently of q , because $s \leq C_nq^{1/d}$ and $r \geq q^2$. The numerator tends to infinity, because $s \geq c_nq^{1/d}$. Equation (6.44) is impossible for arbitrarily large q .

For each q , the choice of $t(q)$, the rational s , the current T , and the modifications precedes the limit in j . The monomial cutoff δ depends only on n , while the required lower bound for divisible j may depend on the fixed data. We choose the very general points after those data are fixed. Thus every simultaneous general-point test above involves at most countably many conditions, and no bound depends on the point. The contradiction disproves the contrary hypothesis and proves Theorem 6.1. $\square$

7 Signed rigidity on simple spaces

The meromorphic section furnished by Theorem 6.1 need not have an effective divisor. We therefore need a boundary argument that retains both its zeroes and its poles. The result below supplies that argument.

Theorem 7.1 (Signed rigidity). *Let (X, D) be a dlt pair on a normal irreducible compact Kähler space, and suppose that the pair has the resolution property of Definition 3.1. Suppose also that X is globally $\mathbb{Q}$ -factorial, that $a(X) = 0$, that X is simple, and that $D = \sum_{i=1}^s D_i$ is reduced. Put $L = K_X + D$, and assume the following.*

(i) *The rational line L is analytically nef and has an actual rational linear equivalence*

$$L \sim_{\mathbb{Q}} G = \sum_{i=1}^s a_i D_i, \quad a_i \in \mathbb{Q}.$$

(ii) *For some real $c > 0$, the pullback of the class $\{L - cD\}$ to a projective resolution with smooth compact Kähler source is pseudo-effective.*

(iii) *The rational holomorphic line $L|_D$ is semiample on the whole reduced analytic space D : some Cartier multiple is generated at every point of D .*

Then L is torsion. More precisely, for some positive integer q , the holomorphic line $\mathcal{O}_X(qL)$ is isomorphic to $\mathcal{O}_X$.

To apply this theorem in the proof of Proposition 2.12, Theorem 6.1 supplies a signed representative of the canonical bundle. After resolving, we enlarge the support of the transformed boundary and this signed canonical divisor to a reduced SNC divisor; the resulting adjoint remains pseudo-effective. Corollary 3.14 gives an ordinary dlt nef model $(X_{\text{nef}}, D_{\text{nef}})$ whose adjoint $L_{\text{nef}} = K_{X_{\text{nef}}} + D_{\text{nef}}$ has a signed representative supported on D_{nef} , supplying the first hypothesis. Theorem 4.1 supplies semiamplessness on the whole reduced D_{nef} . On a common smooth resolution,

simplicity excludes uniruledness, so Ou's criterion [58, Theorem 1.1] and exceptional translation for the canonical comparison give pseudo-effectivity of the pullback of $K_{X_{\text{nef}}} = L_{\text{nef}} - D_{\text{nef}}$. Thus the second hypothesis holds with $c = 1$. Once signed rigidity makes the actual line L_{nef} torsion, the program comparison and Lemma 2.6 let us subtract the added boundary and give the required decomposition for the original adjoint. The final subsection verifies these steps.

The signed construction in [55, Proposition 3.3] is the source of the root and separation operations used below. The lifting method is that of [53, Sections 10–12]. We will establish the analytic form of the signed construction and the change to the lifting argument caused by its residual poles. The numerical argument first singles out positive-dimensional fibers in the positive boundary. We then lift those fibers through every finite boundary neighborhood and deform them to compact subspaces outside the boundary. An argument using the cycle space and Baire's theorem turns these deformations into a covering family, which simplicity excludes.

7.1 The boundary detected by the nef class

If $D = 0$, the signed equivalence already gives $L \sim_{\mathbb{Q}} 0$. We may therefore assume in the constructions below that $D \neq 0$; in particular $\dim X > 0$.

Fix the data of Theorem 7.1, and write

$$G = G_+ - G_-, \quad D_0 = \sum_{a_i=0} D_i,$$

where G_+ and G_- are effective with disjoint prime supports. Choose a positive integer m so that mG_+ , mG_- , mD_0 , and mL are Cartier, the given rational equivalence induces a fixed isomorphism

$$N_0 := \mathcal{O}_X(mG) \simeq \mathcal{O}_X(mL),$$

and $N_0|_D$ is generated. Its sections give a morphism and an actual line isomorphism

$$\phi : D \longrightarrow P = \mathbb{P}^b, \quad N_0|_D \simeq \phi^* \mathcal{O}_P(1). \quad (7.1)$$

Let $n = \dim X$. Choose a projective resolution $\mu : \tilde{X} \rightarrow X$ as in the second hypothesis of the theorem, and a Kähler class ω on $\tilde{X}$. The numerical dimension of the nef pullback is

$$v = \nu(L) := \max\{k \in \{0, \dots, n\} : (\mu^*\{L\})^k \omega^{n-k} > 0\}.$$

The next argument is the Kähler form of [55, Lemma 3.2].

Lemma 7.2. *If $v = 0$, then $D = 0$ and $L \sim_{\mathbb{Q}} 0$. If $v = n > 0$, then $a(X) = n$. If $0 < v < n$ and $r = v - 1$, then*

$$\dim \phi(D_i) \leq r \quad \text{for every } i, \quad \max_{a_i > 0} \dim \phi(D_i) = r,$$

and

$$\dim \phi(\text{Supp } G_+ \cap (\text{Supp } G_- \cup D_0)) < r. \quad (7.2)$$

For $r = 0$, the intersection in Equation (7.2) is empty.

Proof. Choose a Kähler class θ on X . We use Cartier intersection products downstairs, computed on a resolution by projection formula. These products detect v . Indeed, on the chosen resolution the class $h = \mu^*\theta$ is nef and big. Choose $C, \epsilon > 0$ such that $C\omega - h$ is nef and $h - \epsilon\omega$ is pseudo-effective. Pairing these two inequalities successively with products of the nef classes $\mu^*\{L\}$, h , and ω shows that

$$(\mu^*\{L\})^k h^{n-k} > 0 \iff (\mu^*\{L\})^k \omega^{n-k} > 0.$$

We may therefore use θ in all the tests defining v .

Suppose first that $v < n$. The pseudo-effective pullback in the theorem can be paired with nef products, so

$$0 \leq (\{L\} - c\{D\})\{L\}^v \theta^{n-v-1} = -c \sum_i \{D_i\} \{L\}^v \theta^{n-v-1}. \quad (7.3)$$

Each summand on the right is nonnegative. For example, pull the generated line in Equation (7.1) and θ to a resolution of D_i ; their representatives are semipositive and positive at general smooth points, respectively. Thus every summand is zero. When $v = 0$, a nonzero effective divisor has strictly positive θ^{n-1} -degree. Hence $D = 0$, and the signed equivalence gives $L \sim_{\mathbb{Q}} 0$.

Now let $0 < v < n$. For any irreducible effective cycle V in D , the mixed degree $\{L\}^k \theta^{\dim V - k} \cdot V$ is positive exactly when $\dim \phi(V) \geq k$. To see this, resolve V and use a Fubini–Study representative for $m\{L\}|_V$. Its generic rank is $\dim \phi(V)$, and its wedge with the remaining Kähler factors is positive on a nonempty open set exactly in the asserted range. The vanishing in Equation (7.3) therefore gives $\dim \phi(D_i) \leq r$. On the other hand,

$$0 < \{L\}^v \theta^{n-v} = \sum_i a_i \{D_i\} \{L\}^r \theta^{n-v}. \quad (7.4)$$

All the component tests here are nonnegative, so a component with $a_i > 0$ attains dimension r .

It remains to separate its intersections from a general fiber. On $H_{\text{BC}}^{1,1}(\tilde{X}, \mathbb{R})$ consider the form

$$q(\alpha, \beta) = \alpha\beta(\mu^*\{L\})^r h^{n-r-2}, \quad Q_{ij} = q(\mu^*\{D_i\}, \mu^*\{D_j\}).$$

The mixed Kähler Hodge index theorem, followed by approximation of the nef factors by Kähler classes, says that q has at most one positive direction; see [24, Theorems A and C]. Put $l = \mu^*\{L\}$. The definition of v and Equation (7.3) give

$$q(l, l) = 0, \quad q(l, \mu^*\{D_i\}) = 0, \quad q(l, h) = \{L\}^v \theta^{n-v} > 0.$$

Linear algebra now makes q negative semidefinite on $l^\perp$: a positive vector in $l^\perp$, together with a suitable vector in the span of l, h , would give two positive directions. In particular Q is negative semidefinite.

For $i \neq j$, one has $Q_{ij} \geq 0$. This assertion uses the effective intersection cycle of the distinct $\mathbb{Q}$ -Cartier divisors D_i, D_j on X , and projection formula. It does not require their total transforms to intersect effectively. The two Cartier equations have a proper intersection: the dlt ambient space is klt, hence Cohen–Macaulay, after dropping the reduced boundary, and the distinct divisor equations form a regular sequence. The intersection cycle is consequently pure of codimension two, and each of its mixed nef degrees is nonnegative.

Write the coefficient vector as $\mathbf{a} = \mathbf{a}^+ - \mathbf{a}^-$, with nonnegative vectors of disjoint supports. The signed equivalence and orthogonality to l give $Q\mathbf{a} = 0$. Thus

$$Q(\mathbf{a}^+, \mathbf{a}^+) = Q(\mathbf{a}^+, \mathbf{a}^-) \geq 0.$$

Negative semidefiniteness forces equality and $Q\mathbf{a}^+ = 0$. For every j with $a_j \leq 0$, its row is a sum of nonnegative off-diagonal terms:

$$0 = \sum_{a_i > 0} a_i Q_{ji}.$$

Each term is zero. Applying the generated-line degree test to the effective intersection cycles proves Equation (7.2). If $r = 0$, any nonempty intersection would have positive θ^{n-2} -degree, so these intersections are empty.

Finally, if $v = n > 0$, the nef volume criterion makes the rational line on $\tilde{X}$ big [22, Theorem 0.5]. A big holomorphic line on a compact Kähler manifold has maximal Iitaka dimension. Hence $a(\tilde{X}) = n$, and birational invariance gives $a(X) = n$. $\square$

Thus, when $0 < v < n$, a positive component attaining image dimension $r = v - 1$ has general fibers of dimension $(n - 1) - r = n - v > 0$. Equation (7.2) makes those fibers disjoint from $\text{Supp } G_- \cup D_0$. The next construction prepares their normal directions and adjunction for infinitesimal lifting.

7.2 Local roots and residue with residual poles

We work for the rest of the signed argument in the range $0 < v < n$, with $r = v - 1$. Choose $\ell > 1$ divisible by m and by every nonzero integer $|ma_i|$, and put $a = \ell/m$, a positive integer.

Lemma 7.3 (Analytic signed charts). *For each $t \in P$, after shrinking an open neighborhood $U \subset P$, choose a line R_U together with an isomorphism $R_U^\ell \simeq \mathcal{O}_P(1)|_U$. A comparison of two such roots on an overlap is power-compatible if its ℓ -th power commutes with these isomorphisms. If $t \notin \phi(\text{Supp } G_+)$, one may take U disjoint from this closed image and set $Z_U = S = T = \emptyset$; the assertions below are then vacuous for the unique maps from the empty spaces. If $t \in \phi(\text{Supp } G_+)$, there are a Hausdorff normal analytic space Z_U of pure dimension n , considered near a reduced Cartier divisor S of pure dimension $n - 1$, a reduced Weil divisor T with no component in common with S , and maps*

$$\pi : Z_U \longrightarrow X, \quad g : S \longrightarrow U$$

with the following properties.

- (i) *The map g is proper, and $\pi|_S$ is finite over $D_U = \phi^{-1}(U)$, with $g = \phi\pi|_S$. Its image covers the positive components over U , and*

$$\mathcal{O}_S(S) \simeq g^* R_U.$$

The image of $S \cap T$ lies in $\phi(\text{Supp } G_+ \cap (\text{Supp } G_- \cup D_0))$. Near S , the underlying sets satisfy

$$\pi^{-1}(|D|) = |S| \cup |T|.$$

- (ii) *The pair $(Z_U, S + T)$ is log canonical, and there is a fixed actual isomorphism of divisorial sheaves*

$$\tau : \mathcal{O}_{Z_U}(aS) \xrightarrow{\simeq} \omega_{Z_U}(S + T). \quad (7.5)$$

The spaces Z_U and S are Cohen–Macaulay. The support of T is locally set-theoretically principal. Off T , the space Z_U is klt and its canonical sheaf is invertible. The map π is quasi-finite near $S \setminus T$.

- (iii) *Projective log resolutions of the total divisor data can be chosen with Kähler forms on neighborhoods of the compact sets over each compact fiber of g . Power-compatible changes of the boundary root extend to isomorphisms of ambient germs that preserve the full ideal $\mathcal{O}_{Z_U}(-S)$, the reduced divisor T , and Equation (7.5). Resolutions may be chosen functorially for these germ isomorphisms.*

Here the nonempty space Z_U is a neighborhood of the whole compact fiber under consideration; no extension of g to Z_U is asserted.

Proof. The empty branch follows by shrinking away from the closed image $\phi(\text{Supp } G_+)$. We construct the nonempty charts in four stages: take roots and trivialize the remaining adjoint torsion, separate the positive and negative root divisors, retain the root line that controls the normal direction, and pass to an ordinary neighborhood of the compact fiber.

First consider an analytic open set on which the Cartier lines of mG_+ , mG_- , and mD_0 are trivial, and let f_+, f_-, f_0 be the functions representing their sections. Normalize the entire finite space

$$\{y_+^\ell = f_+, \quad y_-^\ell = f_-, \quad y_0^\ell = f_0\},$$

retaining all its components and the lifted action of μ_ℓ^3 that multiplies the three root coordinates. On an overlap, the functions f_+, f_-, f_0 change by holomorphic units. Local ℓ -th roots of those units give comparisons of the finite spaces that lift uniquely to their normalizations; different choices differ by μ_ℓ^3 . The quotient stacks of these normalized charts by μ_ℓ^3 glue to the simultaneous normalized root stack over X , which we denote by $\mathcal{X}_{\text{rt}}$. The zero divisors of the three root sections are denoted by S_+, S_-, S_0 . At a generic prime of multiplicity b , a normalized chart of $y^\ell = x^b$ has ramification index ℓ/b and $\text{ord}(y) = 1$. Here $b = |ma_i|$ or $b = m$, so it divides ℓ . Thus each root divisor is generically reduced and Cartier. A Cartier divisor on a normal space satisfies Serre's condition S_1 , so it is reduced. Log ramification with the full reduced boundary gives a log canonical pair and the rational equivalence

$$K_{\mathcal{X}_{\text{rt}}} + S_+ + S_- + S_0 \sim_{\mathbb{Q}} a(S_+ - S_-).$$

The ambient charts of $\mathcal{X}_{\text{rt}}$ are klt. Off the boundary this follows from the klt complement downstairs and the unramified charts. For a place centered in the Cartier boundary, dropping that boundary increases its log discrepancy by its strictly positive order; this also makes every zero-discrepancy place positive.

To turn the rational adjoint equivalence into an actual isomorphism, consider the integral reflexive sheaf on $\mathcal{X}_{\text{rt}}$

$$\mathcal{F} = (\omega_{\mathcal{X}_{\text{rt}}}(S_+ + S_- + S_0) \otimes \mathcal{O}_{\mathcal{X}_{\text{rt}}}(-a(S_+ - S_-)))^{**}.$$

It is torsion. Choose a periodicity $\mathcal{F}^{[q]} \simeq \mathcal{O}$, take the relative spectrum of its reflexive-power algebra, and normalize; write ρ_{ind} for this finite map, and retain S_+, S_-, S_0 for the pulled-back root divisors on the cover. At codimension one this is a cover $w^q = u$ for a unit u , hence is unramified. Log canonicity, the klt ambient property, and reduced Cartier root divisors persist. The tautological evaluation of this algebra trivializes the reflexive pullback $(\rho_{\text{ind}}^* \mathcal{F})^{**}$: it does so in codimension one, and the identity then extends reflexively. The evaluation is a sheaf morphism on the cover stack itself, so the resulting actual divisorial isomorphism

$$\omega(S_+ + S_- + S_0) \simeq \mathcal{O}(a(S_+ - S_-))$$

on the cover is equivariant on every atlas.

Next separate the positive and negative root divisors. Let $\mathcal{X}_{\text{sep}}$ be the normalization of the blowup of the ideal generated by their equations on the cover, and write p_1 for its map. This blowup embeds in a relative $\mathbb{P}^1$; its normalization still has fibers of dimension at most one. Each exceptional prime therefore lies over a codimension-two positive-negative intersection. At its generic point the original dlt pair is a two-branch SNC pair. After extracting roots of units the normalized Kummer chart there has smooth coordinates $x = u^{\ell/b_+}$, $z = w^{\ell/b_-}$, where b_+ and b_- are the two corresponding multiplicities. On this chart $\mathcal{F}$ is a line and its periodicity cover

is unramified. The two-coordinate blowup calculation consequently applies at every exceptional generic point. If E denotes its exceptional divisor, it gives reduced Cartier divisors

$$E, \quad S'_+ = p_1^* S_+ - E, \quad S'_- = p_1^* S_- - E,$$

with $S'_+ \cap S'_- = \emptyset$. No such two-branch stratum is contained in S_0 , so $p_1^* S_0$ is reduced and has no exceptional component. The full boundary $S'_+ + S'_- + E + p_1^* S_0$ is crepant and reduced. Its codimension-one equality with the pullback adjoint gives the discrepancy equality for every further valuation, hence log canonicity. The ambient charts after this blowup are still klt. Off the full boundary the blowup is an isomorphism to the old klt complement. A place of zero pair discrepancy centered in that boundary gains its strictly positive order when the effective Cartier boundary is dropped, while a place of positive pair discrepancy remains positive. The strict divisor S'_+ is finite over S_+ : before normalization it lies in one section of the projective ratio coordinate, so its proper map has finite fibers, and normalization is finite.

On $\mathcal{X}_{\text{sep}}$, put $S = S'_+$ and, temporarily, $T = S'_- + E + p_1^* S_0$. Write $\pi_{\text{sep}} : \mathcal{X}_{\text{sep}} \rightarrow X$ for the composite map, and define the retained root line

$$\mathcal{R} := \mathcal{O}_{\mathcal{X}_{\text{sep}}}(S'_+ - S'_-), \quad \mathcal{R}^\ell \simeq \pi_{\text{sep}}^* N_0.$$

Near S , the negative strict root section is a unit. The fixed adjoint isomorphism becomes $\omega_{\mathcal{X}_{\text{sep}}}(S + T) \simeq \mathcal{O}_{\mathcal{X}_{\text{sep}}}(aS)$, and the positive root section divided by that negative unit section is a section of $\mathcal{R}$ with zero divisor S . This line and section determine the Cartier divisor S and its normal line; they must survive passage to an ordinary chart.

The pair consisting of $\mathcal{R}$ and its displayed power isomorphism defines a map to the gerbe $\sqrt[\ell]{N_0}$, which parametrizes ℓ -th roots of N_0 . Take the relative coarse space over this gerbe. On an atlas this quotients by the finite relative inertia kernel, the subgroup of each stabilizer acting trivially on $\mathcal{R}$. Thus the quotient retains the root character of $\mathcal{R}$. The deck transformations of the representable periodicity cover are not this inertia kernel and are not removed.

Both $\mathcal{R}$ and its section descend through the kernel, so the reduced image S remains Cartier and its full ideal descends. We retain S, T for the reduced images. A finite-group norm of a local equation of the upstairs T has precisely its image as zero set. Thus the reduced image T is locally set-theoretically principal; it need not be Cartier. All divisorial ramification lies on the full boundary. Log ramification identifies the invariant log dualizing sheaf in codimension one with the downstairs one. Taking reflexive hulls and invariants of the equivariant adjoint isomorphism gives an actual descended isomorphism $\mathcal{O}(aS) \simeq \omega(S + T)$. Its ordinary pullback below is Equation (7.5). Outside $S + T$, the local finite quotient is quasi-étale, so its klt upstairs complement gives a klt downstairs complement by finite discrepancy comparison. Away from T , dropping the Cartier divisor S from the lc pair increases the log discrepancy of every place centered in S by its strictly positive order. Thus the quotient charts are klt away from T , and the same isomorphism makes their canonical sheaves invertible there. Off T the separating blowup is an isomorphism, while the other operations are finite; hence the composite map to X is quasi-finite near $S \setminus T$.

The same quotient preserves Cohen–Macaulayness of both the ambient space and S . Before the quotient the ambient space is klt and hence Cohen–Macaulay, and S is Cartier there. Locally its finite quotient ring, and the quotient ring of S , are invariant direct summands of those finite Cohen–Macaulay rings. A system of parameters downstairs is one upstairs; the vanishing of lower local cohomology upstairs and the invariant projection give its vanishing for the downstairs direct summand. This proves the stated depth.

Removal of the relative inertia kernel also makes the strict positive divisor finite and representable over

$$D \times_P \sqrt[\ell]{\mathcal{O}_P(1)}.$$

Its root line is $\mathcal{O}_S(S)$, and its image covers $\text{Supp } G_+$. The intersection $S \cap T$ maps to the positive–negative or positive–zero intersections: the only new component is the separating exceptional divisor. Before the quotient, the inverse image of $|D|$ is the support of $S'_+ + S'_- + E + p_1^* S_0$. The finite quotient sends this full support to $|S| \cup |T|$, and the equality of underlying sets persists under the ordinary base change below. The image of $S \cap T$ in P consequently has dimension $< r$, by Lemma 7.2.

Finally we realize these stack data on an ordinary analytic neighborhood of the whole compact fiber. We use the dimension-free root-neighborhood result [53, Lemma 10.1]. For a compact analytic subspace F of a Hausdorff analytic space, that lemma extends a specified ℓ -th root of a line on F to a root on a neighborhood. It also extends a specified power-compatible comparison of two roots uniquely as a germ about F . Its proof is the analytic Kummer sequence and continuity of the cohomology of μ_ℓ over neighborhoods of a compact set, in degrees two, one, and zero.

Apply it to $F = \phi^{-1}(t)_{\text{red}} \subset X$ with the root prescribed by R_U in Equation (7.1). Apply the comparison clause also inside D . Properness of ϕ allows a shrink of U for which the comparison is defined along all of D_U : remove the closed image of its complement. Pulling the relative coarse construction across this root over the whole neighborhood of F gives the ordinary Hausdorff space Z_U . Here the chosen root defines a map from that neighborhood to $\sqrt[\ell]{N_0}$, and the relative quotient is representable over the gerbe, so this pullback is an ordinary space. Finite invariant algebras glue its local quotient charts. The preceding finite representable map becomes the finite map $S \rightarrow D_U$, so g is proper. The retained line on S becomes $g^* R_U$, giving $\mathcal{O}_S(S) \simeq g^* R_U$. Normalization and the periodicity cover are finite also in the analytic category. The separating blowup is projective. A common power of its equivariant relative ample line kills the bounded finite stabilizer characters and descends to the coarse quotient; relative ampleness is checked on fibers after the finite pullback. Thus these operations and a projective log resolution admit relative ample metrics. Adding a sufficiently large pullback of the ambient Kähler form gives Kähler forms near the compact sets in question. Finally, power-compatible root comparisons identify the full constructions as ambient germs. Smooth-functorial projective resolution of the same ordered divisor data preserves those identifications. This proves all assertions. $\square$

The residue assertions below are vacuous for an empty chart. Fix a nonempty chart of Lemma 7.3, write $Z = Z_U$, and choose a projective log resolution $p : \widehat{Z} \rightarrow Z$. Write

$$\begin{aligned} D_S &= p^* S, & H &= (D_S)_{\text{red}}, & C &= (p^{-1} T)_{\text{red}}, \\ A &= \sum_{\substack{E \text{ component of } C \\ E \text{ not a component of } H}} E. \end{aligned} \tag{7.6}$$

The reduced divisor $H + A$ has simple normal crossings. The divisor A may meet H ; it has no component in common with H . Thus A retains the components over T whose poles are not already counted by the reduced divisor H . Let $p_H = p|_H : H \rightarrow S$. The maps that will be used in the

residue and lifting arguments are

$$\begin{array}{ccc}
H & \hookrightarrow & \widehat{Z} \\
p_H \downarrow & & \downarrow p \\
S & \hookrightarrow & Z \\
\pi|_S \downarrow & & \downarrow \pi \\
D_U & \hookrightarrow & X \\
\phi \downarrow & & \\
U & &
\end{array}
\quad g = \phi\pi|_S, \quad h = gp_H.$$

Only the support spaces in the left column map to U .

Lemma 7.4 (Residue with residual poles). *For every $i \geq 0$, the fixed adjoint isomorphism gives a split injection*

$$R^i g_* \mathcal{O}_S(aS) \hookrightarrow R^i h_* \omega_H(A). \quad (7.7)$$

The injection and its retraction commute with the root germ comparisons. Under a product with a manifold they are the relative canonical constructions; for absolute canonical sheaves they are tensored with the canonical line of that additional factor.

Proof. The quotient retraction follows the construction of [53, Proposition 10.4]; the residual divisor changes the comparison sheaf, which we now compute. Apply Equation (7.5) to the rational section 1 of $\mathcal{O}_Z(aS)$, and denote the resulting fixed meromorphic adjoint form again by τ . Its pullback has at most a logarithmic pole along $H + A$. This is the log discrepancy inequality for $(Z, S + T)$; away from the inverse boundary, the klt ambient property and integrality of the orders remove a possible pole. We obtain

$$p^* \mathcal{O}_Z(aS) \longrightarrow \omega_{\widehat{Z}}(H + A).$$

Residue on the reduced SNC divisor H gives a morphism

$$\mathcal{O}_S(aS)[0] \longrightarrow R(p_H)_* \omega_H(A) \quad (7.8)$$

in the derived category of S .

The insertion counts a component common to C and H in the logarithmic divisor H . The retraction compares with $\omega_Z(T)$, so it must retain every component over T ; it therefore uses the full divisor C . We claim

$$Rp_* \omega_{\widehat{Z}}(C) = \omega_Z(T)[0]. \quad (7.9)$$

The right side is invertible, since it is $\omega_Z(S + T)(-S)$. A local frame pulls back with at most simple poles over T : subtracting the effective Cartier S from the lc boundary leaves the required lc order inequality. At a prime not over T , the target is klt with invertible canonical sheaf, so its integral order is nonnegative. This proves the inclusion of the right side in $p_* \omega_{\widehat{Z}}(C)$. Conversely, orders at the strict transforms give at most a simple pole on each prime of T and none on any other prime. Normality and reflexivity give the opposite inclusion.

For higher direct images this can be checked locally on Z . Choose an effective Cartier divisor V whose support is T , and a sufficiently small positive rational ϵ so that $C = \lceil \epsilon p^* V \rceil$. There are only

finitely many relevant components after shrinking about a compact fiber. The fractional support is SNC, and ϵp^*V is p -nef and p -big. The latter can be seen directly after shrinking the target to a Stein open. Choose a p -ample Cartier divisor H_p . The coherent sheaf $p_*\mathcal{O}_{\widehat{Z}}(-H_p)$ has generic rank one because p is birational. Cartan's Theorem A supplies a section nonzero at the generic points, giving an effective divisor $B_p \sim -H_p$ on the inverse image. Thus $\epsilon p^*V \sim H_p + (B_p + \epsilon p^*V)$, a relatively ample divisor plus an effective one, which is p -big. Relative Kawamata–Viehweg vanishing for projective analytic morphisms with smooth source [31, Theorem 5.1] gives $R^i p_* \omega_{\widehat{Z}}(C) = 0$ for $i > 0$. This proves Equation (7.9).

Projection formula gives the same comparison after adding $D_S = p^*S$. The quotient sequence therefore identifies

$$\begin{aligned} R(p_D)_* \left(\frac{\omega_{\widehat{Z}}(D_S + C)}{\omega_{\widehat{Z}}(C)} \right) &\simeq \omega_Z(S + T)|_S[0] \\ &\simeq \mathcal{O}_S(aS)[0], \end{aligned}$$

where $p_D : D_S \rightarrow S$ and the quotient is a sheaf on the possibly nonreduced divisor D_S . To justify the displayed derived statement, first push it by the exact closed immersion $S \hookrightarrow Z$. There it is the quotient of the two acyclic comparisons. Closed pushforward detects cohomology sheaves, and canonical truncation yields the statement on S .

There is a quotient morphism

$$\begin{aligned} \omega_H(A) &= \frac{\omega_{\widehat{Z}}(H + A)}{\omega_{\widehat{Z}}(A)} \\ &\longrightarrow \frac{\omega_{\widehat{Z}}(D_S + C)}{\omega_{\widehat{Z}}(C)}. \end{aligned} \tag{7.10}$$

Indeed $H + A \leq D_S + C$ and $A \leq C$. This morphism need not be injective when C and H share a component. Compose its derived pushforward with the preceding isomorphism. The result retracts Equation (7.8): at each generic point of S , its composite is the identity under the fixed residue convention. The composite is an endomorphism of the invertible sheaf $\mathcal{O}_S(aS)$ in degree zero; since S is reduced, agreement at every generic point implies agreement everywhere. Applying Rg_* proves Equation (7.7). All maps use the fixed adjoint isomorphism, divisor ideals, and residue, so they commute with the germ comparisons. Relative canonical forms under products, followed by wedging with the canonical frame of the new factor, give the last assertion. $\square$

7.3 The localized Hodge argument

The target in Equation (7.7) contains the additional poles A . The following version of the filtered SNC calculation incorporates them. In its application, we embed H by the graph of h in $\widehat{Z} \times U$ and use the projection to U as the ambient map. This projection is submersive, while its restriction to the graph is the proper map h ; it requires no extension of h to $\widehat{Z}$. We will use the same graph construction after a change of parameter. The parameter is projective only in the last assertion below; the closed-stratum maps are proper Kähler maps.

Proposition 7.5. *Let $q : \mathcal{V} \rightarrow Y$ be a holomorphic map of complex manifolds, submersive near a reduced closed analytic subspace $i : H \hookrightarrow \mathcal{V}$ of pure dimension d and codimension $c_0 > 0$. Suppose $h = q|_H$ is proper. Assume that locally H is an SNC divisor in a smooth submanifold of codimension $c_0 - 1$, and that a reduced divisor A is simultaneously SNC and transverse to those support equations. Assume that the components H_i and A_j near H have finite global smooth*

indexing, and that every closed stratum $H_I \cap A_J$, with $I \neq \emptyset$, is smooth and proper over Y and carries a global relative Kähler form. Empty or disconnected strata are allowed. Here $H_I = \bigcap_{i \in I} H_i$ and $A_J = \bigcap_{j \in J} A_j$, with $A_\emptyset = \mathcal{V}$.

Use right $\mathcal{D}$ -modules and finite-pole local cohomology, and put

$$\mathcal{K}_A = \mathcal{H}_{[H]}^{c_0}(\omega_{\mathcal{V}})(*A).$$

Its order filtration is generated from $F_0 \mathcal{K}_A = i_* \omega_H(A)$, with $F_p = 0$ for $p < 0$. For $M^j = \mathcal{H}^j_{q+} \mathcal{K}_A$, the filtered direct image is strict at every level, and M^j has a finite filtration by submodules, strict for F , whose quotients are polarizable real pure Hodge modules. In particular,

$$F_0 M^j = R^j h_* \omega_H(A) \hookrightarrow M^j, \quad F_p M^j = 0 \quad (p < 0). \quad (7.11)$$

Define the symbol morphism

$$\sigma : F_0 M^j \otimes T_Y \longrightarrow \mathrm{gr}_1^F M^j, \quad \sigma(m \otimes \xi) = [m\xi].$$

If Y is smooth projective, then for every ample line N on Y ,

$$\mathrm{Hom}_{\mathcal{O}_Y}(N, \ker \sigma) = 0. \quad (7.12)$$

This includes any coherent torsion in the kernel.

Proof. We verify the residual localization in the proof of [53, Proposition 11.1]. We first identify the lowest order step and a filtration by the number of branch poles. Its graded pieces will be dual constant modules on smooth strata, to which proper Kähler direct image applies. The resulting strictness gives the injection at the lowest filtration step, and the graded de Rham complex on a projective base gives the symbol vanishing.

At a point of H , choose simultaneous coordinates

$$(z_1, \dots, z_{c_0-1}, x_1, \dots, x_t, y_1, \dots, y_s, \xi), \quad \mathcal{I}_H = (z_1, \dots, z_{c_0-1}, x_1 \cdots x_t), \quad A = (y_1 \cdots y_s = 0).$$

Let η be the coordinate volume form. The localization Čech complex for the displayed regular support equations has cohomology only in degree c_0 . Localizing this cohomology also in the y 's gives $\mathcal{K}_A$. Successive finite Taylor divisions give its unique additive polar normal forms: finite sums of

$$f(x_{I^c}, y_{J^c}, \xi) \prod_{\nu=1}^{c_0-1} z_\nu^{-u_\nu} \prod_{i \in I} x_i^{-v_i} \prod_{j \in J} y_j^{-w_j} \eta, \quad I \neq \emptyset, \quad (7.13)$$

where every displayed order is positive and J is arbitrary, including empty. The coefficient is independent of exactly the variables occurring negatively. The normal form is an additive statement, not an $\mathcal{O}_{\mathcal{V}}$ -linear decomposition.

The simple fractions, with each displayed order equal to one, are exactly the image of $\omega_H(A)$. More explicitly, their unreduced representatives are

$$\frac{f\eta}{z_1 \cdots z_{c_0-1}(x_1 \cdots x_t)(y_1 \cdots y_s)}.$$

The regular-equation residue identifies their numerators modulo $\mathcal{I}_H$ with the absolute dualizing sheaf of H , twisted by A . This map is injective. If the fraction has no remaining x -pole, Taylor division says that its numerator modulo the z 's is divisible by $x_1 \cdots x_t$. Localization in the y 's

does not enlarge this kernel, because each y_j is a nonzerodivisor modulo $\mathcal{I}_H$. This also proves the identification under changes of coordinates, including the determinant of the normal conormal frame.

Give a term of Equation (7.13) the excess

$$e = \sum_{\nu} (u_{\nu} - 1) + \sum_{i \in I} (v_i - 1) + \sum_{j \in J} (w_j - 1).$$

The right canonical action of a coordinate derivative is minus differentiation of the coefficient of η . A derivative in a denominator variable increases its order by one with a nonzero scalar. Because the coefficient in the normal form is independent of those variables, every term of excess e is obtained from a simple term by e such derivatives. Conversely, order p differentiations produce excess at most p . Thus $F_p \mathcal{K}_A$ consists exactly of the finite sums of excess at most p . This is the good order filtration generated from $i_* \omega_H(A)$.

There is also an increasing filtration W by the total number of branch poles. Intrinsically, its step of index $-d + k - 1$ is the sum of the images of support modules for subunions of the H_i , localized along sublists of the A_j , for which the total number of selected branches is at most k . In the normal form it imposes $|I| + |J| \leq k$. The induced double filtration therefore has

$$\mathrm{gr}_{-d+k-1}^W(\mathcal{K}_A, F) \simeq \bigoplus_{\substack{|I|+|J|=k \\ I \neq \emptyset}} (i_{I,J})_*(\omega_{H_I \cap A_J}, F^{(0)}). \quad (7.14)$$

Here $F_p^{(0)} \omega_{H_I \cap A_J} = 0$ for $p < 0$ and is $\omega_{H_I \cap A_J}$ for $p \geq 0$. Here $A_{\emptyset} = \mathcal{V}$, and empty intersections contribute zero. To check this formula, retain the polar type using exactly I, J . Its normal denominators are the $c_0 - 1$ equations z_{ν} , the $|I|$ selected x 's, and the $|J|$ selected y 's. They are the regular equations of the smooth stratum $H_I \cap A_J$, whose dimension is $d - k + 1$. Right closed transfer has no codimension shift in F , and the excess counts exactly its normal derivative orders. Mayer–Vietoris boundary maps for the x -subunions and the localization boundary for each selected y identify these quotients intrinsically with the support module of that intersection. For the latter assertion, localization modulo the nonlocalized module in one transverse y direction is its first local cohomology in that direction. A fixed order of the component labels fixes the residue signs, so the identifications glue. The smooth- support graded modules are regular holonomic, and their finite extension $\mathcal{K}_A$ is regular holonomic as well.

The filtration W has a real realization. Tensor the finite-pole de Rham comparisons for the regular support coordinates with the open localization comparison for the y coordinates, whose one-coordinate factor is the cohomology of a punctured disk. These comparisons are natural for all sublists. The resulting real support and localization complexes complexify to the de Rham complexes just computed. The latter are perverse by regular holonomicity; exact and faithful complexification gives the same assertion for the real complexes. Their real perverse images then give W . On a graded stratum of dimension $d - k + 1$, the real object is its dual constant object $\mathbb{R}^Hd - k + 1$. Its first right filtration step is zero and its weight is $-d + k - 1$, in agreement with Equation (7.14). Residue signs and constant normalizations preserve its real polarization. This is the real comparison in [53, Section 11.2], with the punctured-disk factors supplying the additional localization boundaries.

The proper-support check also survives localization. If a holomorphic germ f vanishes on reduced H , then

$$f F_p \mathcal{K}_A \subset F_{p-1} \mathcal{K}_A. \quad (7.15)$$

For a generator z_ν of $\mathcal{I}_H$, multiplication reduces its pole order or kills the class. Multiplication by $x_1 \cdots x_t$ also lowers the excess or kills the class, because the polar type always has $I \neq \emptyset$. The y -poles do not affect this argument. In particular every graded stratum stays in the proper support H , and properness is needed only there.

We have now identified the order filtration, its real pure stratum pieces, and the proper support of each level. We next pass these data to the base. Apply Saito's constant-source Kähler direct-image theorem [62, Theorem 1 and Remark 1.2] to each smooth closed stratum in Equation (7.14), using its stipulated relative Kähler form. The resulting degree- j direct image of the step indexed by λ is strict and polarizable real pure of weight $\lambda + j$. In the submersion neighborhood, the level- F_p relative right de Rham complex has term

$$F_{p-e}\mathcal{K}_A \otimes \bigwedge^e T_{\mathcal{V}/Y} \quad \text{in degree } -e.$$

The simultaneous excess and branch bounds show that passing to a W -quotient commutes with taking each such level.

To prove strictness, use the finite W spectral sequence

$$E_1^{-\lambda, j+\lambda} = \mathcal{H}^j q_+ \operatorname{gr}_\lambda^W \mathcal{K}_A \implies \mathcal{H}^j q_+ \mathcal{K}_A.$$

Its first-page terms and their level filtrations are strict pure objects by the preceding application. The differentials are real and filtration preserving, because the support and localization comparisons are real and natural. The first differential preserves weight, hence is a strict morphism of pure Hodge modules; its kernels and cokernels remain pure. A higher differential sends (λ, j) to $(\lambda - k, j + 1)$ for $k \geq 2$ and strictly lowers weight. Such a real filtered map is zero. On distinct strict supports this follows from strict support; on the same dense smooth support a real map preserves both Hodge filtrations, and a source vector of type (p, q) would have image in $F^p \cap \overline{F}^q$ of a pure object of smaller weight, which is zero. Both the unfiltered and each level spectral sequence therefore degenerate at the second page. Their comparison is injective there. A least- W -step argument on the finite abutment proves injection at every F level and strictness of its induced W filtration. This is the strict-direct-image argument of [53, Proposition 11.1], now justified for every graded term with A -poles.

For $p < 0$, the relative complex is zero. For $p = 0$, its only term is $F_0 \mathcal{K}_A = i_* \omega_H(A)$ in degree zero. The level injection gives Equation (7.11).

Suppose finally that Y is smooth projective. Each strict- support pure quotient just obtained is of exactly the type in [53, Lemma 11.2]: a polarizable real pure direct-image piece arising from a dual constant on a smooth Kähler source. That lemma compares its actual right filtration with the Sabbah–Schnell pure component extending the same generic polarized real variation. Its proof uses uniqueness of the intermediate extension and the reconstruction of F from the canonical V -filtration, so it depends only on this pure piece, not on the divisor presentation of its source. It gives the negative-ample vanishing

$$\mathbb{H}^u(Y, N^{-1} \otimes \operatorname{gr}_p^F \operatorname{DR}_Y Q) = 0 \quad (u < 0)$$

for each pure quotient Q , by [61, Theorem 16.3.10]. The finite strict filtration extends this vanishing to M^j . Since its negative F levels vanish, its degree-one graded right de Rham complex is exactly

$$\operatorname{gr}_1^F \operatorname{DR}_Y M^j = \left[F_0 M^j \otimes T_Y \xrightarrow{\sigma} \operatorname{gr}_1^F M^j \right] \quad \text{in degrees } -1, 0.$$

Its negative hypercohomology after tensoring by N^{-1} is $H^0(Y, N^{-1} \otimes \ker \sigma)$. It vanishes, which is Equation (7.12). This calculation does not assume that the kernel is locally free. $\square$

7.4 Lifting every finite boundary neighborhood

Return to the charts and resolution of Lemma 7.3 and Equation (7.6). On an empty chart the sheaves below are zero. On a nonempty chart put $I = \mathcal{O}_Z(-S)$, an invertible ideal. For every integer j and positive integer k , put

$$\mathcal{A}_j = I^j/I^{j+1} = \mathcal{O}_S(-jS), \quad \mathcal{T}_{j,k} = I^j/I^{j+k}.$$

These are sheaves on the underlying topological space of S . Pushforward by g has that meaning for $\mathcal{T}_{j,k}$; it does not require a morphism from a thickening to U . A local frame of R_U^{-1} gives a Laurent graded frame u^j of $\mathcal{A}_j$ for every j . It is a frame on the associated graded, not an extension of a frame to Z .

Proposition 7.6 (All finite lifting orders). *For every root chart, every $j \in \mathbb{Z}$, and every $k \geq 1$, the map of sheaves of complex vector spaces on U*

$$g_*\mathcal{T}_{j,k+1} \longrightarrow g_*\mathcal{T}_{j,k} \tag{7.16}$$

is surjective. The assertion holds on every parameter germ, including germs supported at special parameters.

Proof. The assertion is immediate for an empty chart. We induct on k , simultaneously for all integer j and all nonempty charts. The layer algebra and its connecting maps are the ones in [53, Lemma 12.1]; we recall their construction to specify exactly which obstruction is to be killed.

Assume all smaller orders. The connecting map for

$$0 \longrightarrow \mathcal{A}_{j+k} \longrightarrow \mathcal{T}_{j,k+1} \longrightarrow \mathcal{T}_{j,k} \longrightarrow 0$$

factors uniquely through a map

$$\delta_k : g_*\mathcal{A}_j \longrightarrow R^1g_*\mathcal{A}_{j+k}. \tag{7.17}$$

Indeed the smaller orders make $g_*\mathcal{T}_{j,k} \rightarrow g_*\mathcal{A}_j$ surjective. For $k > 1$, the kernel comes from $g_*\mathcal{T}_{j+1,k-1}$, by the leading-layer sequence. The order $k-1$ in degree $j+1$ lifts this kernel into $g_*\mathcal{T}_{j,k+1}$, so the current connecting map kills it. For $k=1$ the leading map is the identity. Vanishing of all maps in Equation (7.17) is equivalent to the surjectivity at the current order.

The maps δ_k form a degree- k derivation on the Laurent graded algebra $\bigoplus_j g_*\mathcal{A}_j$, with values in its first-cohomology module. On a parameter germ, choose a lift of a leading section x to $g_*\mathcal{T}_{j,k}$, and represent that lift locally by $x_\alpha \in I^j$. Then $x_\beta - x_\alpha \in I^{j+k}$. Their differences modulo I^{j+k+1} represent $\delta_k(x)$. For another degree- l section z , the product difference is

$$x_\beta z_\beta - x_\alpha z_\alpha = x_\alpha(z_\beta - z_\alpha) + z_\alpha(x_\beta - x_\alpha) + (x_\beta - x_\alpha)(z_\beta - z_\alpha).$$

The last term lies in $I^{j+l+2k} \subset I^{j+l+k+1}$. This proves the derivation rule. If $t_1, \dots, t_b$ are coordinates on U , choose simultaneous representatives $G_{i,\alpha}$ of their lifts to $g_*\mathcal{T}_{0,k}$; their differences lie in I^k . Taylor's formula modulo their squared differences gives, for any holomorphic F ,

$$\delta_k(F(t)) = \sum_i F_{t_i}(t) \delta_k(t_i). \tag{7.18}$$

Both formulas hold on germs, without removing torsion in the target, and commute with the prescribed germ comparisons.

Put $m_k = a + k > 0$. An order- k obstruction raises degree by k , whereas the residue injection accepts degree $-a$. We therefore test leading sections x of degree $-m_k = -(a + k)$. Apply the split residue injection to define

$$c(x) = \text{res}_* \delta_k(x), \quad e_i(x) = \text{res}_*(x \delta_k(t_i)) \quad \text{in } R^1 h_* \omega_H(A).$$

Both arguments have degree $-a$, the degree of $\mathcal{O}_S(aS)$ in Equation (7.7). The next construction places these classes in a filtered direct-image module. For the identity graph, the calculation will give the undivided relation $c(x) + \sum_i e_i(x) \partial_{t_i} = 0$. Taking its degree-one symbol gives $\sigma(\sum_i e_i(x) \otimes \partial_{t_i}) = 0$. We will globalize that symbol on a projective coordinate-power cover, where the boundary root is an ample line. The adjugate formula expresses the symbol across the ramification of this cover without dividing by its Jacobian determinant. The symbol vanishing will then kill the obstructions of the parameter coordinates on every stalk; the undivided relation will kill $\delta_k(x)$, and the derivation rule will recover the other degrees.

Let $\psi : U' \rightarrow U$ be a holomorphic map between coordinate opens of dimension b , with coordinates w on U' , transverse to every smooth closed stratum $H_I \cap A_J$. The identity map is allowed. In $\hat{Z} \times U'$, the graph support $H^b = H \times_U U'$ is an SNC divisor in a smooth complete intersection of codimension b . The residual divisor remains transverse. After shrinking about a compact fiber, the finite-component Lemma 10.3 of [53] gives finite smooth indexing of the graph strata, and their relative Kähler forms restrict from the product neighborhood. Complete-intersection adjunction gives the canonical factor

$$\omega_{U'} \otimes \psi^* \omega_U^{-1},$$

whose coordinate frame we denote by dw/dt . This is a ratio of canonical frames, with no inverse Jacobian. Pull the Čech representatives of $c(x), e_i(x)$ to this graph and tensor by that frame. Proposition 7.5 places the resulting classes $\tilde{c}, \tilde{e}_i$ in $F_0 M^b$, where M^b is the degree-one direct-image module of the localized graph support.

Write $J = (\partial \psi_i / \partial w_j)$ and $J^\# = \text{adj}(J)$. Because the adjugate is polynomial in J , the following identity remains meaningful where J is singular:

$$\tilde{c} \det J + \sum_{i,j} (\tilde{e}_i \partial_{w_j}) J_{ji}^\# = 0 \quad \text{in } M^b. \quad (7.19)$$

We verify the change from [53, Lemma 12.2] explicitly. Choose simultaneous representatives of the lifts to length k

$$x_\alpha \in I^{-a-k}, \quad x_\beta - x_\alpha \in I^{-a}, \quad G_{i,\beta} - G_{i,\alpha} \in I^k.$$

After pulling to $\hat{Z}$, put $\eta_\alpha = x_\alpha \tau$, using the meromorphic form defined by Equation (7.5), and put $Q_{i,\alpha} = \psi_i(w) - G_{i,\alpha}$, $P_\alpha = \prod_i Q_{i,\alpha}$. On an overlap write $\Delta_i = G_{i,\beta} - G_{i,\alpha}$. The insertion and $D_S \geq H$ give the precise pole bounds

$$\begin{aligned} \eta_\alpha &\in \omega_{\hat{Z}}(H + A + kD_S), \\ \eta_\beta - \eta_\alpha, \quad \eta_\alpha \Delta_i &\in \omega_{\hat{Z}}(H + A), \\ \eta_\alpha \Delta_i \Delta_l, \quad (\eta_\beta - \eta_\alpha) \Delta_i &\in \omega_{\hat{Z}}(H + A - kD_S) \subset \omega_{\hat{Z}}(A). \end{aligned} \quad (7.20)$$

Thus every quadratic or cross difference has no H -pole, although it may retain a pole on A . Such a term is zero in the support module localized along A .

In that localized graph support module consider the finite generalized fractions

$$S_\alpha = \left[\frac{\eta_\alpha \wedge dw}{P_\alpha} \right].$$

The H -pole is already in the numerator. Each polar class is killed by a power of a reduced equation of H , and each Δ_i is a multiple of that equation. Changing $Q_{i,\alpha}$ to $Q_{i,\beta}$ therefore has a finite geometric expansion on each class. The last line of Equation (7.20) kills all terms beyond the linear terms, and gives the exact equality

$$S_\beta - S_\alpha = \left[\frac{(\eta_\beta - \eta_\alpha) \wedge dw}{P_\alpha} \right] + \sum_i \left[\frac{\eta_\alpha \Delta_i \wedge dw}{Q_{i,\alpha} P_\alpha} \right].$$

Let $C_\bullet$ be the first cochain. Let $E_{i,\bullet}$ be the cochain with the numerator of the i -th summand and only denominator P_α , and let $E_{i,\bullet}^{(l)}$ have its additional denominator $Q_{l,\alpha}$. The simple cochains $C_\bullet, E_{i,\bullet}$ are cocycles representing $\tilde{c}, \tilde{e}_i$; their possible triple-overlap errors are precisely the cross terms killed above. The normal residue frame in this identification is dw/dt .

The numerator of $E_{i,\bullet}$, including its A -poles, is independent of w . Right differentiation therefore gives

$$E_{i,\bullet} \partial_{w_j} = \sum_l J_{lj} E_{i,\bullet}^{(l)}.$$

The preceding fraction equality is $\check{d}S_\bullet = C_\bullet + \sum_i E_{i,\bullet}^{(i)}$. Multiplying by $\det J$ and using $JJ^\# = (\det J)\text{id}$ gives at the cochain level

$$\check{d}(S_\bullet \det J) = C_\bullet \det J + \sum_{i,j} (E_{i,\bullet} \partial_{w_j}) J_{ji}^\#.$$

These cochains are in the end term of the relative right de Rham complex, so a Čech coboundary there is a total coboundary. Passing to the direct image proves Equation (7.19), with $J_{ji}^\#$ placed after the right derivative as displayed. Passing to gr_1^F yields

$$\sigma \left(\sum_{j,i} J_{ji}^\# \tilde{e}_i \otimes \partial_{w_j} \right) = 0. \quad (7.21)$$

For $\psi = \text{id}$ the undivided identity is

$$c(x) + \sum_i e_i(x) \partial_{t_i} = 0. \quad (7.22)$$

For $b = 0$ there are no graph equations, and the same undivided cochain computation gives $c(x) = 0$.

We now prove that the obstructions of the parameter coordinates vanish on every stalk, including an obstruction supported only at a special parameter. Suppose $b > 0$, and fix $t_* \in P$. We will use one compact graph over a projective parameter space, obtained by a parameter change unbranched above t_* . The vanishing of Hom in Equation (7.12) applies to the entire coherent symbol kernel, and the local isomorphism above t_* will detect vanishing on the full germ there. The construction of [53, Proposition 10.5] gives a coordinate- power map, here of exponent ℓ ,

$$\begin{aligned} \psi : P' = \mathbb{P}^b &\longrightarrow P, & [w_0 : \cdots : w_b] &\longmapsto [w_0^\ell : \cdots : w_b^\ell], \\ R = \mathcal{O}_{P'}(1), & & R^\ell &\simeq \psi^* \mathcal{O}_P(1), \end{aligned}$$

in suitably chosen coordinates, unbranched over t_* . We spell out why its geometry also accommodates A . Choose finitely many relatively compact parameter opens covering $\phi(D)$, with the chart resolutions defined on slightly larger opens. Empty charts contribute no strata. Properness of the nonempty supports leaves only finitely many closed smooth strata $H_I \cap A_J$ meeting the corresponding compact sets. A general tuple of coordinate hyperplanes is transverse to the maps of every such stratum, for every subtuple, and avoids t_* . This follows by Sard's theorem applied to the incidence with variable hyperplanes; varying each hyperplane supplies its normal direction. In these coordinates $d\psi$ has image the tangent space to the intersection of its vanishing coordinate hyperplanes. The transversality condition is consequently

$$dh(T_z(H_I \cap A_J)) + d\psi(T_w P') = T_{h(z)} P \quad \text{whenever } h(z) = \psi(w).$$

It gives the required simultaneous regular graph equations and SNC transverse residual divisors.

To obtain a single compact graph support, first form the compact graph $\Xi = D \times_P P' \subset X \times P'$. On it the pullback of N_0 has the specified root R . Lemma 10.1 of [53] extends that root to a neighborhood of Ξ . Perform the full construction of Lemma 7.3 on this neighborhood, including the relative quotient and its ordinary pullback, and use the canonical sheaf relative to P' in the periodicity algebra; the absolute canonical factor is added only when taking graph residues. Resolve the same ordered total divisor data before imposing the graph equations. Near a compact graph slice, the root comparison lemma identifies this construction with the ordinary product of a local chart with a parameter open, as an ambient germ preserving the ideal and the adjoint form. Uniqueness of the comparison as a germ and properness of the graph allow a base shrink on which the identification holds on the entire slice. Normalization thus occurs before ramified graph base change. Smooth-functorial resolution preserves the product identification.

The graph cut H' in this resolved ambient space is compact, has pure dimension $n - 1$, and is locally an SNC divisor in a smooth complete intersection of codimension b . Its ambient projection to P' is submersive near H' . Let A' be the reduced ambient resolved residual divisor, restricted to a neighborhood of H' . Under each product-germ identification it is $A \times U'$; its restriction to the graph complete intersection is the transverse residual cut. The chosen transversality makes A' simultaneously SNC with the graph support equations. All closed strata $H'_I \cap A'_J$ are compact and smooth; splitting their connected components leaves finitely many indices. The relative metric construction in Lemma 7.3 gives one Kähler form near H' , and hence the required forms on every stratum. Thus Proposition 7.5 applies to this single global graph. Write M' for its degree-one module.

Take $x = u^{-m_k}$. On a graph chart choose a frame ϑ of R compatible with the downstairs frame of $\mathcal{O}_P(1)$. The local rule

$$\vartheta^{m_k} \mapsto \sum_{j,i} J_{ji}^\# \tilde{e}_i(u^{-m_k}) \otimes \partial_{w_j} \quad (7.23)$$

defines a global morphism $R^{m_k} \rightarrow F_0 M' \otimes T_{P'}$. Here is the transition check, including the ramification locus. For changes of coordinates let $A_t = \partial t' / \partial t$ and $B_w = \partial w' / \partial w$, and let the root frame change by $\vartheta' = \lambda \vartheta$. Locally λ descends from a holomorphic unit downstairs: its ℓ -th power does, and after choosing a local ℓ -th root of that unit the remaining ratio is a locally constant element of μ_ℓ . Root germ compatibility and Equation (7.18) transform the downstairs column of e_i 's by $\lambda^{m_k} A_t$. No derivative of λ appears, since x multiplies outside $\delta_k(t_i)$. The absolute canonical factor changes by $\det B_w / \det A_t$. Hence

$$\begin{aligned} \tilde{e}' &= \frac{\det B_w}{\det A_t} \lambda^{m_k} A_t \tilde{e}, \quad J' = A_t J B_w^{-1}, \\ (J')^\# \tilde{e}' &= \lambda^{m_k} B_w J^\# \tilde{e}. \end{aligned}$$

The adjugate equality is polynomial in J , so holds also when J is singular. The last formula, together with the inverse change of the vector basis, is exactly the transition of Equation (7.23). The graph classes here are natural Čech pullbacks through ambient germs; no ramified base-change isomorphism for R^1h_* is used.

Equation (7.21) puts the image of this morphism in the first-symbol kernel. Since $R^{m_k} = \mathcal{O}_{\mathbb{P}^b}(a+k)$ is ample, Equation (7.12) makes the morphism zero. At a point above t_* , the map ψ is locally biholomorphic and $J^\#$ is invertible. The ambient germ comparison identifies the graph classes with the downstairs classes there. Thus $e_i(u^{-m_k}) = 0$ as germs at t_* , including any class supported there. The point was arbitrary, so these classes vanish on every chart.

The split injection in Lemma 7.4 and the invertible Laurent frame now give $\delta_k(t_i) = 0$. For every leading section x of degree $-m_k$, the classes $e_i(x)$ vanish. Equation (7.22), the injection in Equation (7.11), and then the split residue injection give $\delta_k(x) = 0$. When $b = 0$, the direct identity $c(x) = 0$ and the same two injections give this conclusion without a graph test. In particular, the derivation rule and characteristic zero give

$$0 = \delta_k(u^{-m_k}) = -m_k u^{-m_k-1} \delta_k(u), \quad \delta_k(u) = 0.$$

For any local $F \in g_*\mathcal{O}_S$, the degree- $-m_k$ section Fu^{-m_k} then gives $0 = \delta_k(Fu^{-m_k}) = u^{-m_k} \delta_k(F)$. Every graded section is Fu^j , so δ_k vanishes in every degree. This proves the current order and completes the induction. $\square$

7.5 Compact deformations and the signed theorem

We now turn the infinitesimal lifting into actual compact subspaces. The role of all the integer layers in Proposition 7.6 is that both the equations of a boundary fiber and a generator of its normal direction can be lifted compatibly. This is the analytic counterpart of the compact-family step in [55, Proposition 5.1].

Lemma 7.7. *Assume $0 < v < n$, set $d = n - v$, and choose a positive component D_i with $\dim \phi(D_i) = r$. There is a nonempty open subset $D_i^\circ \subset D_i$ with the following property. For every $x \in D_i^\circ$, there are a disk Δ about 0, a proper flat analytic family $\mathcal{F} \rightarrow \Delta$ of compact subspaces of pure dimension d , and a finite morphism*

$$\mathcal{F} \longrightarrow X \times \Delta$$

over Δ , such that every nonzero fiber has image disjoint from D , while x is a limit of points in those images as the parameter tends to zero. Its schematic image is flat over the disk with pure d -dimensional fibers, whose fundamental cycles form a bounded family in the cycle space of X after shrinking the disk.

Proof. Choose a general point of the reduced image $\phi(D_i)$, outside the small image in Equation (7.2) and the intersections with distinct component images. Work on one ordinary root chart over a neighborhood U of that point. Its finite map $S \rightarrow D_U$ has a component covering a nonempty open of $D_i \cap D_U$. Shrink to a smooth open Y of $\phi(D_i) \cap U$, remove the images of components that do not dominate it, and remove the nonflat locus of the proper map g . The last removal is proper by analytic generic flatness. The relevant S is now flat over the smooth r -fold Y , and its fibers avoid T . They are compact of pure dimension $n - 1 - r = d > 0$. On the selected component the locus where it is singular or where g has rank less than r is proper. Removing its image, along with the preceding proper bad subsets, under the finite map leaves a nonempty open $D_i^\circ \subset D_i$. Its points have preimages at smooth points of these fiber components.

Fix $x \in D_i^\circ$ and a point above it in the compact fiber $F = g^{-1}(t)$, for $t \in Y$. Work on the single ordinary chart neighborhood of that fiber from Lemma 7.3. Take regular coordinates $t_1, \dots, t_r$ on Y centered at t , extended to local coordinates of its smooth embedding in P . Their pullbacks form a regular sequence on S along F , by flatness. In addition, S is Cohen–Macaulay, so F is Cohen–Macaulay of pure dimension d .

Choose once and for all a Hausdorff open neighborhood $W \subset Z$ of F , disjoint from T , on which π is quasi-finite. Set $I = \mathcal{O}_Z(-S)$. Proposition 7.6 lets us lift the finitely many t_i ’s to compatible germs of sections of $\mathcal{O}_Z/I^q$, $q \geq 1$, along F . It also lifts the conormal frame $u \in I/I^2$ to compatible germs $\tilde{y}_q \in I/I^{q+1}$ along F ; write $\tilde{y}$ for this system. At each order choose simultaneous representatives on a neighborhood of F contained in W . These neighborhoods may shrink with the order. The leading coefficient of $\tilde{y}$ is a unit frame, so it generates I on each finite thickening.

For $R_q = \mathbb{C}[s]/(s^q)$, cut the lifted r equations in the Cartier thickening qS , and map s to $\tilde{y}$. Denote the resulting subspace by F_q . It is flat over the Artin analytic point $\text{Spec } R_q$, with closed fiber F . Before cutting, multiplication by the Cartier generator identifies each successive s -layer with $\mathcal{O}_S$; this is the flatness criterion over R_q . Quotienting by lifts of the closed-fiber regular sequence preserves flatness, by the local flatness criterion. Compatibility of the lifted equations gives compatible reductions of all F_q .

Each of these is an embedded compact deformation in the same fixed neighborhood W . To see why the shrinking representative neighborhoods cause no difficulty, the support of F_q is the fixed compact set F . Its ideal on its representative neighborhood glues with the unit ideal on $W \setminus F$, since the deformation is empty on their overlap away from F . This realizes it as a closed subspace of $W \times \text{Spec } R_q$. The ideal is coherent locally on both opens, and the family is proper over the Artin point because its support is compact.

The Douady space of compact subspaces of W represents proper flat embedded families [25, Section 9]. The compatible F_q ’s thus define a formal arc in its analytic germ at $[F]$. Embed that germ in a finite dimensional analytic coordinate space. The arc is a formal solution of its convergent defining equations. Analytic Artin approximation [3, Theorem 1.2] gives a convergent arc agreeing modulo s^2 . Pull back the universal family and shrink its disk to obtain $\mathcal{F} \subset W \times \Delta$, proper and flat over Δ , with closed fiber F and the prescribed first normal displacement.

The family may be taken to have pure d -dimensional fibers throughout the disk. Here is a local justification. The central fiber is Cohen–Macaulay of dimension d . Flatness over the regular one-dimensional base makes its parameter a nonzerodivisor and gives the depth and dimension of the total local ring as $d + 1$ along that fiber. Cohen–Macaulayness is open for analytic local rings, and properness allows a disk shrink for which it holds on the whole family. Every total component then meets the central fiber after shrinking, and flatness makes it dominate the disk; the dimension formula gives it dimension $d + 1$. Quotient by the nonzerodivisor at each parameter gives Cohen–Macaulay fibers of dimension d . The analytic dimension formula and unmixedness of these local rings give the asserted purity.

Let f be a local equation of S . On $\mathcal{F}$ it vanishes on the central fiber, so flatness writes it as $f = sh$. Agreement modulo s^2 with the formal deformation says that $h|_F$ is a unit: this is precisely the lifted conormal generator. Finitely many such neighborhoods cover the compact F . Properness of $\mathcal{F} \rightarrow \Delta$ lets us shrink the disk so that they cover the entire family and all their h ’s are units; remove the closed image of the complement. Thus a nonzero fiber misses S . The neighborhood W already misses T , and the set equality in Lemma 7.3 gives $\pi^{-1}(|D|) \cap W = |S| \cap W$. Hence the nonzero fiber images miss D .

The induced map $\Pi : \mathcal{F} \rightarrow X \times \Delta$ is proper: the source is proper over Δ and the target is Hausdorff over it, so the graph is closed and the projection is proper. It is quasi-finite by the

choice of W , hence finite. It therefore preserves the dimension of every fiber component. Every total component through a central point dominates the disk by flatness, and its punctured part is dense. Thus every central point, including the point chosen above x , is a limit of points of nonzero fibers of dimension d .

Let $\mathcal{Z} \subset X \times \Delta$ be the schematic image of this finite map. Its algebra is the coherent image of $\mathcal{O}_{X \times \Delta} \rightarrow \Pi_* \mathcal{O}_{\mathcal{F}}$. Multiplication by a nonzero germ from the disk is injective on $\mathcal{O}_{\mathcal{F}}$, by flatness, and remains injective after the exact finite pushforward. It is therefore injective on the image algebra. That algebra is torsion-free, hence flat, over each local discrete valuation ring of the disk. Thus $\mathcal{Z}$ is a proper flat family of subspaces of X . The finite map to its schematic image is surjective, also on each fiber as a map of supports. Each $\mathcal{Z}_s$ consequently has pure dimension d . The Douady-to-cycle morphism, in the form recorded in [26, Section 3.3], makes their fundamental cycles an analytic family in the cycle space of X . The restriction to a smaller closed disk is compact. The volumes of these cycles against a Kähler form are bounded by continuity, proving the last assertion. $\square$

Proof of Theorem 7.1. The assertion is immediate in dimension zero, and the case $D = 0$ follows directly from the signed equivalence. By Lemma 7.2, the case $v = 0$ already gives $L \sim_{\mathbb{Q}} 0$, and $v = n > 0$ contradicts $a(X) = 0$. Suppose, therefore, that $0 < v < n$. Choose the positive D_i in that lemma and put $d = n - v$, so $0 < d < n$. We will contradict simplicity by using the compact deformations in Lemma 7.7.

Let $\mathcal{C}_d(X)$ be the Barlet space of nonzero effective compact d -cycles on X . It is a second-countable analytic space, hence has only countably many irreducible components B_α . For compact Kähler X its connected components, and therefore its irreducible components, are compact [26, Theorem 4.5]. The universal support $\mathcal{U}_\alpha \subset B_\alpha \times X$ is a compact analytic subspace. Take all of its full irreducible components $V_{\alpha\beta}$ that contain an incidence point belonging to an integral d -cycle disjoint from D . There are countably many of these components, since each compact analytic $\mathcal{U}_\alpha$ has finitely many. Their evaluation images

$$W_{\alpha\beta} = \text{pr}_X(V_{\alpha\beta})$$

are closed irreducible analytic subspaces of X , by proper mapping. Each has a point outside D . We use the full compact components here, including their fibers at limiting cycle parameters; the incidence over only the open set of cycles avoiding D would not have a proper evaluation map.

These countably many closed images cover D_i° . Indeed, for $x \in D_i^\circ$, choose the family in Lemma 7.7 and a sequence $x_j \rightarrow x$ in nonzero fiber images. Let C_j be an integral component of the fundamental cycle of the schematic image fiber containing x_j , taken with coefficient one. The cycles C_j avoid D , and their volumes are bounded by the volumes of the total image cycles. Bounded cycles on compact X have relatively compact closure in $\mathcal{C}_d(X)$ [26, Proposition 2.10]. The irreducible components of an analytic space are locally finite, so this compact closure meets only finitely many B_α . After passing to a subsequence, the cycles lie in a fixed B_α ; after a second subsequence, the points (C_j, x_j) lie in a fixed full component $V_{\alpha\beta}$. Its evaluation image is closed, so it contains x . This proves the covering assertion.

The analytic space D_i° is locally compact and Baire. If no $W_{\alpha\beta}$ contained D_i , each intersection with D_i° would be a proper closed analytic subset and hence nowhere dense. The countable cover just proved contradicts Baire. Consequently some $W_{\alpha\beta}$ contains D_i . It also contains a point outside D . An irreducible proper analytic subspace of the irreducible n -fold X that contains the prime divisor D_i must equal D_i , by dimension. It follows that $W_{\alpha\beta} = X$.

Every point of this full incidence component lies on the support of the effective d -cycle at its parameter, including limiting parameters. Thus its surjective evaluation supplies through every

point of X an irreducible compact component of such a cycle. Each has dimension d , strictly between zero and n . This contradicts simplicity. The intermediate range for v is impossible, leaving $L \sim_{\mathbb{Q}} 0$. By the meaning of rational linear equivalence for the actual line L , a positive Cartier multiple is the trivial holomorphic line. $\square$

7.6 The simple case of the induction

We now pass from the original pair to a reduced signed boundary and a nef model, and then subtract the added boundary from the torsion comparison.

Proof of Proposition 2.12. Let (X, B) and $J = K_X + B$ be as in that proposition. A positive-dimensional compact complex curve is projective, so $a(X) = 0$ implies $n \geq 2$. By Theorem 6.1, a positive multiple of K_X has a nonzero meromorphic section. It gives an actual rational equivalence $K_X \sim_{\mathbb{Q}} G_K$ for a signed rational divisor G_K .

Choose a projective log resolution $p_0 : W \rightarrow X$ of the boundary and the support of this divisor. Let B_W be the strict transform of B together with every new exceptional prime with coefficient one. The resolution transfer in Proposition 2.7 gives the actual rational identity

$$K_W + B_W \sim_{\mathbb{Q}} p_0^* J + E_0, \quad E_0 \geq 0 \quad \text{exceptional over } X.$$

The canonical pullback formula transforms G_K to a signed representative of K_W . Enlarge the reduced support of B_W and this representative to a reduced SNC divisor D . Then

$$J_D := K_W + D \sim_{\mathbb{Q}} p_0^* J + E_0 + (D - B_W), \quad E_0 + (D - B_W) \geq 0. \quad (7.24)$$

In particular J_D is pseudo-effective and has a signed rational representative supported on D . Simplicity and algebraic dimension zero persist under these birational modifications.

Apply Corollary 3.14 to (W, D) and J_D . Equation (7.24) supplies pseudo-effectivity, and the signed representative is supported on the floor D . Denote the resulting ordinary dlt nef model by $(X_{\text{nef}}, D_{\text{nef}})$, and put $L_{\text{nef}} = K_{X_{\text{nef}}} + D_{\text{nef}}$. As a working model of the program, X_{nef} is globally strongly $\mathbb{Q}$ -factorial, and the pair has the resolution property of Definition 3.1. The space X_{nef} is simple and compact Kähler with $a(X_{\text{nef}}) = 0$; D_{nef} is reduced, L_{nef} is nef, and its actual signed representative is supported on D_{nef} . Theorem 4.1 makes $L_{\text{nef}}|_{D_{\text{nef}}}$ semiample on the whole reduced floor.

We check the remaining pseudo-effectivity hypothesis of Theorem 7.1, with exactly $c = 1$. Take a common projective resolution $\alpha : V \rightarrow W$, $\beta : V \rightarrow X_{\text{nef}}$, with V smooth compact Kähler. The manifold V is still simple, hence not uniruled. Ou's criterion [58, Theorem 1.1] makes K_V pseudo-effective; this implication uses no assumption about the absence of canonical sections. Since X_{nef} is globally strongly $\mathbb{Q}$ -factorial, $K_{X_{\text{nef}}}$ is a rational line. Write its canonical comparison as an identity of actual rational lines

$$K_V \sim_{\mathbb{Q}} \beta^* K_{X_{\text{nef}}} + F_K, \quad F_K = F_K^+ - F_K^-,$$

where both F_K^+ and F_K^- are effective and β -exceptional. Thus $\beta^*\{K_{X_{\text{nef}}}\} + \{F_K^+\} = \{K_V\} + \{F_K^-\}$ is pseudo-effective. Exceptional translation in Lemma 2.3 removes the effective β -exceptional F_K^+ and proves that $\beta^*\{K_{X_{\text{nef}}}\}$ is pseudo-effective. Since $L_{\text{nef}} - D_{\text{nef}} = K_{X_{\text{nef}}}$ as an actual rational line, this is precisely the resolution hypothesis for $L_{\text{nef}} - D_{\text{nef}}$, with $c = 1$. Theorem 7.1 now gives $L_{\text{nef}} \sim_{\mathbb{Q}} 0$.

On the same common resolution, enlarging it if necessary, the accumulated negative-step comparisons give

$$\alpha^* J_D \sim_{\mathbb{Q}} \beta^* L_{\text{nef}} + F, \quad F \geq 0 \quad \text{exceptional over } X_{\text{nef}}. \quad (7.25)$$

All identities here are identities of actual rational lines. Because L_{nef} is torsion, its pullback is nef with zero class. Lemma 2.3 therefore identifies $F = N(\alpha^* J_D)$.

Put $\mu = p_0 \alpha : V \rightarrow X$ and $Q = \alpha^*(E_0 + D - B_W) \geq 0$. Pulling up Equation (7.24) gives $\mu^* J \sim_{\mathbb{Q}} \alpha^* J_D - Q$, which is pseudo-effective. Apply Lemma 2.6 to Equation (7.25). A positive current for $\mu^* J$, plus $[Q]$, must be the unique current $[F]$ in the larger adjoint class. It follows that $Q \leq F$, and negative-part subtraction gives the actual identity

$$\mu^* J \sim_{\mathbb{Q}} F - Q = N(\mu^* J).$$

The divisor on the right is effective and rational. This is exactly Proposition 2.12 and the required instance of $\mathcal{G}_n$ with torsion positive part. $\square$

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